Particle Physics 2: The Standard Model

Supplementary track, 2010 Winter

Lectures by Leonard Susskind

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About These Notes

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Chapter 1

Forces, Quarks, and the First Hadron Multiplets

1.1 Particles, Fields, and the Missing Third Leg

Waves are configurations of fields, so the two descriptions are closely related in the present discussion. Quantum mechanics says that the energy in such a wave comes in discrete, indivisible units. Those units are its quanta, and the quanta of a field are what we call its particles. With that relation in hand, we have already discussed wave equations, action principles, and the broad properties carried by particles: energy, momentum, spin, and the boson–fermion distinction. Fermions resist occupying the same state; bosons may congregate and condense into one state.

A quantum Lagrangian also records interactions. Products of fields represent vertices at which particles enter and leave. A term with four powers of a field, for example, represents a four-leg interaction vertex. This is how the action principle becomes a compact inventory of possible particle processes.

The next task is to name the particles—not quite one by one, but in families—and give each some personality: its properties, its interactions, and its place in ordinary physics. Before making that list, however, one side of a conceptual triangle is still missing:

- particles,
- fields for which those particles are quanta,
- forces associated with exchange processes involving those same quanta.

The word *elementary* must remain provisional. Every plausible definition of an elementary rather than composite particle eventually encounters a slippery case, and we shall return to that problem. For now, suppose that fundamental fields exist and call their quanta elementary particles. If an electric field gives rise to electric forces and a gravitational field to gravitational forces, then the electron field must also participate in forces. The triangle can close only if force has a particle description as well as a field description.

We shall therefore derive the logic of force twice. First we use classical field energy; then we ask how the same interaction appears when a field is resolved into quanta. Only after that bridge is secure will the particle catalogue begin.

1.2 Force from Field Energy in Classical Electrodynamics

Place two charges in space. One way to find their force is to quote the inverse-square law. A more revealing way is to ask what the charges do to space: they create electric fields, and those fields store energy.

It is useful to separate each charge from the spatial shape of its field,

$$\mathbf{E}_i = q_i \widehat{\mathbf{E}}_i, \tag{1.1}$$

although nothing depends on keeping the hat notation. With both charges present, linearity gives

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2. \tag{1.2}$$

Apart from the choice of units, the electrostatic field energy has the form

$$U_{\text{field}} \propto \int d^3x\, \mathbf{E}^2. \tag{1.3}$$

Substituting the sum of the two fields gives

$$U_{\text{field}} \propto \int d^3x\, (\mathbf{E}_1 + \mathbf{E}_2)^2 \tag{1.4}$$

$$\propto \int d^3x\, (\mathbf{E}_1^2 + \mathbf{E}_2^2 + 2\mathbf{E}_1 \cdot \mathbf{E}_2). \tag{1.5}$$

The first two terms are self-energies. Translating either charge moves its field with it and does not change the corresponding integral. That energy contributes, through $E = mc^2$, to the measured mass of the isolated charged particle. How that self-energy is incorporated into the measured mass belongs to the later subject of renormalization. None of this position-independent energy contributes to the force between the two particles.

The cross-term is different. It belongs to neither particle separately and exists only when both fields are present. If the charges are taken very far apart, their appreciable field regions scarcely overlap. As the charges approach, the overlap changes. The configuration-dependent interaction energy is therefore proportional to

$$U_{\text{int}}(r) \propto q_1 q_2 \int d^3x\, \widehat{\mathbf{E}}_1 \cdot \widehat{\mathbf{E}}_2. \tag{1.6}$$

For two point charges separated by r, evaluating the integral gives the familiar dependence

$$U_{\text{int}}(r) \propto \frac{q_1 q_2}{r}. \tag{1.7}$$

The radial force is its negative gradient,

$$\mathbf{F}_{12} = -\boldsymbol{\nabla}_{\mathbf{r}} U_{\text{int}}, \qquad |\mathbf{F}_{12}| \propto \frac{|q_1 q_2|}{r^2}. \tag{1.8}$$

[1]

The force law has become the derivative of the part of the field energy that knows about relative position. Far apart, the two objects are effectively independent. Bring them together and the field between them is rearranged; the resulting change in energy is the interaction.

But if a field can be resolved into quanta, this cannot be the end of the story. The same force must have a particle-language account.

[1] Editorial clarification: The spoken argument identifies the cross-term with the inverse-square Coulomb force without pausing at the potential. The standard intermediate $U_{\text{int}} \propto 1/r$ is written explicitly here so that its gradient gives the stated $1/r^2$ force.

1.3 Molecular Exchange: Two Protons and One Electron

Before returning to electrodynamics, consider a system in which exchange is easier to picture: two protons and one electron. Exchange is only one of several origins of molecular forces, but this example makes the relation between exchange, energy, and force visible. It is not offered as a substitute for quantum electrodynamics.

If one proton is present and the other is infinitely far away, the electron may occupy the hydrogenic ground state centered on the first proton. Its wave function is localized near that force center.

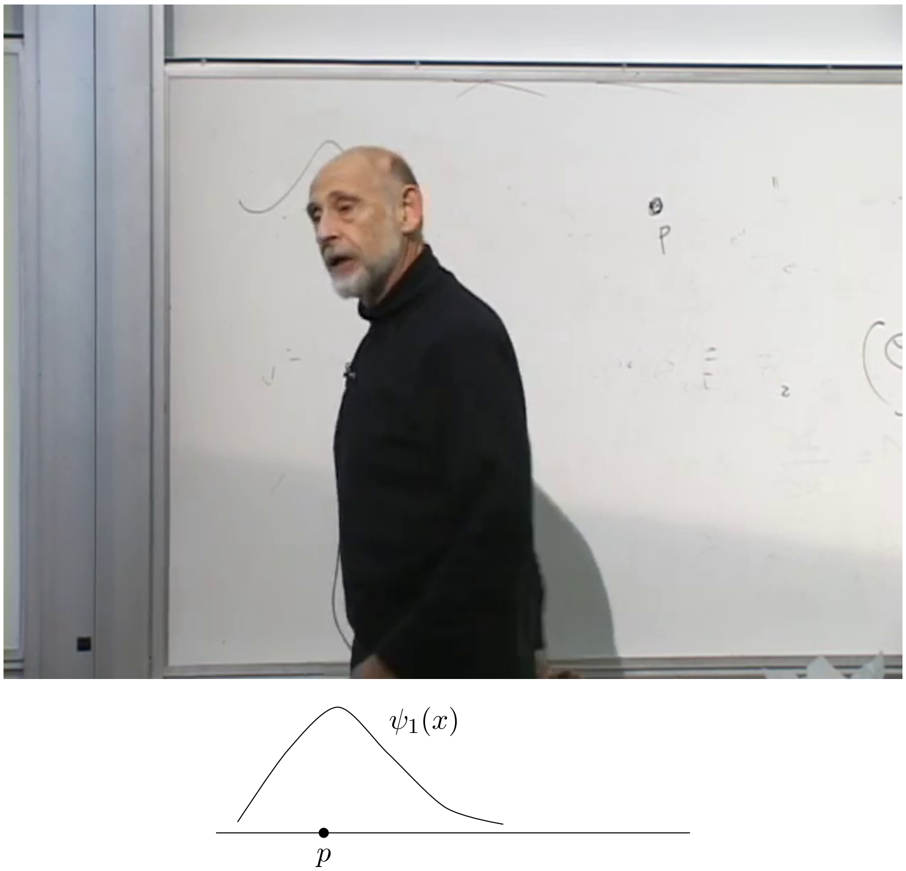

Figure 1.1: A hydrogenic ground-state wave function localized near one proton.

Lecture frame: 00:14:32.

[2]

Now place a second proton at a large but finite separation. By symmetry there is a second localized state with the same one-center energy, this time centered on the right proton.

[3]

At infinite separation the two states are degenerate: localizing the electron on the left or on the right costs the same energy. At finite separation a new process becomes possible. The electron

[2] Editorial clarification: The label $\psi_1(x)$ in the redraw names the localized profile described at 00:14:30–00:14:32; it is not copied from a legible board label.

[3] Editorial clarification: The labels $\psi_1(x)$ and $\psi_2(x)$ identify the two profiles discussed at 00:17:41–00:17:56; the symbols are schematic labels added for legibility.

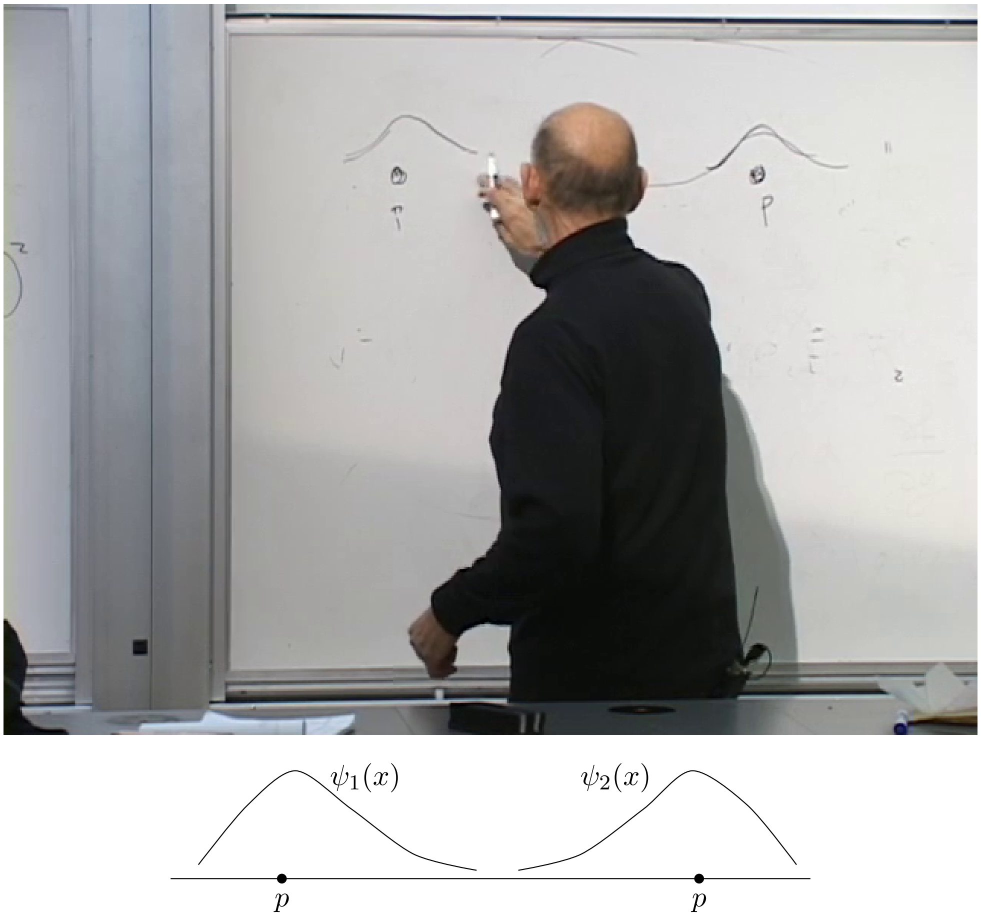

Figure 1.2: The two localized profiles used to form the equilibrium superposition after tunneling is allowed.

Lecture frame: 00:17:54.

can tunnel through the classically forbidden region from one center to the other. Watching for a trajectory through the barrier would disturb the state; "hopping" is a description of quantum evolution, not a film of a classical path.

At equilibrium the electron has equal probability of being found near either proton. Normalization therefore supplies the factor $1/\sqrt{2}$, and the lower equilibrium state is approximately

$$\psi_+(x) \approx \frac{\psi_1(x) + \psi_2(x)}{\sqrt{2}}. \tag{1.9}$$

The approximation sign matters. The true two-center wave function is not exactly a sum of two undisturbed hydrogenic orbitals; it adjusts in the overlap region. The point needed here is that the symmetric equilibrium state lies below the isolated one-center energy. In Hamiltonian language, the left and right configurations are coupled by an off-diagonal matrix element. Allowing the electron to occupy their coherent superposition has lowered the energy.

1.4 From Lowered Energy to an Effective Force

At large proton separation the electron-induced contribution $E_e(r)$ approaches its separated-atom value. At smaller separation, tunneling and superposition lower it. The complete two-proton–one-electron potential must also contain the direct proton repulsion,

$$V_{\text{tot}}(r) = V_{pp}(r) + E_e(r). \tag{1.10}$$

This reconnects the molecular example with the field-energy argument. The wave-function sketches and the qualitative curve for $E_e(r)$ are two views of the same calculation; equilibrium is determined by the sum $V_{\text{tot}}(r)$.

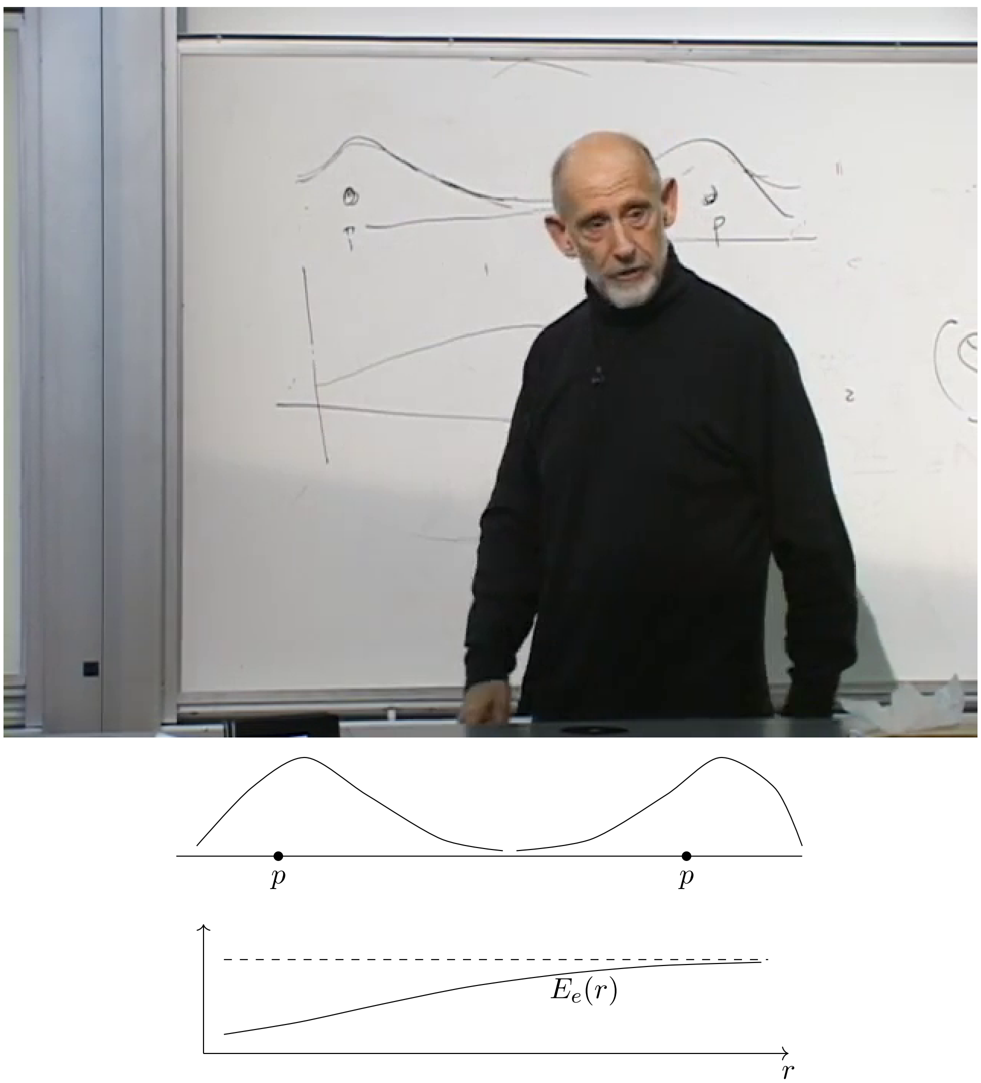

Figure 1.3: Wave-function overlap above and the electron-induced energy below. It rises toward the separated-atom asymptote as r increases; the total proton potential must also include their Coulomb repulsion.

Lecture frame: 00:20:28.

[4]

The two corresponding forces follow from the energy gradients,

$$F_e(r) = -\frac{dE_e}{dr}, \qquad F_{\text{tot}}(r) = -\frac{dV_{\text{tot}}}{dr}. \tag{1.11}$$

The system is therefore pushed toward lower total energy. Where $E_e(r)$ decreases as the protons approach, exchange supplies an attractive contribution; whether a bound separation exists depends on its competition with $V_{pp}(r)$. This is the elementary mechanism behind a covalent bond: the electron is shared because the shared state is energetically favorable.

In the classical calculation, overlap of electric fields changes the energy. Here, two force centers allow a lower quantum state. Both mechanisms produce force because the equilibrium energy depends on configuration.

The exchange language may be summarized by a worldline sketch. It must not be read as an observed trajectory; it represents amplitudes for the electron to be associated with either proton.

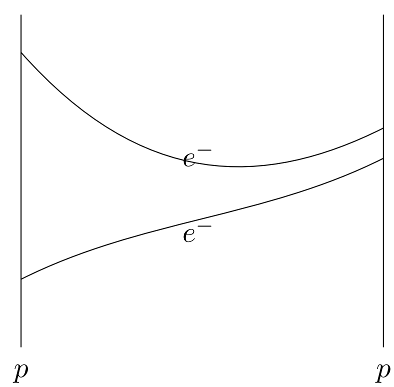

Figure 1.4: Electron exchange between two proton worldlines, understood as a schematic of quantum amplitudes rather than a watched path.

Question from the audience. Can the attraction be understood simply as the electron screening or reducing the proton charge?[a]

Response. No. Charge reduction is not the right picture for this effect. The essential fact is that adding the two localized wave functions produces a state with slightly lower energy than either localized state separately.

[a]Lecture timestamp: 00:23:26.

Question from the audience. Is this what Feynman meant by saying that “light likes light”?[a]

Response. No. That phrase most likely concerns Bose statistics: photons favor common states, whereas electrons obey exclusion. The molecular effect here is tunneling between two centers.

[a]Lecture timestamp: 00:23:51.

The two protons still repel electrostatically. Exchange is shorter-ranged than the Coulomb interaction, so the total potential must be considered rather than either contribution alone. At appropriate

[4]Editorial clarification: The labels $E_e(r)$ and r identify the energy curve and separation discussed at 00:19:12–00:20:38; the surviving board frame does not make those axis labels legible.

separations the molecular ion H_2^+ is nevertheless bound; in a neutral two-electron system the long-range charge repulsion is absent and covalent binding is even more familiar.

Question from the audience. Does placing the electron between the protons mainly reduce their electrostatic repulsion?[a]

Response. It reduces the repulsion a little, but that is not the main effect. Electron exchange is a shorter-range force than the Coulomb repulsion. It is most visible in a neutral two-electron system, where the long-range electrostatic force between the atoms is absent but tunneling still produces an attraction.

[a]Lecture timestamp: 00:24:33.

Question from the audience. Is the positively charged molecular ion H_2^+, with only one electron, actually bound?[a]

Response. Yes. At sufficiently small separation the exchange attraction is strong enough to overcome the proton–proton Coulomb repulsion.

[a]Lecture timestamp: 00:26:06.

Question from the audience. Does the two-center arrangement create a third force?[a]

Response. No new fundamental force is introduced. The Schrödinger equation in the combined potential of the two protons has an equilibrium energy different from the one-center problem; the derivative of that changed energy is the effective force.

[a]Lecture timestamp: 00:26:24.

Question from the audience. Is the energy lowering an example of quantum entanglement?[a]

Response. Not in the ordinary two-particle sense: only one electron is required. The relevant mathematical feature is an off-diagonal Hamiltonian matrix element coupling the left and right states.

[a]Lecture timestamp: 00:27:06.

Question from the audience. What happens to the exchange at high temperature?[a]

Response. At extreme temperature the electron is ionized and forgets the molecule altogether. At a lower excitation energy it may occupy a larger orbital, changing the overlap and hence the force. This is a state-dependent correction, not a new general mechanism.

[a]Lecture timestamp: 00:27:37.

1.5 Photon Exchange and the First Entries of the Table

Can the Coulomb force be told in the same exchange language? A charged particle interacts with the electromagnetic field, whose quanta are photons. Classically we solve for the electric field around the charge. In quantum field theory we resolve that interaction into photon emission and absorption.

A physical charged particle is not merely a bare charge. Its state is a quantum superposition containing a bare electron component, an electron with one photon, an electron with two photons, and so on. In deliberately schematic notation,

$$\begin{aligned} |e_{\text{phys}}\rangle = a_0\,|e\rangle + a_1\,|e + \text{one photon}\rangle \\ + a_2\,|e + \text{two photons}\rangle + \cdots . \end{aligned} \tag{1.12}$$

This notation records only the hierarchy of photon-number components; it does not specify photon momenta or attempt a complete construction of an interacting charged state. The surrounding photons are emitted and reabsorbed rather than escaping as free radiation. Their effects can be measured, although they are not detected in the same way as freely propagating photons. Put a second charge nearby and a photon emitted at one charge can be absorbed at the other. The two dressed states then cease to have an energy equal to the simple sum of two isolated energies.

The classroom analogy is two jugglers, each handling a set of balls. When they are far apart, each juggles independently. Bring them close enough and one may catch a ball thrown by the other. In the Feynman-diagram language, that shared object is the exchanged photon. The analogy is not a microscopic trajectory, but it captures why proximity permits a new process.

We now have two descriptions beyond merely memorizing Coulomb's law. Classical field theory adds the fields and squares the result. Quantum field theory sums amplitudes in which quanta are exchanged. Particles, fields, and forces form a one-to-one-to-one correspondence in this sense. Electrons mediate molecular exchange, photons mediate electromagnetic exchange, and any particle that can be exchanged can contribute to a force in an appropriate setting. Thus the familiar statement that nature has four fundamental interactions is a classification at a broader level; within exchange language, each exchangeable species can generate its own effective force contribution.

It is finally time to name the particles, list their properties, and say which processes they can enter. Writing the entire Standard Model at once would only produce a long memory test, so we shall divide the particles into small groups whose members are simply related and learn what each group can do.

The full catalogue is cumbersome because we do not understand why precisely these particles exist while other imaginable particles do not. Mathematical consistency supplies some relations: once several members of a pattern exist, another may be required to complete it. Those relations reduce the number of independent facts, but only by a modest factor. Many particle types and many parameters remain unexplained. Understanding that unfinished structure is necessary for understanding what later experiments are trying to discover.

The first field–quantum identification is

$$\gamma \leftrightarrow A. \tag{1.13}$$

Here A is the electromagnetic vector potential. The photon is a spin-one boson with zero electric charge, zero baryon number, and zero rest mass. The electron field is conventionally denoted ψ_e; its creation and annihilation pieces act on electron and positron states.

Our table records the name, particle symbol, field, boson or fermion type, electric charge Q, baryon number B, and mass:

The relation between electron and positron is mutual: each is the antiparticle of the other. Calling the negatively charged state "the particle" is convention. We measure electric charge in units of the magnitude of the electron charge, so

$$Q(e^-) = -1, \qquad Q(e^+) = +1. \tag{1.14}$$

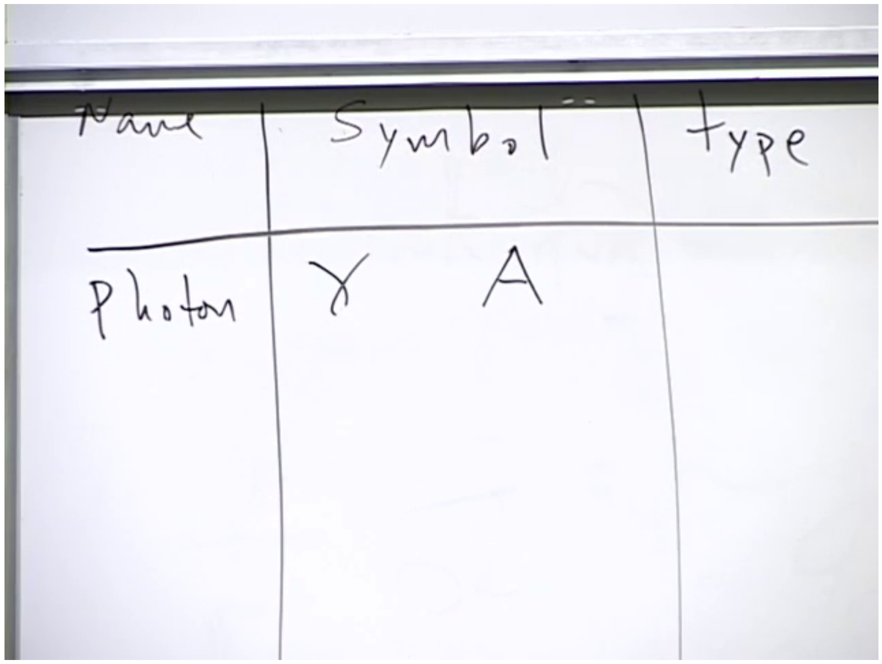

Figure 1.5: The first particle-table entry identifies the photon by γ and its field by the vector potential A.

Lecture frame: 00:39:11.

Name	Symbol	Field	Type	Q	B	Mass
Photon	γ	A	boson	0	0	0
Electron	e^-	ψ_e	fermion	-1	0	0.51 MeV
Positron	e^+	ψ_e	fermion	$+1$	0	0.51 MeV

Table 1.1: The photon and electron–positron entries that begin the particle catalogue.

Mass is measured in energy units by using $E = mc^2$. One electron volt is the energy acquired by a charge e through a potential difference of one volt,

$$1\,\text{eV} = e\,(1\,\text{V}). \tag{1.15}$$

The unit comes naturally from atomic physics: as a rough atomic scale, ionizing hydrogen costs about 13.5 eV. Particle masses require millions or billions of electron volts. The electron mass is

$$m_e c^2 \approx 0.511\,\text{MeV}. \tag{1.16}$$

Question from the audience. Should the electron mass be exactly one half of an MeV?[a]

Response. No. The convenient rounded value 0.51 MeV is not fixed to an exact half by any principle; it is a measured parameter.

[a]Lecture timestamp: 00:44:02.

The elementary QED process has one charged-fermion line entering, one leaving, and one photon line attached. Rearranging or crossing the legs lets the same vertex describe photon emission, absorption, or a positron process. In the noncovariant schematic notation used here, the interaction contains

$$\mathcal{L}_{\text{int}} \sim e\,\psi_e^\dagger \psi_e A. \tag{1.17}$$

Expanding the fields into creation and annihilation operators produces terms that remove an incoming electron, create an outgoing electron, and create or annihilate a photon. The coupling e supplies one factor at the vertex. Thus, for photon emission while an electron is scattered or accelerated,

$$\mathcal{A}_{\text{vertex}} \propto e, \qquad P_{\text{emission}} \propto e^2. \tag{1.18}$$

An electron striking the target in an X-ray tube is one practical setting in which this photon-emission probability matters. Repeated combinations of the same vertex generate the processes of quantum electrodynamics, including the energy shift interpreted as the force between charges.

A schematic vertex is therefore useful.

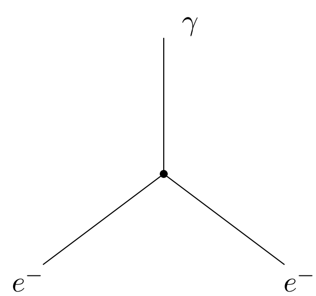

Figure 1.6: Topological schematic of the elementary QED vertex: one charged-fermion line enters, one leaves, and one photon is attached. The straight photon leg is only a simplified drawing convention here.

The exchange contribution to the force joins two such vertices:

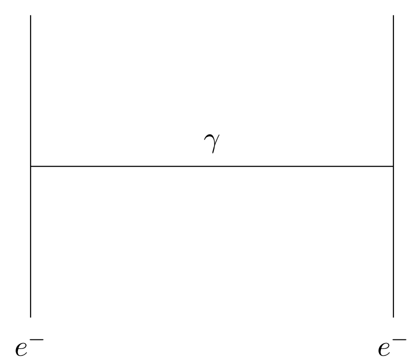

Figure 1.7: Topological schematic of photon exchange, giving the quantum-field-theoretic description of the Coulomb interaction.

With the particle–field–force triangle closed, we can continue the catalogue.

1.6 Quarks, Baryon Number, and the Puzzle of Families

Had this catalogue been written around 1960, protons and neutrons would probably have followed electrons and photons. Modern particle physics inserts quarks first. In the naive constituent picture, protons and neutrons are comparatively large composites made from smaller quark degrees of freedom. The exact meaning of *composite* remains to be examined, but the constituent language is already enormously useful.

All quarks are fermions. They carry baryon number, a quantity introduced in nuclear physics by counting nucleons. A *nucleon* means specifically a proton or neutron. A *baryon* is the broader class that contains nucleons and the similar heavy particles encountered later. A proton and a neutron each have one unit,

$$B(p) = B(n) = 1, \qquad B(\bar{p}) = B(\bar{n}) = -1. \tag{1.19}$$

For a nucleus, this baryon count is exactly the mass number $A = Z + N$, the number of protons plus neutrons. The numerical atomic mass measured in atomic-mass units is only approximately A. Protons may turn into neutrons and neutrons into protons in beta processes, but the total number of nucleons is unchanged. For example,

$$n \longrightarrow p + e^- + \bar{\nu}_e \tag{1.20}$$

changes the proton and neutron counts separately while preserving their sum.

Question from the audience. Do isotopes contradict conservation of proton number plus neutron number?[a]

Response. Different isotopes do contain different neutron numbers, and beta decay can interchange a neutron and a proton. In an ordinary nuclear process the relevant conserved count is the total baryon number, not either species separately.

[a]Lecture timestamp: 00:52:03.

The normalization is conventional. We might have assigned a proton baryon number 2, 7, or π, provided every composite scaled consistently. Once the proton is assigned $B = 1$ and contains three quarks, each quark must carry

$$B(q) = \frac{1}{3}, \qquad B(\bar{q}) = -\frac{1}{3}. \tag{1.21}$$

The fractional value is therefore a historical consequence of normalizing the composite nucleon first.

Question from the audience. Is the number of quarks conserved?[a]

Response. Quark–antiquark pairs may be created or destroyed, so the raw number is not conserved. The quantity corresponding to baryon number is the number of quarks minus antiquarks, divided by three. Empirically it is conserved to extremely high accuracy, although it is not assumed here to be an absolute law.

[a]Lecture timestamp: 00:54:23.

A hypothetical charge-conserving decay such as

$$p \longrightarrow e^+ + \gamma \tag{1.22}$$

would carry baryon number one into a final state with baryon number zero. The channel is an illustrative bookkeeping example, not a proposed decay model. No baryon-number-violating process has been observed. The rough bound quoted here puts the proton lifetime at least in the range of 10^{33}–10^{34} years. One does not watch a single proton for that long; one watches an enormous number of protons for a manageable time, trading lifetime against sample size. Proton decay was nevertheless widely expected, and nothing in the mathematics being discussed here supplied an absolute prohibition. Whether baryon number is exact therefore remained an open experimental question: unobserved, strongly constrained, but not dismissed.

Question from the audience. Does proton–antiproton annihilation violate baryon number?[a]

Response. No. The initial baryon number is $1 + (-1) = 0$, so annihilation into a zero-baryon-number final state conserves it. A lone proton decaying to nonbaryonic products would be the violation.

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[a] Lecture timestamp: 00:56:17.

There are six quark flavors,

$$u,\ d,\ s,\ c,\ b,\ t, \tag{1.23}$$

called up, down, strange, charm, bottom, and top. Nothing points upward about an up quark, and a strange quark is no stranger than its neighbors; the names record history rather than logic. Field notation is not perfectly uniform: one encounters q, Q, ψ_q, or simply the flavor letter u, d, s, c, b, t for both the field and its quantum.

Question from the audience. Are bottom and top the same particles that were once called beauty and truth?[a]

Response. Yes. The less ornate names bottom and top became standard. The naming history carries no dynamical content.

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[a] Lecture timestamp: 00:58:35.

Their electric charges repeat in three pairs:

$$\begin{aligned} Q(u) = Q(c) = Q(t) = +\frac{2}{3}, \\ Q(d) = Q(s) = Q(b) = -\frac{1}{3}. \end{aligned} \tag{1.24}$$

Antiquarks carry the opposite charges and baryon number. Isolated fractional charge is never seen because quarks do not occur as free asymptotic particles; observed hadrons have integer electric charge.

The pairs (d, u), (s, c), and (b, t) are three families repeating the same charge pattern. All six still carry $B = 1/3$.

Question from the audience. Do the heavier quarks also carry baryon number 1/3?[a]

Response. Yes. Every quark carries baryon number 1/3.

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[a] Lecture timestamp: 01:02:02.

The masses show no comparable repetition. Useful order-of-magnitude values are

$$m_u \sim 5\,\mathrm{MeV}, \qquad m_d \sim 10\,\mathrm{MeV}, \tag{1.25}$$

$$m_s \sim 100\,\mathrm{MeV}, \qquad m_c \sim 10^3\,\mathrm{MeV}, \tag{1.26}$$

$$m_b \sim 5\,\mathrm{GeV}, \qquad m_t \sim 170\,\mathrm{GeV}. \tag{1.27}$$

Here 1 GeV = 10^3 MeV; the change of unit keeps the bottom- and top-quark entries readable. The down-quark estimate is far above the electron's 0.511 MeV but far below the proton's roughly 940 MeV, about 1,800 electron masses. Indeed, adding two light up-quark entries and one down-quark entry does not come close to the proton mass; the rest of the hadron's energy must eventually be accounted for.

[5]

Even the ordering within a pair is not repeated: the down quark is heavier than the up, whereas the charm is much heavier than the strange. The top is tens of thousands of times heavier than the up quark, even though their charges and interaction pattern are closely analogous.

There are really two puzzles here. We do not understand why even one quark family exists, although the first family at least supplies the up and down quarks needed for protons, neutrons, and ordinary matter. Nor do we understand why nature repeats that pattern in a second and a third family, or why the repeated particles have masses scattered over so many orders of magnitude. There are ideas, but no accepted explanation that calculates this pattern.

Question from the audience. If the top-quark mass has no fundamental explanation, how was its approximate value anticipated before direct discovery?[a]

Response. The estimate did not derive the mass from a deeper principle. One assumes a candidate mass, computes diagrams in which virtual top quarks affect well-measured processes, and asks which value is consistent with those observations. Indirect sensitivity can locate a parameter without explaining why nature chose it.

[a]Lecture timestamp: 01:05:53.

Question from the audience. If a free quark cannot be isolated, in what sense can a quark be detected?[a]

Response. Evidence for quarks is detectable: their production leaves characteristic energies, directions, and particle jets, and virtual quarks affect other observables. What cannot be prepared is a lone quark separated from all hadronic companions and examined as an isolated asymptotic particle.

[a]Lecture timestamp: 01:08:04.

Question from the audience. Is the pattern of charges $-1/3$ and $+2/3$ derived from theory or merely bookkeeping?[a]

Response. Some of the pattern is required by mathematical consistency, but only some. At this stage the charges are being recorded as facts; the consistency conditions will be discussed only after more of the Standard Model is in place.

[a]Lecture timestamp: 01:09:02.

Only u and d quarks have appreciable abundance in ordinary matter. The heavier flavors decay toward lighter states. The relative abundance of up and down quarks is therefore determined largely by the cosmic and nuclear abundance of protons and neutrons and by their constituent content.

[5]Editorial clarification: The spoken comparison called a 10 MeV down quark about fifty electron masses. With the rough values written here, $10/0.511 \approx 20$; the hierarchy is correct, but the factor fifty is an arithmetic slip.

Question from the audience. What determines the relative abundance of the six quark flavors in the universe?[a]

Response. Stable matter contains essentially up and down quarks; the heavier flavors decay. Free neutrons decay while protons are stable on observed timescales, although neutrons remain abundant when bound in nuclei. Counting up and down quarks then reduces, at this level, to counting those protons and neutrons.

[a]Lecture timestamp: 01:09:49.

1.7 From Quarks to Hadrons: Nucleons, Strange Baryons, and Mesons

Once the quark charges are known, familiar hadrons can be rebuilt. A neutron and a proton each contain three valence quarks,

$$n = ddu, \qquad p = uud. \tag{1.28}$$

Their charges check immediately:

$$Q(n) = Q(d) + Q(d) + Q(u) = -\frac{1}{3} - \frac{1}{3} + \frac{2}{3} = 0, \tag{1.29}$$

$$Q(p) = Q(u) + Q(u) + Q(d) = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = 1. \tag{1.30}$$

Forget photons for a moment. With electromagnetism switched off, proton and neutron look almost like the same object with u and d interchanged. This near-symmetry is not exact because the two quarks do not have exactly equal masses. We shall name and develop the symmetry only after meeting the corresponding light mesons.

Since $m_d > m_u$, the neutron, which contains two down quarks, should receive the larger quark-mass contribution. Electromagnetic self-energy pushes in the opposite direction: a charged proton stores electric-field energy, while a neutron has no net charge. Nature resolves the competition with a slightly heavier neutron. At this rough level of approximation,

$$m_p c^2 \approx 940\,\mathrm{MeV}, \tag{1.31}$$

$$m_n c^2 \approx 940\,\mathrm{MeV}, \tag{1.32}$$

$$(m_n - m_p)c^2 \approx 1.4\,\mathrm{MeV}. \tag{1.33}$$

Question from the audience. What are the proton and neutron masses?[a]

Response. Both are near 940 MeV, with a difference of about 1.4 MeV at this rough level.

[a]Lecture timestamp: 01:15:35.

That small excess is nevertheless enough to permit free-neutron decay. After paying the electron rest energy, the remaining energy can appear as kinetic energy in $n \to p + e^- + \bar{\nu}_e$.

The same constituent logic produces strange baryons. Since s and d have the same electric charge, a down quark may be replaced by a strange quark without changing the charge pattern:

$$uds, \qquad uss, \qquad uus. \tag{1.34}$$

Thus uds and uss are neutral in the examples above, while uus has charge +1. Such states received names including Λ and Ξ; no detailed assignment of those names is needed at this stage. Replacing d by s is qualitatively analogous to substituting one nucleon species for the other in a related nucleus: much of the binding pattern remains, while the constituent change shifts the mass. These states are heavier than ordinary nucleons and decay to lighter baryons together with whatever additional particles are required to carry energy and the other conserved quantum numbers.

Question from the audience. What appears when a strange baryon decays?[a]

Response. A lighter baryon such as a proton or neutron appears, but it cannot be the only product: the excess energy and conserved quantum numbers require additional particles. The detailed decay rules are deferred.

[a]Lecture timestamp: 01:19:02.

The detailed replacement rules are more complicated than this first catalogue, but one safe operation is to replace a flavor by another flavor with the same electric charge. Replacing a down quark by a strange quark adds roughly 90 MeV at this rough constituent level. These baryons are therefore somewhat heavier than protons and neutrons. Because they can decay to the lighter nucleons, they are not abundant in ordinary matter or in the cosmos around us.

Question from the audience. Does every baryon have an antibaryon made by replacing each quark with its antiquark?[a]

Response. Yes. Replacing every constituent by its antiparticle produces the antiproton, antineutron, or corresponding antistrange baryon, with all additive charges reversed.

[a]Lecture timestamp: 01:20:33.

Question from the audience. What if a strange quark is replaced by a bottom quark instead?[a]

Response. The result is a bottom baryon. "Strange baryon" specifically means that an s quark is present; the adjective names the flavor, not a general degree of unfamiliarity.

[a]Lecture timestamp: 01:21:00.

The adjective *strange* arose after physicists had become familiar with protons and neutrons and newly discovered particles seemed unexpected beside them. Once the flavor structure is known, their existence is no stranger than that of any other quark composite.

The constituent masses expose an important warning. Adding two up-quark masses and one down-quark mass gives only a few tens of MeV, nowhere near the proton's roughly 940 MeV. Something substantial is missing from that naive sum. The proton and neutron also contain gluons, and gluons carry energy. In a relativistic bound system the mass of the composite is not obtained by simply adding the listed masses of its constituents.

Question from the audience. Where does the apparently missing proton and neutron mass come from? Do gluons carry mass?[a]

Response. Gluons carry no baryon number in this bookkeeping, but they do carry energy. Their energy is part of why the mass of the proton or neutron is not the sum of the light-quark mass entries.

[a]Lecture timestamp: 01:21:26.

Question from the audience. Is that missing contribution a form of binding energy?[a]

Response. Yes, it may be called a form of binding energy. The warning is that relativistic composite masses do not add in the simple way suggested by adding constituent mass parameters.

[a]Lecture timestamp: 01:22:03.

Question from the audience. Can a composite proton be described in field language as well as particle language?[a]

Response. Yes, but it is much harder. For the constituent bookkeeping being developed here, particle language is far more transparent than trying to combine the spatially extended fields directly.

[a]Lecture timestamp: 01:22:45.

Question from the audience. Ordinary binding energy is negative relative to separated constituents. Why is the nucleon's dynamical contribution a large positive energy?[a]

Response. The usual negative-binding intuition is not true in this case. The explanation belongs to the later treatment of the strong interaction, so for now the question is deferred.

[a]Lecture timestamp: 01:23:32.

Why do three quarks form baryons while two quarks do not form an isolated two-quark hadron? In the early days of quark theory this was a genuine puzzle. One might answer that the observed bound combinations must have electric charge equal to an integer multiple of the electron charge. By itself that sounds circular: it describes the pattern without explaining it. Nevertheless, the rule proved to be pointing in the right direction, and its deeper origin can be understood only after more of the strong interaction is in place. For now it is explicitly a working rule: three quarks can make a baryon, while a quark and an antiquark can make a meson, and the resulting observed composites have integer charge.

Baryons made from three identical light flavors also occur, but their details can wait until after the mesons.

Mesons are $q\bar{q}$ composites. Their baryon number vanishes,

$$B(q\bar{q}) = \frac{1}{3} - \frac{1}{3} = 0. \tag{1.35}$$

The lightest examples use u and d. These flavors are much lighter than the others, are stable or comparatively slow to disappear in ordinary matter, and are abundant in nature. Their low production threshold also means that collisions make them more readily than heavy flavors. For all these reasons, light-quark composites were the first mesons encountered experimentally and are the natural place to begin.

The natural flavor basis is

$$|\bar{u}d\rangle, \qquad \left|u\bar{d}\right\rangle, \qquad |\bar{u}u\rangle, \qquad \left|\bar{d}d\right\rangle. \tag{1.36}$$

No additional restriction is used in this first count: all four light quark–antiquark pairings are retained, and their electric charges are then computed.

The charged states are immediate. Since $Q(\bar{u}) = -2/3$ and $Q(d) = -1/3$,

$$Q(\bar{u}d) = -\frac{2}{3} - \frac{1}{3} = -1, \qquad\qquad \left|\pi^-\right\rangle = |\bar{u}d\rangle. \tag{1.37}$$

Replacing every constituent by its antiparticle gives $u\bar{d}$, so the second state is the antiparticle of the first:

$$Q(u\bar{d}) = +\frac{2}{3} + \frac{1}{3} = +1, \qquad\qquad \left|\pi^+\right\rangle = \left|u\bar{d}\right\rangle. \tag{1.38}$$

The same-flavor states are neutral, but neither receives a separate particle name by itself. Quantum mechanics permits us to superpose them. The two normalized, orthogonal combinations are

$$\left|\pi^0\right\rangle \approx \frac{|\bar{u}u\rangle - \left|\bar{d}d\right\rangle}{\sqrt{2}}, \tag{1.39}$$

$$|\eta\rangle \approx \frac{|\bar{u}u\rangle + \left|\bar{d}d\right\rangle}{\sqrt{2}}. \tag{1.40}$$

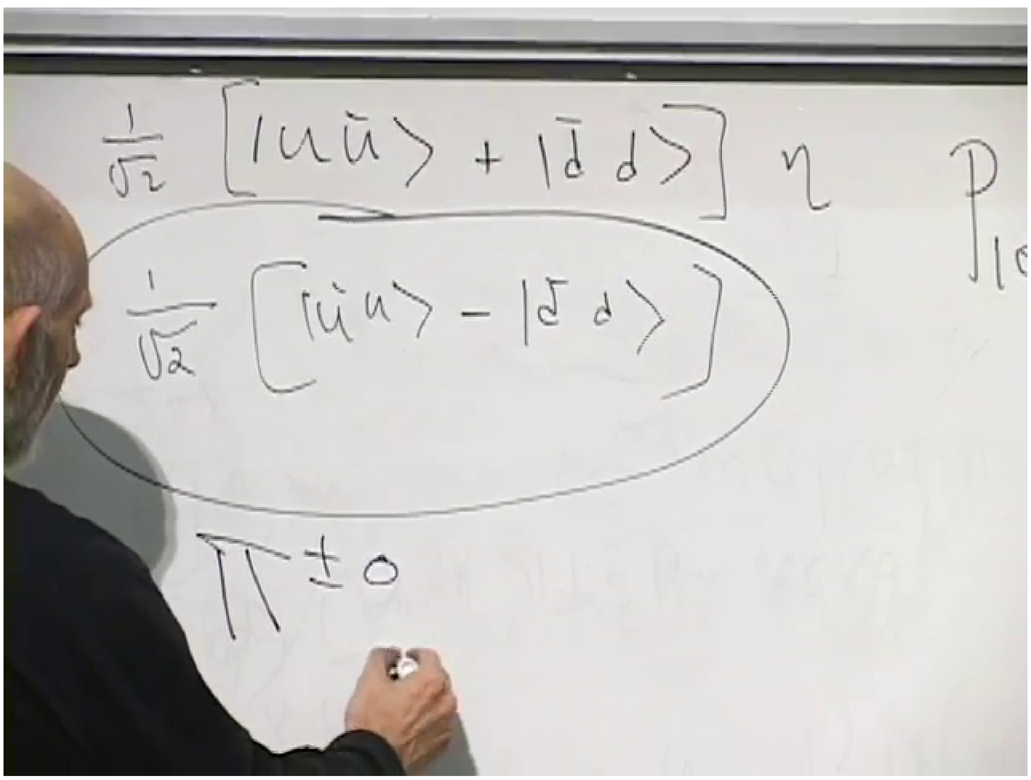

Figure 1.8: The symmetric neutral combination is identified with the η, while the antisymmetric combination is grouped with the pion triplet.

Lecture frame: 01:32:35.

Each neutral state is an entangled quark–antiquark pair, not through ordinary spin but through *upness* and *downness*. If up and down flavor are treated mathematically like the two directions of a spin-half variable, then the sum and difference above play the same role as familiar two-spin superpositions. This internal analogue of spin is called *isospin*. Nothing is literally pointing upward or downward in space; the two directions label u and d flavor.

The three pions π^+, π^0, and π^- form an isospin triplet. Their masses are nearly equal and would coincide more closely if the up and down quarks had exactly the same mass. The orthogonal neutral state is called the η.

The η is substantially heavier than a pion:

$$m_\pi \approx 140\,\mathrm{MeV}, \qquad m_\eta \approx 500\,\mathrm{MeV}. \tag{1.41}$$

Why the η stands apart was obscure for a long time. The explanation is now understood, but it is mathematically abstract and is not developed here.

Question from the audience. Are these mesons relatively long-lived?[a]

Response. Relative to many strongly decaying particles, yes. They do not decay through the strong interaction in the channels under discussion; weak and electromagnetic decays make them longer-lived.

[a] Lecture timestamp: 01:33:59.

Question from the audience. Do the individual neutral states $\bar{u}u$ and $\bar{d}d$ have names of their own?[a]

Response. No. The named particles are the linear combinations formed by their sum and difference; those combinations have definite values.

[a] Lecture timestamp: 01:34:29.

Question from the audience. Is this neutral-state superposition like neutrino oscillation?[a]

Response. It is similar in the limited sense that both involve quantum superpositions. Here the issue is more closely related to the spin combinations of two spin-half constituents and the two ways such states can be entangled. That discussion is deferred to the next lecture.

[a] Lecture timestamp: 01:34:44.

Once the strange quark is admitted, replacing d or $\bar{d}$ by s or $\bar{s}$ gives the four strange or K-meson flavor states

$$\bar{u}s, \qquad u\bar{s}, \qquad \bar{d}s, \qquad d\bar{s}. \tag{1.42}$$

Their charge assignments are

$$Q(\bar{u}s) = -1, \qquad Q(u\bar{s}) = +1, \tag{1.43}$$

$$Q(\bar{d}s) = 0, \qquad Q(d\bar{s}) = 0. \tag{1.44}$$

There are therefore two charged and two neutral kaons.

The first count gives three pions, four kaons, and one η: eight states. But the list has not yet used the $|s\bar{s}\rangle$ pairing. With three flavors and three antiflavors, the complete elementary count is therefore

$$3 \times 3 = 9. \tag{1.45}$$

The missing $s\bar{s}$ state completes the count of nine quark–antiquark flavor combinations.

Chapter 2

Spin, Isospin, Color, and Gluon Self-Interaction

2.1 Quantum Chromodynamics by Analogy with Electrodynamics

Quantum chromodynamics is too large a subject to develop fully in one lecture. The aim here is the basic intuitive picture rather than the deeper mathematics.

Quantum electrodynamics is the theory of electrons and photons. It is also the quantum theory of the Coulomb force. If the nucleus of an atom is treated as a fixed source of electric field, without asking about its motion or internal structure, QED is the theory of the atom.

Quantum chromodynamics has the analogous role for quarks and gluons. It describes the particles themselves, the forces between them, and the composite objects they form. These composites are called *hadrons*. Protons and neutrons are baryons: they carry a net quark number, or equivalently a nonzero baryon number. Mesons are quark–antiquark combinations. Both baryons and mesons also contain gluons.

Gluons are electrically neutral, but they supply the field that binds quarks. In that respect the gluon field plays a role analogous to the electromagnetic field that binds atoms.

Before introducing color and gluons, it is useful to return to the mathematics of spin. The same mathematics reappears almost directly as *isospin*, and a related pattern appears once again in the theory of color. Group theory lies underneath these constructions, but the required ideas can be developed through angular momentum.

2.2 Angular-Momentum Multiplets

The components of angular momentum obey

$$[L_x, L_y] = i\hbar L_z, \tag{2.1}$$

with the cyclic companion relations

$$[L_y, L_z] = i\hbar L_x, \qquad [L_z, L_x] = i\hbar L_y. \tag{2.2}$$

The three components do not commute, so they cannot all be measured simultaneously. A description therefore singles out one axis, conventionally the z-axis, and labels states by the component of angular momentum along that axis.

This choice does not break rotational symmetry in nature. It breaks the symmetry only in the description, by selecting one direction for bookkeeping.

The allowed values of the z-component are quantized and differ by one unit of $\hbar$. Setting $\hbar = 1$ when listing these values, a spin-ℓ multiplet contains

$$m = \ell, \ell - 1, \ldots, -\ell. \tag{2.3}$$

The number of independent states is therefore

$$2\ell + 1. \tag{2.4}$$

The first few multiplets are worth writing explicitly:

$$\ell = 0: \qquad m = 0, \tag{2.5}$$
$$\ell = \frac{1}{2}: \qquad m = \frac{1}{2}, -\frac{1}{2}, \tag{2.6}$$
$$\ell = 1: \qquad m = 1, 0, -1, \tag{2.7}$$
$$\ell = \frac{3}{2}: \qquad m = \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}, \tag{2.8}$$
$$\ell = 2: \qquad m = 2, 1, 0, -1, -2. \tag{2.9}$$

For present purposes, half-integer spin belongs to fermions and integer spin to bosons. An electron is a spin-$\frac{1}{2}$ particle. A spin-zero particle has no angular momentum at all. The photon has spin one, and spin-one states also occur for atoms and nuclei. Atoms and nuclei can likewise have spin $\frac{3}{2}$, and the same counting continues to higher spin.

The notation must distinguish the total spin from its component. The number ℓ labels the multiplet and is also the largest possible z-component. The number m labels a particular state within that multiplet. Thus a spin-one object has three states, whereas a spin-$\frac{1}{2}$ object has two.

2.3 Combining Two Spin-Half Particles

Consider two spin-$\frac{1}{2}$ particles and ignore their orbital angular momentum. Their spin product space contains four basis states:

$$|\uparrow\uparrow\rangle, \qquad |\uparrow\downarrow\rangle, \qquad |\downarrow\uparrow\rangle, \qquad |\downarrow\downarrow\rangle. \tag{2.10}$$

The corresponding z-components are

$$|\uparrow\uparrow\rangle : m = +1, \tag{2.11}$$
$$|\uparrow\downarrow\rangle : m = 0, \tag{2.12}$$
$$|\downarrow\uparrow\rangle : m = 0, \tag{2.13}$$
$$|\downarrow\downarrow\rangle : m = -1. \tag{2.14}$$

The state with both spins up proves that a spin-one multiplet must occur: the maximum z-component is $+1$. A spin-one multiplet has only three states,

$$m = +1, \quad 0, \quad -1. \tag{2.15}$$

But the product space contains four states, including two independent states with $m = 0$. They cannot all belong to a single spin-one multiplet. The extra state cannot be spin two, because a spin-two multiplet would require five states. The four product states must therefore divide into three states of spin one and one state of spin zero.

Which combination of the two $m = 0$ basis states belongs to the spin-one multiplet? Interchange the two particles. The aligned states are unchanged:

$$|\uparrow\uparrow\rangle \longrightarrow |\uparrow\uparrow\rangle, \qquad |\downarrow\downarrow\rangle \longrightarrow |\downarrow\downarrow\rangle. \tag{2.16}$$

By contrast, the mixed product states are exchanged:

$$|\uparrow\downarrow\rangle \longleftrightarrow |\downarrow\uparrow\rangle. \tag{2.17}$$

Their sum is symmetric under the interchange:

$$|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle. \tag{2.18}$$

Normalization supplies a factor of $1/\sqrt{2}$:

$$|1, 0\rangle = \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}. \tag{2.19}$$

Together with the two aligned states, this gives the spin-one triplet:

$$|1, +1\rangle = |\uparrow\uparrow\rangle, \tag{2.20}$$

$$|1, 0\rangle = \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}, \tag{2.21}$$

$$|1, -1\rangle = |\downarrow\downarrow\rangle. \tag{2.22}$$

The remaining orthogonal combination has a minus sign:

$$|0, 0\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}. \tag{2.23}$$

It is antisymmetric under interchange and forms the one-state spin-zero multiplet. Thus

$$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0. \tag{2.24}$$

The symmetric mixed-spin state has $m = 0$, but it does *not* have zero total angular momentum. It belongs to the spin-one triplet. Rotating the coordinate axis mixes it with the $m = \pm 1$ states. Only the antisymmetric combination has $\ell = 0$.

As far as angular momentum is concerned, the singlet is like empty space. It may carry energy, charge, or other quantum numbers, but it carries no angular momentum.

This distinction can be tested by a thought experiment. Ignore every conservation law except conservation of angular momentum and ask which two-particle spin state could simply disappear.

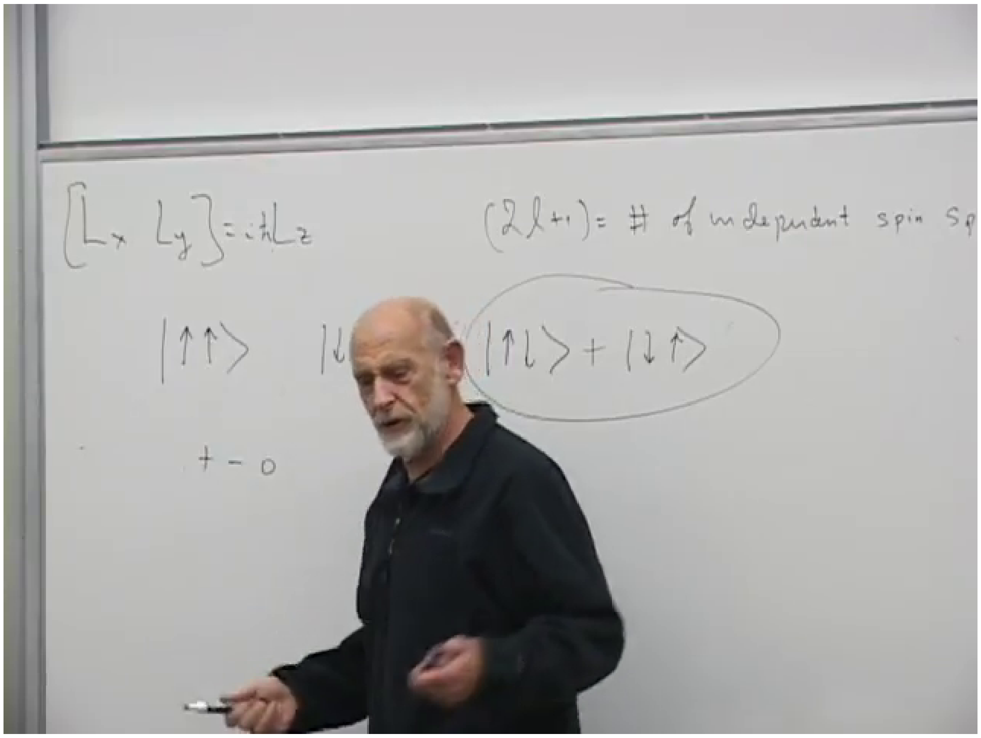

Figure 2.1: Symmetric mixed-spin combination forming the $m = 0$ member of the spin-one triplet.
Lecture frame: 00:15:08.

Neither aligned state could disappear, because each belongs to a spin-one multiplet. The symmetric $m = 0$ state could not disappear either. Although its z-component vanishes, its total spin is still one, with nontrivial components relative to other axes. Only the antisymmetric singlet could disappear without violating angular-momentum conservation.

Question from the audience. Would measuring the symmetric mixed-spin state along the z-axis give $+1$ or -1?[a]

Response. No. Along the z-axis it gives zero. The state nevertheless has total spin one, so a measurement along another axis can give $+1$ or -1.

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[a] Lecture timestamp: 00:19:28.

2.4 Three Spin-Half Particles

Now combine three spin-$\frac{1}{2}$ particles. There are

$$2 \times 2 \times 2 = 8 \tag{2.25}$$

product states. If all three spins point up, the total z-component is $+\frac{3}{2}$. A spin-$\frac{3}{2}$ multiplet must therefore be present. It contains

$$2\left(\frac{3}{2}\right) + 1 = 4 \tag{2.26}$$

states.

Four states remain. They cannot form a second spin-$\frac{3}{2}$ multiplet, because the states with $m = \pm\frac{3}{2}$ have already been used. The remaining states have maximum components only $+\frac{1}{2}$ and $-\frac{1}{2}$. Nor can three half-integer spins make total spin zero. The remaining states must form spin-$\frac{1}{2}$ multiplets.

A spin-$\frac{1}{2}$ multiplet contains two states. Since four states remain, there are two distinct spin-$\frac{1}{2}$ multiplets:

$$\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}. \tag{2.27}$$

This pattern will reappear when the two states called spin up and spin down are replaced by the two quark flavors called up and down.

2.5 Isospin as an Internal Spin

Isospin, or isotopic spin, is an internal analogue of ordinary spin. Ordinary spin is tied to rotations of physical space. Isospin is not. It concerns transformations in an internal mathematical space whose two basic states will be identified with

$$u, \qquad d. \tag{2.28}$$

The terms *up* and *down* are merely flavor labels here. Nothing points upward or downward in ordinary space.

[1]

Why are u and d suitable for this analogy? The natural energy scale of hadron physics is a few hundred MeV. A typical light meson has a mass of a few hundred MeV, and the energy involved in binding three quarks into a nucleon is likewise on that scale. By comparison, the binding energies of ordinary nuclei are only a few MeV. The force holding protons and neutrons together inside a nucleus is therefore being considered on a much smaller energy scale than the dynamics binding quarks inside a hadron.

Among the six quark flavors, u and d are light compared with the hadronic scale. Heavy quarks cost more energy to produce, and hadrons containing them can decay into lighter ones. The most stable light hadrons are consequently made from up and down quarks.

The down quark is heavier than the up quark, roughly by a factor of two in the estimates being used here, but both masses are small compared with hundreds of MeV. To a first approximation they may both be treated as nearly massless on the hadronic scale. More importantly, they may be treated as nearly equal in mass.

This approximate equality leads to a symmetry. If up and down quarks are interchanged, the strong-interaction physics changes very little. The proton,

$$p \sim uud, \tag{2.29}$$

and the neutron,

$$n \sim udd, \tag{2.30}$$

provide the simplest example: they differ by exchanging u and d, and their masses are very close.

[1] Editorial clarification: A brief transition between the definition of internal spin space and the following hadronic mass-scale argument is not recoverable with confidence. No additional physical claim is inferred from it.

The interchange

$$u \longleftrightarrow d \tag{2.31}$$

is therefore treated mathematically like

$$|\uparrow\rangle \longleftrightarrow |\downarrow\rangle. \tag{2.32}$$

For ordinary spin, the transformation corresponds to a rotation of an axis in space. For isospin, it is a rotation in an internal space. The mathematics is analogous even though the physical meanings are different.

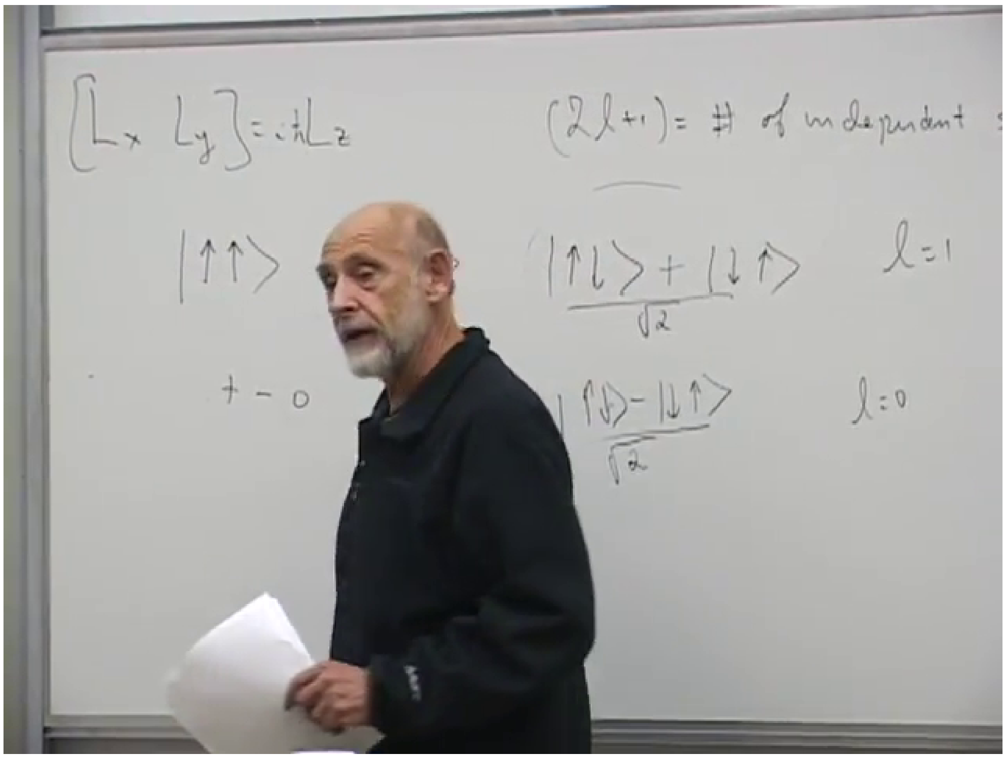

Figure 2.2: Triplet and singlet decomposition of two spin-$\frac{1}{2}$ states, the model for the corresponding isospin construction.

Lecture frame: 00:27:42.

The prefix *iso* comes from *isotope*. In the older nuclear description, proton and neutron were regarded as two members of one internal doublet. Interchanging a proton and neutron in a nucleus changes one isotope into another. The quark model later traced this proton-neutron interchange to the underlying interchange of u and d.

Question from the audience. Did isospin predate the quark model?[a]

Response. Yes. Isospin was introduced before quarks and is associated historically with Heisenberg. The quark model later supplied its u- and d-flavor interpretation.

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[a] Lecture timestamp: 00:30:19.

Isospin is not an exact symmetry of nature. The up and down quarks do not have exactly equal masses, and they have different electric charges:

$$Q(u) = +\frac{2}{3}, \qquad Q(d) = -\frac{1}{3}. \tag{2.33}$$

Even if their masses were equal, electromagnetism would still distinguish them.

For small hadronic and nuclear systems, electromagnetic effects are weak compared with the strong interaction. In a large nucleus, electric repulsion can become important; for a proton, neutron, or a small nucleus such as helium, it is a comparatively small correction to the strong-interaction scale. The isospin approximation therefore consists of two steps:

1. neglect the u–d mass difference;
2. turn off their electromagnetic distinction and retain only the strong interaction.

Within that approximation, the transformation between u and d becomes a symmetry mathematically analogous to ordinary spin symmetry.

2.6 The Nucleon Isospin Doublet

A single quark cannot be isolated and examined as a free laboratory object. The simplest observed baryons contain three quarks. The three-spin counting argument therefore has a direct isospin analogue: three isospin-$\frac{1}{2}$ quarks can form total isospin $\frac{3}{2}$ or $\frac{1}{2}$.

Begin with total isospin $\frac{1}{2}$. Such a multiplet has

$$2\left(\frac{1}{2}\right)+1=2 \tag{2.34}$$

states. They are the proton and neutron:

$$p\sim uud, \qquad n\sim udd. \tag{2.35}$$

Their charges are

$$Q(p)=\frac{2}{3}+\frac{2}{3}-\frac{1}{3}=+1, \tag{2.36}$$
$$Q(n)=\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0. \tag{2.37}$$

Interchanging u and d interchanges proton and neutron. They form an isospin-$\frac{1}{2}$ doublet.

To discuss exchange symmetry, temporarily label three quark slots by $1,2,3$. These labels should not be mistaken for fixed quark positions; the quarks move. A schematic proton factor is

$$d_1(u_2u_3-u_3u_2). \tag{2.38}$$

The purpose of the expression is to emphasize fermionic exchange antisymmetry. It is not a complete proton wavefunction. The full state must combine flavor, ordinary spin, spatial dependence, and—as will shortly become essential—color.
[2]

[2]Editorial clarification: The subscripts are temporary slot or one-particle labels. Two identical u flavor states do not form the usual antisymmetric isospin singlet, so the displayed factor by itself supports no isospin-zero conclusion. It is retained only as a local slot-antisymmetrization step; the proton's total isospin follows from its complete three-quark state.

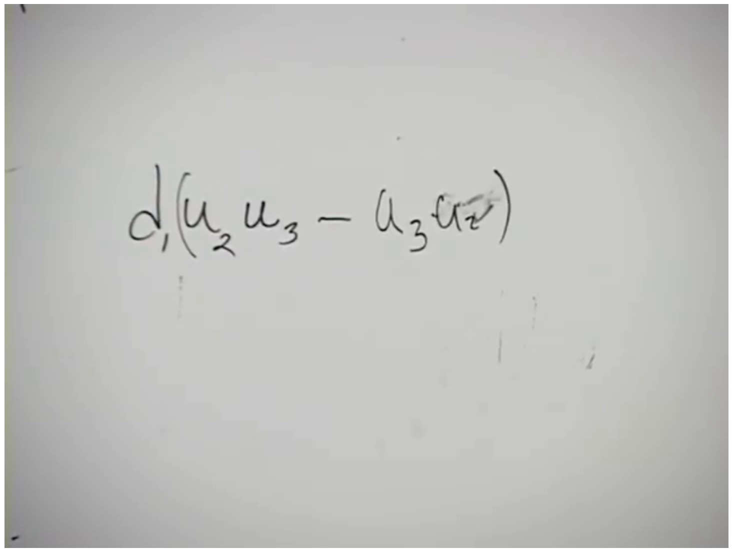

Figure 2.3: Antisymmetrized two-u-quark factor in a schematic proton state.
Lecture frame: 00:35:43.

The limited comparison with the earlier spin construction is that quantum states may be combined symmetrically or antisymmetrically under an exchange of labels. That comparison alone does not determine the isospin carried by the displayed uu factor.

The proton has both ordinary spin $\frac{1}{2}$ and isospin $\frac{1}{2}$. The same is true of the neutron. These are different quantum numbers: ordinary spin concerns physical rotations, while isospin concerns the u–d internal symmetry.

2.7 The Delta Quartet

Three light quarks can also form total isospin $\frac{3}{2}$. The resulting four-state multiplet is the Delta quartet:

$$\begin{aligned} \Delta^{++} &\sim uuu, \quad \Delta^{+} \sim uud, \\ \Delta^{0} &\sim udd, \quad \Delta^{-} \sim ddd. \end{aligned} \tag{2.39}$$

The Delta states have ordinary spin $\frac{3}{2}$ as well as isospin $\frac{3}{2}$. In the states with maximum components, the three ordinary spins are aligned and the three isospins are aligned.

A composition such as uud does not by itself determine whether the state is a proton or a Δ^{+}. The same flavor content can be combined into different total-spin and total-isospin states.

Question from the audience. What distinguishes the uud member of the Delta quartet from a proton?[a]

Response. Their ordinary spins differ. The proton has spin $\frac{1}{2}$, whereas the Delta state has spin $\frac{3}{2}$. The quarks are also combined into different total-isospin states.

[a]Lecture timestamp: 00:41:08.

Question from the audience. Are the middle members of the Delta quartet isospin-$\frac{1}{2}$ states?[a]
Response. No. Their isospin components are $+\frac{1}{2}$ and $-\frac{1}{2}$, but their total isospin is $\frac{3}{2}$. A total-isospin-$\frac{3}{2}$ multiplet necessarily contains four component states.

[a] Lecture timestamp: 00:41:52.

The four isospin components correspond to

$$I_3 = \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}. \tag{2.40}$$

Their electric charges are

$$Q(\Delta^{++}) = +2, \tag{2.41}$$
$$Q(\Delta^{+}) = +1, \tag{2.42}$$
$$Q(\Delta^{0}) = 0, \tag{2.43}$$
$$Q(\Delta^{-}) = -1. \tag{2.44}$$

The middle two charges happen to equal the proton and neutron charges, but the states remain distinct because their spin and isospin couplings differ.

Approximate isospin symmetry makes the four Delta masses nearly equal. The small u–d mass difference produces only a small splitting within the quartet. The common Delta mass scale is roughly

$$m_\Delta \sim 1200\,\text{MeV}, \tag{2.45}$$

whereas the proton and neutron are near

$$m_N \sim 940\,\text{MeV}. \tag{2.46}$$

2.7.1 Decay of the Δ^{++}

The Delta states are heavy enough to decay into a nucleon and a pion. Consider

$$\Delta^{++} \sim uuu. \tag{2.47}$$

As the constituents separate, a quark–antiquark pair can be produced. An attempted $u\bar{u}$ insertion would leave a three-u baryon in the final state and does not provide a lighter allowed channel with the available energy. Producing a $d\bar{d}$ pair instead permits the constituents to regroup as

$$uuu + d\bar{d} \longrightarrow uud + u\bar{d}. \tag{2.48}$$

The two final composites are

$$uud = p, \qquad u\bar{d} = \pi^+, \tag{2.49}$$

so

$$\Delta^{++} \longrightarrow p + \pi^+. \tag{2.50}$$

Charge conservation is immediate:

$$Q(p) + Q(\pi^+) = 1 + 1 = 2 = Q(\Delta^{++}). \tag{2.51}$$

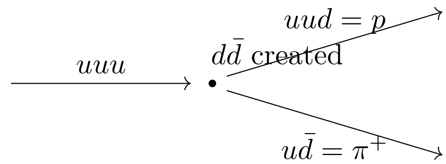

Figure 2.4: Constituent regrouping in the decay $\Delta^{++} \to p + \pi^+$.

The rough mass balance also permits the decay:

$$1200\,\text{MeV} > 940\,\text{MeV} + 140\,\text{MeV}. \tag{2.52}$$

There is enough energy left for kinetic energy in the final state.

All four Delta states can decay to a proton or neutron together with a pion. Their lifetimes are extremely short. An order-of-magnitude estimate is the time light takes to cross a proton:

$$\tau_\Delta \sim \frac{10^{-13}\,\text{cm}}{10^{10}\,\text{cm/s}} \sim 10^{-23}\,\text{s}. \tag{2.53}$$

They nevertheless live long enough to be identified as distinct objects produced in particle collisions.

Question from the audience. Does the written order of the u and d symbols matter in this constituent bookkeeping?[a]

Response. Not for the purpose at hand. The important information is the flavor content and the way the full quantum state is combined.

[a]Lecture timestamp: 00:48:49.

2.8 The Pauli Puzzle and the Need for Color

The Δ^{++} exposes a contradiction. It consists of three up quarks. In a spin-$\frac{3}{2}$ state their ordinary spins can all point in the same direction. Their isospins are also identical because they are all u. If spin, flavor, and the lowest-energy spatial state were the whole description, the three quarks would be identical fermions occupying the same state. The Pauli exclusion principle forbids this.

Question from the audience. How can the repeated up quarks be fermions if they appear to occupy the same state?[a]

Response. They cannot be identical in every quantum number. The Δ^{++} shows that quarks require an additional label beyond ordinary spin, flavor, and the spatial state. That additional label is color.

[a]Lecture timestamp: 00:48:58.

The new label has three possible values:

$$r, \qquad g, \qquad b, \tag{2.54}$$

called red, green, and blue. The words have nothing to do with visible color. They are arbitrary names for three internal states. Other names could have been used; what matters is that there are three distinct possibilities.

The red, green, and blue versions of a given quark flavor have the same mass and the same ordinary observable properties. They are nevertheless distinct quantum states. A Δ^{++} can therefore contain one red u, one green u, and one blue u, so the three fermions do not occupy the same complete state.

The six flavors are

$$u,\ d,\ c,\ s,\ t,\ b. \tag{2.55}$$

This ordering is not by mass. The up quark is the lightest and the down quark comes next. The charm quark is much heavier than the strange quark, and the top quark is vastly heavier than the bottom quark. Each flavor comes in three colors. A quark therefore carries at least two distinct internal labels:

$$\text{flavor} \quad \text{and} \quad \text{color}. \tag{2.56}$$

Could the three quarks have been distinguished by position or momentum instead? The Delta puzzle concerns the lowest-energy state, for which the relevant spatial and momentum structure cannot supply three distinct labels without changing the energy and symmetry properties. The missing distinction must be a new internal degree of freedom.

There is an instructive analogy with the role of spin in atomic physics. Two electrons that appear to occupy the same orbital state can still differ by their spin state. In the same structural way, quarks that agree in flavor, ordinary spin, and spatial state can differ by color.

[3]

Yoichiro Nambu was associated with recognizing that quarks required this additional degree of freedom. The physical point is independent of its name: three quark colors resolve the Pauli contradiction. The relevant exclusion principle is the Pauli principle, not the Heisenberg uncertainty principle.

[4]

The same color construction applies to the proton and neutron. Their states are not built by assigning one permanently identifiable color to each written flavor symbol. A physical baryon is a quantum superposition of the possible red, green, and blue assignments, combined with the appropriate exchange symmetries.

Question from the audience. Is color an ordinary physical property that can be measured directly?[a]

Response. The three colors have the same mass and the same ordinary properties. Their distinction is revealed relationally, through allowed combinations and collision processes. Experiments show that three distinct color states are present even though an isolated colored quark cannot be prepared for direct inspection.

---

[a] Lecture timestamp: 00:56:55.

[3] Editorial clarification: The precise historical route involving helium spectroscopy, the Pauli principle, and Uhlenbeck and Goudsmit was explicitly treated as uncertain. The analogy is retained without asserting that this was the exact chronology by which spin was discovered.

[4] Editorial clarification: The precise attribution and later history of the name *color* were left uncertain, so no firmer chronology is asserted here. A spoken reference to the Heisenberg uncertainty principle is corrected to the Pauli exclusion principle, as required by the surrounding fermion-statistics argument.

Question from the audience. Is color conserved?[a]

Response. Yes. Color is conserved, but the observable particles under discussion always have zero color, so it is a somewhat unusual-looking conservation law.

[a]Lecture timestamp: 00:57:38.

[5]

Question from the audience. Does the uud proton by itself really require all three colors? It seems sufficient merely to distinguish the two u quarks.[a]

Response. That example visibly requires some additional distinction, but it does not make the number three unavoidable. The Δ^{++}, with three up quarks of parallel ordinary spin and parallel isospin, makes the need for three distinct labels unambiguous.

[a]Lecture timestamp: 00:57:58.

This is why the Delta played such an important role. The proton and neutron have more complicated spin and flavor symmetries, so the contradiction can be obscured by the details of their wavefunctions. In the Δ^{++}, the conflict is immediate:

$$uuu, \qquad \uparrow\uparrow\uparrow, \qquad \text{one lowest spatial state.} \tag{2.57}$$

A new three-valued label is required. The existence of three color states is also confirmed independently by hadron-collision data.

A symmetry label can be meaningful even when its alternatives are equivalent in isolation. Without a chosen spatial axis, two spin directions are physically equivalent. A second object can provide a reference axis and reveal their relation. Similarly, a proton or neutron picks a direction in isospin space, and comparisons with other states reveal whether internal orientations are parallel or antiparallel. Color is subtler, but its physical content likewise appears through relations, correlations, and interaction rules rather than through visible coloring.

One could instead imagine abandoning the Pauli principle. That would require a new mathematical framework: ordinary quantum field theory classifies particles as fermions or bosons and provides no established third option that solves this problem. QCD keeps the fermionic nature of quarks and introduces color.

Question from the audience. What observable corresponds to the hidden distinction required by the Pauli principle?[a]

Response. That question is much subtler than the familiar position, momentum, or isospin observables. Its full answer requires more of the color dynamics and is deferred.

[a]Lecture timestamp: 01:01:08.

Question from the audience. Was the fermionic character of quarks inferred from the spin assignments of hadrons?[a]

Response. Yes.

[a]Lecture timestamp: 01:01:43.

[5]Editorial clarification: More precisely, asymptotic hadrons are color singlets. The phrase “zero color” is not an abelian net charge obtained by numerically adding red, green, and blue.

[6]

Question from the audience. How can one associate a particular color with one quark in a proton?[a]

Response. One does not assign a fixed color to a permanently identifiable constituent. The proton is a quantum superposition of all allowed red, green, and blue assignments, with the spin, flavor, spatial, and color factors symmetrized or antisymmetrized appropriately. The Δ^{++} is useful because its parallel spin and flavor structure exposes the need for color without requiring the full proton wavefunction.

[a] Lecture timestamp: 01:01:55.

Color is to QCD what electric charge is to electrodynamics: it is the quantum number that controls the interaction. QCD gives a highly accurate account of experimental data from hadron collisions. The ultimate justification for the framework is that its consequences agree with experiment.

2.9 Photons and Gluons

Quarks are the fermionic matter particles of QCD, analogous in this limited respect to electrons in QED. What binds them?

In electrodynamics, an electron is a source of electromagnetic field. In particle language it emits and absorbs photons, and photon exchange produces forces between charged particles. A quark is similarly a source of the gluon field and can emit and absorb gluons.

Gluons resemble photons in several important respects. They are massless spin-one particles and have photon-like polarization states. Their momentum and polarization are ordinary dynamical properties. The crucial difference is that gluons carry color information and therefore interact with one another.

Photons carry no electric charge. At the primitive QED level, one photon does not emit another photon, and two light beams pass through one another without interacting. There are very small higher-order photon-photon interactions involving electron loops, but these are secondary effects rather than a direct photon self-coupling. In empty space, the elementary QED vertex is simply an electron emitting or absorbing a photon.

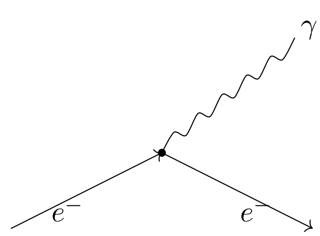

Figure 2.5: Elementary QED vertex: an electron emits or absorbs a photon.

Reversing the direction of a fermion line turns the electron interpretation into a positron interpretation. Emission, absorption, and crossed positron processes are all rearrangements of this one vertex.

[6] Editorial clarification: The wording of the two preceding audience questions is partly obscured. Their subjects and the brief responses are secure, but no more detailed formulation is inferred.

For bookkeeping, one can pretend that a photon has some of the properties of an electron–positron pair. The pair is electrically neutral, and aligned electron and positron spins can form total spin one. With an external interaction supplying the required energy and momentum, a photon can produce an electron–positron pair. None of this means that a photon is literally made from an electron and a positron. The fictitious-pair picture contributes little to QED itself, but its color analogue is useful in QCD.

2.10 Color–Anticolor Labels for Gluons

Represent the three color states of a quark as a column:

$$q = \begin{pmatrix} r \\ g \\ b \end{pmatrix}. \tag{2.58}$$

Represent the corresponding antiquark labels as a row:

$$\bar{q} = \begin{pmatrix} \bar{r} & \bar{g} & \bar{b} \end{pmatrix}. \tag{2.59}$$

A gluon is not a quark–antiquark bound state. Nevertheless, its color bookkeeping behaves as if it carried one color and one anticolor. The possible labels can therefore be arranged as

$$\begin{pmatrix} r\bar{r} & r\bar{g} & r\bar{b} \\ g\bar{r} & g\bar{g} & g\bar{b} \\ b\bar{r} & b\bar{g} & b\bar{b} \end{pmatrix}. \tag{2.60}$$

At first sight this gives nine combinations.

Question from the audience. Are the diagonal color–anticolor entries indistinguishable?[a]

Response. Not simply. The diagonal entries are distinct; the point is that their sum is not an independent quantum state.

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[a]Lecture timestamp: 01:13:37.

There are actually only eight independent gluon states. The three diagonal labels are not simply indistinguishable from one another, but the particular diagonal sum

$$r\bar{r} + g\bar{g} + b\bar{b} \tag{2.61}$$

does not furnish an additional independent gluon of QCD. The deeper explanation of this reduction is postponed. For the present color-flow arguments, it is useful to work with the nine apparent labels and remember that one diagonal combination is absent.

2.11 The Quark–Gluon Vertex

A quark can emit a gluon and emerge with either the same or a different color. If a quark enters with color a and leaves with color b, the emitted gluon carries the bookkeeping label $a\bar{b}$:

$$a \longrightarrow b + (a\bar{b}). \tag{2.62}$$

For example,

$$r \longrightarrow g + (r\bar{g}). \tag{2.63}$$

The red color line continues into the gluon, while the green line continues through the outgoing quark and reverses into an antigreen line in the gluon.

The quark need not change color. The same rule also allows

$$r \longrightarrow r + (r\bar{r}), \tag{2.64}$$

or

$$r \longrightarrow b + (r\bar{b}). \tag{2.65}$$

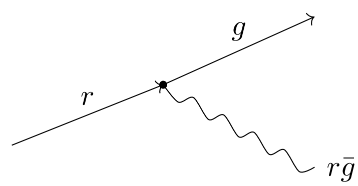

Figure 2.6: A red quark becomes green while emitting an $r\bar{g}$ gluon.

This is one primitive building block of QCD.

Question from the audience. Can a gluon be emitted only when the quark changes color?[a]

Response. No. The outgoing quark may have the same color or a different color. In either case, the emitted gluon's color–anticolor label makes the color lines continuous.

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[a]Lecture timestamp: 01:17:15.

2.12 The Three-Gluon Vertex

QCD has another primitive interaction with no elementary analogue in QED. A gluon can interact directly with other gluons.

Begin with an $r\bar{b}$ gluon. In color-flow language it may split as

$$r\bar{b} \longrightarrow r\bar{g} + g\bar{b}. \tag{2.66}$$

The red line continues into $r\bar{g}$, the antiblue line continues into $g\bar{b}$, and the green-antigreen pair joins the two outgoing gluons. This gives a three-gluon vertex.

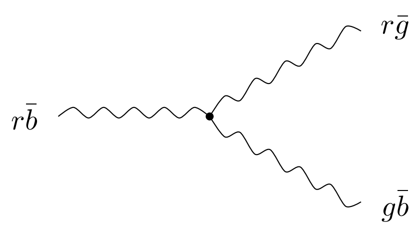

Figure 2.7: Three-gluon color flow: $r\bar{b} \to r\bar{g} + g\bar{b}$.

There is no need to memorize every allowed combination. Follow each color line through the diagram. When a fermion line reverses its time direction, reinterpret it as the corresponding antiparticle or anticolor line.

2.13 Gluon Exchange and Forces

A gluon produces forces by being exchanged, just as a photon produces electromagnetic forces by being exchanged between charged particles.

For example, a blue quark and an antiblue antiquark can exchange a $g\bar{b}$ gluon and emerge as green and antigreen. A color-flow assignment satisfying both vertices is[7]

$$b + \bar{b} \longrightarrow g + \bar{g}. \tag{2.67}$$

At the antiquark vertex,

$$\bar{b} \longrightarrow \bar{g} + (g\bar{b}), \tag{2.68}$$

while at the quark vertex,

$$b + (g\bar{b}) \longrightarrow g. \tag{2.69}$$

The exchanged gluon carries the color change from one fermion line to the other.

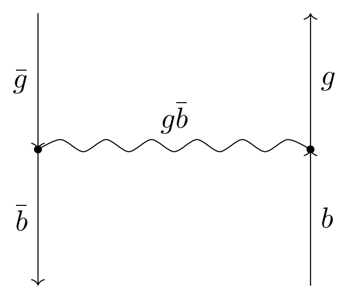

Figure 2.8: Exchange of a $g\bar{b}$ gluon changes a blue–antiblue pair into a green–antigreen pair; antiquark arrows point opposite to increasing time.

The same interaction permits forces between gluons. A consistent realization is[8]

$$(r\bar{b}) + (b\bar{r}) \longrightarrow (r\bar{g}) + (g\bar{r}), \tag{2.70}$$

with a $g\bar{b}$ gluon exchanged between the two external gluons:

$$r\bar{b} \longrightarrow r\bar{g} + g\bar{b}, \tag{2.71}$$

$$b\bar{r} + g\bar{b} \longrightarrow g\bar{r}. \tag{2.72}$$

[7] The net color change $b + \bar{b} \longrightarrow g + \bar{g}$ is explicit, but the local color labels are corrected several times. The exchanged label $g\bar{b}$ is reconstructed by requiring continuous color flow at both vertices.

[8] The lecture establishes gluon exchange between gluons and the follow-the-lines rule, but the local labels in the diagram are repeatedly revised. The displayed assignment is one consistent realization of the final rule rather than a secure transcription of every intermediate label.

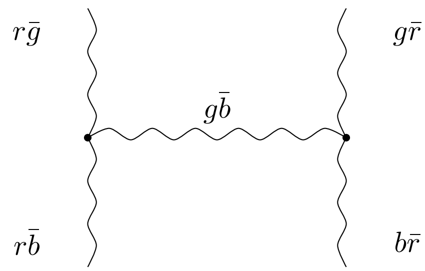

Figure 2.9: Exchange of a $g\bar{b}$ gluon between two gluons, with each color and anticolor line continuous through the vertices.

2.14 Gluon Self-Interaction and Nonlinear Fields

Gluon self-interaction changes the character of QCD. Quarks can bind by exchanging gluons and form hadrons, but gluons can also bind by exchanging gluons. QCD can therefore contain composite objects made entirely from gluonic degrees of freedom, with no quarks. There is no corresponding elementary bound state in QED made only from photons.

The field language gives the same conclusion. Two electromagnetic waves in empty space pass through one another at the elementary level. Two gluon waves interact and deform one another. Even different portions of one gluon field configuration can exert forces on each other. The gluon field is therefore nonlinear.

Question from the audience. Must the same numbers of red, green, and blue labels appear on each side of a diagram?[a]

Response. The practical rule is to follow every color line through the diagram. That continuity is color conservation. If a fermion line turns around in time, its label becomes the corresponding antiparticle or anticolor label.

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[a]Lecture timestamp: 01:24:59.

Question from the audience. Are all the color and anticolor labels in the first version of the diagram consistent?[a]

Response. No. Several labels must be corrected. In a consistent assignment, the red, antiblue, and green lines each continue through the vertices, with a reversed fermion arrow interpreted as an antiparticle.

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[a]Lecture timestamp: 01:26:00.

Question from the audience. Is an intermediate gluon line being treated as a quark–antiquark pair?[a]

Response. Only in its color bookkeeping. A gluon is not literally a quark–antiquark composite, but it carries the corresponding color and anticolor properties.

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[a]Lecture timestamp: 01:27:58.

At this stage the elementary rules are compact:

- quarks emit and absorb gluons;
- gluons interact directly with gluons;

- every color and anticolor line must continue through a vertex;
- reversing a fermion line in time changes it into the corresponding antiparticle line.

Question from the audience. Are gluons still massless when they are represented by all these color–anticolor lines?[a]

Response. Yes. Gluons remain massless. The following qualification concerns what is meant by the mass assigned to a quark inside a hadron.

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[a] Lecture timestamp: 01:28:13.

Confinement and the structure of hadrons come next: why isolated color is never observed and how quarks and gluons organize themselves inside composite particles. First, it is useful to clarify what is meant by the mass of a constituent quark.

2.15 Probe Scale and the Meaning of Quark Mass

Imagine a small object of mass m attached to a soft, deformable piece of material—a dangling piece of chewing gum will do. Apply a very gentle force, so slowly that the soft material has time to adjust and the entire system moves together. The measured inertia is then the inertia of the small object plus the attached material.

Now shake the small object at very high frequency. The soft material does not have time to follow the rapid motion. It remains nearly stationary while the small core responds. The measured high-frequency inertia is then much closer to m alone.

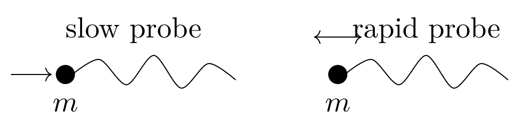

Figure 2.10: A slow probe moves the core and its soft attachment together, whereas a rapid probe initially measures mainly the core response.

This is a classical analogy. The inferred mass depends on the frequency of the applied force because different parts of the system have time to participate at different frequencies.

A quark inside a hadron is surrounded by strongly interacting gluonic material and by the other constituents. If one could pull very slowly on a quark, the entire hadron would respond. The inertia encountered would then be of order the nucleon mass,

$$m_N \sim 930\text{–}940\,\mathrm{MeV}. \tag{2.73}$$

If the quark is struck very hard and probed over a very short time, the initial response is associated with the much smaller mass parameter conventionally assigned to a light quark, roughly

$$m_{u,d} \sim 5\text{–}10\,\mathrm{MeV} \tag{2.74}$$

in the estimates being used here.

Thus the phrase "the mass of a quark" requires a specification of the scale of the probe. A gentle, low-frequency disturbance moves the extended composite; a hard, high-frequency disturbance initially resolves a much smaller part of it. This analogy is not a derivation of renormalization, but it motivates why parameters used to describe particles depend on the frequencies and wavelengths of the interactions in which they are measured. Such scale-dependent parameters are called running parameters or running constants.

Question from the audience. Is this frequency-dependent mass merely a restatement of an uncertainty relation such as $\Delta x\, \Delta t$?[a]

Response. No. The example is already a classical phenomenon. A small mass attached to something soft accelerates one way under a rapid force and another way under a slow force because different amounts of the attached material have time to respond.

[a]Lecture timestamp: 01:33:32.

Question from the audience. Does the earlier discussion of the electron's electric charge involve the same kind of scale dependence?[a]

Response. Yes. The general lesson is that the parameters used to describe particles depend on the frequencies and wavelengths of the interactions in which they are probed. They are called running constants.

[a]Lecture timestamp: 01:34:04.

Question from the audience. Does the same qualification apply to the fractional charge assigned to a quark?[a]

Response. It does. One must define carefully what is meant by the charge of a quark. At this level it is quoted relative to the proton charge; a fuller definition is a separate discussion.

[a]Lecture timestamp: 01:34:36.

[9]

Question from the audience. Why does the color bookkeeping always use a quark and an antiquark? Why could two quarks not form the corresponding object?[a]

Response. That is an important question, but its answer belongs to the next discussion of allowed color combinations and confinement.

[a]Lecture timestamp: 01:34:55.

Question from the audience. If individual quarks cannot be isolated, how were fractional charge, color, and the quark model inferred?[a]

Response. The picture emerged gradually from many clues together with apparent contradictions. Hadron patterns suggested quark constituents; the Delta states exposed a conflict with Fermi statistics; and the permanent confinement of quarks supplied another peculiar fact. The quark proposal appeared in the early 1960s, Nambu had an important early version of the color idea, and many physicists assembled and tested the full structure over roughly the following decade. It was not the result of one observation or one person.

[a]Lecture timestamp: 01:35:17.

[9]Editorial clarification: The fixed fractional electric-charge quantum number and the scale-dependent interaction coupling are distinct. Running of the coupling does not change the $+\frac{2}{3}$ or $-\frac{1}{3}$ flavor labels themselves.

Question from the audience. Why are there only eight gluons rather than the nine apparent color–anticolor combinations?[a]

Response. The detailed explanation of why the diagonal sum is not an additional gluon is deferred.

[a]Lecture timestamp: 01:37:24.

Question from the audience. Is this discussion of QCD ignoring electrons?[a]

Response. Yes. QCD is being isolated in the same way that a discussion of QED may temporarily ignore quarks. The separate sectors must ultimately be combined into a larger structure containing quarks, electrons, photons, and the other particles. Some parts of that larger synthesis have been accomplished and some remain incomplete. Physics often proceeds by isolating a subsystem, understanding it, and then putting the pieces back together.

[a]Lecture timestamp: 01:37:33.

Chapter 3

Rotations, Spinors, and Color Symmetry

3.1 Rotations as a Group

Rotations of space are symmetries, and they act on many kinds of objects. They act on rigid bodies, on velocity vectors, on vector fields, and on quantum states. An ordinary vector is the simplest place to begin because its transformation can be pictured directly.

Rotations can also be combined. Rotate an object about one axis through one angle, and then rotate it about a second axis through a second angle. The final orientation is equivalent to a single rotation about some third axis through some third angle. Thus the product of two rotations is another rotation.

This product is an operation on transformations, not an addition of the two axis vectors. The first rotation changes the object on which the second acts. If R_1 is performed first and R_2 second, the combined operation is written R_2R_1. It is itself some R_3, even though finding the axis and angle of R_3 may be inconvenient. This closure under composition is the first reason to regard rotations as a group rather than merely as a list of geometrical motions.

It is useful first to describe one rotation geometrically. Choose a unit vector $\hat{\boldsymbol{n}}$ along the rotation axis and an angle θ. We adopt the right-hand convention. A left-handed rotation through θ can equally be written as a right-handed rotation through $-\theta$, or as a right-handed rotation about $-\hat{\boldsymbol{n}}$.

A direction on the unit sphere requires two parameters, such as longitude and latitude. The angle about that direction supplies a third. Rotations therefore form a three-parameter family. We shall not yet worry about the global identifications caused by angles differing by 2π; the local parameter count is what matters here.

The axis-angle description has some redundancy. Reversing the axis while reversing the angle gives the same operation, and sufficiently large angles can be reduced modulo a full turn. None of that changes the fact that near the identity a rotation has three continuously variable parameters. That local count will later be matched against the number of independent parameters in a matrix representation.

The group properties can now be checked in turn:

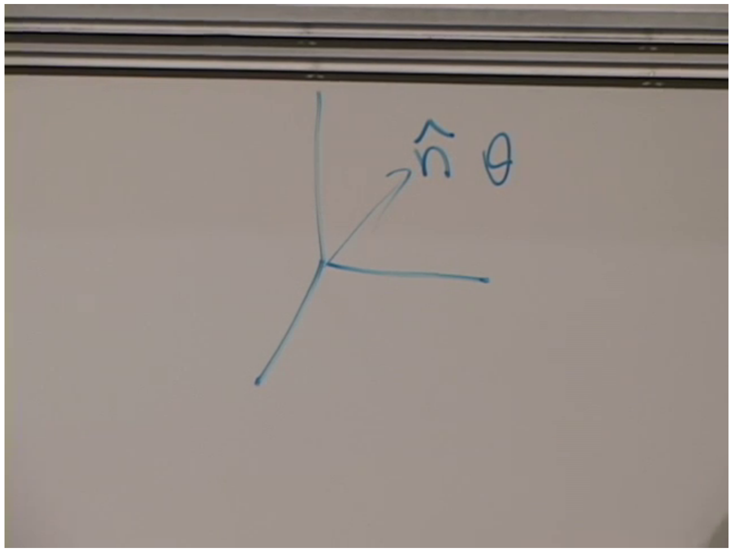

Figure 3.1: An axis $\hat{\boldsymbol{n}}$ and an angle θ specify a spatial rotation.
Lecture frame: 00:02:34.

1. *Closure.* Performing two rotations in succession produces another rotation.
2. *Identity.* Doing nothing is the identity rotation.
3. *Inverse.* A rotation through θ about $\hat{\boldsymbol{n}}$ is undone by a rotation through $-\theta$ about the same axis.
4. *Associativity.* Three successive rotations satisfy

$$(R_1R_2)R_3 = R_1(R_2R_3).$$

Rotations are not commutative. A rotation about the x-axis followed by one about the y-axis generally gives a different orientation from the same two operations in the reverse order:

$$R_x(\alpha)R_y(\beta) \neq R_y(\beta)R_x(\alpha). \tag{3.1}$$

The difference becomes second order when both angles are very small, which is why infinitesimal rotations can look almost commutative. The full group is nevertheless non-abelian.

The noncommutativity can be checked without any equations. Take a book or a box, mark one face as the front, and turn it by ninety degrees about the x-axis and then by ninety degrees about the y-axis. Return to the original orientation and reverse the order. The marked face ends in a different direction. For small angles the discrepancy is harder to see, but it has not vanished; it begins at the product of the two small angles.

Question from the audience. Are these rotations sequential rather than simultaneous?[a]

Response. Yes. Group multiplication means one operation followed by another. The order of those successive rotations is precisely what can fail to commute.

[a]Lecture timestamp: 00:04:45.

This is our first non-abelian group. The eventual application is not the color of an ordinary object but the red, green, and blue color states of a quark. That is a three-state internal system, and its interesting transformations mix the three colors. Ordinary rotations provide a familiar model of how a continuous group acts through matrices.

A group by itself is abstract. To use it in physics we must say what objects it acts on. A emphrepresentation assigns an operator or matrix to each group element so that multiplication of group elements is reproduced by multiplication of the assigned matrices. Different physical objects can carry different representations of the same group. Vectors, spin-one states, and spin-one-half states will provide three examples.

3.2 The Three-Dimensional Vector Representation

Let an ordinary vector have components

$$\boldsymbol{V} = (V_x, V_y, V_z), \qquad V_i, \quad i = 1, 2, 3. \tag{3.2}$$

One may rotate the physical vector while holding the axes fixed, or rotate the coordinate axes while holding the vector fixed. The conventions differ, but in either description the components are mixed by a 3×3 matrix. For every axis-angle pair $(\hat{\boldsymbol{n}}, \theta)$, write

$$V_i' = R_{ij}(\hat{\boldsymbol{n}}, \theta) V_j, \tag{3.3}$$

where a repeated index is summed.

The repeated index j is a dummy label. Equation (3.3) means explicitly

$$V_i' = R_{i1} V_1 + R_{i2} V_2 + R_{i3} V_3 \tag{3.4}$$

for each output component i. The free index i labels which component is being computed; the repeated j is summed and may be renamed without changing the equation. Keeping those two roles separate prevents most index errors in the length-preservation argument.

A true rotation preserves length. For the moment both V_i and R_{ij} are real, so

$$V_i' V_i' = V_i V_i \tag{3.5}$$

for every vector. To expose the matrix condition, write the same transformation with different dummy indices:

$$V_i' = R_{ij} V_j, \qquad V_i' = R_{ik} V_k. \tag{3.6}$$

Multiplying and summing over i gives

$$V_i' V_i' = R_{ij} V_j R_{ik} V_k \tag{3.7}$$

$$= R_{ij} R_{ik} V_j V_k. \tag{3.8}$$

This equals $V_j V_j$ for every choice of V only if

$$R_{ij} R_{ik} = \delta_{jk}. \tag{3.9}$$

In matrix notation,

$$R^T R = \mathbf{1}, \tag{3.10}$$

and hence

$$R^{-1} = R^T. \tag{3.11}$$

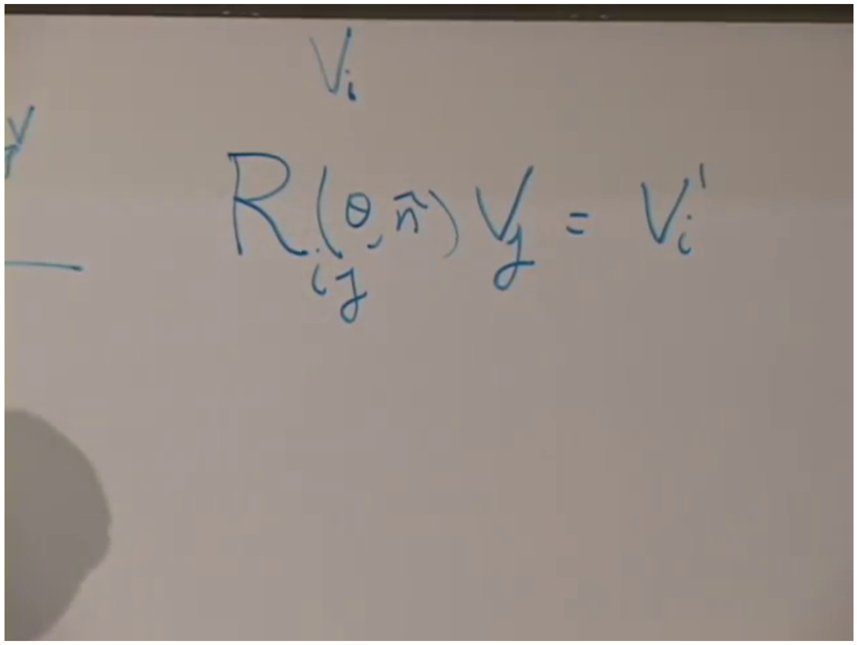

Figure 3.2: A rotation matrix maps the original vector components to the rotated components.
Lecture frame: 00:07:41.

Transposition simply interchanges rows and columns.

Why must the coefficient be δ_{jk}? The Kronecker delta sets $j = k$ under the sum:

$$\delta_{jk} V_j V_k = V_1^2 + V_2^2 + V_3^2. \tag{3.12}$$

If $R_{ij} R_{ik}$ differed from δ_{jk}, some direction or some cross term $V_j V_k$ would change the length. Since the equality is required for every possible vector, not merely for one chosen vector, the matrix coefficient itself must equal the identity.

Equation (3.10) is the defining orthogonality condition for this real matrix representation. Multiplying the matrices reproduces the multiplication table of spatial rotations. This is not the only representation of the rotation group; it is merely the one most easily visualized.

The inverse relation has an immediate geometrical meaning. A rotation can be undone without solving a new matrix equation: exchange rows and columns. If R_1 and R_2 represent two rotations, then $R_2 R_1$ represents their successive action and is again orthogonal. Thus the matrix operations preserve the same closure, identity, inverse, and associative structure established geometrically.

3.3 Spin One and the Vector Representation

The three-dimensional representation also acts on quantum states. A spin-one particle has three possible spin projections,

$$m = +1, \quad 0, \quad -1. \tag{3.13}$$

Its spin state is therefore a three-component column of probability amplitudes. Rotating the particle, or changing the orientation of a magnetic field used to define the spin axis, mixes those three amplitudes by a three-dimensional representation of the rotation group.

The number of spin states follows the familiar rule $2s+1$. Spin zero has one state, spin one-half has two, spin one has three, and spin three-halves has four. The dimension of the state vector therefore changes with the spin. The same physical rotation must be represented by matrices of different sizes when it acts on these different state spaces.

For a spin-one particle, choose the basis

$$|1,+1\rangle, qquad|1,0\rangle, qquad|1,-1\rangle. \tag{3.14}$$

A state is

$$|\psi\rangle = c_+|1,+1\rangle + c_0|1,0\rangle + c_-|1,-1\rangle. \tag{3.15}$$

Rotating the apparatus does not create a fourth spin state; it mixes the three coefficients c_+, c_0, c_- among themselves. The corresponding rotation operator must therefore be a 3×3 matrix.

For spin one, this representation transforms in the same way as an ordinary spatial vector. This is the precise meaning of calling a spin-one particle a *vector particle*: its three spin amplitudes and the three components of a vector obey the same rotation law. The quantum state need not be imagined as a literal arrow; the statement concerns how its components mix.

This correspondence is not accidental. A three-vector and a spin-one state carry equivalent three-dimensional representations of the rotation group. The words emphvector particle summarize that transformation property. They do not claim that a quantum polarization state is a tiny rigid arrow hidden inside the particle.

3.4 Discrete and Continuous Symmetries

Before turning to spin one-half, it helps to distinguish two broad kinds of groups. A reflection gives a simple discrete example. There is the identity and there is the reflection P; one either flips or does not flip. By contrast, a phase

$$e^{i\phi} \tag{3.16}$$

depends continuously on ϕ, and a spatial rotation depends continuously on its axis and angle. Such continuous groups are called *Lie groups*, after Sophus Lie; the name is pronounced *Lee*. Continuous symmetries will be the main concern, although discrete symmetries remain important.

For the two-element reflection group,

$$P^2 = 1. \tag{3.17}$$

There is no continuously adjustable amount of reflection between 1 and P. A phase group is different: between $e^{i\phi_1}$ and $e^{i\phi_2}$ lie continuously many other phases. Spatial rotations have the same continuous character, but unlike $U(1)$ they are non-abelian.

Spin zero gives the simplest representation of the rotation group. A spin-zero state is unchanged by every rotation, so every group element is represented by the one-dimensional identity:

$$D^{(0)}(R) = 1. \tag{3.18}$$

The group itself still has infinitely many rotations; it is the representation on the spin-zero state that is trivial.

This distinction between a group and one of its representations is essential. The rotation group does not collapse to a one-element group whenever it acts on a scalar. Rather, every one of its elements is represented by the same number 1 on that particular scalar state.

3.5 A General Spin-One-Half State

A spin-one-half system has two basis states. Relative to a chosen z-axis,

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.19}$$

A general state is a two-component spinor,

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}, \qquad |\alpha_1|^2 + |\alpha_2|^2 = 1. \tag{3.20}$$

The entries are complex. The choice $(\alpha_1, \alpha_2) = (1, 0)$ is spin up, and $(0, 1)$ is spin down. A general normalized combination describes a spin polarized along some other direction.

The words emphup and emphdown have meaning only after an axis has been chosen. Here that axis is z. A different measuring axis has a different pair of basis states, and a state that is definite up along one axis generally appears as a superposition in another basis. Rotating the physical spin or rotating the reference axes changes the coefficients accordingly.

Two complex amplitudes contain four real parameters. Normalization removes one and the physically irrelevant overall phase removes another, leaving two real parameters. That is exactly the number needed to choose a direction on a sphere. We need not derive the explicit map between (α_1, α_2) and that direction in order to study rotations.

The overall phase is removed because

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \quad \text{and} \quad e^{i\chi} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \tag{3.21}$$

represent the same physical ray. Normalization fixes the total probability, not a direction in ordinary Euclidean space. After both identifications, the remaining two parameters may be regarded as the polar and azimuthal angles of the spin-polarization direction.

When the spin is rotated, its state becomes

$$\alpha_i' = U_{ij}\alpha_j, \qquad \begin{pmatrix} \alpha_1' \\ \alpha_2' \end{pmatrix} = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}. \tag{3.22}$$

Thus spatial rotations possess a two-dimensional matrix representation acting on spinors. There is a global two-valuedness in this representation that we defer; locally, the matrices have the transformation properties needed below.

The transformed pair (α_1', α_2') must again be a possible quantum state. Therefore the first question about U is not its explicit axis-angle formula but the condition that it preserve the normalization of every input spinor.

3.6 Probability Preservation and Unitary Matrices

A rotation must map a normalized state to another normalized state:

$$\alpha_i^* \alpha_i = \alpha_i'^* \alpha_i' = 1. \tag{3.23}$$

Substituting Equation (3.22),

$$\alpha_i'^* \alpha_i' = (U_{ij}\alpha_j)^*(U_{ik}\alpha_k) \tag{3.24}$$
$$= \alpha_j^* U_{ij}^* U_{ik}\alpha_k. \tag{3.25}$$

The analogue of the real-vector condition is therefore

$$(U^T)^* U = \mathbf{1}. \tag{3.26}$$

The transpose followed by complex conjugation is the Hermitian conjugate,

$$U^\dagger = (U^T)^*, \tag{3.27}$$

so the normalization-preserving condition is

$$U^\dagger U = \mathbf{1}. \tag{3.28}$$

The extra complex conjugation is the only essential change from the real vector calculation. For a real vector, $V_i V_i$ is a length. For a spinor, $\alpha_i^* \alpha_i$ is a probability. Complex conjugation is required to make that probability real and nonnegative, so transposition is replaced by Hermitian conjugation.

Equation (3.28) also gives the inverse immediately:

$$U^{-1} = U^\dagger. \tag{3.29}$$

The inverse rotation therefore preserves the same probability norm. As with ordinary rotation matrices, products of unitary matrices remain unitary, so the representation respects group multiplication.

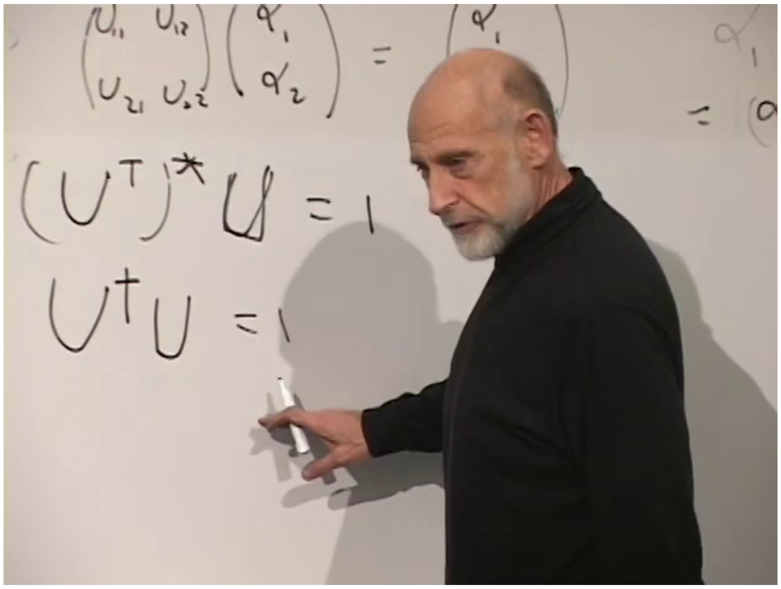

Figure 3.3: The spinor rotation matrix obeys $U^\dagger U = \mathbf{1}$.
Lecture frame: 00:27:41.

Matrices satisfying Equation (3.28) are unitary. A general 2×2 complex matrix begins with eight real parameters. Unitarity is a 2×2 matrix equation that supplies four independent real constraints, leaving four parameters. A spatial rotation has only three. One more restriction is needed to isolate the spinor representation of rotations from all of $U(2)$.

The four constraints can also be seen by viewing the two columns of U as two complex vectors. Each column must have unit norm, providing two real conditions, and the columns must be orthogonal, providing one complex or two real conditions. Altogether that is four. The remaining fourth parameter in $U(2)$ will turn out to be an overall determinant phase rather than a new spatial rotation.

3.7 Determinant One and $SU(2)$

The additional condition is

$$\det U = 1. \tag{3.30}$$

It is compatible with group multiplication because

$$\det(AB) = \det A \, \det B. \tag{3.31}$$

The product of two determinant-one unitary matrices again has determinant one, and the identity has determinant one. These matrices form a closed subgroup.

The inverse is also in the subgroup. If $\det U = 1$, then

$$\det U^{-1} = \frac{1}{\det U} = 1. \tag{3.32}$$

Thus imposing determinant one does not remove the identity, spoil closure, or lose inverses. It consistently selects a group with one fewer continuous parameter.

Unitarity alone implies

$$\det(U^\dagger U) = (\det U)^* \det U = |\det U|^2 = 1. \tag{3.33}$$

It does not force $\det U$ to be $+1$ or -1. A general unitary determinant may be any phase:

$$\det U = e^{i\phi}, \qquad \det U^\dagger = e^{-i\phi}. \tag{3.34}$$

Condition (3.30) fixes this otherwise free phase and removes one real parameter.

Question from the audience. Does $U^\dagger U = \mathbf{1}$, without the determinant-one condition, force $\det U^\dagger$ to equal $\det U$ or to be only $+1$ or -1?[a]

Response. No. The two determinants are complex conjugates. They can be $e^{i\phi}$ and $e^{-i\phi}$, whose product is one. The special-unitary restriction then fixes both to one.

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[a] Lecture timestamp: 00:31:21.

The unitary $n \times n$ matrices form $U(n)$. For $n = 1$, a unitary matrix is simply the phase $e^{i\phi}$, giving $U(1)$. The determinant-one subgroup is the *special unitary group*,

$$SU(n) = \{U \in U(n) : \det U = 1\}. \tag{3.35}$$

The two-component spinor is acted on by $SU(2)$, a three-parameter group.

The notation records both restrictions: U means unitary, while S means special, namely determinant one. These special-unitary groups occur repeatedly because they preserve a complex inner product while excluding the independent common phase carried by the determinant.

Locally, $SU(2)$ has the same rotation algebra and multiplication structure needed to rotate spin-one-half states. Globally, one must remember the deferred subtlety: $SU(2)$ is the double cover of the ordinary rotation group $SO(3)$, so two elements of $SU(2)$ correspond to one spatial rotation. [1]

3.8 Infinitesimal Rotations and the Pauli Matrices

Question from the audience. Where do the spin matrices, or Pauli matrices, enter this construction?[a]

Response. They appear when a group element is taken very close to the identity, corresponding to a rotation through a very small angle. They are the generators of those infinitesimal rotations.

[a]Lecture timestamp: 00:33:16.

Begin with

$$U = \mathbf{1} + \epsilon M + O(\epsilon^2), \tag{3.36}$$

where ϵ is real and small. Then

$$U^\dagger = \mathbf{1} + \epsilon M^\dagger + O(\epsilon^2), \tag{3.37}$$

and unitarity gives

$$U^\dagger U = \mathbf{1} + \epsilon(M^\dagger + M) + O(\epsilon^2). \tag{3.38}$$

To first order,

$$M^\dagger = -M. \tag{3.39}$$

Thus the matrix M in Equation (3.36) is anti-Hermitian.

Only terms linear in ϵ are being retained. The product $\epsilon^2 M^\dagger M$ is discarded because it is of second order. This is the advantage of working close to the identity: nonlinear group conditions become linear conditions on their generators.

It is more convenient to extract a factor of i:

$$U = \mathbf{1} + i\epsilon H + O(\epsilon^2). \tag{3.40}$$

Complex conjugation changes i to $-i$, so

$$U^\dagger = \mathbf{1} - i\epsilon H^\dagger + O(\epsilon^2). \tag{3.41}$$

Now

$$U^\dagger U = \mathbf{1} + i\epsilon(H - H^\dagger) + O(\epsilon^2), \tag{3.42}$$

and first-order unitarity requires

$$H^\dagger = H. \tag{3.43}$$

With the factor of i exposed, the generator is Hermitian.

[1]Editorial clarification: The double-cover formulation makes precise the two-valuedness mentioned but not developed in the source.

Multiplication by i has not changed the allowed infinitesimal matrices; it has only changed the convention used to name the generator. Physics usually chooses Hermitian generators because their eigenvalues are real and because the associated observables, such as angular momentum, are Hermitian.

The determinant-one condition supplies tracelessness[2]. For a small 2×2 matrix A,

$$\det(\mathbf{1} + A) = \det \begin{pmatrix} 1 + A_{11} & A_{12} \\ A_{21} & 1 + A_{22} \end{pmatrix}. \tag{3.44}$$

Expanding,

$$\det(\mathbf{1} + A) = (1 + A_{11})(1 + A_{22}) - A_{12}A_{21} \tag{3.45}$$
$$= 1 + A_{11} + A_{22} + O(A^2) \tag{3.46}$$
$$= 1 + \operatorname{tr} A + O(A^2). \tag{3.47}$$

Applying this to $A = i\epsilon H$,

$$\det U = 1 + i\epsilon \operatorname{tr} H + O(\epsilon^2). \tag{3.48}$$

Therefore $\det U = 1$ requires

$$\operatorname{tr} H = 0. \tag{3.49}$$

The trace condition removes precisely the generator proportional to the identity matrix. A general Hermitian 2×2 matrix has four real parameters: two real diagonal entries and one complex off-diagonal entry. Its trace fixes one real combination, leaving three. This matches the three axes about which an infinitesimal spatial rotation can be made.

The infinitesimal generators of $SU(2)$ are thus traceless Hermitian 2×2 matrices. The three Pauli matrices form a basis:

$$\begin{aligned} \sigma_x &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \\ \sigma_z &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned} \tag{3.50}$$

Every generator can be written

$$H = a_x\sigma_x + a_y\sigma_y + a_z\sigma_z. \tag{3.51}$$

A small rotation about the x-axis selects σ_x; one about the y-axis selects σ_y; and a rotation about any other axis selects the corresponding linear combination. In units with $\hbar = 1$, the spin operators are

$$S_i = \frac{\sigma_i}{2}. \tag{3.52}$$

For example, an axis halfway between x and y uses a generator proportional to

$$\frac{\sigma_x + \sigma_y}{\sqrt{2}}. \tag{3.53}$$

The direction of the coefficient vector (a_x, a_y, a_z) selects the rotation axis, while its magnitude combines with ϵ to set the small angle. This is the connection between the spin matrices encountered in angular-momentum commutators and finite rotation operators. In particular,

$$[S_x, S_y] = iS_z \tag{3.54}$$

[2] Editorial clarification: The spoken determinant expansion briefly garbles the subscripts. The corrected linear term is $A_{11} + A_{22} = \operatorname{tr} A$; off-diagonal products begin at quadratic order.

and its cyclic partners encode the noncommutativity already seen geometrically. The standard finite rotation follows by exponentiating the infinitesimal generator[3],

$$U(\hat{\boldsymbol{n}}, \theta) = \exp\left(-\frac{i\theta}{2}\,\hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma}\right). \tag{3.55}$$

3.9 A Three-State Color Space

Combining representations can wait. We now leave spatial spin and return to quark color. A quark still has spin one-half, but its color is an independent three-state degree of freedom:

$$|r\rangle, \qquad |g\rangle, \qquad |b\rangle. \tag{3.56}$$

A general color state is

$$\begin{aligned} |q\rangle_{\text{color}} &= \alpha_r|r\rangle + \alpha_g|g\rangle + \alpha_b|b\rangle, \\ |\alpha_r|^2 + |\alpha_g|^2 + |\alpha_b|^2 &= 1. \end{aligned} \tag{3.57}$$

Equivalently,

$$q = \begin{pmatrix} \alpha_r \\ \alpha_g \\ \alpha_b \end{pmatrix}. \tag{3.58}$$

The squared magnitudes are the probabilities for red, green, and blue in a chosen color basis. The same triplet notation can assemble three quark field operators,

$$\psi = \begin{pmatrix} \psi_r \\ \psi_g \\ \psi_b \end{pmatrix}. \tag{3.59}$$

Spin and color must be kept conceptually separate. The spin label tells us how the state transforms under rotations of physical space. The color label tells us how it transforms under an internal symmetry that does not rotate the laboratory. A quark state carries both labels, even though the present discussion suppresses spin in order to concentrate on color.

Writing three field operators in one column is more than bookkeeping. It makes it possible to state the symmetry in one matrix equation. A transformation may turn a red field into a linear combination of red, green, and blue fields, just as a spatial rotation turns one vector component into a linear combination of the other components.

One can first imagine discrete transformations that exchange two colors. The larger symmetry contains continuous mixtures of all three. A simple interchange matrix may need an accompanying phase to make its determinant equal to one; the exchange is then embedded in the special-unitary group.

For two basis states the bare exchange is represented by

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \det X = -1. \tag{3.60}$$

[3] Editorial clarification: The exponential finite-rotation formula is the standard completion of the infinitesimal construction; the source derives only the generator conditions and their Pauli-matrix basis.

Multiplying the whole matrix by i gives

$$\det(iX) = i^2 \det X = 1. \tag{3.61}$$

The discrete exchange can therefore appear as a particular element of the larger continuous group once the phase is chosen consistently.

Question from the audience. Does a matrix that merely interchanges two basis states have the wrong determinant for $SU(2)$?[a]

Response. The bare interchange has determinant -1. An appropriate factor of i supplies a determinant-one representative, so the exchange can be included within the continuous special-unitary transformations.

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[a] Lecture timestamp: 00:43:46.

3.10 $SU(3)$ and the Eight Color Generators

The continuous color transformation is

$$\begin{aligned} q' = Uq, \quad U^\dagger U = \mathbf{1}, \\ \det U = 1, \qquad U \in SU(3). \end{aligned} \tag{3.62}$$

The mathematics is the same special-unitary construction used for spinors, but now U is 3×3. This is not a rotation of physical space. It is a rotation in the internal color space.

Unitarity preserves the color-state normalization:

$$q^\dagger q = |\alpha_r|^2 + |\alpha_g|^2 + |\alpha_b|^2. \tag{3.63}$$

Determinant one removes the common $U(1)$ phase direction from the continuous mixing. The resulting matrices can perform simple exchanges, relative phase changes, and genuinely continuous superpositions while remaining in one closed group.

The parameter count is immediate. A 3×3 complex matrix has nine complex entries, or eighteen real parameters. The matrix equation $U^\dagger U = \mathbf{1}$ imposes nine real conditions. Determinant one removes one more:

$$\dim SU(3) = 18 - 9 - 1 = 8. \tag{3.64}$$

Thus $SU(3)$ has eight independent infinitesimal generators.

The nine real unitarity conditions may again be understood in terms of the columns of U. Three complex columns must each have unit norm, and every pair must be orthogonal. These conditions remove nine of the original eighteen real parameters. The determinant phase removes the tenth, leaving eight. This is the three-state analogue of the $8 - 4 - 1 = 3$ count for $SU(2)$.

The appearance of eight is the same fact that ultimately produces eight gluon species. Saying that there are not nine gluons because the determinant restriction removed one direction is only a mnemonic. More precisely, the nine color-anticolor matrices decompose into one color-singlet direction, proportional to the identity, and eight traceless directions. The eight traceless generators form the adjoint color structure carried by the QCD gluons.[4]

[4] Editorial clarification: The singlet-plus-octet formulation is the standard precise version of the source's intentionally heuristic determinant remark; its gauge-field construction is deferred.

At this stage the relation to eight gluons is a signpost, not yet a complete derivation. To obtain gauge particles one must promote the color symmetry to a local one and determine the compensating fields. What the parameter count has already shown is that there are eight independent directions in which an $SU(3)$ color transformation can begin to move away from the identity.

3.11 Color Symmetry Constrains QCD

The basic symmetry statement is that the physics is invariant under the color transformation in Equation (3.62). This is the central internal symmetry of quantum chromodynamics. Mixing the color components is analogous in form to mixing spin-up and spin-down amplitudes, and those spinor transformations are in turn related to rotations of spatial directions, but the three spaces must not be confused.

Calling color a symmetry means that changing the color basis does not change physical predictions. Red, green, and blue are convenient names for three basis vectors, not three observable pigments. A simultaneous $SU(3)$ rotation of the color components must leave the form of the laws unchanged. This invariance is not an optional aesthetic feature; it is the organizing principle of QCD.

Symmetry restricts the Lagrangian. In the earlier $U(1)$ example, invariance under a common phase transformation constrained allowed terms to contain balanced factors of a field and its conjugate. In QCD, every term must likewise be invariant under the $SU(3)$ color transformation. That requirement has consequences for interactions, conserved quantities, and the form of the gluon field.

Suppose a proposed term carries an exposed color index. Under Equation (3.62) that index rotates, so the term cannot by itself be invariant. Allowed terms must contract color indices into invariant combinations or use gauge fields whose transformations compensate the change. This is the practical force of the symmetry principle: it rules out arbitrary couplings and organizes the interactions that remain.

The full construction of QCD requires one further step: the color transformation must be allowed to vary from point to point, and the gauge field must compensate that local change.[5] That development belongs to the next stage. For now the essential structure is in place: a color triplet acted on by a special-unitary three-state symmetry with eight generators.

The calculation has gone as far into group theory as is needed for the moment. We began with rotations that can be drawn in ordinary space, passed to the matrices that act on vectors and spinors, and then reused the same logic in an internal color space. The next step can return to particle physics with the mathematical framework already available, rather than asking the formalism to carry the entire physical story at once.

[5]Editorial clarification: Localizing the color symmetry is the standard gauge-theory completion of the global $SU(3)$ discussion; it is not developed in this source lecture.

Chapter 4

Color Representations, Hadrons, and Confinement

A brief opening interlude. Before returning to group theory, allow one shameless commercial. *The Black Hole War* tells the story of the long argument with Stephen Hawking over whether black holes destroy quantum information. It also records some of the cruelty peculiar to theoretical physicists: not only Schrodinger's cat, but Susskind's fish and their unfortunate trip down the toilet. With that duty discharged, we return to Alice and Bob. Alice is still suffering, because Bob is still making her learn group theory.

Overview. A symmetry becomes physical through its action on states. We begin with rotations to introduce representations and generators, then repeat the construction for color $SU(3)$. Color representations combine into hadrons, and their dynamics explain why isolated color is never observed.

4.1 The Group and the Things It Acts On

There are two ideas in every symmetry discussion. First there is an abstract group: a collection of operations with a multiplication law, an identity, and inverses. Second there are the objects on which those operations act. The two ideas can be distinguished, but in physics they cannot be separated. A symmetry matters because it transforms physical states, fields, coordinates, or observables.

Question from the audience. A group has been described both as an abstract multiplication table and as matrices acting on physical objects. Can those two parts be dissociated? [a]

Response. They are different levels of description, but they are completely related. A rotation exists as an abstract operation before we specify what it rotates. Physics then asks how that same operation acts on the states and quantities relevant to a particular system. The matrices supply that concrete action while preserving the abstract multiplication law.

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[a] Lecture timestamp: 00:01:58.

Spatial rotations provide the cleanest example. Classically, the same rotation may act on a particle's coordinates, on the direction of a magnet, or on any other vector. In quantum mechanics, states live in a linear vector space, so a rotation is represented by a linear operator on that space. If g denotes an abstract group element and $D(g)$ its matrix representative, consistency means

$$D(g_1 g_2) = D(g_1)D(g_2), \qquad D(e) = \mathbf{1}. \tag{4.1}$$

The matrices need not look like the abstract operations, but their multiplication must reproduce the group multiplication. [1]

The dimension of the matrix is determined by the state space, not by the abstract group alone. If we retain only the spin of one electron, there are two orthogonal possibilities, spin up and spin down, and a state is a two-component column vector. Rotations therefore act through 2×2 matrices. For two spin-one-half objects there are four basis states,

$$|\uparrow\uparrow\rangle\,, \qquad |\uparrow\downarrow\rangle\,, \qquad |\downarrow\uparrow\rangle\,, \qquad |\downarrow\downarrow\rangle\,, \tag{4.2}$$

and the induced action on the combined space is four-dimensional.

The familiar sequence is

spin	number of states	matrix dimension
0	1	1×1
$\frac{1}{2}$	2	2×2
1	3	3×3
$\frac{3}{2}$	4	4×4

(4.3)

and so forth. A spin-zero state furnishes the trivial representation: every rotation leaves it alone. The one-by-one matrix is simply 1. A spin-one state has three components, and its rotation matrices are the familiar three-dimensional vector rotations. More generally, spin j has $2j + 1$ states.

The representation can also be infinite-dimensional. If we describe the full orbital motion of an electron rather than only its spin, a state contains an amplitude for every position, or equivalently for every momentum. The rotation operator then acts on an infinite-dimensional space of wavefunctions. Thus one and the same abstract rotation group has trivial, two-dimensional, three-dimensional, and infinite-dimensional representations. This is the fact we will reuse when the discussion moves from spin to color.

4.2 Why Spin Rotations Are Special Unitary

Return now to a single electron spin and discard its orbital motion. A rotation is represented by a 2×2 unitary matrix U,

$$U^\dagger U = \mathbf{1}, \qquad U^{-1} = U^\dagger. \tag{4.4}$$

The inverse transformation must undo the rotation, and unitarity preserves the norm, hence every probability.

A general 2×2 complex matrix contains four complex entries, or eight real parameters. Equation (4.4) imposes four independent real conditions, leaving four parameters. An ordinary spatial rotation has only three: two parameters choose the axis and one chooses the angle. The unitary matrix therefore contains one phase too many. The additional restriction is

$$\det U = 1. \tag{4.5}$$

[1] Editorial clarification: The homomorphism equation writes explicitly the correspondence between group multiplication and matrix multiplication described verbally in the source.

The count is then

$$8 - 4 - 1 = 3. \tag{4.6}$$

The determinant-one restriction is stable under multiplication because

$$\det(AB) = \det A \, \det B. \tag{4.7}$$

If two matrices each have determinant one, so does their product. It is also compatible with unitarity. Taking the determinant of Equation (4.4) gives

$$1 = \det(U^\dagger U) = \det(U^\dagger) \det U = (\det U)^* \det U. \tag{4.8}$$

Consequently the determinant of a unitary matrix lies on the unit circle,

$$\det U = e^{i\phi}. \tag{4.9}$$

Question from the audience. Does unitarity require the determinant to be real? [a]

Response. No. Unitarity requires only unit modulus. The determinant may be any pure phase $e^{i\phi}$. For a two-dimensional matrix, multiplying U by $e^{-i\phi/2}$ removes that common phase and produces a determinant-one matrix.

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[a] Lecture timestamp: 00:12:10.

Indeed, if

$$V = e^{-i\phi/2} U, \tag{4.10}$$

then V remains unitary and

$$\det V = e^{-i\phi} \det U = 1. \tag{4.11}$$

The determinant-one unitary matrices form $SU(2)$.

There is one fine point. It is tempting to say that $SU(2)$ is simply the group of rotations in three-dimensional space, but the correspondence is two-to-one. The matrices U and $-U$ act differently on a spinor, yet they correspond to the same rotation of an ordinary vector. Thus

$$SU(2)/\{\mathbf{1}, -\mathbf{1}\} \simeq SO(3). \tag{4.12}$$

For the present discussion the local multiplication structure and the three generators are what matter, but the double cover explains the two-valuedness of spin-one-half rotations.

4.3 Infinitesimal Transformations and Their Generators

A finite rotation can be built from many very small rotations. A group element close to the identity is called infinitesimal, and knowing all infinitesimal directions is enough, in a connected group, to reconstruct finite transformations by repeated multiplication. More importantly for physics, the operators that generate these infinitesimal transformations are the conserved quantities associated with the symmetry.

Write a near-identity element as

$$U = \mathbf{1} + i\epsilon T, \qquad |\epsilon| \ll 1. \tag{4.13}$$

The factor of i is a convenient convention. Substituting into unitarity and discarding terms of order ϵ^2 gives

$$(\mathbf{1} + i\epsilon T)(\mathbf{1} - i\epsilon T^\dagger) = \mathbf{1}, \tag{4.14}$$
$$i\epsilon(T - T^\dagger) = 0, \tag{4.15}$$
$$T = T^\dagger. \tag{4.16}$$

The generator is Hermitian, as a quantum-mechanical observable must be.

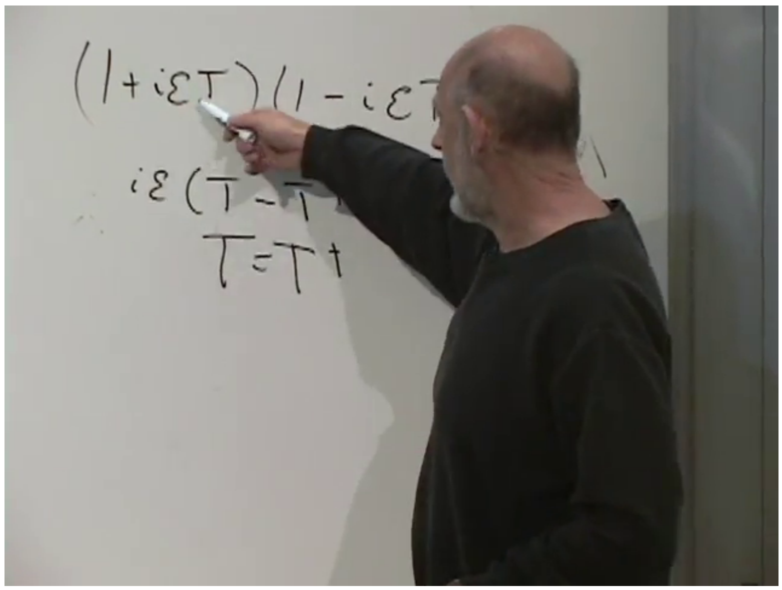

Figure 4.1: The near-identity unitarity calculation ending in $T = T^\dagger$.
Lecture frame: 00:23:28.

For spatial rotations there are three linearly independent generators because there are three independent infinitesimal axes. A small rotation about an arbitrary direction is assembled from small rotations about x, y, and z. The corresponding generators are the angular momentum operators J_x, J_y, J_z. Their commutators encode the noncommutativity of rotations,

$$[J_i, J_j] = i\epsilon_{ijk} J_k. \tag{4.17}$$

Repeated infinitesimal rotations then build a finite one.

We have used unitarity but have not yet used determinant one. For any small matrix m,

$$\det(\mathbf{1} + \epsilon m) = 1 + \epsilon \operatorname{tr} m + O(\epsilon^2). \tag{4.18}$$

Applying Equation (4.18) to Equation (4.13),

$$\det U = 1 + i\epsilon \operatorname{tr} T + O(\epsilon^2). \tag{4.19}$$

The special-unitary condition therefore requires

$$\operatorname{tr} T = 0. \tag{4.20}$$

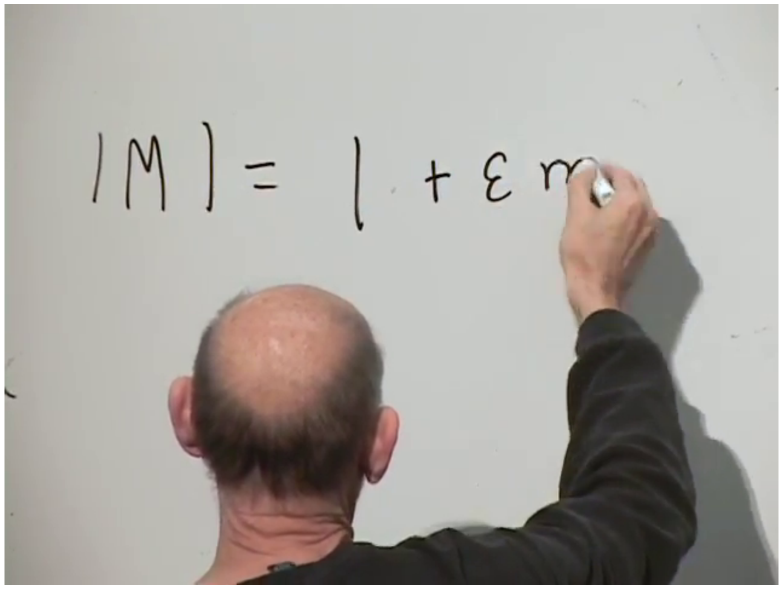

Figure 4.2: The first-order determinant expansion as it begins on the board.

Lecture frame: 00:21:51.

The generators of $SU(2)$ are precisely the traceless Hermitian 2×2 matrices. That vector space has dimension three. A standard basis is supplied by the Pauli matrices,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \\ \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{4.21}$$

For a spin-one-half representation one may take $J_i = \sigma_i/2$, so a finite rotation through angle θ around $\hat{\boldsymbol{n}}$ is

$$U(\hat{\boldsymbol{n}}, \theta) = \exp(-i\theta\, \hat{\boldsymbol{n}} \cdot \boldsymbol{\sigma}/2)\,. \tag{4.22}$$

[2]

Question from the audience. So there are three independent traceless Hermitian 2×2 matrices? [a]

Response. Yes. They are the three Pauli matrices. For 3×3 special-unitary matrices the corresponding count is eight, and a conventional basis is given by the eight Gell-Mann matrices.

[a]Lecture timestamp: 00:24:07.

Question from the audience. Is that eight the same "eight" as the eightfold way and the eight gluons? [a]

Response. The mathematics is the same kind of $SU(3)$ representation theory, but one must specify which $SU(3)$ symmetry is acting. The historical eightfold-way flavor symmetry is not color $SU(3)$. The eight gluons belong to the adjoint representation of color $SU(3)$.

[a]Lecture timestamp: 00:24:43.

[2]Editorial clarification: The finite exponential is the standard completion of the infinitesimal-generator discussion. The source emphasizes repeated small rotations rather than writing this formula explicitly.

4.4 From SU(2) to Color SU(3)

Now increase the matrices from 2×2 to 3×3. A general complex 3×3 matrix has nine complex entries, hence eighteen real parameters. The condition $U^\dagger U = \mathbf{1}$ supplies nine real constraints, and $\det U = 1$ removes one more. Therefore

$$18 - 9 - 1 = 8. \tag{4.23}$$

There are eight independent infinitesimal directions and eight traceless Hermitian generators. Unlike $SU(2)$, this group is not the rotation group of ordinary three-dimensional space. It is an internal symmetry acting on color.

Before assigning those representations to quarks, recall how composite representations arise. Put two spin-one-half systems on top of one another and ignore their orbital motion. There are four product states, but a rotation does not mix all four irreducibly. Three combinations transform as spin one,

$$|1,1\rangle = |\uparrow\uparrow\rangle\,, \tag{4.24}$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \tag{4.25}$$

$$|1,-1\rangle = |\downarrow\downarrow\rangle\,, \tag{4.26}$$

while the fourth transforms as spin zero,

$$|0,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \tag{4.27}$$

Thus

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{3} \oplus \mathbf{1}. \tag{4.28}$$

The aligned states give the $m = \pm 1$ members of the triplet, the symmetric zero-projection state completes it, and the antisymmetric combination is the singlet. This is the general logic behind adding angular momenta and behind combining color representations.

The same singlet–triplet distinction occurs whenever two spin-one-half degrees of freedom are coupled. In atomic hydrogen the electron and proton spins form hyperfine singlet and triplet states. Ortho- and para-hydrogen use the same representation theory for the nuclear spins of the two protons in an H_2 molecule. [3]

4.4.1 The Triplet and the Anti-Triplet

A single quark has three color amplitudes. In a chosen basis write

$$q = \begin{pmatrix} q_r \\ q_g \\ q_b \end{pmatrix}, \qquad q' = Uq, \qquad U \in SU(3). \tag{4.29}$$

The three entries may be state amplitudes for one quark or three field operators that create red, green, and blue quarks. The group mixes the three components. A state that is initially red generally

[3] Editorial clarification: The source uses "ortho and para-hydrogen" while referring to proton–electron spins in a hydrogen atom. The mathematical decomposition is correct; the molecular and atomic names are distinguished here for precision.

becomes a superposition of red, green, and blue after a color transformation. This fundamental three-dimensional representation is called the **3**.

For an anti-quark, the group acts through the complex conjugate of the quark representation. Taking the complex conjugate of Equation (4.29) gives

$$\bar{q}' = U^* \bar{q}. \tag{4.30}$$

This representation is denoted $\bar{\mathbf{3}}$. The $U(1)$ analogy is immediate: if an electron field acquires $e^{i\theta}$, a positron field acquires $e^{-i\theta}$. Quark and anti-quark color transformations are the non-Abelian version of that conjugate pairing.

Near the identity, if

$$U = \mathbf{1} + i\epsilon T, \tag{4.31}$$

then

$$U^* = \mathbf{1} - i\epsilon T^*. \tag{4.32}$$

Thus the anti-triplet generators are $-T^*$, the conjugates with the opposite sign. This is the precise sense in which anti-quarks carry opposite color charges.

Question from the audience. What happens to the complex-conjugate representation in the $SU(2)$ case? [a]

Response. The $SU(2)$ doublet is a special case: its complex conjugate is equivalent to the original doublet after a fixed change of basis involving the antisymmetric epsilon tensor. For $SU(3)$, the **3** and $\bar{\mathbf{3}}$ are genuinely inequivalent representations.

[a] Lecture timestamp: 00:36:19.

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[4]

4.5 Combining Color Representations

The symbol $\otimes$ does not mean ordinary multiplication of the numbers written beside it. It means that two state spaces are combined. The resulting tensor-product space is then reorganized into pieces that do not mix with one another under the group action.

4.5.1 Two Quarks: $\mathbf{3} \otimes \mathbf{3}$

Two quarks have nine color basis states: red–red, red–green, red–blue, and the six remaining ordered pairs. Organize the two color indices into a symmetric and an antisymmetric part. The symmetric part has six independent components; the antisymmetric part has three. Under $SU(3)$,

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}. \tag{4.33}$$

The anti-triplet may be represented explicitly by contracting two quark indices with the invariant epsilon tensor,

$$(qq)^k_{\bar{3}} = \frac{1}{\sqrt{2}} \epsilon^{kij} q_i q_j. \tag{4.34}$$

[4] Editorial clarification: The mathematically precise term for the $SU(2)$ fundamental representation is pseudoreal, not real. The response preserves the source's intended distinction while correcting the terminology.

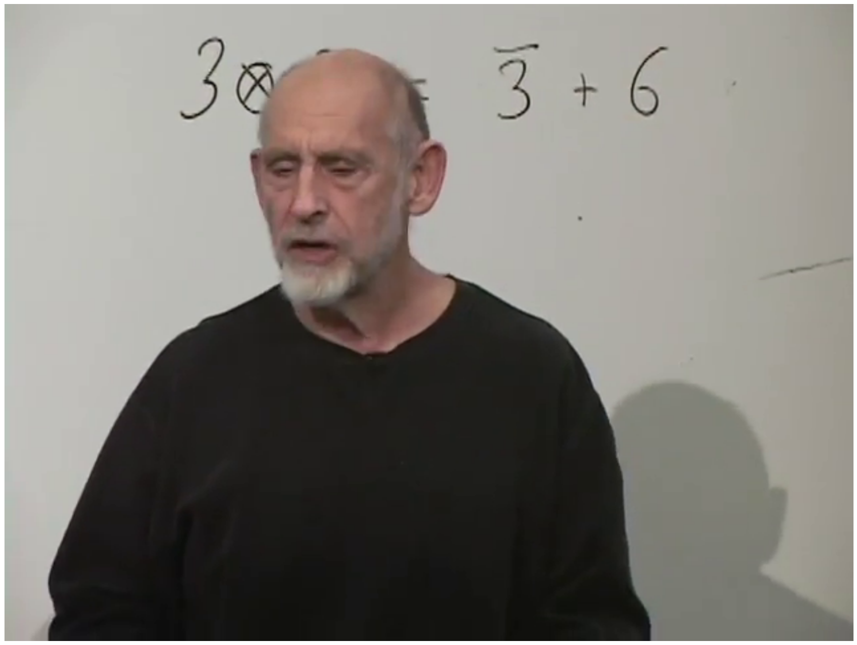

Figure 4.3: The two-quark product split into an anti-triplet and a sextet.
Lecture frame: 00:39:33.

This object has three components and transforms like an anti-quark. The six symmetric combinations form the sextet. The statement is not that two quarks become an anti-quark dynamically; it is that one sector of their color wave function has the same transformation law as an anti-quark.

4.5.2 A Quark and an Anti-Quark: $\mathbf{3} \otimes \bar{\mathbf{3}}$

A quark–anti-quark color state can be viewed as a 3×3 matrix $M_i{}^j = q_i \bar{q}^j$. Every such matrix splits uniquely into its trace and traceless parts,

$$M = \frac{\operatorname{tr} M}{3}\mathbf{1} + \left(M - \frac{\operatorname{tr} M}{3}\mathbf{1}\right). \tag{4.35}$$

The trace is a single invariant component and the traceless part has eight components. Therefore

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}. \tag{4.36}$$

Up to normalization, the singlet is

$$|1\rangle_{q\bar{q}} = \frac{1}{\sqrt{3}}(|r\bar{r}\rangle + |g\bar{g}\rangle + \left|b\bar{b}\right\rangle). \tag{4.37}$$

It cannot rotate into anything else because it is invariant. The other eight linear combinations mix among themselves. They furnish the adjoint representation, the same representation in which the eight generators transform.

4.5.3 Three Quarks

Combine a third quark with Equation (4.33). The anti-triplet branch gives

$$\bar{\mathbf{3}} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8}, \tag{4.38}$$

so the three-quark product contains a singlet. In full,

$$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{10}. \tag{4.39}$$

The color singlet is the fully antisymmetric red–green–blue combination,

$$|1\rangle_{qqq} = \frac{1}{\sqrt{6}}\epsilon^{ijk} |q_i q_j q_k\rangle . \tag{4.40}$$

Because ϵ^{ijk} vanishes whenever two indices coincide, every nonzero term contains one red, one green, and one blue quark, with alternating signs under interchange.

Question from the audience. Are the non-singlet combinations merely unlikely, so that their probabilities can be ignored? [a]

Response. No. The issue is energy, not a small probability. An isolated non-singlet would carry a color field to arbitrarily large distance and, in the confining regime, would require unbounded energy. The dynamics behind that statement will be developed below.

[a] Lecture timestamp: 00:46:02.

4.5.4 The Gluon Octet

The gluon field transforms like the octet in Equation (4.36). With respect to color it carries one triplet index and one anti-triplet index, with the singlet trace removed. Schematically,

$$q \sim \mathbf{3}, \qquad \bar{q} \sim \bar{\mathbf{3}}, \qquad G \sim \mathbf{8}. \tag{4.41}$$

These symbols name transformation laws; particles are not literally equal to the integers three and eight.

The ninth quark–anti-quark color combination, proportional to the identity, is a singlet and does not mix with the octet. Omitting it is therefore mathematically consistent: an $SU(3)$ gauge field is traceless and has eight components rather than the nine of a $U(3)$ gauge field.

Question from the audience. Why does nature omit the extra singlet gluon? [a]

Response. The immediate group-theory answer is that the singlet is a separate invariant sector, so the octet closes consistently without it. The deeper reason for choosing $SU(3)$ rather than $U(3)$ is not derived here; it is part of the gauge structure inferred from QCD phenomenology.

[a] Lecture timestamp: 00:49:00.

4.6 Color Singlets as Hadrons

The free, finite-mass particles of the strong interaction are color singlets. This does not make quarks or gluons unreal; it means they do not occur as isolated asymptotic particles. The elementary singlet-building patterns are

$$qqq, \qquad q\bar{q}, \qquad GG \text{ and higher gluonic products.} \tag{4.42}$$

They lead respectively to baryons, mesons, and glueballs. Products of several already colorless objects are of course colorless again, so more complicated multihadron states can be built from these ingredients.

Question from the audience. What are the three-quark and quark–anti-quark singlets called, and can gluons also make singlets? [a]

Response. A three-quark singlet is a baryon; a quark–anti-quark singlet is a meson. Gluons can also combine into color singlets. The simplest such strongly interacting states are called glueballs, and more than two gluons may participate.

[a]Lecture timestamp: 00:51:38.

Three spin-one-half quarks always combine to an overall half-integer spin, although orbital angular momentum can change which half-integer occurs. Quark–anti-quark mesons have integer spin. Purely gluonic glueballs also have integer spin.

For two gluons the first board statement is only a count,

$$8 \times 8 = 1 + 63. \tag{4.43}$$

There are sixty-four color states in all; one combination is a singlet and the remaining sixty-three belong to nonsinglet sectors.

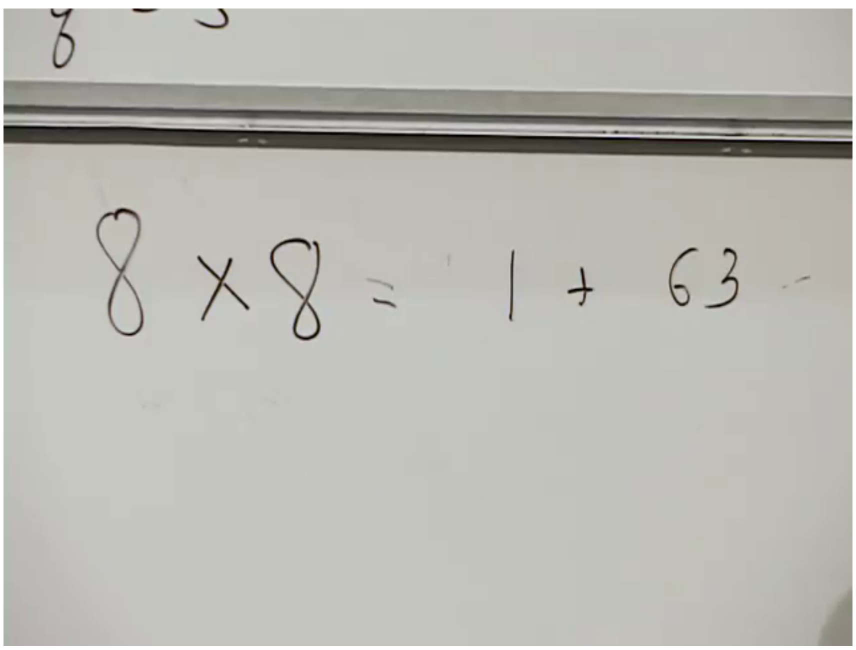

Figure 4.4: The board's two-gluon state count: one singlet among sixty-four states.
Lecture frame: 00:53:44.

The complete irreducible decomposition is

$$\mathbf{8} \otimes \mathbf{8} = \mathbf{1} \oplus \mathbf{8}_S \oplus \mathbf{8}_A \oplus \mathbf{10} \oplus \overline{\mathbf{10}} \oplus \mathbf{27}. \tag{4.44}$$

The dimensions add correctly:

$$1 + 8 + 8 + 10 + 10 + 27 = 64. \tag{4.45}$$

The two octets are distinct invariant subspaces, one symmetric and one antisymmetric under exchange of the two adjoint factors. The singlet is the two-gluon color channel from which a glueball state can be formed. [5]

4.6.1 Why Observable Electric Charges Are Integral

The quark charges come in two types,

$$Q_{\text{up-like}} = +\frac{2}{3}, \qquad Q_{\text{down-like}} = -\frac{1}{3}. \tag{4.46}$$

Three quarks therefore have an integral total charge: $-1, 0, +1,$ or $+2$, depending on how many are up-like. A quark and an anti-quark likewise have charge 0 or ± 1. Gluons carry no electric charge. Thus the basic color singlets built from the observed quark representations have integer electric charge, even though their constituents carry fractional charge.

Question from the audience. What is the electric charge of the strange quark? [a]

Response. It is down-like, so its charge is $-1/3$. The three families repeat the same charge pattern: (u, d), (c, s), and (t, b), with charges $+2/3$ and $-1/3$ in each pair.

[a] Lecture timestamp: 00:56:08.

Color singletness and integral electric charge are closely linked for this particle content, but they are not identical mathematical statements. The integrality also uses the particular Standard Model electric-charge assignments just written.

Question from the audience. Is a color singlet an entangled state, and does that have the same significance as an entangled pair in a Bell experiment? [a]

Response. The singlet color wavefunction is certainly entangled: it is a coherent superposition, not a product of separately definite colors. In principle its constituents may be separated, but confinement makes that separation increasingly expensive and eventually produces new hadrons. It is therefore not a convenient system for a Bell-style manipulation, even though the state is entangled.

[a] Lecture timestamp: 00:56:58.

4.7 What the Tensor-Product Equations Mean

At this point the multiplication tables deserve a more concrete explanation. Let ψ_i and ϕ_j be operators that create two spin-one-half objects; they may describe two species, such as an electron and a proton, or two copies of the same species when the appropriate statistics are included. Each index takes two values. The four products

$$\psi_\uparrow\phi_\uparrow, \quad \psi_\uparrow\phi_\downarrow, \quad \psi_\downarrow\phi_\uparrow, \quad \psi_\downarrow\phi_\downarrow \tag{4.47}$$

span the tensor-product space.

[5] Editorial clarification: The board first gives only $8 \times 8 = 1 + 63$. The later classroom exchange supplies the listed irreducible dimensions; bars and symmetry labels are restored here in standard $SU(3)$ notation.

If a group element acts as U on each factor, it acts as $U \otimes U$ on the product. In components,

$$\psi_i \phi_j \longmapsto U_i{}^k U_j{}^\ell \psi_k \phi_\ell. \tag{4.48}$$

This is a four-dimensional representation, but it is reducible. The antisymmetric combination

$$\frac{1}{\sqrt{2}}(\psi_\uparrow \phi_\downarrow - \psi_\downarrow \phi_\uparrow) \tag{4.49}$$

maps into itself, while the three symmetric combinations map among themselves. In the basis adapted to those combinations, every product representation matrix takes block-diagonal form,

$$D_{2\otimes 2}(g) \sim \begin{pmatrix} 1 & 0 \\ 0 & D_3(g) \end{pmatrix}. \tag{4.50}$$

Question from the audience. What operation is represented by an equation such as $\mathbf{3} \otimes \mathbf{3} = \bar{\mathbf{3}} \oplus \mathbf{6}$? [a]

Response. The left-hand side forms the composite state space. The group acts on both factors and therefore mixes the product basis states. The right-hand side says that a change of basis separates those states into invariant sets: three antisymmetric combinations that transform as an anti-triplet and six symmetric combinations that transform as a sextet. The plus sign means a direct sum of alternatives, not ordinary numerical addition.

[a] Lecture timestamp: 00:57:51.

Question from the audience. How do we know the nine states split as $3+6$, or $1+8$, rather than, say, $2+7$? [a]

Response. Dimension counting alone cannot decide the split. One must know which irreducible representations $SU(3)$ possesses and construct invariant subspaces, for example by symmetrizing indices, antisymmetrizing them, or separating trace from traceless parts. A full classification takes more group theory; here we use only the representations needed for quarks and gluons.

[a] Lecture timestamp: 01:04:39.

The invariant epsilon tensor makes the singlet construction transparent. For $SU(2)$,

$$\epsilon_{ij} U^i{}_k U^j{}_\ell = (\det U)\epsilon_{k\ell} = \epsilon_{k\ell}. \tag{4.51}$$

Therefore $\epsilon_{ij}\psi^i\phi^j$ is invariant. For $SU(3)$, the three-index version gives

$$\epsilon_{ijk} U^i{}_a U^j{}_b U^k{}_c = (\det U)\epsilon_{abc} = \epsilon_{abc}, \tag{4.52}$$

which proves the invariance of the three-quark state in Equation (4.40).

Question from the audience. Can one show explicitly why the singlet does not transform? [a]

Response. Yes. Contract all fundamental indices with the fully antisymmetric epsilon tensor. Under the transformation, the product of matrices multiplies epsilon by $\det U$. For a special-unitary matrix that determinant is one, so the contracted state is unchanged. In $SU(2)$ this proves the two-spin singlet; in $SU(3)$ it proves the red–green–blue three-quark singlet.

[a] Lecture timestamp: 01:07:18.

Question from the audience. What is the full decomposition of two gluon octets? [a]

Response. It is the decomposition in Equation (4.44): one singlet, two octets, a decuplet, an anti-decuplet, and a twenty-seven-dimensional representation. There is exactly one color-singlet combination of the two octet indices.

[a]Lecture timestamp: 01:09:24.

Question from the audience. What is the corresponding full decomposition for three quarks? [a]

Response. The physically essential answer is that it contains one singlet. Completing the arithmetic gives Equation (4.39): one singlet, two octets, and one decuplet, for a total of $1 + 8 + 8 + 10 = 27$ states.[b]

[a]Lecture timestamp: 01:11:51.

[b]Editorial clarification: The lecturer does not recall the complete $3 \otimes 3 \otimes 3$ decomposition during the exchange. The standard completion is supplied explicitly and is labeled here rather than attributed to the spoken answer.

The same idea scales to the two-gluon space. In the naive product basis, $SU(3)$ acts through 64×64 matrices. In a basis adapted to the irreducible components, those matrices are block diagonal,

$$\begin{aligned} D_{8\otimes 8}(g) \sim \operatorname{diag}\big(1, D_{8_S}(g), D_{8_A}(g), \\ D_{10}(g), D_{\overline{10}}(g), D_{27}(g)\big). \end{aligned} \tag{4.53}$$

No block mixes with another, although states within each block mix among themselves.

Question from the audience. Is tensor-product decomposition just a change to a basis in which the large representation matrix becomes block diagonal? [a]

Response. Exactly. For two spins the four-dimensional matrix becomes a one-dimensional singlet block and a three-dimensional triplet block. For two gluons the sixty-four-dimensional matrix becomes blocks of dimensions $1, 8, 8, 10, \overline{10},$ and 27. Each block is closed under every group transformation.

[a]Lecture timestamp: 01:12:02.

Question from the audience. Is there a name for one of those closed pieces inside the full tensor-product space? [a]

Response. It is an invariant subspace carrying a subrepresentation. When it cannot be decomposed further, it carries an irreducible representation, or irrep. The full tensor product is the direct sum of those irreducible components.[b]

[a]Lecture timestamp: 01:14:55.

[b]Editorial clarification: The source leaves the requested term unresolved in the room. Standard representation-theory terminology is supplied in the response.

Question from the audience. When the nonsinglet tens and other representations are called nonexistent, does that mean they should be ignored altogether? [a]

Response. They are legitimate color sectors and matter in intermediate calculations, but they do not appear as isolated finite-energy particles in the laboratory. Confinement, not a failure of representation theory, prevents them from becoming free asymptotic states.

[a]Lecture timestamp: 01:15:36.

4.8 From Nambu's Color Idea to Confinement

The color idea predates the final formulation of QCD. Yoichiro Nambu already understood in the early 1960s that a hidden three-valued quantum number could solve a striking problem: quarks carry fractional electric charge, yet no free fractionally charged particles are observed. If physical states are forced to be color singlets, quarks occur in the integral-charge combinations described above. The remaining question is dynamical: what forces all finite-energy free states into singlets?

Begin with the electromagnetic analogy. A charge and an opposite charge attract. Bringing them together lowers the energy; pushing like charges together requires work. The field itself stores positive energy,

$$E_{\text{field}} = \frac{1}{2}\int d^3x\, \boldsymbol{E}^2 \tag{4.54}$$

in simple electrostatic units. A net charge leaves field extending far outside the system, while a tightly bound neutral pair has only a short-range dipole field. Neutralization therefore removes field energy from the exterior.

Imagine increasing the electromagnetic coupling enormously. The field-energy penalty for net charge would become so large that low-energy objects would overwhelmingly be neutral, and extracting a charged constituent could become prohibitively expensive. Nambu transferred this intuition to color: the eight color charges are sources of eight gluon fields, so a large color-field energy could forbid isolated nonsinglets.

That intuition is essentially right, but QCD contains a decisive new feature. The photon carries no electric charge, whereas the gluon carries color. The photon is not a source of the classical electromagnetic field, and Maxwell's equations are linear: two electromagnetic waves superpose and pass through one another. A gluon transforms in the color octet, so gluons source and interact with other gluons. Even different parts of a single gluon field interact. The non-Abelian gauge field is therefore nonlinear.

4.8.1 Flux Tubes and a Linear Potential

At long distance the gluon self-interaction prevents color flux from spreading out like an ordinary electric dipole field. The flux is squeezed into a tube joining a quark to an anti-quark. The field strength and cross-sectional area of a long tube are approximately fixed, so the tube stores an approximately constant energy per unit length. If σ denotes this string tension, then

$$V(R) \simeq \sigma R + \text{constant} \qquad (R \text{ large}). \tag{4.55}$$

This is stronger than the Coulomb potential, whose magnitude falls as $1/R$. Pulling the color sources farther apart does not weaken the force indefinitely; it lengthens the tube and keeps adding energy. A lone color charge in an infinite volume would require a tube extending without end and therefore unbounded energy. That is the operational content of saying that only color singlets have finite-energy asymptotic states. [6]

[6]Editorial clarification: The string-tension notation and explicit long-distance potential make quantitative the source's statement that a uniform flux tube carries uniform energy per unit length.

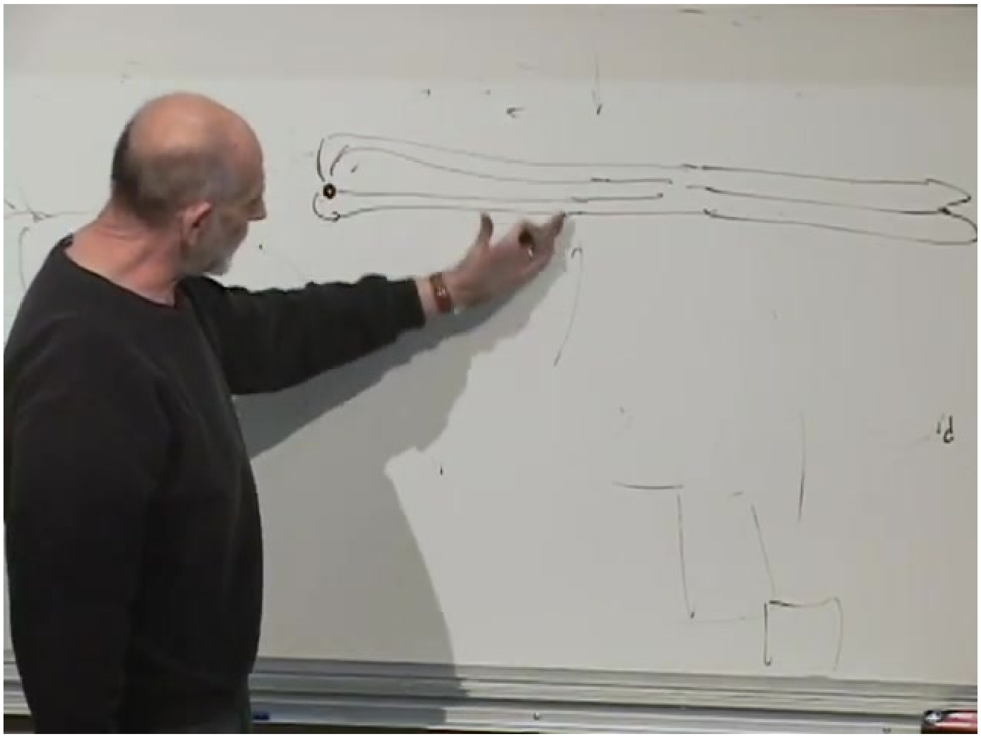

Figure 4.5: Color flux bundled into a long tube between a quark and an anti-quark.
Lecture frame: 01:31:35.

4.8.2 String Breaking and Jets

Suppose a hard collision strikes one quark inside a meson. The quark begins to move away, but the flux tube stretches behind it. It may lose its kinetic energy and be pulled back. More often, once the stored field energy is large enough, the vacuum produces a new quark–anti-quark pair. The new anti-quark terminates the flux attached to the escaping quark, while the new quark terminates the flux attached to the original anti-quark. One stretched singlet becomes two shorter singlets,

$$(q\bar{q})_{\text{stretched}} \longrightarrow (q\bar{q}) + (q\bar{q}). \tag{4.56}$$

Question from the audience. If a struck quark has enough energy to keep stretching the tube, what else can happen besides its being pulled back? [a]

Response. The tube can break by quark–anti-quark pair creation. Its energy becomes the mass and kinetic energy of the new pair. The endpoints reconnect into two singlets, so the process creates additional mesons rather than an isolated quark.

[a]Lecture timestamp: 01:33:50.

If one of those partons still carries a large momentum, the process repeats. The result is a cascade of color-singlet hadrons collimated around the original parton direction: a jet. This conversion of energetic quarks and gluons into hadrons is called hadronization. The initiating colored object is inferred from the jet, never observed by itself.

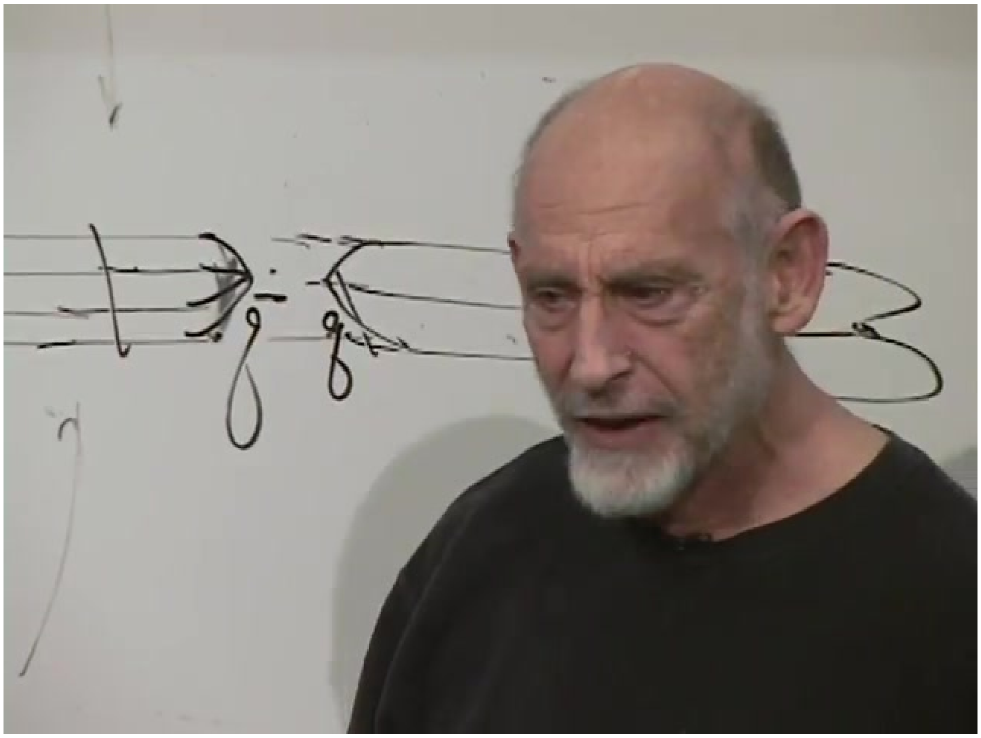

Figure 4.6: Pair creation breaks one stretched flux tube into two color-singlet tubes.
Lecture frame: 01:34:52.

Question from the audience. Does string breaking produce only mesons, or can other hadrons appear in the jet? [a]

Response. Other hadrons can certainly appear. The only color requirement is that the final observable objects be singlets. Mesons are the simplest picture for one break, but baryons, antibaryons, and more complicated singlet combinations can also be produced.

[a]Lecture timestamp: 01:35:20.

Three quarks can terminate their flux in a color-singlet junction. The three-pronged picture is the dynamical counterpart of the antisymmetric epsilon tensor in Equation (4.40): no net color flux escapes from the complete baryon.

4.9 Gauge Fields, Conserved Charges, and the Standard Model

At the present level, a gauge theory may be recognized as a theory in which a conserved charge sources a photon-like field. Electric charge sources the electromagnetic field; the eight color charges source the eight gluon fields. The deeper formulation makes the symmetry local and introduces a connection that compares internal states at neighboring spacetime points, but that construction is not needed for the confinement picture developed here. [7]

Symmetry, conserved charge, and flux are tightly connected. If field lines may end only on charges,

[7]Editorial clarification: The local-symmetry formulation is the standard deeper meaning of gauge theory alluded to but deliberately postponed in the source.

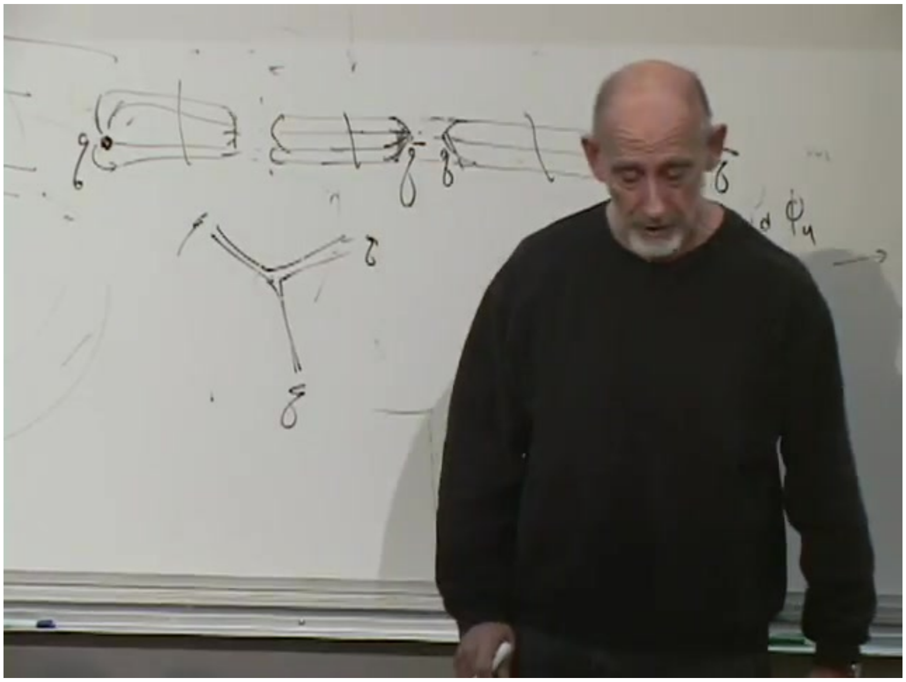

Figure 4.7: A three-pronged color-flux junction for a baryon, below the meson string-breaking sketch.

Lecture frame: 01:36:12.

then a source cannot simply disappear while its flux remains attached to nothing. Nor can all of its distant field disappear instantaneously, because field changes propagate causally. In local form the same physics is expressed by a continuity equation,

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{j} = 0, \tag{4.57}$$

which says that charge can leave a region only by flowing through its boundary. The field-line argument is the geometric picture of that local conservation law.

We can now locate the result inside the Standard Model. Color contributes the eight generators of $SU(3)$ and their eight gluon fields. The Abelian sector provides one generator and one gauge field. Three more generators belong to an additional $SU(2)$, not to ordinary spatial angular momentum. Before symmetry breaking the gauge structure is

$$SU(3)_c \times SU(2)_L \times U(1)_Y, \tag{4.58}$$

with $8 + 3 + 1 = 12$ generators. The next task is to understand the electroweak $SU(2)$, then the Higgs field and the way the observed photon, $W^{\pm}$, and Z emerge. [8]

The central chain is now complete. Abstract groups act through representations; infinitesimal generators are the conserved charges; quarks, anti-quarks, and gluons occupy the **3**, $\bar{\mathbf{3}}$, and **8**; tensor products identify the singlet hadrons; and the nonlinear gluon field turns singlet selection into the physical phenomenon of confinement.

[8] Editorial clarification: The source counts the photon together with the eight gluons and describes the remaining factor as $SU(2) \times U(1)$. In the unbroken Standard Model the Abelian factor is hypercharge $U(1)_Y$; the photon appears only after electroweak mixing.

Chapter 5

Gauge Fields, Interaction Scales, and Charged Weak Currents

Quantum electrodynamics and quantum chromodynamics look very different in the laboratory, yet both are gauge theories. In fact, nearly every interaction in the Standard Model is organized this way. The word *gauge* has a deeper meaning involving local symmetry, but we can postpone that machinery. For the moment we need a minimal physical description: a gauge theory contains Maxwell-like fields, conserved sources, symmetry transformations, gauge bosons, and rules for emitting and absorbing those bosons.

5.1 Maxwell Theory as the Prototype

5.1.1 Fields and potentials

The electromagnetic field has six components: three components of electric field and three components of magnetic field. The same information can be encoded, with some redundancy, in a four-vector potential

$$A_\mu = (A_0, \boldsymbol{A}), \tag{5.1}$$

where A_0 is the electrostatic potential and $\boldsymbol{A}$ is the spatial vector potential. In the electrostatic limit the electric field is obtained from the spatial gradient,

$$\boldsymbol{E} = -\boldsymbol{\nabla} A_0, \qquad \boldsymbol{F} = q\boldsymbol{E}, \tag{5.2}$$

while the magnetic field is

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}. \tag{5.3}$$

The sign in the first equation depends on the standard definition of the potential; the board discussion is concerned mainly with the fact that fields are derivatives of potentials. [1]

Thus electrodynamics admits two equivalent descriptions: one in terms of $\boldsymbol{E}$ and $\boldsymbol{B}$, and one in terms of A_μ. The latter contains only four displayed components because different potentials can describe the same physical fields. That redundancy is the first hint behind the word gauge, though we will not use it formally yet.

[1] Editorial clarification: For time-dependent fields the complete relation is $\boldsymbol{E} = -\boldsymbol{\nabla} A_0 - \partial_t \boldsymbol{A}$. The lecture deliberately works at the simpler electrostatic level here.

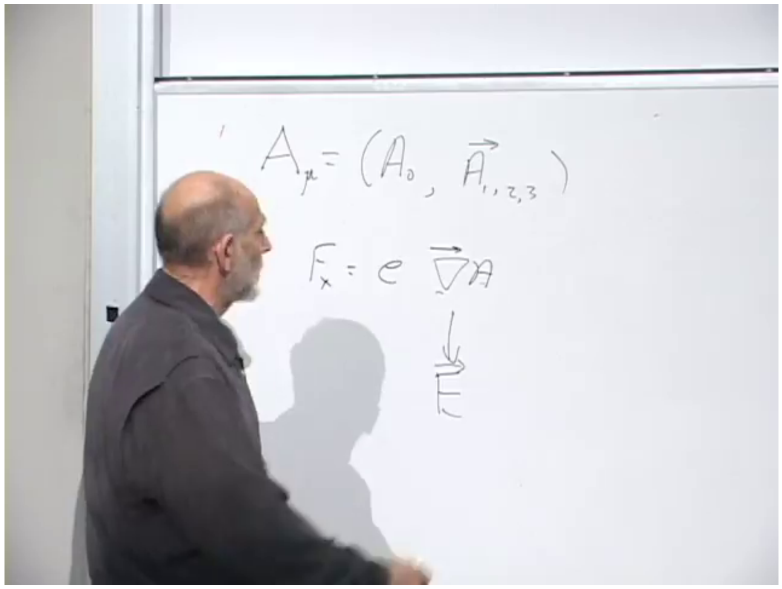

Figure 5.1: The electromagnetic four-potential and its derivative relation to the field.
Lecture frame: 00:03:33.

5.1.2 Waves and polarization

Maxwell's equations support waves. Take a plane wave moving along a unit vector $\hat{\boldsymbol{k}}$. At every point the electric and magnetic fields are transverse:

$$\boldsymbol{E} \cdot \hat{\boldsymbol{k}} = 0, \qquad \boldsymbol{B} \cdot \hat{\boldsymbol{k}} = 0, \qquad \boldsymbol{E} \cdot \boldsymbol{B} = 0. \tag{5.4}$$

In units with $c = 1$, their magnitudes are equal,

$$|\boldsymbol{E}| = |\boldsymbol{B}|. \tag{5.5}$$

For a linearly polarized wave, the direction of $\boldsymbol{E}$ specifies the polarization. The entire pattern propagates at the speed of light.

Quantization changes the language but not the polarization structure. A quantum of the electromagnetic wave is a photon. Drawing a photon as a point is inevitably misleading, but it is useful to imagine that the point carries a small transverse pointer recording its polarization. A quarter-wave plate or another optical element can rotate that polarization. A static Coulomb field may likewise be pictured, only roughly, as arising from repeated emission and absorption of photons.

5.1.3 Sources, flux, and conservation

Electromagnetic waves can propagate through empty space, but the field also has sources: electric charges and currents. A positive point charge produces an outward radial electric field proportional to $1/r^2$. The number of field lines piercing any sphere around the charge is independent of the sphere's radius. Mathematically,

$$\oint_{S^2} \boldsymbol{E} \cdot d\boldsymbol{S} = \frac{Q_{\text{enc}}}{\epsilon_0}, \tag{5.6}$$

in SI units.

This is more than a convenient way to calculate the field. Electric field lines may end only on charge. If one isolated charge simply disappeared, its field would either have to disappear everywhere at once, violating causality, or a front of missing field would have to propagate outward with field lines ending on empty space. Neither is allowed. The geometry of flux therefore makes charge conservation natural: charge can move, and opposite charges can annihilate, but net charge cannot vanish locally without a compensating current.

The corresponding local statement is the continuity equation

$$\partial_t \rho + \boldsymbol{\nabla} \cdot \boldsymbol{J} = 0. \tag{5.7}$$

[2]

5.1.4 A working definition of gauge theory

We can now state the operational pattern. A gauge theory has:

- electric-like and magnetic-like fields;
- waves that, when interactions are weak enough, resemble light waves;
- sources analogous to electric charge;
- a Gauss law tying flux to those sources;
- a conservation law and an associated symmetry;
- quanta of the field, called gauge bosons; and
- a coupling constant controlling emission and absorption.

The source need not be electric charge. As a deliberately unphysical example, imagine gauging baryon number. Protons and neutrons both have baryon number +1, so they would repel through a new Coulomb-like force; a proton and an antiproton would attract. Such a theory is mathematically imaginable, but nature supplies no observed long-range gauge field coupled to baryon number. Conservation by itself does not guarantee a gauge force.

Electric charge does come with a symmetry. For a charged matter field,

$$\psi(x) \longmapsto e^{i\theta} \psi(x). \tag{5.8}$$

The phase symmetry and charge conservation are two sides of the same structure. Later we can make θ depend on spacetime and see why the potential A_μ is required. Here the global transformation is enough to identify the conserved charge.

5.2 QCD as the Non-Abelian Example

5.2.1 Triplets, anti-triplets, and the gluon matrix

In quantum chromodynamics a quark field carries a color index $i \in \{r, g, b\}$ and transforms in the fundamental representation of $SU(3)$:

$$q_i' = U_{ij} q_j, \qquad U^\dagger U = \mathbf{1}, \qquad \det U = 1. \tag{5.9}$$

[2]Editorial clarification: Equation (5.7) is the standard local mathematical form of the causal flux argument developed verbally in the lecture.

This is a rotation in a three-dimensional complex color space, not a rotation of ordinary position space. The three-component representation is also called the defining representation.

Antiparticle fields are represented by the complex-conjugate variables. Taking the complex conjugate of Equation (5.9) gives

$$q_i'^* = U_{ij}^* q_j^*. \tag{5.10}$$

Thus quarks transform as $\mathbf{3}$, whereas antiquarks transform as $\bar{\mathbf{3}}$.

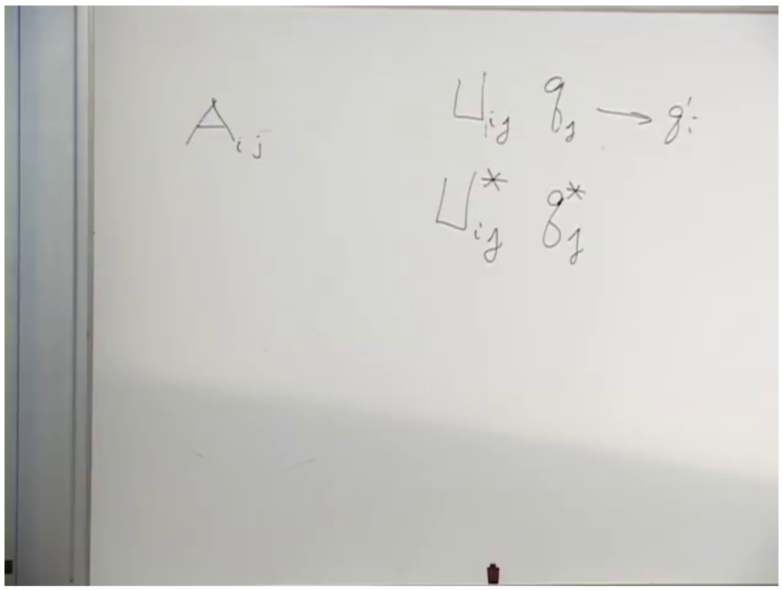

Figure 5.2: The fundamental transformation and its complex-conjugate anti-triplet transformation.

Lecture frame: 00:18:47.

Question from the audience. How many U matrices are there – seven, eight, or nine? [a]

Response. There are infinitely many group elements U. The finite number being counted is the number of independent generators, which is eight for $SU(3)$. A general 3×3 matrix has nine matrix positions, but special unitarity leaves eight continuous directions.

---

[a] Lecture timestamp: 00:19:37.

The gluon field may be written as a traceless matrix A_{ij}. One index transforms like a quark and the other like an antiquark, so its color structure is that of

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}. \tag{5.11}$$

QCD retains the traceless octet,

$$A_{ii} = \mathrm{tr}\, A = 0, \tag{5.12}$$

and discards the singlet. That is why QCD has eight gluons rather than nine. More generally, $SU(N)$ has

$$N^2 - 1 \tag{5.13}$$

generators and the same number of gauge-boson fields.

The photon is neutral under electric $U(1)$: rotating the phase of an electron does not rotate the photon into a different photon. Gluons are different. A gluon carries a color and an anticolor, so it transforms in the adjoint representation. A red–antiblue combination, for example, is not a color singlet even though it contains one color and one anticolor.

5.2.2 Index flow and self-interaction

The interaction rule can be read by following indices. Schematically, a quark of color i can become a quark of color j while emitting a gluon carrying the corresponding color flow:

$$q_i \longrightarrow q_j + A_{ij}. \tag{5.14}$$

The precise placement of i and j depends on the arrow convention; the invariant content is that every index entering a vertex must continue through another line. No color index is lost.

This rule applies in any $SU(N)$ gauge theory. Fundamental matter has one index, antifundamental matter has one conjugate index, and a gauge boson has one of each. The physical names can change, but the index structure remains.

Because a gluon itself carries color, it can emit another gluon. An A_{ij} gluon can split through an intermediate index k, schematically,

$$A_{ij} \longrightarrow A_{ik} + A_{kj}. \tag{5.15}$$

This is not a complete Yang–Mills vertex, but it displays the color flow. Nothing analogous occurs for photons in ordinary electrodynamics. Gluons therefore exert forces on gluons, and non-Abelian field dynamics is much more complicated than Maxwell's linear theory.

One qualitative consequence is confinement. Color flux lines cannot end, and their self-interaction draws them into a tube joining a quark to an antiquark. The tube carries an approximately constant energy per unit length:

$$V(R) \simeq \sigma R. \tag{5.16}$$

Pulling the endpoints farther apart therefore stores more energy instead of liberating isolated color. This is the gooey piece of glue picture: the field itself prevents colored constituents from escaping as free particles.

5.3 Couplings and the Meaning of Strong or Weak

5.3.1 Charge as an emission amplitude

A gauge theory still needs a coupling constant. In QED the electron charge e is the coefficient associated with an electron–photon vertex. It is an amplitude, and probabilities are squares of amplitudes. Operationally, imagine stopping a fast electron abruptly in a cathode-ray tube. The probability for radiating a photon is proportional to e^2, with conventional factors absorbed into the fine-structure constant,

$$\alpha = \frac{e^2}{4\pi\hbar c} \simeq \frac{1}{137}. \tag{5.17}$$

In natural units $\hbar = c = 1$, the coupling is dimensionless. A one-percent scale for an elementary emission makes electromagnetism weak in the perturbative sense: most stopped electrons do not each release a spray of hard photons.

5.3.2 Natural units connect energy, time, and distance

Set

$$c = 1, \qquad \hbar = 1. \tag{5.18}$$

Then time and distance have the same units, while energy has inverse units:

$$E = \hbar\omega \longrightarrow E = \omega \sim \frac{1}{t}, \qquad x \sim t \sim \frac{1}{E}. \tag{5.19}$$

Large energies probe short distances and short times.

For orientation,

$$1\ \mathrm{eV}^{-1} = \frac{\hbar c}{1\ \mathrm{eV}} \simeq 1.97 \times 10^{-7}\ \mathrm{m}. \tag{5.20}$$

An atomic length of order 10^{-10} m therefore corresponds roughly to an inverse kiloelectronvolt. These are order-of-magnitude conversions, not precision spectroscopy.

5.3.3 The hierarchy inside an atom

For a typical atom, take

$$R_{\mathrm{atom}} \sim 10^{-10}\ \mathrm{m}. \tag{5.21}$$

Light crosses this distance in

$$t^{\mathrm{atom}}_{\mathrm{transit}} \sim 10^{-18}\ \mathrm{s}. \tag{5.22}$$

An electron, however, moves at only a small fraction of the speed of light in a light atom. In the Bohr picture $v/c \sim \alpha$, so a typical orbital time is longer:

$$t^{\mathrm{atom}}_{\mathrm{orbit}} \sim 10^{-16}\ \mathrm{s}. \tag{5.23}$$

An excited atom takes longer still to radiate:

$$t^{\mathrm{atom}}_{\mathrm{decay}} \sim 10^{-9}\ \mathrm{s} \tag{5.24}$$

as a broad lecture-scale estimate. There are two suppressions in the physical picture. The electromagnetic force produces only modest acceleration, and an accelerated electron has another small emission probability controlled by α. The small fine-structure constant therefore creates a hierarchy among transit, orbital, and radiative-decay times.

5.3.4 The absence of hierarchy in a hadron

A hadron is an atom-like bound system made of quarks and gluons, but its scale is much smaller:

$$R_{\mathrm{hadron}} \sim 10^{-15}\ \mathrm{m}, \qquad t^{\mathrm{hadron}}_{\mathrm{transit}} \sim 10^{-23}\ \mathrm{s}. \tag{5.25}$$

Quarks cross the hadron on essentially the same time scale, and strongly decaying excited hadrons also live for roughly

$$t^{\mathrm{hadron}}_{\mathrm{orbit}} \sim t^{\mathrm{hadron}}_{\mathrm{decay}} \sim 10^{-23}\ \mathrm{s}. \tag{5.26}$$

A struck hadron may vibrate and emit a pion almost as quickly as causality allows. There is no seven- or nine-order-of-magnitude separation like the one inside an atom.

Question from the audience. Does special relativity imply that a hadron cannot decay substantially faster than its light-crossing time? [a]

Response. That is a fair way to state the order-of-magnitude bound. Strong decays occur about as fast as the size of the system permits; there is no small coupling or other large suppression slowing them down.

[a]Lecture timestamp: 00:47:15.

The corresponding strong coupling is therefore much larger than α. At an illustrative hadronic scale the lecture quotes

$$\alpha_{\text{QCD}} \sim 0.2, \tag{5.27}$$

much closer to unity than 1/137. This is the dynamical origin of the older name *strong interactions*. [3]

Question from the audience. Is the strong analogue simply called the strong-force constant? [a]

Response. It is conventionally called α_{QCD}, or more often α_s. The electromagnetic one is usually just α, though α_{QED} can be used when both appear in the same discussion.

[a]Lecture timestamp: 00:48:17.

Question from the audience. Would a paper distinguish a bare electromagnetic coupling from the measured, shielded value? [a]

Response. Yes. One speaks of a bare coupling and a renormalized coupling. Vacuum polarization makes the effective coupling depend on the scale at which it is measured.

[a]Lecture timestamp: 00:48:38.

5.4 The Puzzle of Weak Decay

Weak interactions were named from their very long decay times. They are usually slower than strong decays and can even be slower than atomic radiation. The oldest example is nuclear beta decay.

5.4.1 Neutron beta decay

A free neutron decays through

$$n \longrightarrow p + e^- + \bar{\nu}_e. \tag{5.28}$$

The lecture uses a lifetime of roughly twelve minutes, absurdly long compared with 10^{-23} s. Modern precision values do not matter for the argument.

Part of the slowness is kinematic. Roughly,

$$\begin{aligned} &m_n \simeq 940 \text{ MeV}, \quad m_p \simeq 939 \text{ MeV}, \\ &m_e \simeq 0.5 \text{ MeV}. \end{aligned} \tag{5.29}$$

[3]Editorial clarification: The QCD coupling runs with energy, so α_s is not one fixed number. The quoted value is an illustrative scale-dependent value, as the lecture intends.

Only a tiny fraction of the neutron's rest energy is available as kinetic energy. If the neutron were lighter than the sum of the final rest masses, decay would be impossible. If it lay only infinitesimally above threshold, the available phase space would be tiny and the decay would be slow. But phase space alone cannot explain a lifetime measured in minutes.

5.4.2 The charged pion removes the threshold excuse

Consider also

$$\pi^- \longrightarrow e^- + \bar{\nu}_e \tag{5.30}$$

and its charge-conjugate process. The pion has ample energy above the electron channel, yet its weak lifetime is still of nanosecond order rather than hadronic 10^{-23}-second order. A similar channel with a muon will soon become important. The puzzle is therefore intrinsic to the weak interaction, not merely an accident of the neutron's near-degenerate masses.

[4]

The term beta decay originates historically from calling emitted electrons beta rays. The pion and neutron examples involve the same charged-current interaction even though their kinematics differ sharply.

5.5 A Table of Quarks and Leptons

5.5.1 Color acts vertically

Arrange the six quark flavors in three paired columns,

$$(u, d), \qquad (c, s), \qquad (t, b), \tag{5.31}$$

and place a red, green, and blue copy of every quark in separate rows. Color $SU(3)$ acts vertically: it rotates red, green, and blue versions of an up quark into one another, and performs the same color rotation simultaneously on down, charm, strange, top, and bottom. It does not turn an up quark into a down quark.

5.5.2 Weak isospin acts horizontally

The weak symmetry acts in the other direction. It mixes each neighboring pair

$$Q_1 = \begin{pmatrix} u \\ d \end{pmatrix}, \qquad Q_2 = \begin{pmatrix} c \\ s \end{pmatrix}, \qquad Q_3 = \begin{pmatrix} t \\ b \end{pmatrix}. \tag{5.32}$$

The up-type quarks have charge $+2/3$, and the down-type quarks have charge $-1/3$, so the charge changes by one unit across each pair. [5]

[4] Editorial clarification: The charged-pion lifetime is about 26 ns, and the dominant channel is $\pi^- \to \mu^-\bar{\nu}_\mu$. The electron channel is helicity suppressed. The lecture first uses the electron channel because its particle bookkeeping is simplest, then returns to the dominant muon channel.

[5] Editorial clarification: The source momentarily says that a down quark has charge $-2/3$. The surrounding subtraction and every subsequent process require the standard value $-1/3$, used here.

Because the symmetry acts on doublets, the natural group is $SU(2)$. It acts on every color copy of each quark doublet without changing color.

There is also a fourth row of particles that do not carry color and therefore do not interact with gluons: the leptons. They form three more doublets,

$$L_e = \begin{pmatrix} \nu_e \\ e^- \end{pmatrix}, \qquad L_\mu = \begin{pmatrix} \nu_\mu \\ \mu^- \end{pmatrix}, \qquad L_\tau = \begin{pmatrix} \nu_\tau \\ \tau^- \end{pmatrix}. \tag{5.33}$$

The neutrinos are electrically neutral; the electron, muon, and tau have charge -1. Again the two members differ by one unit of electric charge. The electron, muon, and tau have the same gauge quantum numbers but different masses, just as the three generations of up-type or down-type quarks repeat the same charge pattern.

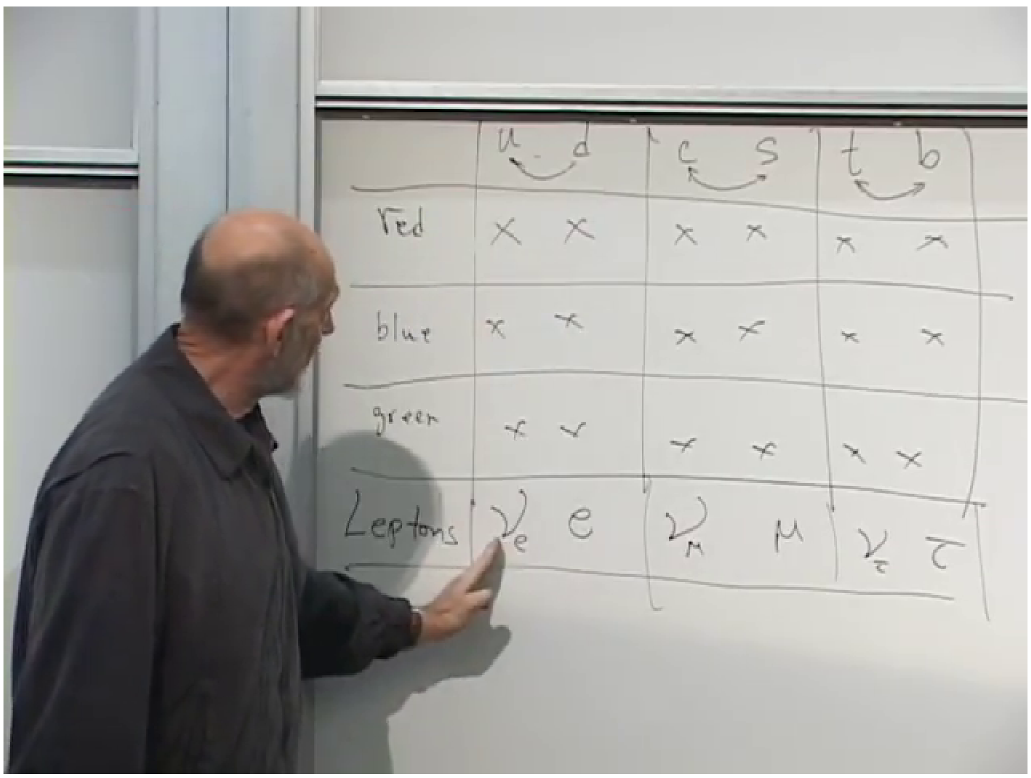

Figure 5.3: The board table: color acts among the red, blue, and green rows, whereas weak $SU(2)$ links the paired flavor columns and the lepton row.

Lecture frame: 01:03:35.

Calling leptons a fourth color is only an analogy at this stage. They fill a fourth row in the weak-doublet table, but ordinary QCD $SU(3)$ does not rotate quarks into leptons.

5.6 Constructing the Charged Weak Bosons

5.6.1 Quantum numbers of the charged bosons

Gauge bosons carry the transformation properties needed to move from one member of a doublet to the other. Begin with the lepton pair $e^-\bar{\nu}_e$. It has electric charge -1, so the corresponding charged gauge boson is called W^-:

$$W^- \sim e^- \bar{\nu}_e. \tag{5.34}$$

The same quantum numbers arise from a down quark and an anti-up quark:

$$W^- \sim d\bar{u}. \tag{5.35}$$

The symbol $\sim$ means has the same relevant gauge quantum numbers as. It does not mean that the W is literally a bound state of these particles. The antiparticle has the conjugate combinations,

$$W^+ \sim e^+ \nu_e \sim \bar{d}u. \tag{5.36}$$

The charged weak vertices include

$$e^- \longrightarrow \nu_e + W^-, \qquad \mu^- \longrightarrow \nu_\mu + W^-, \tag{5.37}$$

$$d \longrightarrow u + W^-, \qquad s \longrightarrow c + W^-, \tag{5.38}$$

with analogous transitions for the third generation when kinematically allowed. The gauge boson performs dynamically the same horizontal mixing that the symmetry performs on the doublet.

Question from the audience. Charge conservation fixes the emitted boson to have charge -1, but why must the electron, down-quark, and strange-quark transitions all involve the same particle? [a]

Response. That is an empirical fact encoded by the common weak symmetry. Nature could consistently have assigned a different W-type boson to every family or to quarks and leptons separately. Instead, one $SU(2)$ acts simultaneously on all the doublets, and the same charged gauge bosons mediate all these transitions.

[a] Lecture timestamp: 01:12:56.

5.6.2 Reading and reversing a vertex

A single vertex represents a family of processes. Reversing an incoming line to an outgoing line changes a particle into its antiparticle, and reversing a W^- line gives a W^+. One can therefore read a vertex as emission, absorption, decay, annihilation, or the inverse reaction, provided charge, momentum, and all relevant quantum numbers are followed consistently.

Question from the audience. Can the arrows and lines in these weak vertices be flipped? [a]

Response. Yes. A line may be crossed from the final state to the initial state or vice versa, but a particle then becomes its antiparticle. The same local vertex consequently describes many differently oriented reactions.

[a] Lecture timestamp: 01:14:51.

Question from the audience. Does writing a W^- with strange and charm lines mean that the boson is literally a strange quark bound to an anti-charm quark? [a]

Response. No. The line bookkeeping says only that the W^- carries the quantum-number change between those columns. If the diagram is read as a decay, a sufficiently energetic W^- can produce a strange quark and an anti-charm quark; it is not composed of them.

[a] Lecture timestamp: 01:15:43.

Question from the audience. Can an electron and an antineutrino make a W, which then produces a quark pair? [a]

Response. Exactly. If enough center-of-mass energy is available, the initial leptons can form the charged-current intermediate state and that state can emerge through an allowed quark channel. The two vertices are the same ones used in decay, read in another orientation.

[a]Lecture timestamp: 01:17:22.

5.6.3 The missing third boson

An $SU(2)$ group has three generators, hence three gauge fields. So far we have displayed only W^+ and W^-. A third, electrically neutral field must be associated with diagonal combinations such as electron–positron and neutrino–antineutrino quantum numbers. It is deliberately left lurking in the background. Its relation to the physical Z boson and photon requires electroweak mixing, which comes later.

Question from the audience. Were the weak and strong forces suspected at different historical times? [a]

Response. Their experimental roots appeared together in early radioactivity. Alpha decay revealed nuclear and strong-interaction physics, beta decay revealed the weak interaction, and gamma decay emitted electromagnetic photons. Becquerel's era already contained all three clues, long before their modern field theories existed.

[a]Lecture timestamp: 01:24:11.

5.7 Worked Charged-Current Decays

5.7.1 Pion decay

The negatively charged pion has valence-quark content

$$\pi^- = d\bar{u}. \tag{5.39}$$

Those quantum numbers can annihilate into a W^-, which can then produce a charged lepton and the corresponding antineutrino:

$$\pi^- \longrightarrow W^{-*} \longrightarrow e^- + \bar{\nu}_e, \tag{5.40}$$
$$\pi^- \longrightarrow W^{-*} \longrightarrow \mu^- + \bar{\nu}_\mu. \tag{5.41}$$

The star reminds us that the intermediate W is virtual. The second channel is the dominant one.

Question from the audience. Is this really a Feynman diagram even though the time direction was drawn horizontally rather than vertically? [a]

Response. Yes. The orientation of time on the page is a convention. Once one direction is declared to be time and all arrows are interpreted consistently, the drawing carries the same interaction information.

[a]Lecture timestamp: 01:20:21.

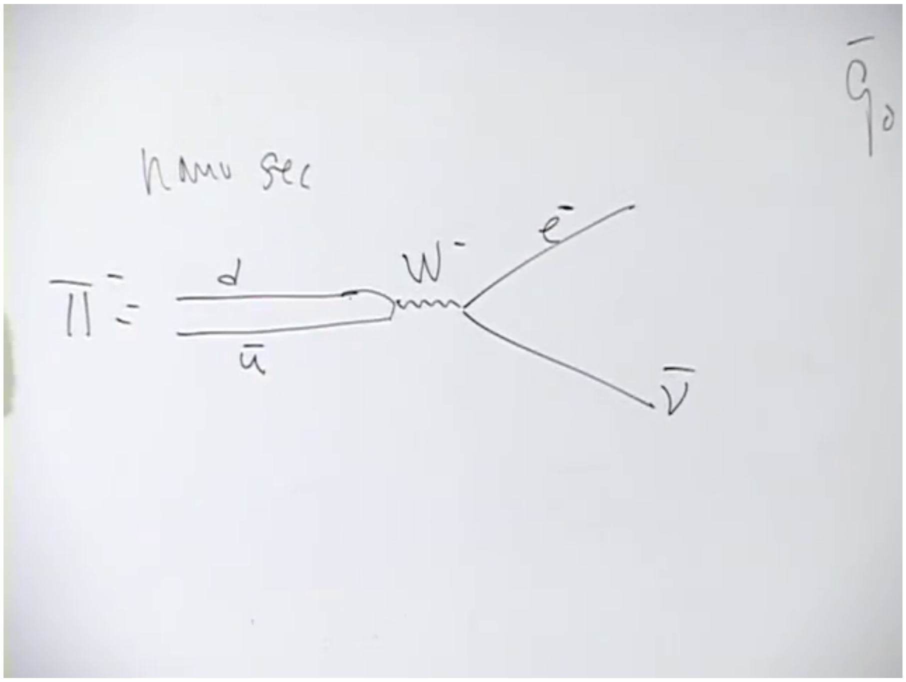

Figure 5.4: A $d\bar{u}$ pion annihilates into a virtual W^-, followed by a charged lepton and its matching antineutrino.

Lecture frame: 01:19:38.

Question from the audience. Could the diagram be read more like a reversible chemical reaction than as an irreversible one-way decay? [a]

Response. The inverse process is allowed. A pion produced in empty space decays because the outgoing particles separate and do not ordinarily recombine. But if a muon and antineutrino are fired together with the right energy and momentum, there is a nonzero amplitude to form the pion through the reverse interaction.

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[a]Lecture timestamp: 01:21:35.

5.7.2 Energy and a virtual intermediate state

For a pion at rest,

$$m_\pi \simeq 140 \text{ MeV}, \qquad m_\mu \simeq 106 \text{ MeV}, \qquad m_\nu \approx 0 \tag{5.42}$$

at the precision relevant here. The remaining energy becomes kinetic energy of the outgoing particles. Momentum conservation requires

$$\boldsymbol{p}_\mu + \boldsymbol{p}_{\bar{\nu}} = 0, \tag{5.43}$$

so the two particles leave back to back in the pion rest frame.

Question from the audience. How can energy be conserved when the intermediate W is far heavier than the pion? [a]

Response. Only the initial and final particles must be on their mass shell. Four-momentum is conserved at every vertex, but an internal virtual line need not obey $E^2 = \boldsymbol{p}^2 + m_W^2$. The larger its off-shell mismatch, the more strongly its propagator suppresses the process.[b]

[a] Lecture timestamp: 01:25:08.

[b] Editorial clarification: The lecture uses the traditional energy–time uncertainty language and says the intermediate state may violate energy briefly. The modern precise statement is the one in the response: energy-momentum is conserved, while a virtual internal line need not satisfy the real-particle mass-shell condition.

5.7.3 Muon decay

A muon can first change into its own neutrino by emitting a virtual W^-:

$$\mu^- \longrightarrow \nu_\mu + W^{-*}. \tag{5.44}$$

The virtual boson then produces an electron and electron antineutrino:

$$\mu^- \longrightarrow \nu_\mu + e^- + \bar{\nu}_e. \tag{5.45}$$

This is a purely leptonic process. The muon is about two hundred times heavier than the electron, so the final rest masses leave ample kinetic energy.

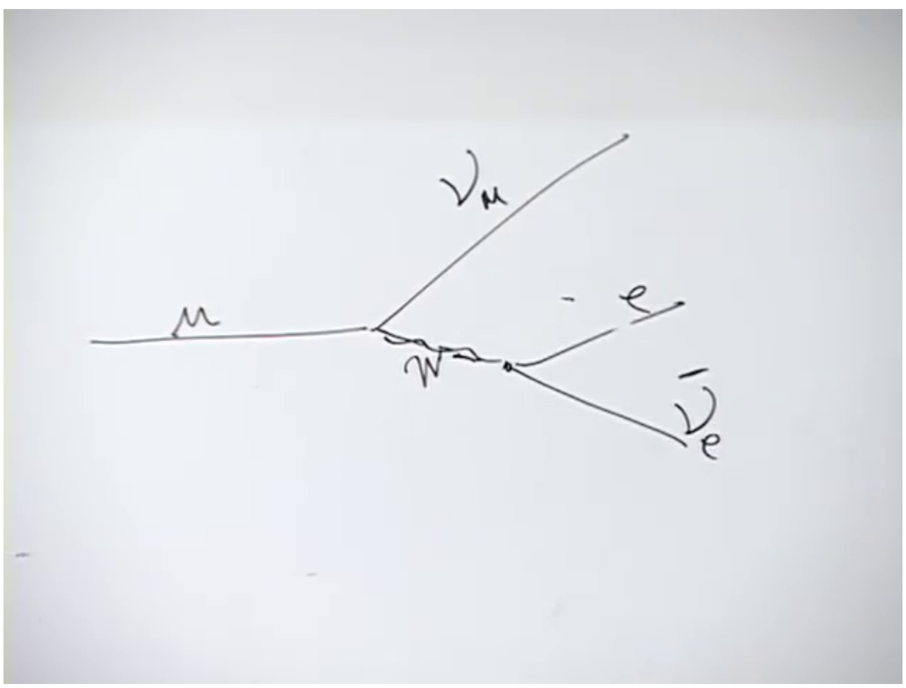

Figure 5.5: Muon decay through a virtual charged weak boson.

Lecture frame: 01:29:24.

Question from the audience. Is the neutral particle emitted with the electron an antineutrino? [a]

Response. Yes. The full final state contains one muon neutrino, one electron antineutrino, and the electron. Keeping the flavor labels prevents the two neutral particles from being confused.

[a]Lecture timestamp: 01:29:06.

The electron cannot reverse this decay in empty space because it is lighter than the muon. The tau is heavier and can decay through analogous weak channels. Stability here is partly an accident of the mass ordering, not a difference in the underlying vertex.

Question from the audience. Is a linear accelerator effectively increasing the electron's mass so that otherwise forbidden reactions can run in the opposite direction? [a]

Response. It does not change the electron's invariant rest mass, but it supplies kinetic energy. Sufficient center-of-mass energy can create heavier final particles, so the proposed physical idea is correct once mass is replaced by available collision energy.

[a]Lecture timestamp: 01:30:23.

5.7.4 Neutron beta decay at quark level

The neutron and proton valence compositions are

$$n = udd, \qquad p = uud. \tag{5.46}$$

One down quark in the neutron makes the charged-current transition

$$d \longrightarrow u + W^{-*}, \tag{5.47}$$

turning udd into uud. The virtual boson then decays:

$$W^{-*} \longrightarrow e^- + \bar{\nu}_e. \tag{5.48}$$

Combining the two vertices reproduces

$$n \longrightarrow p + e^- + \bar{\nu}_e. \tag{5.49}$$

The opposite quark transition $u \to d + W^+$ would leave a three-down-quark baryon, but that available state is too heavy for a free neutron to produce. The actual down-to-up route is the energetically allowed one.

Question from the audience. Could a neutron decay into a proton, a muon, and the corresponding antineutrino instead? [a]

Response. Not for a free neutron at rest: the neutron–proton mass difference is far smaller than the muon mass. With enough additional collision or excitation energy, the same weak vertex can participate in a channel that produces a muon. The prohibition is kinematic, not a missing interaction.

[a]Lecture timestamp: 01:32:38.

5.8 The Unresolved Suppression

The charged weak vertices explain which processes can occur, but not yet why they are so slow. One tempting answer would be an extraordinarily tiny weak coupling. That answer is wrong: the dimensionless weak gauge coupling is of roughly electromagnetic size. The suppression must come from something else.

The missing ingredient is already visible in the virtual propagator: the W is very massive. At energies far below m_W, charged-current amplitudes are suppressed roughly by powers of $1/m_W^2$. That connection will be developed after the Higgs mechanism and electroweak symmetry breaking explain how the gauge bosons acquire mass. [6]

Question from the audience. When we say that the theory has this symmetry, do we mean that its Lagrangian is invariant under the corresponding operations? [a]

Response. Exactly. The statement that the theory has the symmetry is shorthand for invariance of the Lagrangian, or equivalently the action, under those transformations. Writing the full Lagrangian is being postponed, not replaced by a different definition.

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[a]Lecture timestamp: 01:34:14.

The route is now complete: Maxwell theory supplies the prototype; QCD shows how non-Abelian gauge bosons carry and exchange charge; coupling constants explain the contrast between atomic and hadronic time scales; and weak $SU(2)$ organizes quarks and leptons into doublets connected by $W^{\pm}$. The remaining task is to understand why these gauge-mediated processes are weak in practice even though their underlying coupling is not exceptionally small.

[6]Editorial clarification: The lecture preserves this as the puzzle for the next meeting. The heavy-W explanation is included here as a forward connection, not as a claim that its derivation occurred in this lecture.

Chapter 6

Massive Weak Exchange, Virtual Particles, and Broken Symmetry

The Standard Model group becomes much less mysterious when we read it as a set of allowed motions on the fermion table. Color transformations move vertically among red, green, and blue. Charged weak transformations move horizontally within quark and lepton doublets. Electromagnetic phase transformations supply a third operation. The corresponding gauge bosons are not decorations added after the symmetries have been found: emitting a gauge boson physically performs the same change of state that a generator of the symmetry describes.

That observation sets up the main question of this lecture. A weak vertex has a coupling comparable in size to an electromagnetic vertex, yet beta decay is enormously slower than a typical strong or electromagnetic process. The missing ingredient is propagation. At low momentum, exchanging a massive W costs a factor of order $1/m_W^2$ in the amplitude. We shall develop that statement first in distance space and then in momentum space, follow its consequences through a long discussion of real and virtual particles, and finally turn to the first qualitative model of spontaneous symmetry breaking.

6.1 Reading Gauge Symmetry from the Fermion Table

6.1.1 Color acts vertically

Arrange the six quark flavors in three weak families,

$$(u, d), \qquad (c, s), \qquad (t, b), \tag{6.1}$$

and give every quark a red, green, or blue color label. The resulting quark table has three color rows and six flavor columns. Color $SU(3)$ mixes the three entries in any one flavor column. For example, one generator may turn a red up quark into a green up quark. In particle language the same operation is accompanied by emission of a gluon carrying the compensating color and anticolor.

The same color transformation must act on every flavor. The gluon that changes a red up quark into a green up quark is the same species of gluon that changes a red top quark into a green top quark. A genuine symmetry operation does not apply to the first generation while ignoring the others. Thus color motion is vertical and universal:

$$q_r \longleftrightarrow q_g \longleftrightarrow q_b, \qquad q \in \{u, d, c, s, t, b\}. \tag{6.2}$$

It changes color but not flavor.

This is the characteristic relation between a gauge group and its gauge bosons. The infinitesimal generators describe transformations among states, and emission or absorption of the corresponding bosons realizes those transformations as physical processes.

6.1.2 Weak transformations act horizontally

The charged weak interaction moves in the other direction. An up-type quark can emit a charged weak boson and become its down-type partner without changing color:

$$u_b \longrightarrow d_b + W^+. \tag{6.3}$$

Exactly the same W^+ occurs in

$$c_g \longrightarrow s_g + W^+, \qquad t_r \longrightarrow b_r + W^+. \tag{6.4}$$

The labels r, g, b are spectators. Gluons do not move horizontally across the table, and charged W bosons do not move vertically through color.

Add the electromagnetic phase symmetry $U(1)$, under which every charged field acquires a phase. The first-pass group structure is then

$$SU(3) \times SU(2) \times U(1). \tag{6.5}$$

The cross signs mean that these are independent factors: one acts on color, one on weak doublets, and one by a phase. This is the useful rough picture, but it is not yet the complete electroweak story.[1]

6.1.3 The same pattern includes leptons

Below the three color rows place a colorless lepton row:

$$\begin{array}{c|cc|cc|cc} & u & d & c & s & t & b \\ \hline Q & +\frac{2}{3} & -\frac{1}{3} & +\frac{2}{3} & -\frac{1}{3} & +\frac{2}{3} & -\frac{1}{3} \\ \text{lepton partner} & \nu_e & e^- & \nu_\mu & \mu^- & \nu_\tau & \tau^- \\ Q & 0 & -1 & 0 & -1 & 0 & -1 \end{array} \tag{6.6}$$

Within every adjacent pair the electric charges differ by one unit. That is why the same charged weak boson can connect u to d, c to s, t to b, and a charged lepton to its neutrino. The W vertex cares about the appropriate quantum numbers and charge difference, not about whether the fermion is a quark or a lepton.

[1] Editorial clarification: The exact Standard Model gauge group is conventionally written $SU(3)_c \times SU(2)_L \times U(1)_Y$. The weak $SU(2)_L$ acts only on left-handed doublets, and the final electromagnetic $U(1)_{\text{em}}$ emerges from $SU(2)_L \times U(1)_Y$ after symmetry breaking. The lecture explicitly warns that its first-pass identification of the last two factors is not yet fully consistent; later lectures supply the missing chiral and electroweak structure.

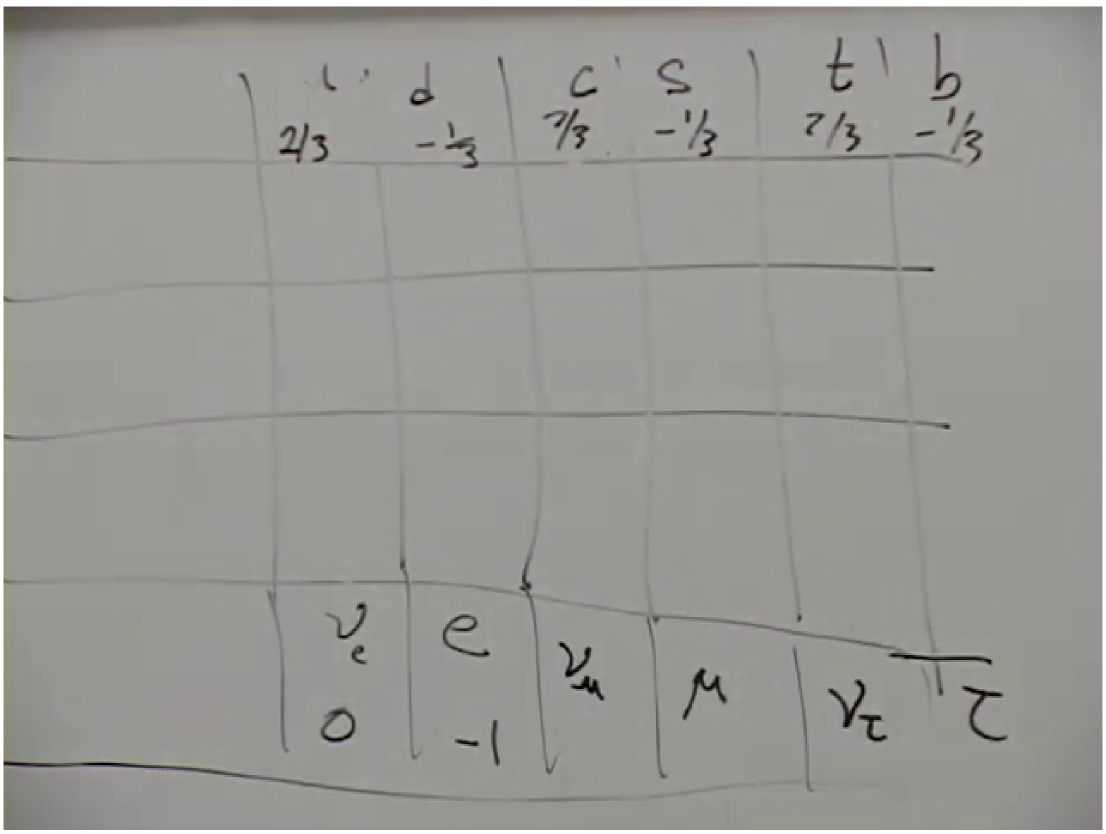

Figure 6.1: The completed fermion table: three quark-color rows above the colorless lepton row, with the quark charges and three lepton doublets visible.

Lecture frame: 00:18:55.

The muon has the same charge and known intrinsic quantum numbers as the electron, but is about two hundred times heavier. The tau is heavier still. The neutrino masses are very small and were not known in detail in the lecture's discussion. No principle has yet emerged here that explains the observed pattern of family masses.

Question from the audience. Why are leptons, baryons, and mesons called by those names? [a]

Response. The historical Greek roots meant roughly light, heavy, and intermediate. This mass-based naming became misleading. The muon was once called the mu meson because its mass lay between the electron and proton masses, but its interactions show that it is a lepton, much more like an electron than a pion. Today a meson means a quark–antiquark hadron and is a boson; the muon is not a meson.

[a]Lecture timestamp: 00:19:02.

6.2 The Puzzle of Weakness

6.2.1 A concrete charged-current process

Use neutron beta decay as the working diagram. The valence quark compositions are

$$n = udd, \qquad p = uud. \tag{6.7}$$

One down quark makes the transition

$$d \longrightarrow u + W^-, \tag{6.8}$$

and the charged boson produces

$$W^- \longrightarrow e^- + \bar{\nu}_e. \tag{6.9}$$

Together,

$$n \longrightarrow p + e^- + \bar{\nu}_e. \tag{6.10}$$

The other two valence quarks carry through as spectators. At the first vertex the neutron becomes a proton; at the second, the same weak interaction that connects a lepton doublet converts the W^- into the observed electron and electron antineutrino.

6.2.2 Three couplings, three very different impressions

In QED the electron charge e is the vertex amplitude. Squaring it gives the characteristic emission probability, with conventional factors collected in

$$\alpha = \frac{e^2}{4\pi} \simeq \frac{1}{137} \tag{6.11}$$

in natural units. The strong interaction has a coupling g_s, and at a typical hadronic scale its analogue α_s may be of order 0.2. That is much larger than 1/137, so gluon emission is common when a quark is struck.

Let g_w denote the charged weak coupling. Its square is not fantastically small; at the order-of-magnitude level used here it is comparable to the electromagnetic coupling. Yet a free neutron waits roughly fifteen minutes before decaying, while a strong decay may occur on a hadronic crossing time. The historical estimate in the lecture uses the older rough figure of twelve minutes, but either number makes the same puzzle unavoidable.[2]

> If g_w is not extraordinarily small, what makes weak processes weak?

The answer cannot lie at the vertices alone. Every Feynman diagram also contains propagators, the factors that describe the intermediate motion from one vertex to another.

6.3 A Massive Propagator in Distance Space

6.3.1 Vertices are not the whole diagram

The beta-decay diagram contains two weak vertices, contributing two powers of g_w, and a W line joining them. Denote the position-space propagator by $\Delta(x-y)$, where x is the emission event and y the absorption or conversion event. It is the amplitude kernel associated with carrying the field disturbance between those events.

The exact relativistic propagator contains more structure than we need. Its important qualitative feature is its dependence on mass. At short spacelike distance it has a power-law singularity, schematically

$$\Delta(d) \sim \frac{1}{d^2}. \tag{6.12}$$

A massless field retains a power-law tail. A massive field, by contrast, is exponentially suppressed once the separation becomes large compared with its Compton scale. The graph therefore bends sharply downward.

6.3.2 Natural units locate the bend

Set

$$c = \hbar = 1. \tag{6.13}$$

Then

$$[m] = [E] = [p] = L^{-1}. \tag{6.14}$$

The familiar relations explain each equality: $E = mc^2$ identifies mass and energy, while $p\,x$ has the dimensions of $\hbar$, so momentum and distance have inverse units. The only length that can be formed from the mass m is therefore

$$d_m \sim \frac{1}{m}. \tag{6.15}$$

Restoring constants gives

$$d_m \sim \frac{\hbar}{mc}, \tag{6.16}$$

the Compton wavelength. A heavier exchanged particle has a shorter effective range.

[2]Editorial clarification: The modern free-neutron mean lifetime is about 8.8×10^2 s, with ongoing experimental tension between major measurement methods. The lecture's rough lifetime is retained as the pedagogical scale, not as a precision value.

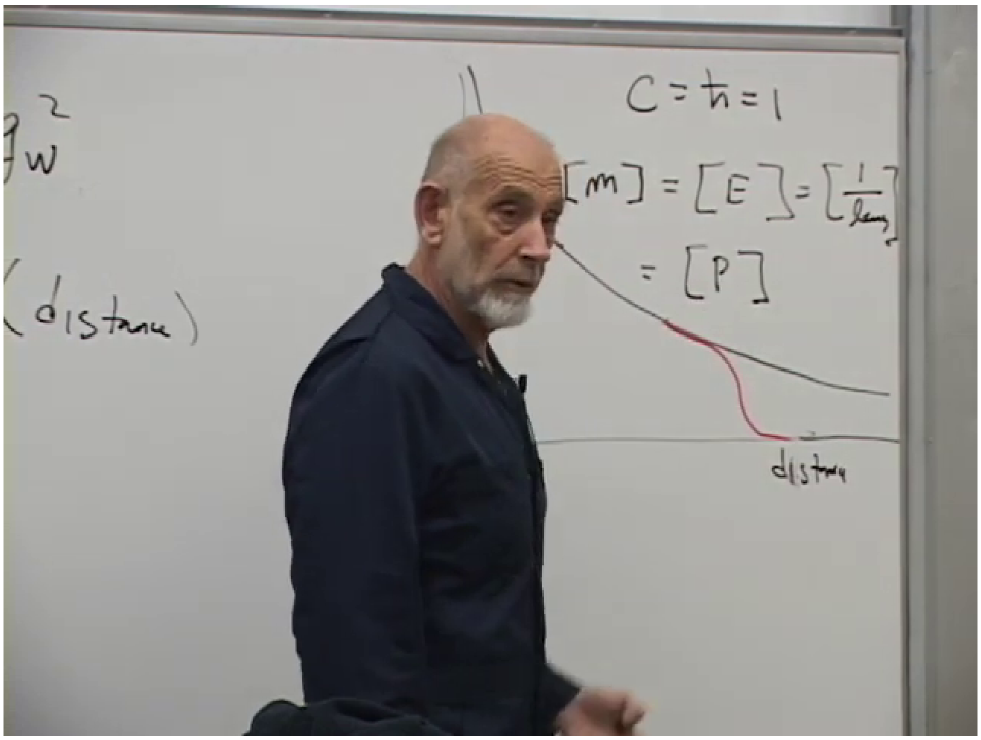

Figure 6.2: Natural-unit dimensions and the qualitative propagator falloff. The black curve continues with a power-law tail; the red massive curve turns down near the inverse-mass scale.

Lecture frame: 00:27:19.

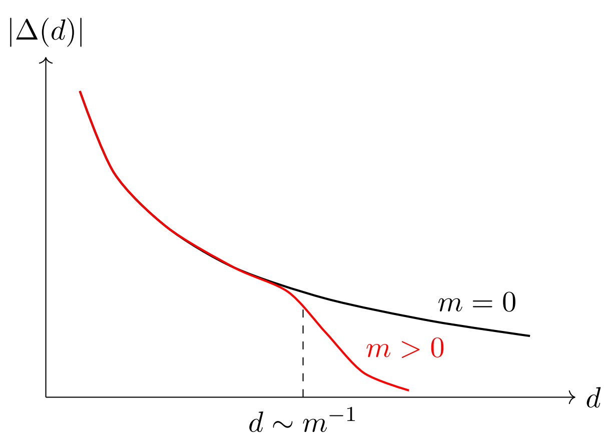

Figure 6.3: A clean schematic of the board argument. It is not an exact formula for a four-dimensional propagator; it isolates the mass-controlled range.

Question from the audience. What is plotted on the vertical axis? [a]

Response. The size of the propagator, denoted by Δ. Its square contributes to a probability after the complete amplitude, external states, and phase-space factors are included. The graph is meant to show that the amplitude for massive exchange over a long distance shuts off rapidly.[b]

[a]Lecture timestamp: 00:28:30.

[b]Editorial clarification: A propagator by itself is not a directly measurable probability amplitude for a particle following a classical path. The lecture's language is a useful range estimate: the propagator is one factor in the full scattering or decay amplitude.

6.4 Fourier Transform and the Fermi Limit

6.4.1 Experiments specify momenta

An experiment usually measures external momenta rather than the two spacetime points at which an internal line begins and ends. In beta decay, the momentum carried by the W is inferred from the difference between the neutron and proton momenta, and it equals the total momentum of the electron and antineutrino. Position-space and momentum-space propagators are related by a Fourier transform.

$$k = p_n - p_p = p_e + p_{\bar{\nu}_e}. \tag{6.17}$$

In the positive-denominator convention used on the board,

$$\widetilde{\Delta}_W(k) = \frac{1}{k^2 + m_W^2}, \qquad \widetilde{\Delta}_\gamma(k) = \frac{1}{k^2}. \tag{6.18}$$

The massless photon propagator becomes large at small momentum transfer. The massive W propagator never exceeds a scale of order $1/m_W^2$. For beta decay,

$$|k^2| \ll m_W^2, \tag{6.19}$$

so the details of k^2 are negligible and

$$\widetilde{\Delta}_W(k) \simeq \frac{1}{m_W^2}. \tag{6.20}$$

Question from the audience. Should the relativistic k^2 contain a minus sign between energy and spatial momentum? [a]

Response. Yes. With a mostly-minus Minkowski metric, $k^2 = (k^0)^2 - \boldsymbol{k}^2$, and the scalar denominator is conventionally $k^2 - m_W^2 + i\epsilon$. The board is using the equivalent positive spacelike or Euclidean magnitude. In low-energy beta decay the transferred four-momentum is tiny compared with m_W, so either notation gives the same $1/m_W^2$ suppression up to an irrelevant sign in the amplitude.

[a]Lecture timestamp: 00:35:11.

6.4.2 The low-energy amplitude

Two weak vertices and one propagator give

$$\mathcal{A}_{\text{weak}} \sim g_w\, \widetilde{\Delta}_W(k)\, g_w \tag{6.21}$$

$$\sim \frac{g_w^2}{k^2 + m_W^2} \simeq \frac{g_w^2}{m_W^2}. \tag{6.22}$$

Squaring the amplitude produces the characteristic suppression

$$|\mathcal{A}_{\text{weak}}|^2 \sim \frac{g_w^4}{m_W^4}, \tag{6.23}$$

before the kinematic and phase-space factors appropriate to the particular decay are included. The weak interaction is weak at nuclear energies not because its elementary gauge coupling is absurdly small, but because its mediator is heavy.

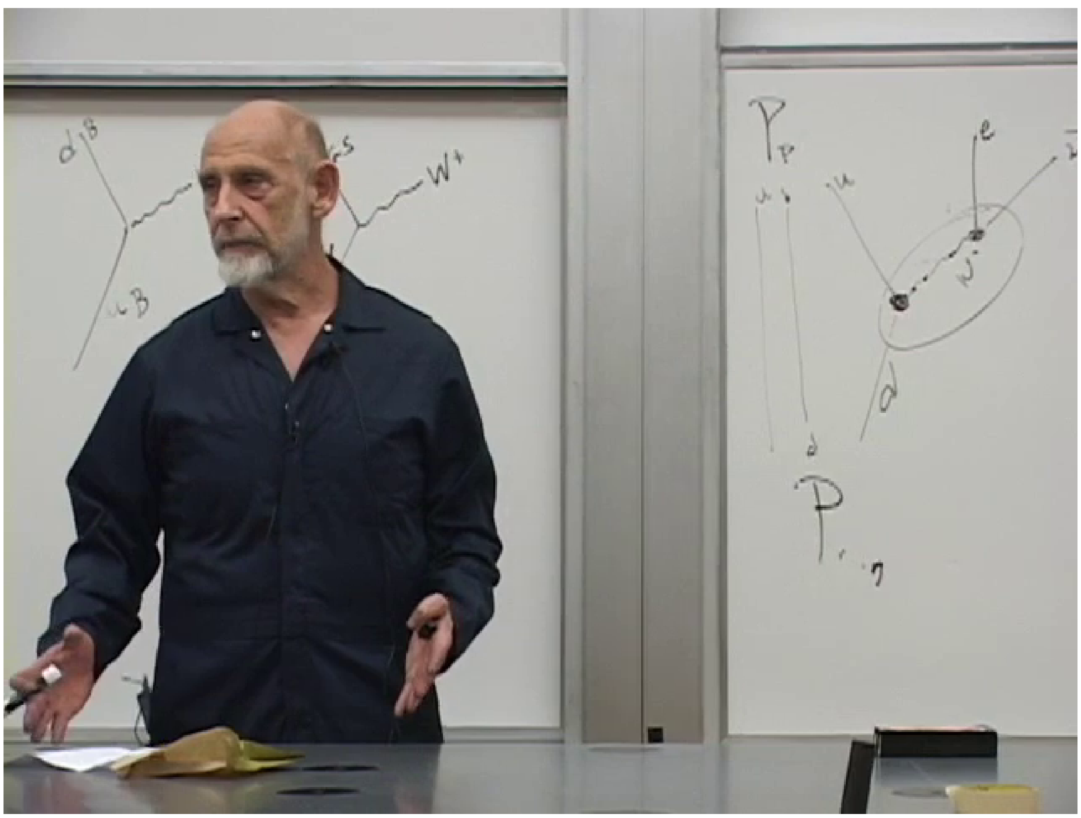

Figure 6.4: The neutron–proton weak process and the momentum transfer carried by the internal line, as discussed while fixing the propagator sign.

Lecture frame: 00:35:40.

Numerically m_W is roughly one hundred proton masses at the accuracy of this lecture. A factor of m_W^{-2} in the amplitude becomes m_W^{-4} in a rate, so even a coupling of ordinary gauge-theory size produces a very long lifetime.

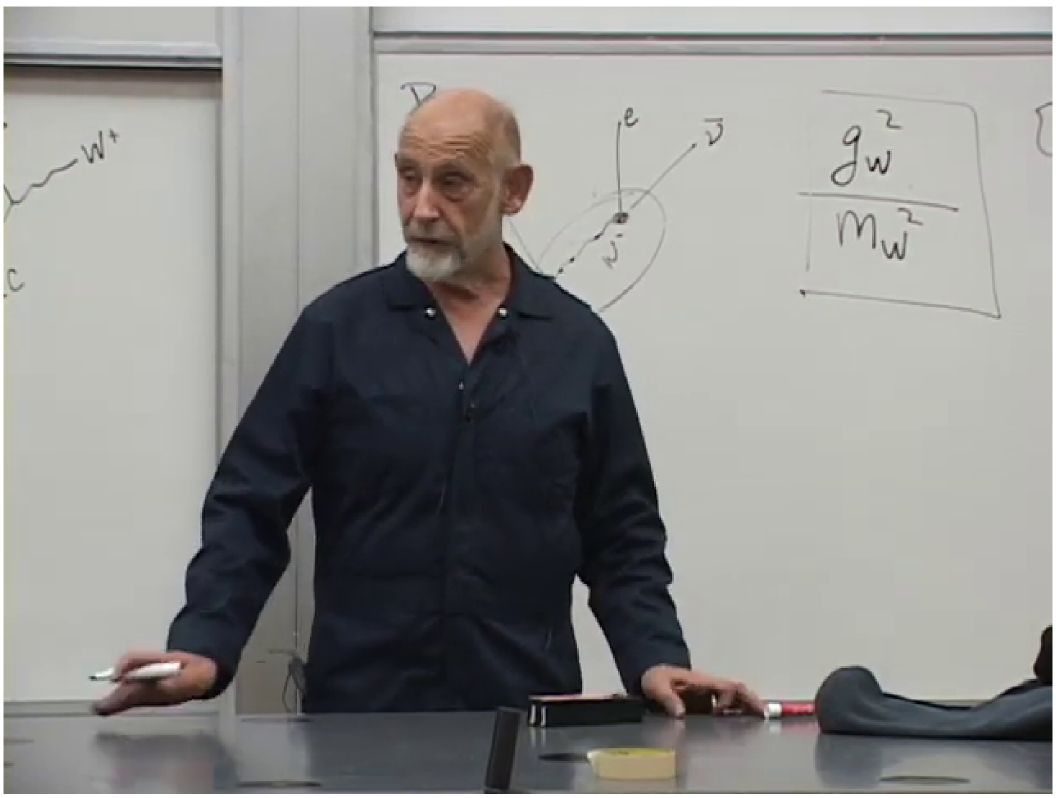

Figure 6.5: The beta-decay topology beside the boxed low-energy estimate g_w^2/m_W^2.

Lecture frame: 00:37:25.

Question from the audience. Are we talking about a suppression of one in a million or one in a billion, and how was the W mass originally known? [a]

Response. The lecture's rough counting points toward the billion-scale smallness of a low-energy weak probability relative to an unsuppressed hadronic process. Historically, low-energy experiments did not separately measure g_w and m_W. They measured the small combination g_w^2/m_W^2, called the Fermi constant. Once the gauge coupling was tied to the electroweak theory, that combination predicted a very heavy mediator; later accelerators produced the W directly and measured its mass.

[a] Lecture timestamp: 00:38:11.

At momenta far below m_W, every experiment sees nearly the same constant:

$$G_F \sim \frac{g_w^2}{m_W^2}. \tag{6.24}$$

This is why Fermi could describe beta decay in the 1930s as a pointlike four-fermion interaction, long before the mediator was observed. As collision energies increased, experiments became sensitive to the momentum dependence hidden in the propagator, confirming that the effective constant was the low-energy shadow of massive exchange.[3]

Historically, the idea that weak interactions might be mediated by a heavy particle emerged in the mid-twentieth century; the lecture mentions Julian Schwinger among the theorists who explored that possibility and notes that the letter W probably stood for weak. Low-energy data already implied that such a mediator would have to be extremely heavy on the nuclear scale, long before accelerators had enough energy to make it as a real particle.

6.4.3 Three independent controls on a rate

A classroom question about nuclear binding separates three logically different controls on a transition:

1. vertex coupling constants;
2. propagators, including masses and momentum dependence;
3. available initial-to-final energy and phase space.

Binding can close a decay channel because the final state would have greater energy than the initial state. That is a kinematic prohibition, not a further reduction of g_w or a change in m_W. When a channel remains open but has little available energy, its phase space can still be strongly suppressed.

[3] Editorial clarification: With standard electroweak normalization at tree level,

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8m_W^2}.$$

The lecture deliberately suppresses numerical and spinor factors and keeps only the scaling $G_F \sim g_w^2/m_W^2$. The W and Z were directly discovered at CERN in 1983; the lecture's passing mid-to-late-1970s dating is too early for the direct observation.

Question from the audience. Does the field or binding of a nearby proton suppress beta decay through the same propagator mechanism? [a]

Response. Not in the example being discussed. Binding matters because it changes the energies of the initial and final nuclear states. A process cannot occur if the final state is energetically inaccessible. Couplings, propagators, and available energy are three separate ingredients.

[a]Lecture timestamp: 00:41:13.

Question from the audience. Are the coupling constants really constant? [a]

Response. Only at a fixed scale. Quantum corrections make effective couplings run with momentum or distance. The present argument treats each coupling at the scale relevant to the process; scale dependence will be developed later.

[a]Lecture timestamp: 00:42:55.

6.5 Virtual Exchange without Abandoning Conservation Laws

6.5.1 Why the heavy line can appear

Ordinary neutron decay does not contain enough energy to produce a free, on-shell W. The W in the diagram is internal and off shell. The lecture introduces this fact with the traditional energy–time estimate

$$\Delta E\, \Delta t \lesssim \hbar : \tag{6.25}$$

an intermediate configuration whose classical energy appears wrong by ΔE can contribute only over a time of order $\hbar/\Delta E$.

The tunneling analogy makes the intended picture vivid. Put an alpha particle in a nuclear potential well. Its energy lies below the top of the barrier, so classically it cannot escape. Quantum mechanically the wavefunction extends through the barrier, and the particle can emerge at a point on the far side with the same total energy it had initially. If we try to describe the interior of the barrier as an ordinary classical trajectory, the potential energy seems too large and the kinetic energy would have to be negative. That failed classical picture is analogous to trying to regard the internal W as a real particle.

In the beta-decay picture the neutron may fluctuate into a proton plus an internal W^-, but the heavy line must be removed quickly. It may be reabsorbed, restoring the neutron, or it may terminate at the second vertex and produce the much lighter electron and antineutrino. The full initial and final energies agree.[4]

6.5.2 Real W Bosons

A sufficiently energetic accelerator can produce a real W. It then has its own proper lifetime, unrelated to a putative uncertainty time for the virtual line. A boosted real W lives longer in the

[4]Editorial clarification: Energy and momentum are conserved at every Feynman vertex. Virtual particles do not literally borrow energy and repay it later. The precise statement is that an internal line is integrated over four-momenta that need not satisfy the real-particle mass-shell relation $k^2 = m_W^2$. The uncertainty-principle and tunneling language is retained because it is the lecture's explanatory route, but the off-shell statement is the modern QFT formulation.

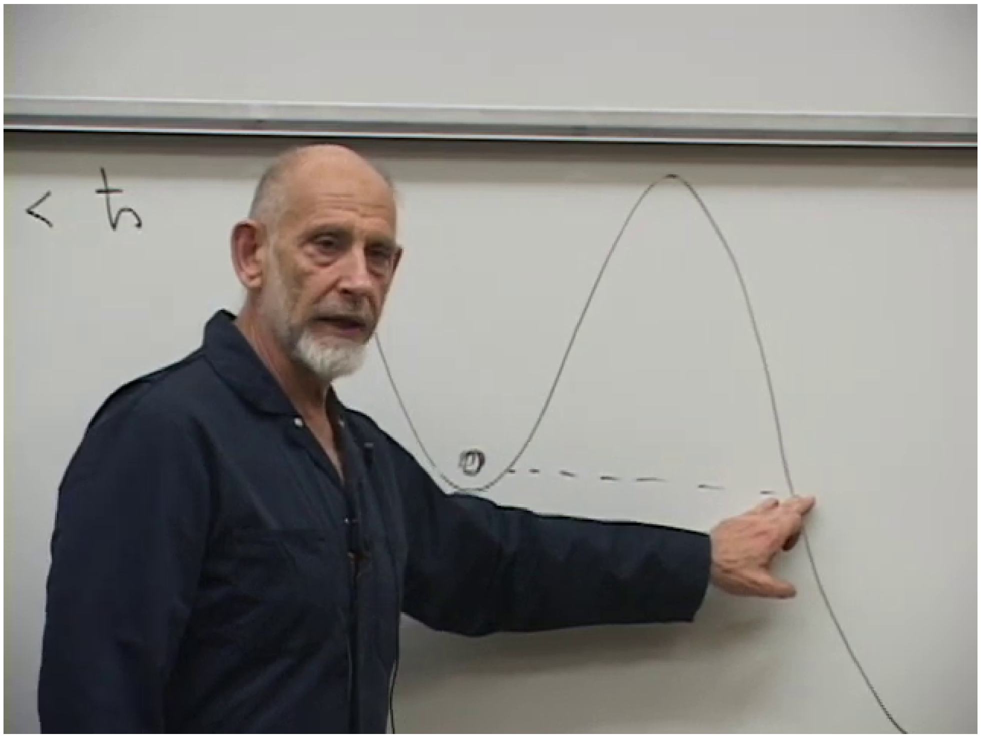

Figure 6.6: The tunneling analogy: the initial point and the marked point on the far slope have equal total energy, while the intervening barrier is classically forbidden.

Lecture frame: 00:46:00.

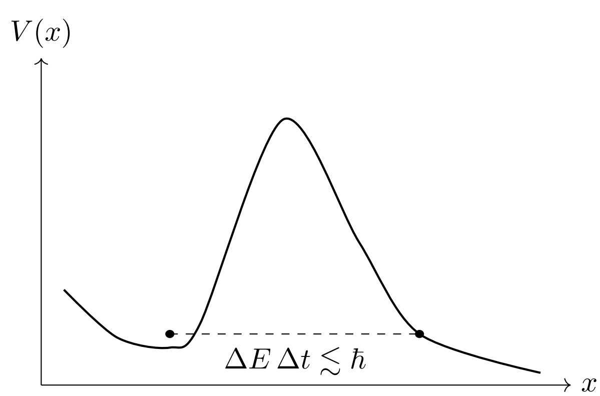

Figure 6.7: A readable reconstruction of the barrier and equal-energy points.

laboratory by ordinary relativistic time dilation. In a two-body leptonic decay of a W at rest,

$$W^- \longrightarrow e^- + \bar{\nu}_e, \tag{6.26}$$

the electron and antineutrino carry large, nearly back-to-back kinetic energies, each of order $m_W/2$. The rest energy has not disappeared; it has become final-state energy. Muon and tau channels, as well as quark channels, are also available.

Question from the audience. How long does a real W live, and can its lifetime be computed from the same coupling? [a]

Response. Yes. One calculates each allowed decay amplitude, squares it, integrates over final-state phase space, and sums the partial widths. The inverse total width gives the lifetime. The transcript's spoken numerical estimate is garbled, but the qualitative point is that the proper lifetime is extremely short and a moving W is time-dilated in the usual way.[b]

[a]Lecture timestamp: 00:49:49.

[b]Editorial clarification: Using the measured width $\Gamma_W \simeq 2.1\,\text{GeV}$ gives $\tau_W = \hbar/\Gamma_W \simeq 3.2 \times 10^{-25}\,\text{s}$. This modern number resolves the corrupted numerical exchange without attributing it to the lecture.

Question from the audience. How does a squared amplitude become a rate and hence a lifetime? [a]

Response. The squared matrix element is combined with the density of available final states, conservation delta functions, and normalization factors to obtain a probability per unit time, the decay width Γ. The survival probability is $e^{-\Gamma t}$ in units with $\hbar = 1$, so $\tau = 1/\Gamma$, or $\tau = \hbar/\Gamma$ with units restored. A small weak width therefore means a long weak lifetime.

[a]Lecture timestamp: 00:50:52.

Question from the audience. Can one assign a velocity to a particle tunneling across the barrier? [a]

Response. One can divide an inferred distance by an inferred transit time, but that quantity should not be interpreted as the velocity of a classical particle or a signal. Attempting to measure a trajectory inside the barrier changes the state and destroys the original tunneling experiment.

[a]Lecture timestamp: 00:52:42.

6.6 What Does It Mean to Observe a Virtual Particle?

6.6.1 External particles and internal lines

Real particles are the incoming and outgoing asymptotic states that can be prepared or detected far from the interaction region. Virtual particles are internal lines in a perturbative representation of the amplitude. They are not additional hidden detector tracks.

Question from the audience. Is there a clear definition of the difference between a real and a virtual particle? [a]

Response. A real particle is an external on-shell state entering or leaving the interaction region. A virtual particle is an internal off-shell line in a Feynman diagram. The latter is part of the mathematical expansion, not an independently observed object.

[a]Lecture timestamp: 00:55:05.

To resolve a process lasting only Δt, a probe needs frequencies of order $1/\Delta t$ and hence energy of order $\hbar/\Delta t$. A photon sharp enough to resolve an alpha particle beneath a high barrier can supply the energy needed to lift it into the barrier region. Likewise, a probe energetic enough to resolve the short-lived internal W can create a real W. The attempted observation turns the original process into a different experiment.

The lecture phrases this as a recurring quantum-mechanical frustration: the intervention required to check the forbidden intermediate picture supplies the very energy that makes the apparently forbidden configuration real.

6.6.2 Fluctuations of a proton

In perturbation theory a proton state contains amplitudes for configurations such as

$$p \longleftrightarrow n + W^+, \tag{6.27}$$

and a neutron similarly has

$$n \longleftrightarrow p + W^-. \tag{6.28}$$

If no other interaction intervenes, the internal boson is reabsorbed. A hard probe can instead leave a W among the outgoing particles. One may call that act knocking out a fluctuation or creating a W. The language differs; the measurable statement is the same final-state amplitude.

Question from the audience. When a hard probe produces an outgoing W, did it knock out a particle that was already there or create a new one? [a]

Response. Either picture can organize the mathematics. One time-ordered description says that the proton fluctuated to $n + W^+$ and the probe removed the neutron before reabsorption; another says the collision created the final W. What is invariant is the calculated probability for the observed final state, not a unique classical story about what existed before measurement.

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[a] Lecture timestamp: 01:01:20.

Increasing the collision energy does not expose a fixed inventory of objects that had been hidden inside the target. It opens many possible final states: one W, several bosons, quark jets, hadrons, and other particles allowed by the conserved quantum numbers. Striking even an elementary field hard enough can produce protons and many other composite objects. What comes out need not have been present beforehand as a constituent. The lecture pushes the point to comic extremes: if energy and quantum numbers allow it, even tables, chairs, or a hydrogen atom are not logically excluded as fantastically unlikely final arrangements. The joke carries a serious warning against defining elementarity by a naive inventory of collision debris.

6.6.3 Vacuum fluctuations and temperature

A fluctuation attached to a proton is not the same as a vacuum diagram. The first is a component of an interacting proton state; a vacuum fluctuation has no external proton line. In either case, a sufficiently energetic probe may produce real particles corresponding to an allowed final state. Two energetic photons, for example, can produce an electron–positron pair when the center-of-mass energy exceeds threshold.

Question from the audience. Are these reactions dependent on temperature? [a]

Response. The elementary amplitude does not require a thermal bath. Temperature matters statistically: a high-temperature system contains many particles with high kinetic energy, so threshold-crossing collisions become common. Heating a system containing a bound alpha particle may also supply enough energy for over-the-barrier escape in addition to quantum tunneling. A few isolated energetic particles constitute a beam; many thermally distributed energetic particles constitute a hot bath.

[a]Lecture timestamp: 01:05:17.

Question from the audience. Is the W emitted in a proton fluctuation a fluctuation of the vacuum? [a]

Response. No. The internal W in $p \leftrightarrow n + W^+$ is a fluctuation of the interacting proton state. A closed virtual process with no external proton is a vacuum fluctuation. They obey the same field-theory rules but belong to different amplitudes.

[a]Lecture timestamp: 01:06:56.

Question from the audience. Why is the antineutrino required at the leptonic W vertex? [a]

Response. It supplies the conserved lepton quantum numbers and is part of the basic charged-current vertex connecting e, ν_e, and W. The bar distinguishes the outgoing electron antineutrino from a neutrino.

[a]Lecture timestamp: 01:08:51.

6.7 How Far Can Energy Conversion Be Pushed?

The class next drives the rule that energy can create particles to an intentionally absurd limit. Could a coffee bean become a chocolate cake? There is no practical process of that kind, but relativistic quantum theory does not impose a rule that final matter must resemble the initial object. It requires enough energy, the correct conserved charges, and a nonzero amplitude. A kilogram has an energy of order

$$Mc^2 \sim 9 \times 10^{16} \text{ J}, \tag{6.29}$$

so any microscopic collision that assembled a particular kilogram-scale object would be beyond fantastically improbable.

The Planck mass,

$$m_{\mathrm{P}} = \sqrt{\frac{\hbar c}{G}} \simeq 2.2 \times 10^{-5} \text{ g}, \tag{6.30}$$

is not a maximum possible mass. It is the scale at which gravity becomes strong for a quantum localized to its Compton wavelength. A kilogram contains roughly 5×10^7 Planck masses.

Question from the audience. If collisions exceed the Planck scale, can arbitrarily heavy objects such as chairs or planets emerge? [a]

Response. At trans-Planckian center-of-mass energy concentrated in a small region, gravitational collapse is the overwhelmingly natural outcome: a black hole. A highly organized macroscopic object would be an extraordinarily rare fluctuation. The lecture invokes time reversal: because a chair can fall into a black hole, the reverse amplitude is not exactly zero. Its probability is suppressed by an enormous entropy factor, so possibility must not be confused with practical likelihood.

[a] Lecture timestamp: 01:10:37.

This excursion reinforces the earlier warning. Quantum field theory predicts amplitudes between specified initial and final states. A picturesque claim that every product was literally contained in the target before impact is not part of the prediction.

6.8 Bosons, Their Interactions, and Their Statistics

6.8.1 Photons can scatter

Being a boson does not mean being noninteracting. Boson or fermion specifies exchange statistics; interaction vertices are separate dynamical facts. Classical light beams normally pass through one another because QED has no tree-level four-photon vertex and low-energy photon–photon scattering is extremely small. Nevertheless, two photons can interact through a virtual electron–positron loop:

$$\gamma + \gamma \longrightarrow (e^- e^+)_{\text{virtual}} \longrightarrow \gamma + \gamma. \tag{6.31}$$

When the center-of-mass energy is high enough, they may instead create a real pair,

$$\gamma + \gamma \longrightarrow e^- + e^+. \tag{6.32}$$

Optical photons have energies of a few electron volts, while the pair threshold is of order $2m_e c^2 \approx 1.02\,\text{MeV}$. The virtual process is therefore heavily suppressed at ordinary optical energies, which is why crossing light beams appear not to notice each other.

Question from the audience. Do virtual W bosons interact with other bosons simply because they are bosons? [a]

Response. They can interact, but not merely because they obey Bose statistics. Gauge vertices determine the dynamical interactions. Statistics instead determines how multiparticle amplitudes behave when identical particles are exchanged. Photons interact weakly through charged virtual particles, and the electroweak theory also contains direct non-Abelian W-boson self-interactions.

[a] Lecture timestamp: 01:12:47.

6.8.2 Bose enhancement is not a force

Fermion versus boson statistics becomes conspicuous when several identical particles occupy nearby quantum states. Bosons favor multiple occupation; fermions exclude identical occupation. In a typical high-energy collision, particles fly into well-separated momentum states, so this statistical effect may be less obvious than it is in a laser or condensate.

In principle, W bosons possess the same Bose enhancement. The lecture jokes that one could imagine a system that lases W bosons. Their enormous mass and ultrashort lifetime make such a device fantastically impractical, but the statistics is indeed bosonic.

Question from the audience. What color is a proton or neutron? [a]

Response. A physical hadron is a color singlet, often called colorless or white. The shorthand says that a baryon contains one red, one green, and one blue quark. More precisely, its color wavefunction is the antisymmetric superposition

$$|\text{singlet}\rangle = \frac{1}{\sqrt{6}}\epsilon_{abc}|q^a q^b q^c\rangle,$$

not a classical assignment of a permanent color to each named quark.

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[a] Lecture timestamp: 01:18:41.

Question from the audience. Will Bose condensation enter this course? [a]

Response. Yes, through the Higgs phenomenon. A nonzero Higgs field throughout the vacuum is analogous in important ways to a bosonic condensate. The analogy will be developed after spontaneous symmetry breaking has been introduced.[b]

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[a] Lecture timestamp: 01:19:25.

[b] Editorial clarification: The Higgs vacuum expectation value is not literally an ordinary gas of on-shell Higgs particles occupying one state. Condensation is an illuminating analogy for a coherent bosonic field and a chosen vacuum, not a complete microscopic identification.

6.9 The Missing Neutral Weak Boson

Return now to the group table. If the horizontal weak symmetry is $SU(2)$, then it has

$$\dim SU(2) = 3 \tag{6.33}$$

generators and therefore three gauge fields. The charged combinations W^+ and W^- account for only two. A third, neutral field is required. It is closely related to the observed Z boson, but it is not simply the photon. The precise neutral combinations involve both the third $SU(2)_L$ field and the $U(1)_Y$ field. That mixing, together with the correct definition of electric charge, is deliberately postponed to the next lecture.

6.10 Spontaneous Symmetry Breaking

6.10.1 A symmetric choice can have an unstable indecision

Imagine a circular dinner table at which each place setting has a utensil to the left and one to the right. Before anyone moves, left and right are equivalent. Once one guest reaches to the right, every neighboring choice is forced and the whole table settles into the right-handed arrangement. The rules were symmetric, but the realized configuration selected one of two possibilities. The story is only an analogy, yet it isolates the combination we need: symmetry, multiple equivalent outcomes, and an instability that lets an arbitrarily small disturbance select one outcome globally.

6.10.2 The heads–tails lattice

For a cleaner model, put a two-state variable $s_i = \pm 1$ at every site of a lattice. Call $+1$ heads and -1 tails. Let neighboring sites prefer to align:

$$H = -J \sum_{\langle ij \rangle} s_i s_j, \qquad J > 0. \tag{6.34}$$

Parallel neighbors contribute lower energy than antiparallel neighbors. There are therefore two degenerate ground states,

$$\begin{aligned} s_i &= +1 \quad \text{for every } i, \\ s_i &= -1 \quad \text{for every } i. \end{aligned} \tag{6.35}$$

The Hamiltonian is invariant under the global flip $s_i \to -s_i$, but either chosen ground state is not.

Now fix one very distant boundary spin to heads. In an ordinary short-range system we might expect one far-away degree of freedom to have no influence here. In the ordered phase, however, that boundary condition selects the all-heads vacuum. Once selected, a single tail in the sea of heads costs energy because it disagrees with its neighbors. Local excitations are defined relative to the chosen ground state even though the underlying equations retain the heads–tails symmetry.

6.10.3 Explicit breaking and domain walls

Explicit breaking would add a local bias,

$$H_{\text{explicit}} = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i, \qquad h \neq 0. \tag{6.36}$$

Then heads and tails do not have equal energy even at one isolated site, and there is only one preferred vacuum. In spontaneous breaking $h = 0$: the local rules are symmetric and the two vacua remain degenerate, but the system occupies one of them.

Insist instead that the far-left boundary is all heads and the far-right boundary all tails. The lowest-energy configuration must interpolate between the two vacua. The interface is a domain wall.

Question from the audience. What distinguishes spontaneous symmetry breaking from ordinary explicit symmetry breaking? [a]

Response. With explicit breaking the local Hamiltonian already assigns different energies to heads and tails, so there is no pair of symmetry-related ground states. With spontaneous breaking the equations retain the symmetry, the vacua are degenerate, and boundary conditions or an arbitrarily small disturbance select one. The possibility of a domain wall interpolating between distinct vacua is a diagnostic of the latter situation.

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[a] Lecture timestamp: 01:28:36.

The model is deliberately primitive, but it contains the essential logic that will be transferred to the Higgs field: a symmetric theory, an unstable or degenerate vacuum structure, a selected ground state, and physical consequences determined by that choice.

6.11 Where the Lecture Leaves Us

The fermion table turns the Standard Model group into a set of physical directions. Color $SU(3)$ moves vertically through red, green, and blue; charged weak transformations move horizontally

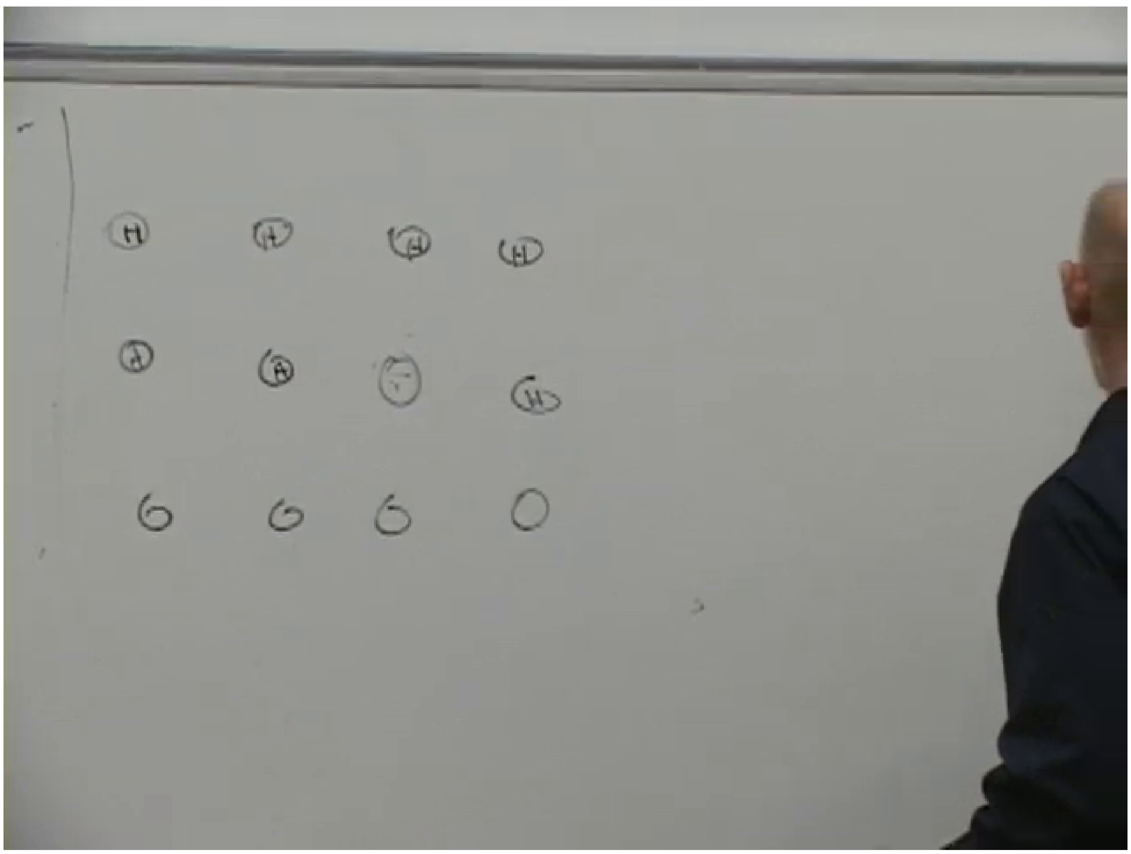

Figure 6.8: The heads–tails lattice divided by a dashed domain wall. Opposite boundary conditions force the system to contain an interface between the two degenerate ordered states.

Lecture frame: 01:30:35.

through quark and lepton doublets; a phase symmetry supplies the preliminary $U(1)$ factor. Gauge boson emission realizes these group transformations as interactions.

The weakness of beta decay is then traced not to an exceptionally tiny g_w, but to the heavy mediator. In distance space, mass cuts off the propagator beyond the Compton scale $\hbar/(m_W c)$. In momentum space,

$$\mathcal{A}_{\text{weak}} \sim \frac{g_w^2}{k^2 + m_W^2} \simeq \frac{g_w^2}{m_W^2} \tag{6.37}$$

at nuclear momenta. Old low-energy experiments measured this combination as the Fermi constant.

The extended classroom discussion clarifies what this formula does and does not mean. Internal W lines are virtual and off shell; energy–momentum is still conserved. A hard enough probe can produce a real W, but then it has created a different final-state process. Collision products are not a literal list of pre-existing constituents. Temperature changes the population of energetic projectiles, photons can interact through charged virtual particles, and Bose statistics is distinct from a dynamical force.

Finally, $SU(2)$ requires a missing neutral gauge field related to the Z, and the heads–tails lattice supplies the first model of spontaneous symmetry breaking. The next step is to combine those two unfinished threads: electroweak mixing and the Higgs mechanism.

Chapter 7

Broken Symmetry, Goldstone Bosons, and Gauge Invariance

We want to understand how a photon-like particle can acquire a mass. That is the destination, but it is not a sensible starting point. A direct mass term for a gauge field appears to destroy the very gauge invariance that makes the theory consistent. Before we can see the escape route, we need two ideas in working form:

1. spontaneous symmetry breaking, especially for a continuous symmetry;
2. gauge invariance, in which the symmetry operation may vary from one spacetime point to another.

Neither idea can be replaced by a slogan. We shall build the first from a magnet, translate it into scalar field theory, and discover the massless Goldstone mode. We shall then take the same $U(1)$ symmetry and make it local, which forces us to introduce a vector potential and a covariant derivative. Only at the end will the two threads be placed side by side. Their actual union—the Higgs phenomenon—is the subject of the next lecture.

7.1 A Symmetric Law with an Asymmetric Ground State

7.1.1 The two-state magnet

Begin with a lattice of small magnets. Each spin can point up or down, and neighboring spins lower their energy by pointing in the same direction. On a three-dimensional cubic lattice a spin has six nearest neighbors, but the precise coordination number is not important. What matters is the rule

$$E_{\parallel} < E_{\text{antiparallel}}. \tag{7.1}$$

The microscopic rule is unchanged if every spin is reversed:

$$\uparrow \longleftrightarrow \downarrow . \tag{7.2}$$

Nevertheless, the lowest-energy configurations are not individually symmetric. They are

$$\{\uparrow\uparrow\uparrow\cdots\}, \qquad \{\downarrow\downarrow\downarrow\cdots\}. \tag{7.3}$$

The equations allow both, but a cold macroscopic sample must choose one.

This choice is extraordinarily sensitive. Fix one spin very far away to point up. Its neighbors prefer to follow it, their neighbors prefer to follow them, and the information propagates through the sample. A perturbation that is arbitrarily small compared with the total system can select the orientation of the whole ground state. That is the instability meant by *spontaneous* symmetry breaking.

Question from the audience. Can a neutral particle orient itself in a magnetic field? [a]

Response. Yes. Electric neutrality does not imply the absence of a magnetic moment. The neutron is neutral but has a magnetic moment because of its internal charged constituents. A magnetic field can therefore distinguish its spin orientations.

[a] Lecture timestamp: 00:05:14.

Question from the audience. If the system changes orientation, does the change occur everywhere at once? Can several reorientation waves coexist? [a]

Response. The change propagates. One region tips its neighbors, rather like a three-dimensional array of dominoes, and the disturbance travels as a wave. Different regions can begin choosing independently, so several fronts and several domains may be present before one orientation wins. We shall return to precisely these waves when the symmetry becomes continuous.

[a] Lecture timestamp: 00:05:41–00:06:38.

7.1.2 Explicit breaking is a different theory

Now place the sample in an external magnetic field pointing upward. An isolated spin already has lower energy when it points up. The transformation $\uparrow \leftrightarrow \downarrow$ no longer leaves the energy unchanged: the preference has been inserted into the law itself. This is *explicit* symmetry breaking.

The difference can be tested without knowing how the sample was prepared. Take a very large specimen, hold its left boundary up, and hold its right boundary down. If a bulk magnetic field explicitly favors up, the right-hand constraint penetrates only a short distance. A macroscopic down region would pay an energy proportional to its volume, so the bulk cannot remain divided into two equally good phases.

Without the external bias, up and down have the same bulk energy. The lowest-energy constrained state then contains an up domain, a down domain, and a finite transition layer between them. The interface need not sit at the geometrical center, but there is no volume pressure forcing one phase to erase the other. This interface is a *domain wall*, the characteristic defect of this broken discrete symmetry.

Question from the audience. Does the argument require interactions only between nearest neighbors? [a]

Response. No. A second-neighbor interaction that also favors alignment changes details but not the conclusion. More interesting behavior appears if different ranges compete—nearest neighbors prefer alignment while next-nearest neighbors prefer anti-alignment, for example. That produces frustration and more complicated ordered phases, but it does not alter the basic distinction being made here.

[a] Lecture timestamp: 00:18:18.

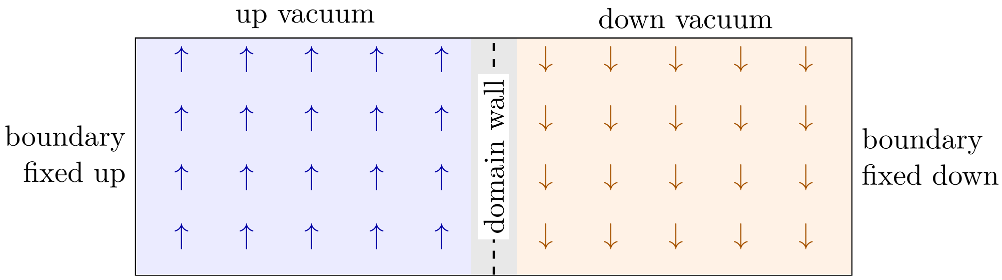

Figure 7.1: The boundary test for spontaneous breaking of a discrete symmetry. Both bulk orientations have the same energy, so a wall can separate macroscopic domains.

Question from the audience. Does *spontaneous* refer to the large derivative inside the wall?[a]

Response. No. It refers to the fact that the underlying energy has no up–down preference while the selected ground state does. The dramatic response to an arbitrarily small perturbation reveals an instability. The gradient merely tells us how much the interface costs.

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[a] Lecture timestamp: 00:19:05.

Question from the audience. Was the material already in one of the two states, or was its state initially unknown?[a]

Response. That depends on its history. At high temperature thermal motion can leave up and down spins mixed with nearly zero average magnetization. As the material cools, patches choose different orientations, domain walls form, and the domains coarsen. Chance and tiny environmental perturbations determine which vacuum finally occupies most of the sample.

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[a] Lecture timestamp: 00:19:40.

7.2 The Same Argument as Scalar Field Theory

Replace the spins by a real scalar field $\phi(\boldsymbol{x}, t)$. With metric and normalization conventions fixed once and for all, write

$$\begin{aligned} \mathcal{L} &= \frac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi - V(\phi), \\ \mathcal{E} &= \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi). \end{aligned} \tag{7.4}$$

The relativistic shorthand in $\mathcal{L}$ contains the positive time-derivative term and negative spatial-gradient term; the Hamiltonian energy density has both contributions positive.

Choose an even potential with two degenerate minima,

$$V(\phi) = \frac{\lambda}{4}\left(\phi^2 - f^2\right)^2, \qquad \lambda > 0. \tag{7.5}$$

The Lagrangian is invariant under $\phi \mapsto -\phi$, while either uniform vacuum

$$\phi(\boldsymbol{x}) = +f, \qquad \text{or} \qquad \phi(\boldsymbol{x}) = -f \tag{7.6}$$

chooses one side. A potential with a unique minimum at $\phi = 0$, by contrast, has a symmetric vacuum and is not spontaneously broken.

Fix $\phi = +f$ on one side of a sample and $\phi = -f$ on the other. The field cannot jump without cost, because

$$E_{\text{wall}} = \int dx \left[\frac{1}{2} \left(\frac{d\phi}{dx} \right)^2 + V(\phi) - V(f) \right]. \tag{7.7}$$

An abrupt jump makes $d\phi/dx$ enormous. The minimum-energy wall therefore spreads the change over a finite width, balancing gradient energy against the potential cost of passing between the minima. The familiar kink solution for the potential in Eq. (7.5) is

$$\phi_{\text{wall}}(x) = f \tanh \left[f \sqrt{\frac{\lambda}{2}} \, (x - x_0) \right], \tag{7.8}$$

an explicit realization of the smooth transition described at the board. [1]

Tilt the double well slightly and one minimum becomes lower. The Lagrangian itself is then asymmetric. A wall can still appear temporarily, but the higher-energy or false vacuum exerts a pressure on it; the wall moves and the favored phase expands. This is the field-theory version of explicit breaking by an external magnetic field.

So far the symmetry has been discrete. Particle physics is dominated by continuous groups such as $U(1)$, $SU(2)$, and $SU(3)$. A continuous symmetry changes the defect story and introduces a new kind of particle.

7.3 A Complex Field and Global $U(1)$

7.3.1 Rotations in an internal plane

Let

$$\phi = \phi_R + i\phi_I, \qquad \phi^* = \phi_R - i\phi_I, \qquad \phi^*\phi = \phi_R^2 + \phi_I^2. \tag{7.9}$$

The ϕ_R and ϕ_I axes form an internal field space, not two directions in ordinary space. A complex scalar is simply two real scalar fields packaged so that rotations in this internal plane are manifest.

Use the Lagrangian

$$\mathcal{L} = \partial_\mu \phi^* \, \partial^\mu \phi - V(\phi^*\phi). \tag{7.10}$$

For a complex field the conventional factor $1/2$ is absent, because this single product already contains the kinetic terms of both real components. The transformation

$$\phi'(x) = e^{i\theta}\phi(x), \qquad \phi^{*\prime}(x) = e^{-i\theta}\phi^*(x) \tag{7.11}$$

rotates every field value through the same angle. Here θ is constant in space and time. The prime labels the transformed field; it is not a derivative.

Because θ is constant,

$$\partial_\mu \phi' = e^{i\theta}\partial_\mu \phi, \qquad \partial_\mu \phi^{*\prime} = e^{-i\theta}\partial_\mu \phi^*. \tag{7.12}$$

The phases cancel in the kinetic term, and $\phi^{*\prime}\phi' = \phi^*\phi$. Thus every potential that depends only on $\phi^*\phi$ has the same symmetry. A term depending on ϕ_R alone would pick a direction in the internal plane and break it explicitly.

[1] Editorial clarification: The lecture argues from the shape of the potential and the gradient term but does not solve the field equation. Equation (7.8) is included as the standard solution of that stated model, making the qualitative wall concrete without changing the argument.

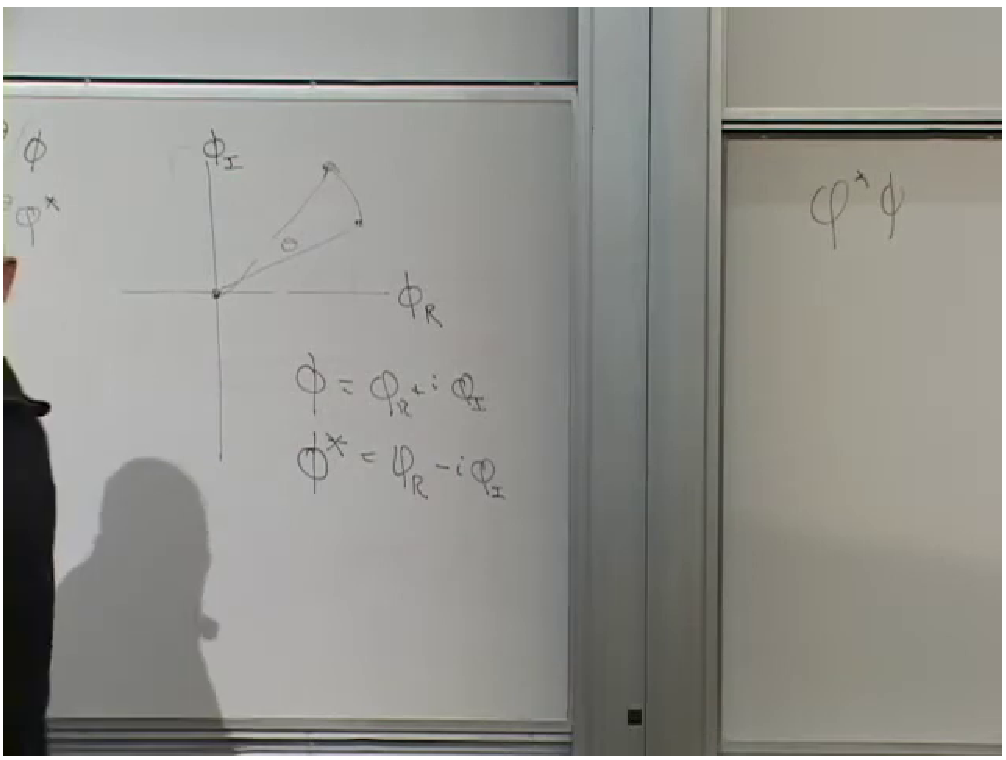

Figure 7.2: The complex field on the internal (ϕ_R, ϕ_I)-plane. The board records the decomposition of ϕ and ϕ^* and the invariant radius $\phi^*\phi$.

Lecture frame: 00:33:33.

7.3.2 A point minimum and a circle of minima

There are two qualitatively different symmetric potentials. The first has its only minimum at the origin, for example

$$V_{\text{unbroken}} = m^2 \phi^* \phi, \qquad m^2 > 0. \tag{7.13}$$

Its graph is a rotationally symmetric paraboloid. The vacuum $\phi = 0$ stays fixed under every rotation, so the symmetry is unbroken.

Now choose

$$V_{\text{broken}} = -a\, \phi^* \phi + b(\phi^* \phi)^2, \qquad a, b > 0. \tag{7.14}$$

The negative quadratic term drives the field away from the origin, while the positive quartic term prevents it from running to infinity. Writing $r^2 = \phi^* \phi$, minimization gives

$$\frac{dV}{d(r^2)} = -a + 2br^2 = 0, \qquad r^2 = f^2 = \frac{a}{2b}. \tag{7.15}$$

The minima therefore form a circle. The potential is perfectly rotationally symmetric, yet a uniform ground state must choose one point on that circle.

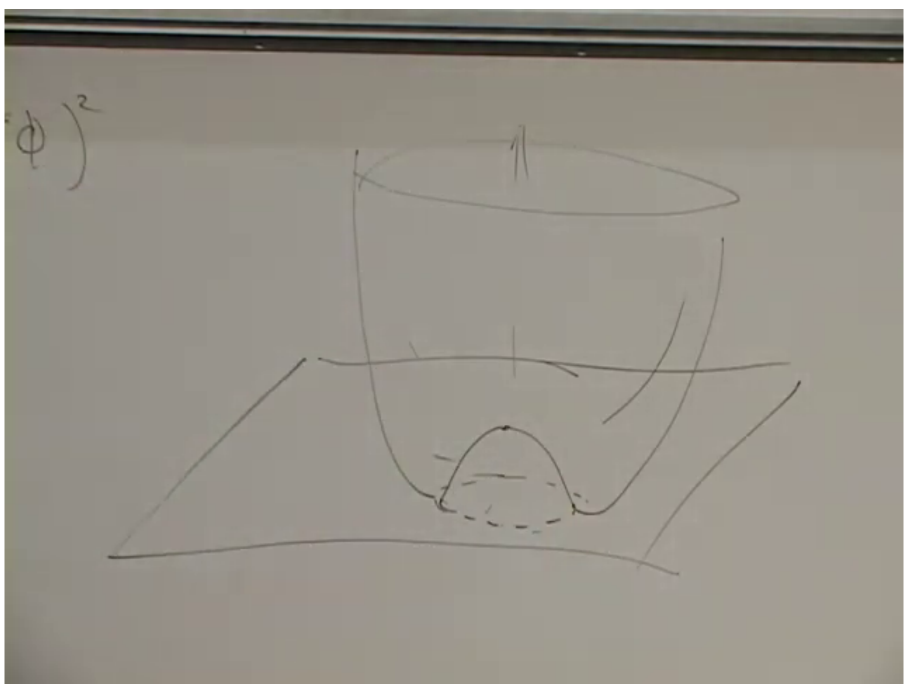

Figure 7.3: The symmetry-breaking potential drawn as a surface of revolution. The origin is unstable and the degenerate minima form the circular trough.

Lecture frame: 00:37:25.

Think of the field as a little arrow in this internal plane at every point of ordinary space. The potential asks every arrow to have length f. The gradient energy asks nearby arrows to point in nearly the same direction. The ground state consequently has the same angle everywhere, but no angle is preferred by the law. That is continuous spontaneous symmetry breaking.

7.4 From a Domain Wall to a Smooth Twist

Repeat the boundary experiment. At the left boundary fix the field at a vacuum point A; at the right fix it at another vacuum point B. A localized jump would make the gradient large. Unlike the discrete case, however, the field can remain at the bottom of the potential while changing continuously around the circle.

At low energy write the field as $\phi = f e^{i\alpha(x)}$. In one spatial dimension the relevant energy is

$$E[\alpha] = f^2 \int_0^L dx \left(\frac{d\alpha}{dx}\right)^2 . \tag{7.16}$$

For fixed endpoint angles, variation gives

$$\frac{d^2\alpha}{dx^2} = 0, \qquad \alpha(x) = \alpha_A + \frac{\alpha_B - \alpha_A}{L}\, x. \tag{7.17}$$

The preferred configuration spreads the twist uniformly through the sample. Its energy scales as

$$E_{\text{twist}} = f^2 \frac{(\alpha_B - \alpha_A)^2}{L}, \tag{7.18}$$

so it can be made arbitrarily cheap by increasing L.

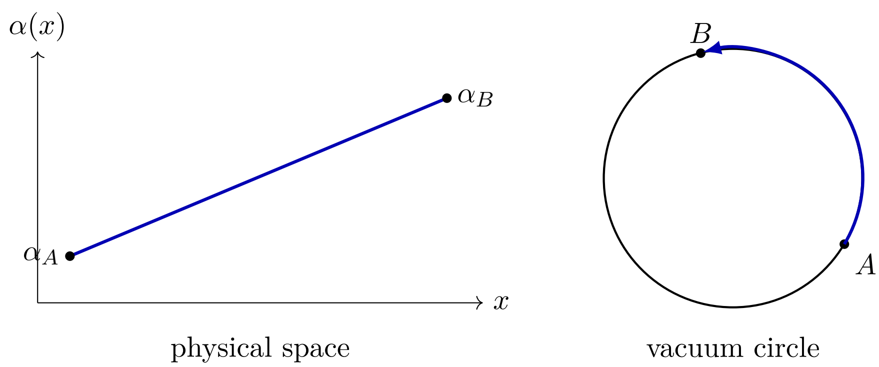

Figure 7.4: The same low-energy configuration in two descriptions. The angle varies linearly across physical space while the field follows the short arc of the vacuum circle in internal space.

Question from the audience. There can be two ways around the circle. Which path does the field take, and can it become trapped on the longer one? [a]

Response. For generic endpoints the shorter angular route has the smaller gradient and therefore the lower energy. Diametrically opposite endpoints make the two simplest routes degenerate. A configuration prepared by slowly moving an endpoint may retain an extra winding and become trapped behind an energy barrier, but for a specified pair of ordinary endpoints the untwisted shortest route is the ground state.

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[a] Lecture timestamp: 00:41:54.

Question from the audience. Could a path leave the bottom of the trough and cut across the interior, becoming shorter overall?[a]

Response. In the low-energy, long-sample regime it does not pay to climb the radial potential. Keeping the radius at its vacuum value and varying the angle linearly minimizes the effective energy, as the calculus-of-variations calculation above shows. For a finite system with a specific potential one can minimize radial and angular motion together, but the Goldstone conclusion concerns the arbitrarily long-wavelength limit in which the smooth motion along the trough wins.

[a]Lecture timestamp: 00:43:32.

Question from the audience. What changes if the circular trough is tilted so that one angle is lower?[a]

Response. Then the symmetry is explicitly broken. One point becomes the true ground state, motion along the circle acquires a restoring force, and the angular excitation is no longer exactly massless.

[a]Lecture timestamp: 00:44:33.

7.5 Mass as the Cost of a Zero-Momentum Oscillation

The smooth twist reveals a field whose energy comes only from gradients. To understand why that means a massless particle, compare the relativistic dispersion relation

$$\omega^2 = \boldsymbol{k}^2 + m^2. \tag{7.19}$$

A wave $e^{i(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}$ carries momentum $\boldsymbol{k}$. If $m = 0$, its energy tends to zero as the wavelength tends to infinity: $\omega = |\boldsymbol{k}| \to 0$. If $m \neq 0$, a spatially homogeneous mode $\boldsymbol{k} = 0$ still oscillates with frequency $\omega = m$. The mass is the irreducible zero-momentum energy.

This same distinction is visible in the Lagrangian. A term

$$-\frac{1}{2}m^2\varphi^2 \tag{7.20}$$

supplies a restoring force even for a homogeneous displacement. A field with only derivative terms has no such restoring force: make its wavelength longer, and its gradient energy approaches zero.

7.5.1 Polar variables separate the two modes

Write the broken complex field as

$$\phi(x) = \rho(x)e^{i\alpha(x)}. \tag{7.21}$$

Both ρ and α are fields. Differentiation gives

$$\partial_\mu\phi = e^{i\alpha}\left(\partial_\mu\rho + i\rho\,\partial_\mu\alpha\right), \tag{7.22}$$

$$\partial_\mu\phi^* = e^{-i\alpha}\left(\partial_\mu\rho - i\rho\,\partial_\mu\alpha\right). \tag{7.23}$$

The cross terms cancel, leaving

$$\partial_\mu\phi^*\,\partial^\mu\phi = (\partial_\mu\rho)(\partial^\mu\rho) + \rho^2(\partial_\mu\alpha)(\partial^\mu\alpha). \tag{7.24}$$

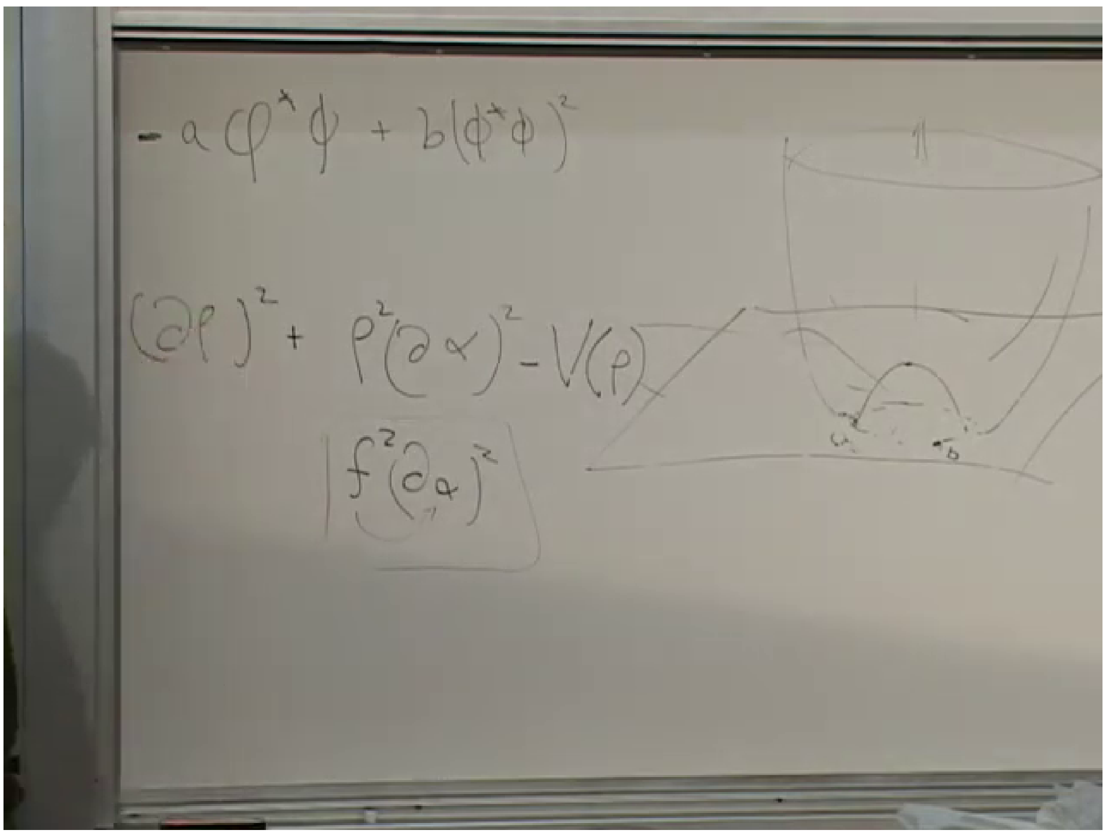

Figure 7.5: The polar-field Lagrangian, its boxed low-energy angular term, and the symmetry-breaking potential. Radial motion climbs the potential; angular motion follows the trough.

Lecture frame: 00:56:40.

Since $\phi^*\phi = \rho^2$,

$$\mathcal{L} = (\partial\rho)^2 + \rho^2(\partial\alpha)^2 - V(\rho^2). \tag{7.25}$$

The potential contains ρ but not α. That absence is the symmetry in its most useful form.

For a very low-energy excitation the field remains close to $\rho = f$. Then

$$\mathcal{L}_{\text{angular}} \simeq f^2(\partial_\mu\alpha)(\partial^\mu\alpha). \tag{7.26}$$

Define the canonically normalized angular field, up to the convention used for factors of 1/2, by

$$\beta = \sqrt{2}\, f\alpha, \qquad \mathcal{L}_{\text{angular}} = \frac{1}{2}(\partial_\mu\beta)(\partial^\mu\beta). \tag{7.27}$$

There is no β^2 term. The quanta of this angular field are massless. They are the Goldstone bosons.

The radial field behaves differently. Displace ρ from f everywhere at once and it rolls back, overshoots, and oscillates about the trough. If $\rho = f + h$, expansion of Eq. (7.14) produces

$$V(f+h) = V(f) + \frac{1}{2}m_h^2h^2 + \cdots, \tag{7.28}$$

so h is massive. Geometrically, the angular direction is flat and the radial direction is curved.

Question from the audience. If the field is displaced radially, does it simply drop back down, and what happens next?[a]

Response. With no angular motion it falls toward the minimum, overshoots, and oscillates radially. A homogeneous oscillation has zero spatial momentum but nonzero frequency; that frequency is the mass of the radial excitation. A homogeneous angular displacement, in contrast, lands on another vacuum and has no restoring force.

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[a] Lecture timestamp: 00:57:10.

This is the content of the Goldstone principle in the present example: spontaneously breaking a continuous global symmetry produces a massless mode for each independent flat direction of the vacuum manifold. [2]

7.6 Spin Waves, Strings, and Sound

The theorem is abstract, but its mechanism is familiar. In an isotropic ferromagnet all spins may be rotated together without changing the energy. Let the orientation vary very slowly across the sample and one obtains a spin wave. Its cost vanishes with its wave number, so it is the Goldstone mode of broken spin-rotation symmetry.

A long elastic string supplies another picture. Ignore gravity and mount the ends so that the whole string may be translated vertically. A uniform vertical translation costs no energy. Let the translation vary slowly along the string and it becomes a long-wavelength transverse wave. The uniform symmetry operation has been promoted to a gentle position-dependent deformation.

[2] Editorial clarification: The statement assumes the usual conditions of relativistic field theory. In nonrelativistic many-body systems the counting and dispersion of Nambu–Goldstone modes can be subtler, but the lecture's long-wavelength conclusion remains the right one for this relativistic $U(1)$ model.

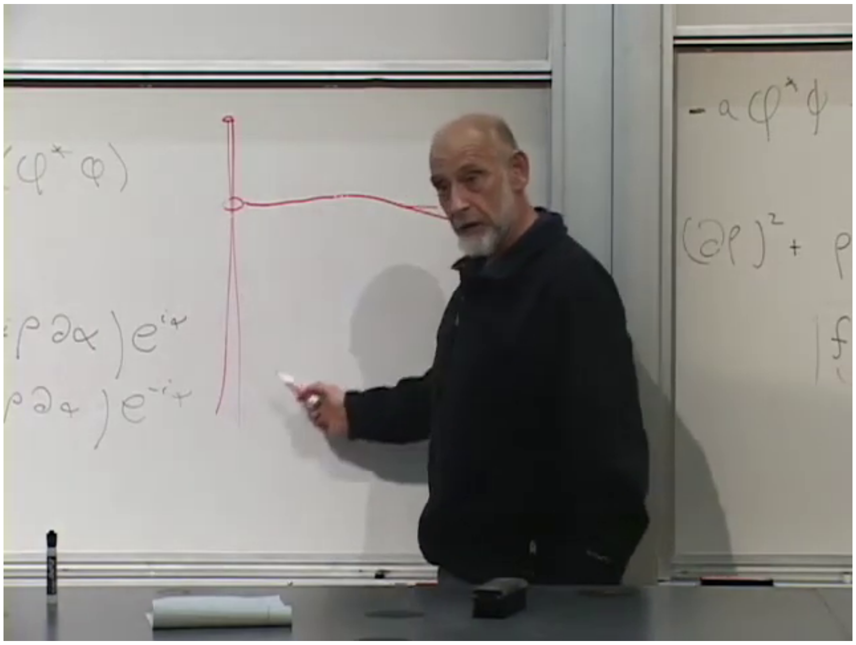

Figure 7.6: The slowly varying string profile drawn during the Goldstone-mode analogy. The physical point is the small slope of a long-wavelength deformation, not the vertical placement of the curve.

Lecture frame: 01:04:19.

Question from the audience. Can the string carry oscillation energy while having zero momentum?[a]

Response. One must distinguish vertical mechanical momentum from wave momentum along the string. A standing wave can have zero net momentum because it is a superposition of modes with $+k$ and $-k$, while still carrying energy. In the field-theory analogy, momentum refers to the wave number along the axis; the vertical displacement stands for the internal field coordinate.

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[a]Lecture timestamp: 01:04:41.

A slinky gives the longitudinal version. Translate the entire idealized infinite slinky along its axis and nothing changes. Translate neighboring parts by slightly different amounts and the result is a gentle compression wave. In a solid or fluid, long-wavelength sound similarly arises from collective displacements and density variations. Spin waves, transverse string waves, and sound are not the same microscopic excitation, but they share the decisive pattern: a uniform continuous operation costs no energy, and its slowly varying version becomes a soft mode. [3]

The terminology matters enough for a brief classroom joke: these are *Goldstone* bosons, named for Jeffrey Goldstone, not gauge bosons. The lecture notes that Goldstone was a friend of Susskind's and then pivots to the genuinely different subject of gauge invariance.

[3]Editorial clarification: In a crystal, acoustic phonons are conventionally described as Nambu–Goldstone modes of spontaneously broken continuous translations. The string and slinky are mechanical analogies used in the lecture to expose the same derivative-only energy structure.

7.7 From Global Phase Symmetry to Gauge Symmetry

The $U(1)$ phase symmetry has already appeared in the course as the symmetry associated with conservation of charge. Charged particle fields are complex, and multiplying such a field by a common phase leaves the global theory unchanged. The word electric may be used for intuition, though the charge could represent another $U(1)$ quantum number. The real photon is massless; the eventual target is a photon-like field such as the massive Z.

The new question is whether

$$\phi'(x) = e^{i\theta(x)}\phi(x) \tag{7.29}$$

can be a symmetry when θ varies arbitrarily through spacetime. The function $\theta(x)$ is a transformation parameter, not an additional dynamical field. We test the proposal by differentiating:

$$\partial_\mu\phi' = e^{i\theta}\left[\partial_\mu\phi + i(\partial_\mu\theta)\phi\right]. \tag{7.30}$$

The extra term was absent for constant θ. For the conjugate,

$$\partial_\mu\phi^{*\prime} = e^{-i\theta}\left[\partial_\mu\phi^* - i(\partial_\mu\theta)\phi^*\right]. \tag{7.31}$$

Their product is

$$\begin{aligned}\partial_\mu\phi^{*\prime}\partial^\mu\phi' = \partial_\mu\phi^*\partial^\mu\phi i\left(\phi\,\partial_\mu\phi^* - \phi^*\partial_\mu\phi\right)\partial^\mu\theta \\ + \phi^*\phi\,\partial_\mu\theta\,\partial^\mu\theta.\end{aligned} \tag{7.32}$$

The potential remains invariant because $\phi^*\phi$ does, but the ordinary kinetic term fails. A global symmetry has not automatically become a local one.

Question from the audience. In the Mexican-hat picture, does this failure mean that there are bumps around the circular trough?[a]

Response. No. The potential can remain perfectly flat in the angular direction. The added cost comes from gradients when the phase differs from one spacetime point to another. It is not a corrugation of the trough; it is the ordinary derivative term detecting spatial variation.

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[a] Lecture timestamp: 01:19:37.

7.7.1 The compensating vector field

Nature's photon, W, Z, and gluon interactions are organized by local gauge symmetries, not merely by rigid transformations. To make the local phase operation a symmetry, enlarge the theory by a four-vector field A_μ. In electromagnetism A_0 is the scalar potential and the three spatial components form the vector potential. Their derivatives build the electric and magnetic fields.

Choose the internally consistent convention

$$\phi' = e^{i\theta}\phi, \qquad A'_\mu = A_\mu - \partial_\mu\theta, \qquad D_\mu = \partial_\mu + iA_\mu. \tag{7.33}$$

The board changes a sign while checking the cancellation; Eq. (7.33) fixes all three signs together. The gauge coupling and charge have temporarily been absorbed into A_μ. [4]

[4] Editorial clarification: With explicit charge q and coupling g, one may write $D_\mu = \partial_\mu + igqA_\mu$ and $A'_\mu = A_\mu - (gq)^{-1}\partial_\mu\theta$. The lecture suppresses these constants to keep the cancellation visible.

For the conjugate field,

$$D_\mu \phi^* = \partial_\mu \phi^* - iA_\mu \phi^*. \tag{7.34}$$

Now

$$\begin{aligned} D'_\mu \phi' &= \left(\partial_\mu + iA'_\mu\right) e^{i\theta}\phi \\ &= e^{i\theta}\left[\partial_\mu \phi + i(\partial_\mu\theta)\phi + i(A_\mu - \partial_\mu\theta)\phi\right] \\ &= e^{i\theta} D_\mu \phi. \end{aligned} \tag{7.35}$$

The unwanted derivative of θ is canceled by the transformation of A_μ. Likewise,

$$D'_\mu \phi^{*\prime} = e^{-i\theta} D_\mu \phi^*. \tag{7.36}$$

Their product is invariant. The adjective *covariant* means that the derivative transforms in the same simple way as the field itself.

7.8 Dynamics of the Gauge Field

Introducing A_μ only to repair a derivative would leave it without its own dynamics. The gauge-invariant field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{7.37}$$

It is antisymmetric, so $F_{\mu\mu} = 0$. Its time–space components contain the electric field, while its space–space components contain the magnetic field. In familiar three-vector language the electromagnetic energy density is proportional to $\boldsymbol{E}^2 + \boldsymbol{B}^2$, whereas the Lorentzian Lagrangian is proportional to $\boldsymbol{E}^2 - \boldsymbol{B}^2$, with signs depending on the metric convention.

Question from the audience. What are the energy and Lagrangian of the electromagnetic field?[a]

Response. The energy is built from $E^2 + B^2$; the Lagrangian uses $E^2 - B^2$. Covariantly the latter is written as a conventional multiple of $F_{\mu\nu}F^{\mu\nu}$. The antisymmetric tensor is the four-dimensional curl of the vector potential.

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[a] Lecture timestamp: 01:33:20.

The field strength is itself gauge invariant:

$$\begin{aligned} F'_{\mu\nu} &= \partial_\mu(A_\nu - \partial_\nu\theta) - \partial_\nu(A_\mu - \partial_\mu\theta) \\ &= F_{\mu\nu} - \partial_\mu\partial_\nu\theta + \partial_\nu\partial_\mu\theta = F_{\mu\nu}. \end{aligned} \tag{7.38}$$

The final equality follows because ordinary mixed partial derivatives commute. A standard gauge-invariant Lagrangian is therefore

$$\boxed{\mathcal{L} = (D_\mu\phi)^* D^\mu \phi - V(\phi^*\phi) - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}} \tag{7.39}$$

up to the sign and normalization conventions chosen for the metric.

This construction promotes a global $U(1)$ symmetry to a local one, but it does so at the price—or, physically, with the gift—of a new vector field. That field is the electromagnetic potential in the Abelian example.

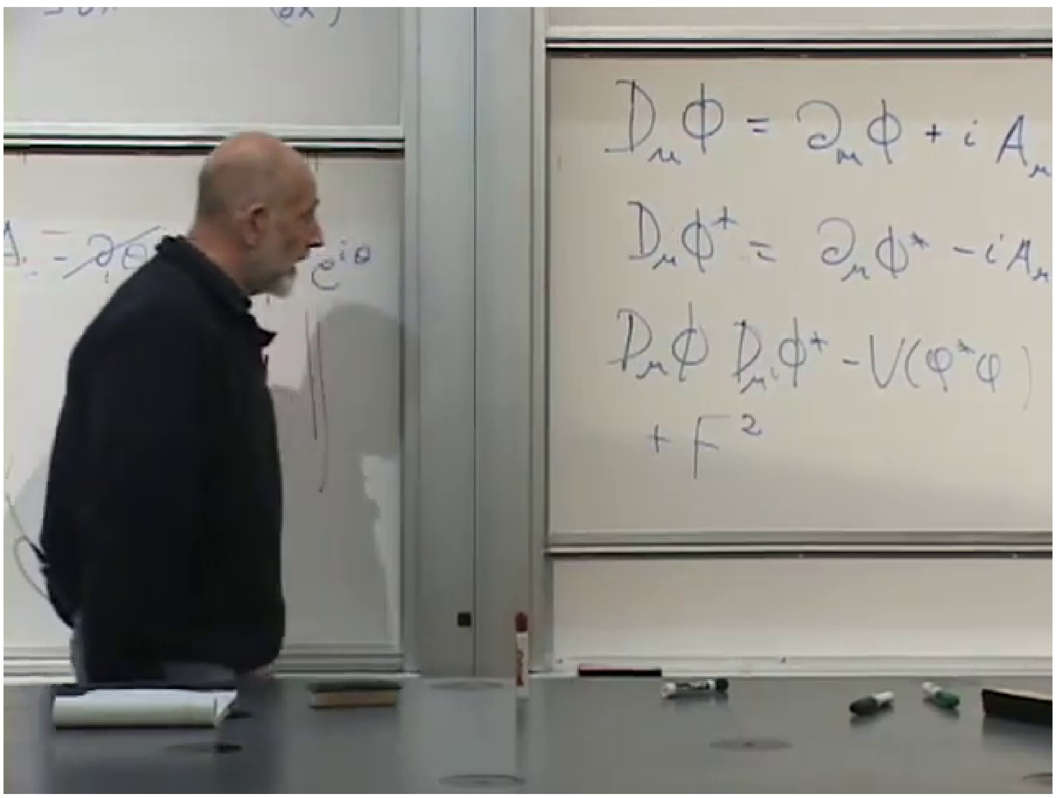

Figure 7.7: The covariant derivatives and the assembled gauge-invariant Lagrangian on the board. Numerical factors are suppressed there so that the transformation structure remains visible.

Lecture frame: 01:38:45.

Question from the audience. The Lagrangian we began with used ordinary derivatives. Why are we entitled to replace them by covariant derivatives and call the result the real theory?[a]

Response. We are constructing a different, enlarged theory: charged matter plus a gauge field. Local gauge invariance determines their coupling. The resulting quantum theory is a coherent mathematical structure and agrees with how nature organizes the photon and weak gauge fields. Simply coupling a relativistic vector field inconsistently can introduce unphysical polarizations and violations of probability, so the gauge principle is not a cosmetic rewriting of the old free scalar theory.

[a]Lecture timestamp: 01:42:55.

Question from the audience. So adding the new field is the essential justification for the modification?[a]

Response. Yes. Without A_μ, the derivative of a locally rotated field contains an uncanceled $\partial_\mu\theta$. With A_μ and its transformation law, the cancellation works. In geometric language A_μ is a connection on a bundle; that deeper mathematics is real and useful, but the algebra above contains the part needed here.

[a]Lecture timestamp: 01:44:57.

7.9 Why a Bare Gauge-Boson Mass Is Forbidden

For an ordinary bosonic field, a mass is represented by a quadratic term. It is tempting to add

$$\Delta\mathcal{L}_{\text{Proca}} = \frac{1}{2}m^2 A_\mu A^\mu. \tag{7.40}$$

But under $A_\mu \mapsto A_\mu - \partial_\mu\theta$,

$$A'_\mu A^{\mu\prime} = A_\mu A^\mu - 2A^\mu\partial_\mu\theta + \partial_\mu\theta\,\partial^\mu\theta, \tag{7.41}$$

so Eq. (7.40) is not gauge invariant. We cannot place a mass into this gauge theory by hand while retaining its local symmetry.

This is the apparent impasse. Gauge invariance explains the massless photon and protects the theory from unphysical degrees of freedom, but the W and Z are massive. The escape cannot be an explicit vector mass. It must arise from the fields already present in a gauge-invariant Lagrangian.

Question from the audience. Does introducing gauge invariance itself produce the Goldstone boson?[a]

Response. No. The Goldstone boson arose when a continuous *global* symmetry was spontaneously broken, before the vector potential was introduced. Once the symmetry is local, a varying phase can be compensated by A_μ. In the broken gauge theory the angular mode is no longer an independent massless particle; it supplies the additional polarization of the now-massive vector field. The next lecture will derive that statement rather than merely announce it.

[a]Lecture timestamp: 01:45:51.

Question from the audience. Does any of this explain why rivers meander? [a]

Response. No direct connection is known here. Fluid eddies and even the shapes of organisms can certainly be described with symmetry methods, and one should never be astonished to find field-theory ideas in unexpected places, but the lecture claims no gauge theory of river meanders.

[a]Lecture timestamp: 01:47:03.

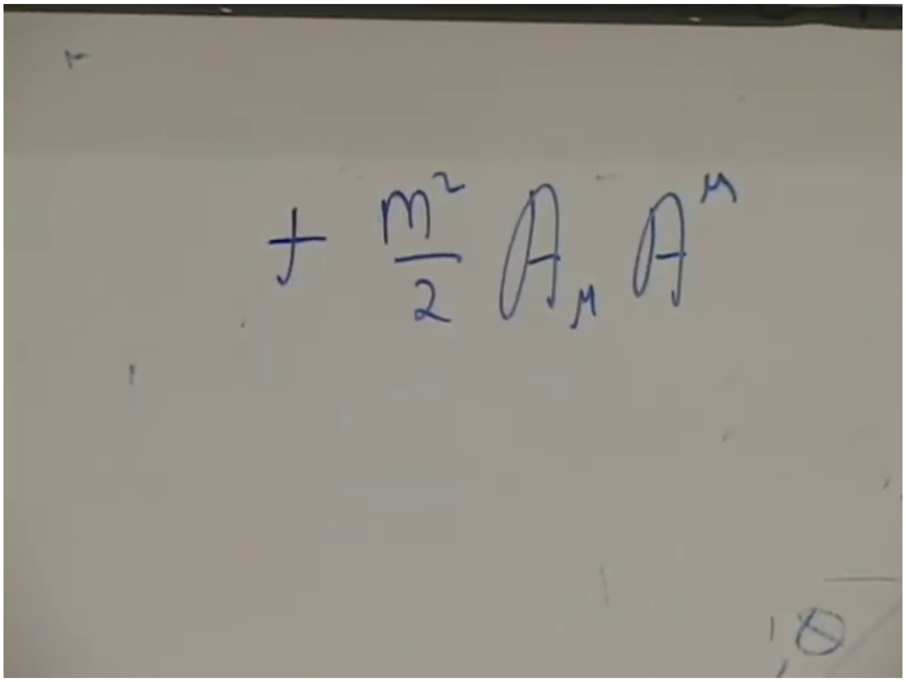

Figure 7.8: The proposed bare vector mass $\frac{1}{2}m^2 A_\mu A^\mu$. Its failure under $A_\mu \mapsto A_\mu - \partial_\mu\theta$ is the obstruction that sets up the Higgs mechanism.

Lecture frame: 01:39:48.

7.10 Where the Lecture Stops

We now have the two ingredients promised at the beginning. A continuous global symmetry with a circle of vacua has a flat angular direction, and its slow variation is a massless Goldstone field. A local symmetry requires a gauge connection and covariant derivatives, and it forbids the direct vector mass in Eq. (7.40).

The next step is to use the broken potential of Eq. (7.14) inside the gauge-invariant theory of Eq. (7.39). Then two things happen together: the separate Goldstone particle disappears from the physical spectrum, and the vector field acquires a mass. In the usual pedagogical language the gauge symmetry is spontaneously broken; more precisely, the gauge redundancy remains while the theory enters a Higgs phase with a massive vector excitation. [5]

That is the Higgs phenomenon. We have not yet performed it. This lecture has done the necessary harder work of making the problem precise enough that the solution, when it comes, will be unavoidable.

[5] Editorial clarification: A local gauge symmetry is a redundancy of description rather than an ordinary physical symmetry acting between distinct observable states. The lecture uses the standard shorthand of spontaneous gauge-symmetry breaking. The gauge-invariant physical statement is the rearrangement of degrees of freedom in the Higgs phase.

Chapter 8

The Higgs Mechanism: How a Gauge Boson Acquires Mass

We now join the two ideas developed in the previous lecture. A complex scalar field with a broken global $U(1)$ symmetry contains a massive radial oscillation and a massless angular oscillation. A locally invariant $U(1)$ theory contains a massless gauge field. When the scalar is charged and its equilibrium magnitude is nonzero, these statements cannot simply be added together. The angular mode and the gauge field combine: the gauge field becomes massive, the apparent Goldstone boson ceases to be an independent particle, and the radial Higgs mode remains.

The photon will be our model gauge boson, but not our physical conclusion. Electromagnetic $U(1)$ is not broken in the vacuum, so the real photon is massless. The same mechanism acts on the $W^{\pm}$ and Z in the electroweak theory, where the gauge group and scalar multiplet are more complicated. The abelian model exposes the mechanism without hiding it under group theory.

Question from the audience. Did the previous discussion mean that the Higgs mechanism gives a mass to the real photon?[a]

Response. No. We ask what would happen *if* the $U(1)$ associated with the photon were in a broken phase. That hypothetical photon would become massive. The real electromagnetic vacuum is not in that phase; the application to particle physics is instead the mass of the W and Z.

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[a] Lecture timestamp: 00:00:23.

The symmetry associated with electric-charge conservation is the phase rotation

$$\phi(x) \longmapsto e^{i\theta}\phi(x), \tag{8.1}$$

an internal rotation called $U(1)$. A superconductor provides a real analogue of the broken phase. Inside it, the electromagnetic field has a finite penetration depth and behaves in important respects like a massive field. Outside it, the ordinary massless photon reappears.

8.1 What Mass Means for a Field

8.1.1 Dispersion and the zero-momentum gap

It is worth deciding precisely what we mean by mass before introducing the Higgs field. For a wave,

$$E = \hbar\omega, \qquad p = \hbar k, \tag{8.2}$$

so an (E, p) graph and an (ω, k) graph contain the same information up to factors of $\hbar$. In empty space a massless relativistic excitation obeys

$$E = c|p|, \qquad \omega = c|k|. \tag{8.3}$$

The two signs of k distinguish opposite directions of propagation. The essential point is that ω tends to zero with k.

A massive relativistic particle obeys

$$E^2 = c^2p^2 + m^2c^4. \tag{8.4}$$

In units $c = \hbar = 1$,

$$\omega^2 = k^2 + m^2. \tag{8.5}$$

At $k = 0$, the energy is $E = mc^2$, or simply m in natural units. Thus mass is the nonzero energy gap of the zero-momentum mode.

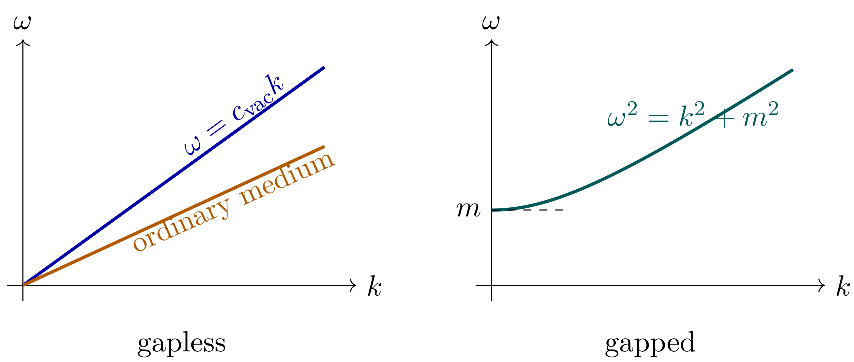

Figure 8.1: Changing the propagation speed changes the slope of a gapless dispersion relation. Giving the excitation a mass opens a nonzero gap at $k = 0$.

An ordinary transparent material may lower a wave's phase velocity without creating this relativistic mass gap. At a boundary the incident frequency is fixed, while the wave number adjusts to the medium's dispersion relation. Changing the slope is not the same as lifting the intercept. [1]

[1] Editorial clarification: A material can have several electromagnetic branches, and a plasma or resonant medium can possess a nonzero plasma-frequency gap. The lecture's comparison is specifically between an ordinary refractive branch and the superconducting response. More generally, gaplessness means $\omega(k) \to 0$ as $k \to 0$; exact linearity is not required in every nonrelativistic medium.

Question from the audience. Does a photon acquire mass whenever it travels through matter, for example through a prism?[a]

Response. No. Refraction can change the relation between frequency and wave number, and therefore the propagation speed, while the branch still passes through $\omega = 0$ at $k = 0$. A superconducting phase instead gives the electromagnetic response a finite inverse penetration length, the analogue of a mass.

[a]Lecture timestamp: 00:03:02.

Question from the audience. For light of a fixed frequency entering glass, is it the wave number that changes?[a]

Response. Yes. Time-translation invariance at a stationary interface keeps the frequency fixed. Because the dispersion curve has a different slope in the glass, the same ω corresponds to a different k, and therefore a different wavelength.

[a]Lecture timestamp: 00:09:20.

The slope

$$v_g = \frac{d\omega}{dk} \tag{8.6}$$

is the group velocity of a wave packet. If that slope varies with k, different Fourier components move at different speeds and the packet disperses. This is why the relation $\omega(k)$ is called the *dispersion relation*.

Question from the audience. What is the name of the curve, and what does its slope represent?[a]

Response. The curve is the dispersion relation. Its derivative $d\omega/dk$ is the group velocity. A constant slope gives the same velocity to all wavelengths; a varying slope makes a wave packet spread. For the present purpose, the decisive distinction is whether the curve reaches the origin or has a gap.

[a]Lecture timestamp: 00:11:02.

8.1.2 A homogeneous oscillation is a particle at rest

The mode $k = 0$ has infinite wavelength: the field is displaced by the same amount everywhere. Release a homogeneous displacement. If the field oscillates with a nonzero frequency, the corresponding quantum has a nonzero rest energy and is massive. If there is no restoring force in that direction, the frequency vanishes and the mode is massless.

This statement is classical. It does not require spatially separated particles to be in a quantum superposition. A homogeneous classical field configuration is simply one field value repeated throughout space.

Question from the audience. If an entire field is shifted together, is that the same as quantum entanglement?[a]

Response. No. A homogeneous classical field is one classical configuration. Entanglement correlates distinct quantum alternatives, such as up–down and down–up spin configurations. Preparing the same field value at many points is not by itself entanglement, nor does it imply an instantaneous causal response.

[a]Lecture timestamp: 00:10:04.

For a canonically normalized real field,

$$\mathcal{L} = \frac{1}{2}\partial_\mu\varphi\,\partial^\mu\varphi - V(\varphi), \tag{8.7}$$

expand about an equilibrium φ_0:

$$V(\varphi_0+\eta) = V(\varphi_0) + \frac{1}{2}V''(\varphi_0)\eta^2 + \cdots . \tag{8.8}$$

The linear term vanishes at the minimum, and the field equation gives

$$\omega^2 = \boldsymbol{k}^2 + m^2, \qquad m^2 = V''(\varphi_0). \tag{8.9}$$

Mass is therefore the curvature of the potential in the direction of motion. A flat direction has zero curvature and yields a massless mode.

8.2 The Broken Complex Scalar

8.2.1 Radial and angular motion

Write a complex scalar in Cartesian and polar coordinates:

$$\phi = \phi_R + i\phi_I = \rho e^{i\alpha}, \qquad \phi^* = \phi_R - i\phi_I = \rho e^{-i\alpha}. \tag{8.10}$$

The complex plane is an *internal* field space. At every physical spacetime point x, the pair $(\phi_R(x), \phi_I(x))$ specifies one point in that internal plane. Neither internal axis is a direction in the laboratory.

The scalar kinetic term becomes

$$\begin{aligned}\partial_\mu\phi^*\,\partial^\mu\phi &= \partial_\mu\phi_R\,\partial^\mu\phi_R + \partial_\mu\phi_I\,\partial^\mu\phi_I \\ &= \partial_\mu\rho\,\partial^\mu\rho + \rho^2\partial_\mu\alpha\,\partial^\mu\alpha.\end{aligned} \tag{8.11}$$

A globally $U(1)$-invariant potential depends on $\phi^*\phi = \rho^2$, not on α. For example,

$$V(\phi) = \lambda\left(\phi^*\phi - \frac{f^2}{2}\right)^2, \qquad \lambda > 0. \tag{8.12}$$

Its minima form a circle. The law is invariant under a common phase rotation, but choosing one equilibrium phase selects one point on the circle.

Set

$$\phi(x) = \frac{f + h(x)}{\sqrt{2}}e^{i\pi(x)/f}. \tag{8.13}$$

Near the selected vacuum, h measures radial displacement and π measures tangential displacement with canonical dimensions. Substitution in Eq. (8.11) gives

$$\partial_\mu\phi^*\,\partial^\mu\phi = \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}\left(1 + \frac{h}{f}\right)^2(\partial_\mu\pi)^2. \tag{8.14}$$

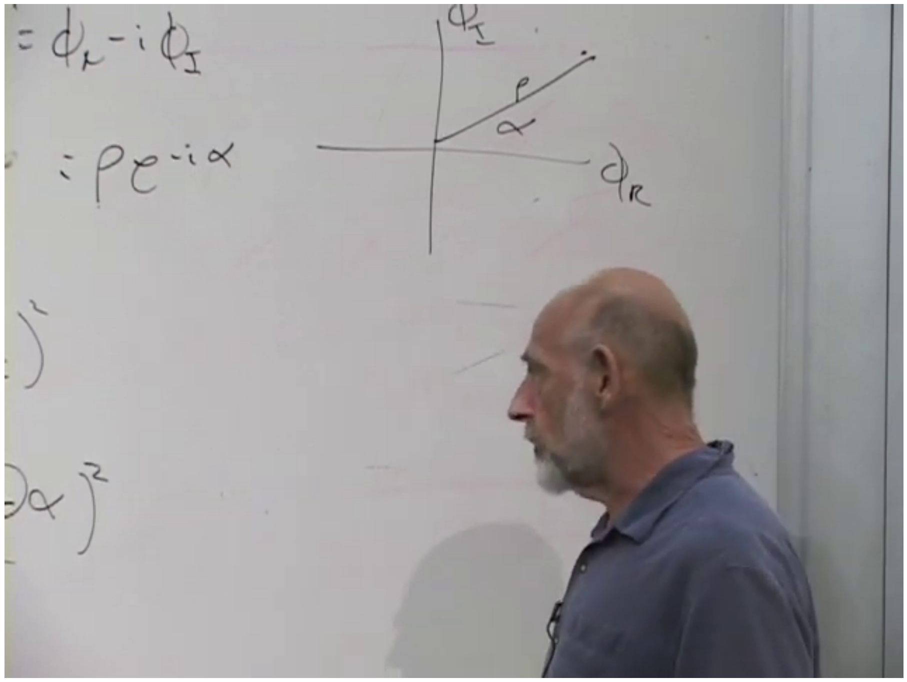

Figure 8.2: The complex scalar represented in its internal (ϕ_R, ϕ_I)-plane. The radius is ρ, and α is the internal phase.

Lecture frame: 00:24:41.

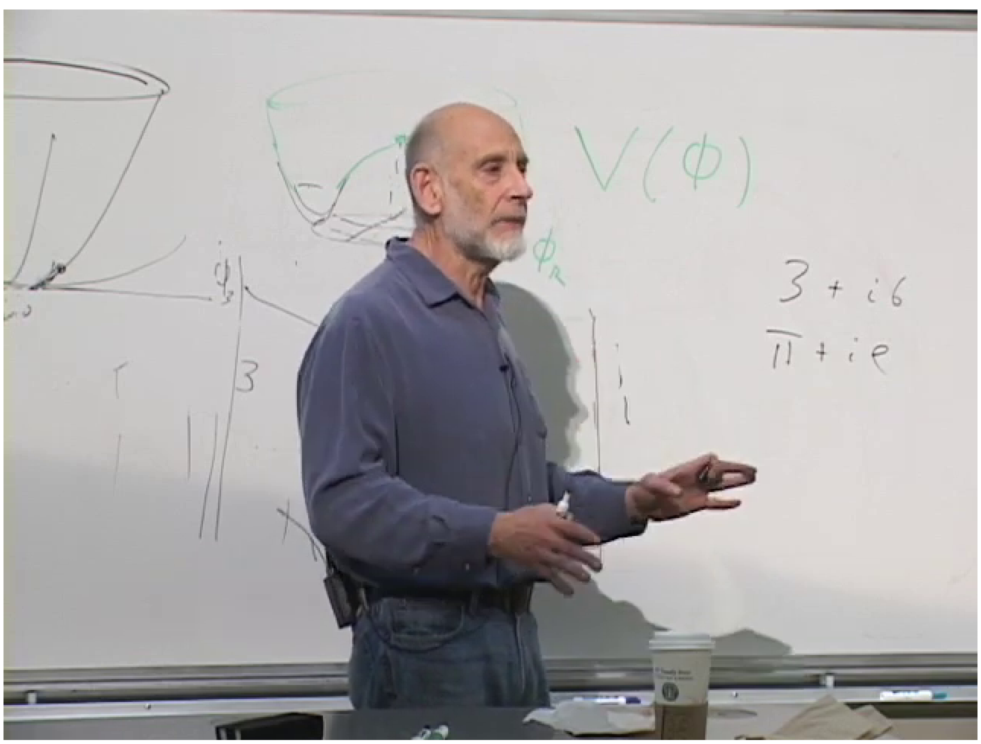

Figure 8.3: The board's radial potential sketches. The potential is symmetric under rotations of the internal field value, even though the field itself need not sit at the origin.

Lecture frame: 00:34:22.

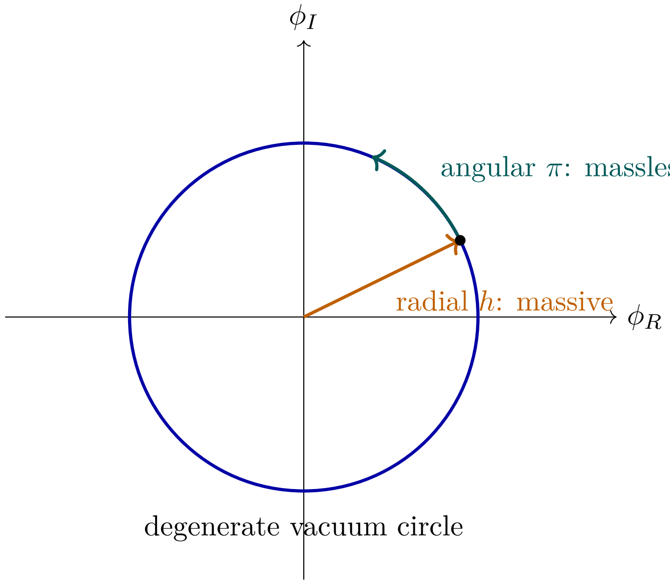

Figure 8.4: The global broken theory has curvature normal to the vacuum circle but no curvature along it. The radial Higgs mode is massive; the angular Goldstone mode is massless.

The potential in Eq. (8.12) supplies

$$m_h^2 = 2\lambda f^2, \qquad m_\pi^2 = 0. \tag{8.15}$$

At energies far below m_h, radial motion is expensive, so we may temporarily freeze $h = 0$. The remaining Lagrangian contains only $\frac{1}{2}(\partial\pi)^2$, with no π^2 term. This is the Goldstone boson.[2]

Question from the audience. Is this the only spontaneous symmetry breaking in the Standard Model?[a]

Response. The electroweak Higgs phase is the established symmetry-breaking mechanism relevant here. Its actual symmetry is $SU(2)_L \times U(1)_Y$, not this single $U(1)$, and QCD has additional vacuum structure such as chiral symmetry breaking. The abelian example is a deliberately reduced model of the gauge-boson mechanism.

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[a]Lecture timestamp: 00:16:39.

Question from the audience. Does the Higgs give mass to every particle, including itself?[a]

Response. There is a yes-and-no distinction. Electroweak symmetry forbids bare masses for the W, Z, and Standard Model fermions; their masses appear after the Higgs field takes a nonzero equilibrium magnitude. The radial Higgs mass comes from the curvature of its own potential. Other hypothetical particles can possess symmetry-allowed masses unrelated to this mechanism.

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[a]Lecture timestamp: 00:17:11.

The lecture associates such symmetry-allowed masses with potentially very heavy, difficult-to-observe particles and mentions dark matter as a possible example. That is a motivation, not a mass

[2]Editorial clarification: At two points the spoken explanation reverses the names and says that eating the Goldstone gives the Higgs boson its mass. The board calculation and the remainder of the lecture establish the intended result: radial curvature already gives the Higgs mode a mass, while the gauge boson acquires the Goldstone mode and becomes massive.

prediction. [3]

Question from the audience. Is ρ a physical radius, or is it the field value? What exactly is radially symmetric?[a]

Response. At each spacetime point the field value is a point in an auxiliary complex plane, and $\rho = |\phi|$ is the distance of that point from the origin. The *potential* is radially symmetric in this internal plane: all points with the same ρ have the same potential energy. This radius is not a distance in ordinary space.

[a]Lecture timestamp: 00:33:08.

Question from the audience. Could a sufficiently violent radial kick carry the field over the central maximum to the other side of the potential? [a]

Response. In principle a local configuration with enough energy can cross the barrier. Moving a homogeneous macroscopic field over it would require an enormous energy, and a generic violent disturbance would shed energy into many field modes rather than arrive coherently at the opposite point. The low-energy approximation freezes ρ precisely because such radial excursions are inaccessible there.

[a]Lecture timestamp: 00:36:00.

8.3 Why the Complex Plane Is Not Spacetime

The distinction between physical and internal coordinates deserves the lecture's full pause. A scalar field is called scalar because it is invariant under rotations of physical coordinate axes. Two real scalar fields $\phi_R(x)$ and $\phi_I(x)$ remain two scalars under those rotations. We may package them into one complex number, but this bookkeeping does not turn them into components of a spatial vector.

A global $U(1)$ transformation rotates the *values* of the field:

$$\begin{aligned} \phi(x) &\longmapsto e^{i\theta}\phi(x), \\ \rho(x) &\longmapsto \rho(x), \\ \alpha(x) &\longmapsto \alpha(x) + \theta. \end{aligned} \tag{8.16}$$

The same constant θ is used at every spacetime point. Consequently, $\partial_\mu \alpha$ is unchanged. This is a symmetry of both the potential and the ordinary kinetic term. Once θ is allowed to depend on x, the derivative detects the variation, and the argument changes.

8.4 Making the Phase Rotation Local

8.4.1 The ordinary derivative fails

Let

$$\phi'(x) = e^{i\theta(x)}\phi(x). \tag{8.17}$$

Then

$$\partial_\mu \phi' = e^{i\theta}\left(\partial_\mu \phi + i\,\partial_\mu \theta\,\phi\right). \tag{8.18}$$

[3]Editorial clarification: The dark-matter mass scale remains unknown. The lecture's suggestion of a particle around a thousand proton masses is historical speculation, not an experimentally established property of dark matter.

The extra term proportional to $\partial_\mu\theta$ prevents $\partial_\mu\phi$ from transforming like ϕ. Therefore $\partial_\mu\phi^*\partial^\mu\phi$ is not invariant under arbitrary local phase rotations.

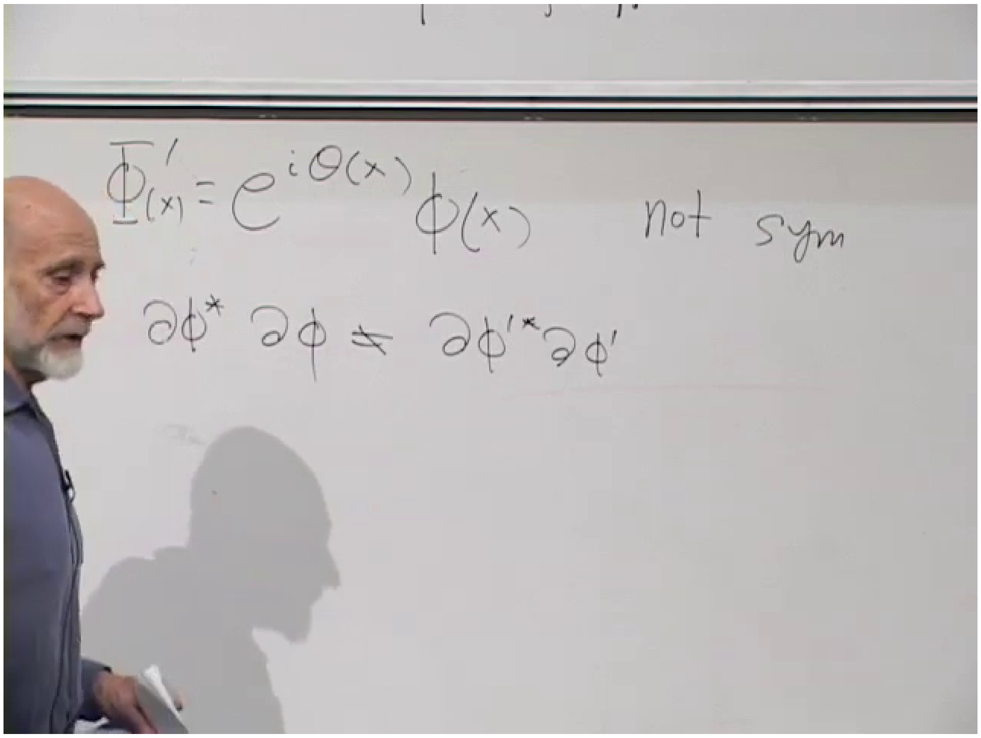

Figure 8.5: The board records the local phase transformation and the failure of the ordinary derivative term to remain invariant.

Lecture frame: 00:46:49.

Introduce a vector potential and fix one convention consistently:

$$D_\mu = \partial_\mu + igA_\mu, \qquad A'_\mu = A_\mu - \frac{1}{g}\partial_\mu\theta. \tag{8.19}$$

Now

$$\begin{aligned} D'_\mu\phi' &= \left(\partial_\mu + igA'_\mu\right)e^{i\theta}\phi \\ &= e^{i\theta}\left[\partial_\mu\phi + i(\partial_\mu\theta)\phi + igA_\mu\phi - i(\partial_\mu\theta)\phi\right] \\ &= e^{i\theta}D_\mu\phi. \end{aligned} \tag{8.20}$$

The unwanted terms cancel. The conjugate derivative is

$$(D_\mu\phi)^* = (\partial_\mu - igA_\mu)\phi^*, \tag{8.21}$$

and its opposite phase cancels in the product

$$(D_\mu\phi)^* D^\mu\phi. \tag{8.22}$$

Question from the audience. Was the gauge-field transformation written with the opposite sign in the previous lecture?[a]

Response. A simultaneous reversal of the sign in D_μ and in the transformation law is merely a convention. With $D_\mu = \partial_\mu + igA_\mu$ and $\phi' = e^{i\theta}\phi$, consistency requires $A'_\mu = A_\mu - g^{-1}\partial_\mu\theta$. The alternative convention works equally well if every sign is changed together.

[a] Lecture timestamp: 00:48:04.

Question from the audience. Does the use of an upper-case or lower-case form of ϕ on the board mark a different field?[a]

Response. No distinction is intended. The changing letter shape is handwriting, not an additional operation or a second scalar. The meaningful distinctions are the prime, the complex-conjugation star, and the spacetime index.

[a] Lecture timestamp: 00:55:02.

Question from the audience. Is complex conjugation the same operation as raising or lowering the Lorentz index in the kinetic term? [a]

Response. No. The star conjugates the internal complex field. Lorentz indices are raised with the spacetime metric: $D^\mu = g^{\mu\nu}D_\nu$. A Lorentz scalar needs the contraction $(D_\mu\phi)^* D^\mu\phi$; both operations are present and serve different purposes.

[a] Lecture timestamp: 00:56:50.

8.4.2 The gauge field has its own dynamics

The field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{8.23}$$

is gauge invariant because the two mixed second derivatives of θ cancel. Its time–space components contain the electric field, and its space–space components contain the magnetic field. The scalar version of electrodynamics is therefore

$$\mathcal{L} = (D_\mu\phi)^* D^\mu\phi - V(\phi^*\phi) - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \tag{8.24}$$

The charged field here is bosonic. It is not the electron field, which is a fermion. A charged helium nucleus or a Cooper-pair condensate illustrates how a bosonic charged degree of freedom can arise.

Before symmetry breaking, the gauge term contains derivatives of A_μ but no direct mass term. One might try to add

$$\frac{1}{2}m_A^2 A_\mu A^\mu, \tag{8.25}$$

but under Eq. (8.19),

$$A_\mu A^\mu \longmapsto \left(A_\mu - \frac{1}{g}\partial_\mu\theta\right)\left(A^\mu - \frac{1}{g}\partial^\mu\theta\right), \tag{8.26}$$

which is not equal to $A_\mu A^\mu$. A bare gauge-boson mass is forbidden by the local symmetry.

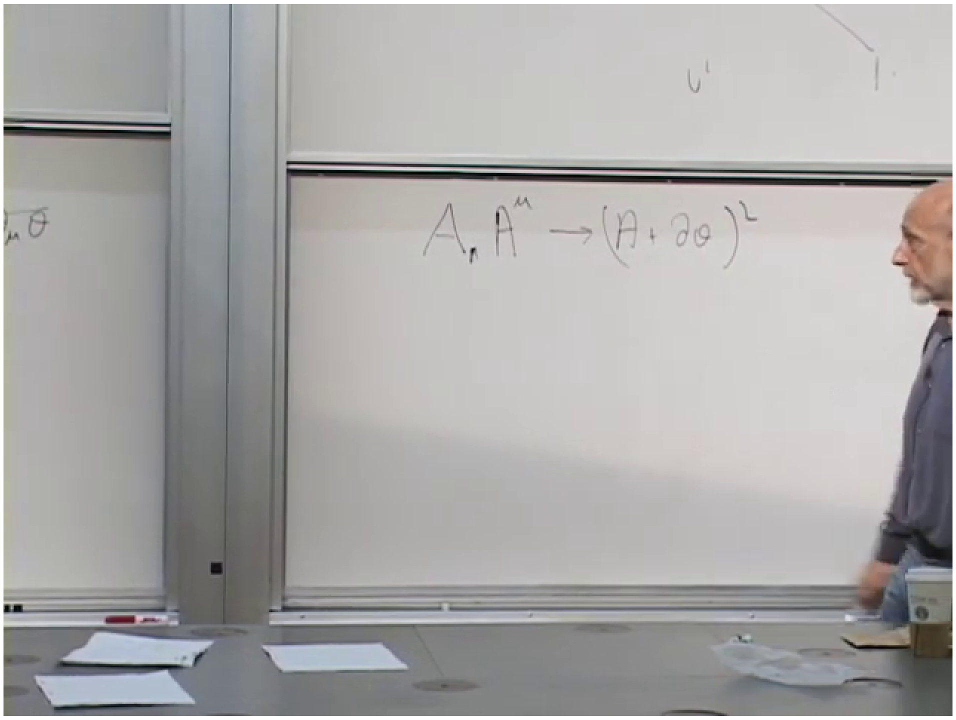

Figure 8.6: A verified board state for the obstruction: under a gauge transformation, $A_\mu A^\mu$ becomes a square containing $\partial_\mu \theta$, so a bare vector mass is not invariant.

Lecture frame: 01:07:30.

8.5 The Higgs Mechanism

8.5.1 The covariant derivative produces the mass

Return to the broken potential and use the canonically normalized parametrization

$$\phi(x) = \frac{f + h(x)}{\sqrt{2}} e^{i\pi(x)/f}. \tag{8.27}$$

Its covariant derivative is

$$D_\mu \phi = \frac{e^{i\pi/f}}{\sqrt{2}} \left[\partial_\mu h + i(f+h) \left(\frac{\partial_\mu \pi}{f} + gA_\mu \right) \right]. \tag{8.28}$$

Consequently,

$$\begin{aligned}(D_\mu \phi)^* D^\mu \phi = \frac{1}{2}(\partial_\mu h)^2 \\ + \frac{1}{2}(f+h)^2 \left(\frac{\partial_\mu \pi}{f} + gA_\mu \right) \\ \times \left(\frac{\partial^\mu \pi}{f} + gA^\mu \right).\end{aligned} \tag{8.29}$$

The decisive term appears before any gauge choice:

$$\frac{1}{2} g^2 f^2 A_\mu A^\mu. \tag{8.30}$$

Comparing it with the Proca form $\frac{1}{2}m_A^2 A_\mu A^\mu$, we find

$$\boxed{m_A = gf}. \tag{8.31}$$

The same expansion also produces interactions $g^2 f h A_\mu A^\mu$ and $\frac{1}{2}g^2 h^2 A_\mu A^\mu$. The vector mass is therefore not an arbitrary violation of gauge symmetry. It comes from a gauge-invariant scalar kinetic term evaluated in a phase with nonzero equilibrium magnitude f.

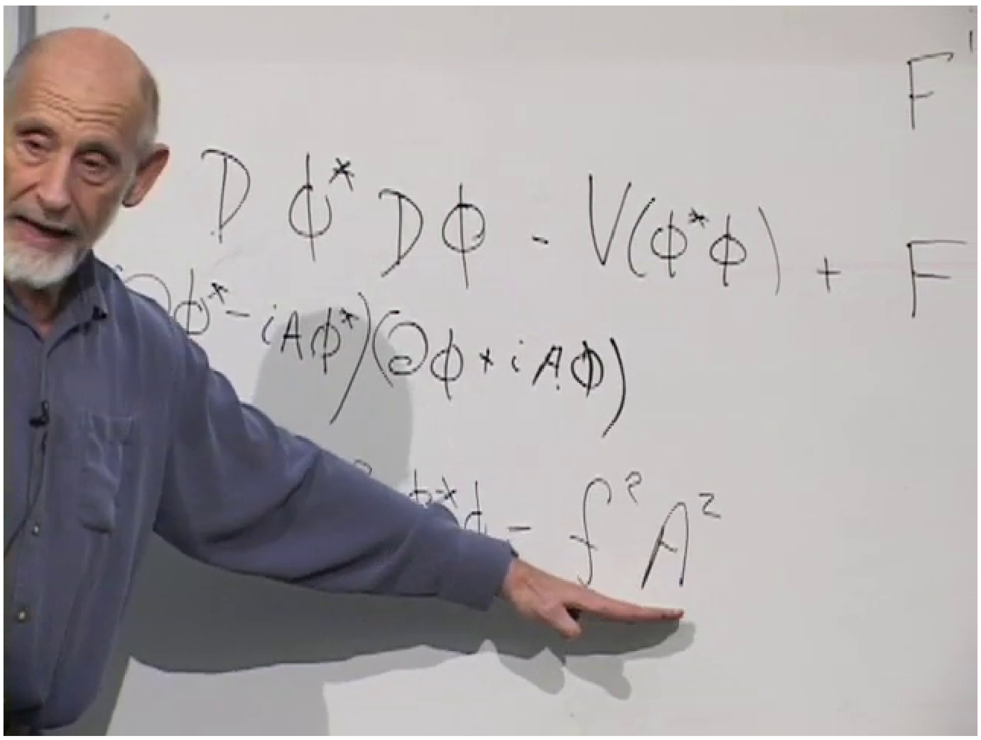

Figure 8.7: The board isolates the $f^2 A^2$ contribution inside $(D\phi)^*(D\phi)$. Numerical factors depend on how the scalar field and the gauge coupling are normalized; with Eqs. (8.13) and (8.19), the result is $m_A = gf$.

Lecture frame: 01:10:10.

The lecture absorbs g and several factors of $\sqrt{2}$ into its board notation, so it briefly alternates among f, $2f$, and $\sqrt{2}f$ while identifying the mass. Those factors have no invariant meaning until the kinetic and potential normalizations are stated. The robust result is $m_A \propto gf$; Eq. (8.31) gives the factor for the convention used here.

8.5.2 The phase is absorbed

Under a gauge transformation the phase variable changes as

$$\pi'(x) = \pi(x) + f\,\theta(x). \tag{8.32}$$

Choose

$$\theta(x) = -\frac{\pi(x)}{f}. \tag{8.33}$$

Then $\pi' = 0$. This is unitary gauge, in which

$$\phi'(x) = \frac{f + h(x)}{\sqrt{2}} \tag{8.34}$$

is real, and Eq. (8.29) reduces to

$$(D_\mu \phi')^* D^\mu \phi' = \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2} g^2 (f+h)^2 A'_\mu A'^\mu. \tag{8.35}$$

The angular Goldstone field has disappeared from the list of independent variables, while the vector mass remains.

The lecture first demonstrates this in the low-energy limit $\rho = f$, where

$$\begin{aligned} \phi &= f e^{i\alpha}, \\ D_\mu \phi &= i f e^{i\alpha} \left(\partial_\mu \alpha + g A_\mu\right), \\ (D_\mu \phi)^* &= -i f e^{-i\alpha} \left(\partial_\mu \alpha + g A_\mu\right). \end{aligned} \tag{8.36}$$

Their product is

$$f^2 \left(\partial_\mu \alpha + g A_\mu\right)\left(\partial^\mu \alpha + g A^\mu\right). \tag{8.37}$$

Choosing $\theta = -\alpha$ leaves $g^2 f^2 A'_\mu A'^\mu$.

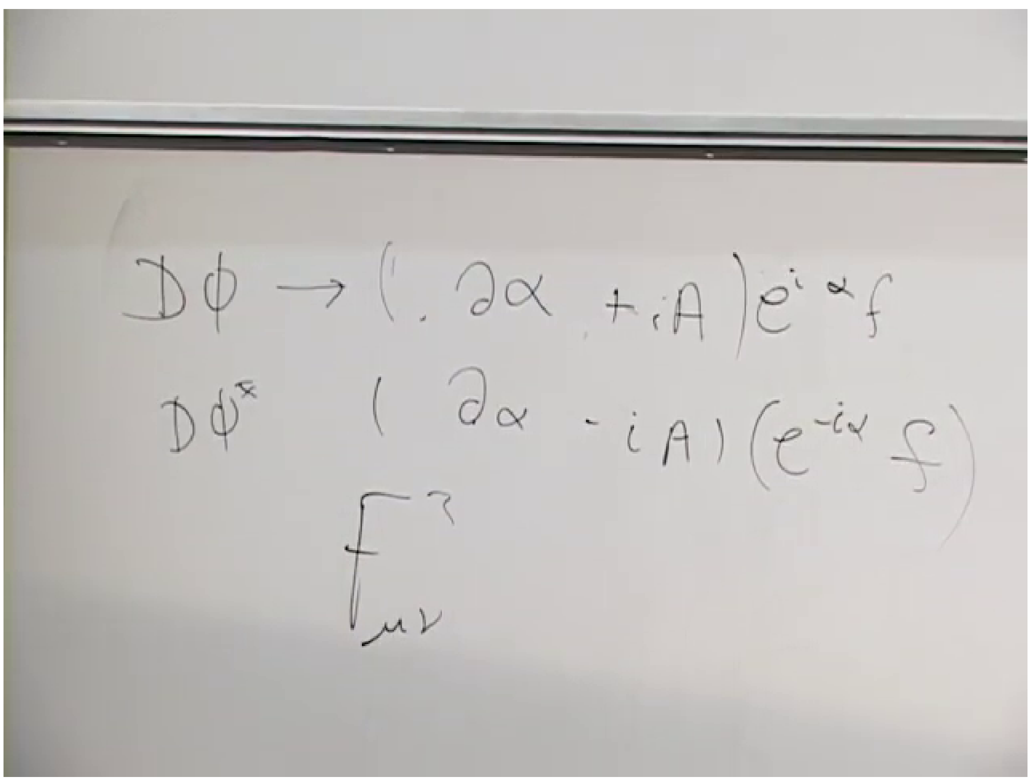

Figure 8.8: The complete frozen-radius board state: the covariant derivatives contain the phase derivative and vector potential together. The lecture next chooses $\theta = -\alpha$, eliminating the phase and leaving the vector mass term.

Lecture frame: 01:16:20.

Question from the audience. Can the whole Lagrangian be rewritten in variables that make the disappearance of the angular mode explicit? [a]

Response. Yes. In polar variables the phase appears only through the gauge-invariant combination $\partial_\mu\alpha+gA_\mu$. A gauge transformation with $\theta = -\alpha$ sets the phase to zero everywhere that this gauge is well-defined. The scalar kinetic term then becomes a vector mass term, while the gauge-invariant field-strength term is unchanged.

[a] Lecture timestamp: 01:13:49.

Question from the audience. Are oscillations in α indistinguishable from gauge oscillations?[a]

Response. In the gauged theory, a pure change of α that is compensated by the corresponding change of A_μ is a gauge redescription, not a distinct physical wave. The invariant quantity is $\partial_\mu\alpha + gA_\mu$. The physical mode that was represented by the global Goldstone field reappears as the longitudinal polarization of the massive vector.

[a] Lecture timestamp: 01:18:02.

It is common to say that a local gauge symmetry is spontaneously broken. Operationally this language is useful: after choosing a gauge, the scalar field has a nonzero vacuum value and the spectrum contains a massive vector. Strictly, however, gauge transformations relate redundant descriptions, not distinct physical states. [4]

8.6 Where the Goldstone Degree of Freedom Went

Nothing physical can vanish merely because we changed variables. Count the degrees of freedom before and after entering the Higgs phase.

Beforehand:

$$\underbrace{2}_{\text{massless vector}} + \underbrace{2}_{\text{complex scalar}} = 4. \tag{8.38}$$

A massless vector has two transverse polarizations. The complex scalar has two real components, radial and angular.

Afterward:

$$\underbrace{3}_{\text{massive vector}} + \underbrace{1}_{\text{radial Higgs}} = 4. \tag{8.39}$$

A massive vector can be brought to rest. In its rest frame no spatial direction is privileged, so it must have three polarization states. When it moves, two are transverse and one is longitudinal. The Goldstone mode has become that longitudinal polarization.

[4] Editorial clarification: A local gauge redundancy is not broken in the same gauge-invariant sense as a global symmetry. The precise statement is that the theory is in a Higgs phase. Gauge-fixed perturbation theory describes that phase by a nonzero scalar expectation value and the resulting massive vector spectrum.

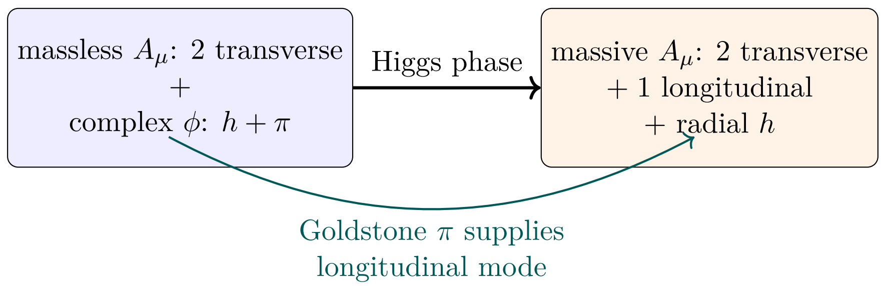

Figure 8.9: The Higgs mechanism rearranges, rather than destroys, physical degrees of freedom.

Question from the audience. How can the angular degree of freedom disappear without violating degree-of-freedom counting?[a]

Response. It does not disappear physically. A massless vector has two helicities because it cannot be brought to rest. A massive vector has three polarizations, including a longitudinal one. The Goldstone mode becomes that third polarization, leaving the total count unchanged.

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[a] Lecture timestamp: 01:19:13.

The lecture offers a useful kinematic test. Imagine a vector particle with mass, slow it to rest, and accelerate it in a new direction. A polarization that was transverse to the original motion can become longitudinal relative to the new motion. Likewise, if a circularly polarized massive particle is carried through rest and sent in the opposite direction, its helicity label reverses. A strictly massless particle cannot be taken through this experiment, because no inertial frame can bring it to rest. Its two helicities therefore form a consistent physical spectrum.

8.7 The Superconducting Analogue

In a conventional superconductor, electrons form Cooper pairs. A pair is a boson carrying electric charge $2e$, and the condensate is described by a charged complex order parameter. When its magnitude is nonzero, the combination of its phase and the electromagnetic vector potential produces the Anderson–Higgs mechanism. Static magnetic fields decay over the London penetration depth rather than extending freely through the bulk. In this precise medium-dependent sense, the photon behaves as though it has acquired a mass.

This example also clarifies the opening distinction. The electromagnetic field is not permanently converted into a different elementary particle. The massive propagation law belongs to the collective state inside the material; outside, the vacuum photon remains massless.

8.8 What Has Been Established

The argument can now be compressed without losing its logic:

1. A broken global continuous symmetry has a flat angular direction and therefore a massless Goldstone mode.
2. Local phase invariance requires a gauge potential and replaces ∂_μ by D_μ.

3. A bare $A_\mu A^\mu$ mass violates the local transformation law.
4. When the charged scalar has nonzero equilibrium magnitude f, the gauge-invariant term $|D_\mu \phi|^2$ contains $\frac{1}{2} g^2 f^2 A_\mu A^\mu$.
5. The phase can be removed by a gauge choice, and its physical degree of freedom becomes the longitudinal polarization of the massive vector.
6. The radial Higgs excitation remains, with mass fixed by the curvature of the scalar potential.

The next task is to give fermions their masses. Electrons and quarks cannot use the vector mechanism directly; their Higgs couplings are Yukawa interactions. Once the scalar settles at f, those interactions become fermion mass terms.

The broad statement that familiar particles would be massless without electroweak symmetry breaking requires one final qualification. The elementary quark and charged-lepton masses, and the W and Z masses, do vanish with the Higgs expectation value. Most of the proton and neutron mass, however, arises dynamically from QCD confinement and strong-field energy rather than from the quarks' Higgs masses. That distinction does not alter the Higgs mechanism derived here; it keeps its domain of application precise.

Chapter 9

Fermion Masses: Chirality, Yukawa Couplings, and Majorana Neutrinos

Overview. We first rebuild the gauge-boson Higgs mechanism in a toy $U(1)$ theory, then ask the parallel question for fermions. A Dirac mass mixes left and right chiral fields, but the weak interaction assigns them different charges and therefore forbids a bare mass. A Yukawa coupling to the Higgs field respects the symmetry; after the Higgs field acquires a vacuum value, the same interaction becomes a fermion mass and a measurable Higgs–fermion coupling. The final part follows this logic into neutrino mass, possible heavy unprotected particles, cosmic-ray kinematics, relativistic inertia, and gravity.

The purpose of the opening recap is not merely to repeat the previous lecture. We need one idea from it in a form that can be carried over to fermions: a mass is a nonzero energy for an excitation at zero momentum. For a field, zero momentum means infinite wavelength, or a displacement that is the same everywhere. A derivative-only theory cannot detect a uniform displacement. A term without derivatives can.

9.1 The Gauge-Boson Story in One Pass

9.1.1 Derivative energy and a massless vector

The model gauge field is A_μ, with field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{9.1}$$

The board writes the gauge kinetic term schematically as F^2. With a standard normalization it is

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \tag{9.2}$$

Every appearance of A_μ in (9.2) is differentiated. Consequently,

$$A_\mu(x) \longmapsto A_\mu(x) + c_\mu, \qquad \partial_\nu c_\mu = 0, \tag{9.3}$$

does not change $F_{\mu\nu}$. A homogeneous vector potential therefore costs no field-strength energy. In the language of the lecture, the zero-momentum mode has no gap and the gauge boson is massless.

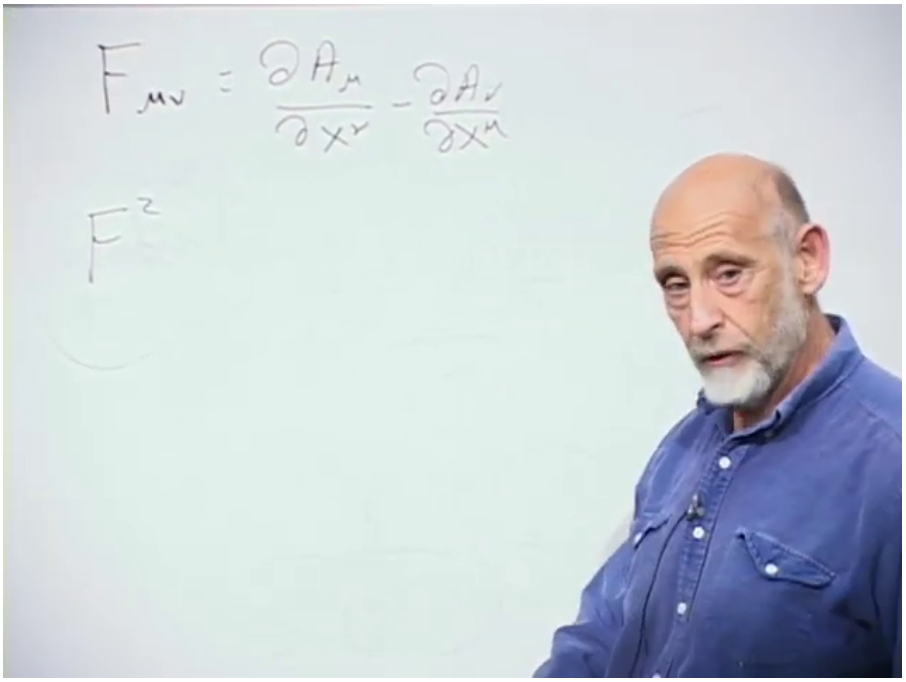

Figure 9.1: The field-strength definition and the board's F^2 shorthand for the derivative energy of the gauge field.

Lecture frame: 00:02:39.

This $U(1)$ field is a deliberately simple stand-in. The physical target is the $W^\pm$ and Z, not the electromagnetic photon. The unbroken electromagnetic symmetry leaves the photon massless; the broken electroweak symmetry gives masses to the weak gauge bosons.

9.1.2 Freezing the Higgs radius

Introduce a charged complex scalar and write it in polar form,

$$\phi(x) = \rho(x)e^{i\alpha(x)}. \tag{9.4}$$

Its Mexican-hat potential is minimized on a circle of nonzero radius. Calling that radius f, the radial direction is comparatively stiff. For the first pass we freeze it and retain only the phase:

$$\phi(x) \simeq fe^{i\alpha(x)}. \tag{9.5}$$

Choose the convention

$$\begin{aligned} D_\mu &= \partial_\mu + igA_\mu, \\ \phi &\longmapsto e^{i\theta}\phi, \\ A_\mu &\longmapsto A_\mu - \frac{1}{g}\partial_\mu\theta. \end{aligned} \tag{9.6}$$

The lecture suppresses g; keeping it visible makes the dimensions and the eventual mass transparent. Acting on the frozen field gives

$$\begin{aligned} D_\mu\phi &= (\partial_\mu + igA_\mu)fe^{i\alpha} \\ &= i(\partial_\mu\alpha + gA_\mu)fe^{i\alpha}. \end{aligned} \tag{9.7}$$

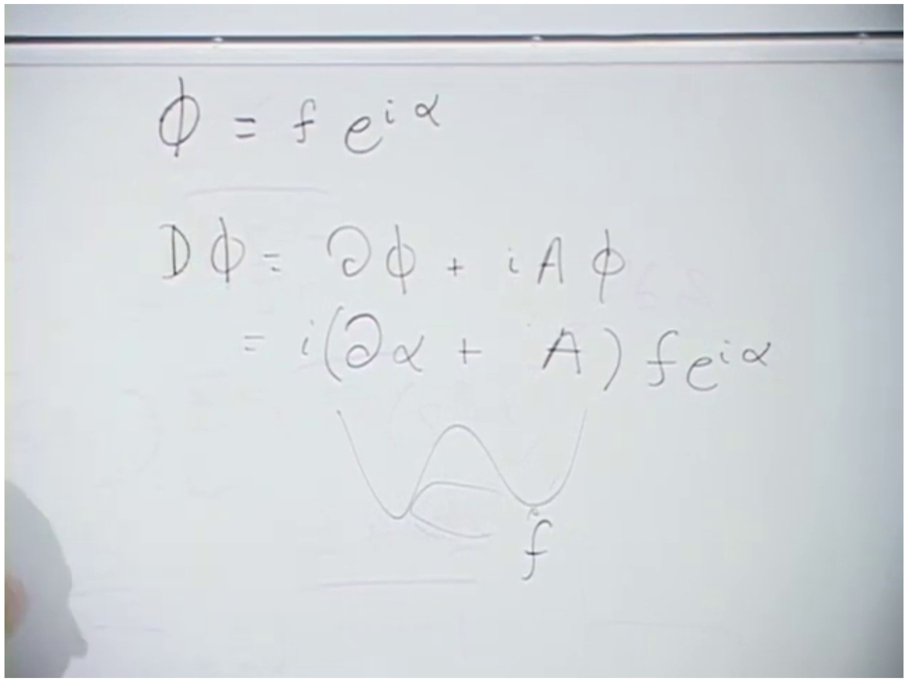

Figure 9.2: The frozen Higgs radius, its covariant derivative, and the small broken-potential sketch used in the recap.

Lecture frame: 00:06:42.

Multiplication by the complex conjugate removes the phase and also turns i times $-i$ into $+1$:

$$|D_\mu\phi|^2 = f^2(\partial_\mu\alpha + gA_\mu)(\partial^\mu\alpha + gA^\mu). \tag{9.8}$$

Question from the audience. Should the square of the covariant derivative carry a minus sign because the derivative produced a factor of i? [a]

Response. No. The kinetic term is the product with the complex conjugate. One factor contributes i, the other $-i$, and $i(-i) = 1$. The sign convention for D_μ may be reversed consistently, but the modulus square is positive in the corresponding energy expression.

[a] Lecture timestamp: 00:07:39.

The combination in (9.8) is itself a shifted gauge field. Choose $\theta = -\alpha$, so that the scalar phase disappears, and define

$$A'_\mu = A_\mu + \frac{1}{g}\partial_\mu\alpha. \tag{9.9}$$

Then

$$|D_\mu\phi|^2 = g^2 f^2 A'_\mu A'^\mu. \tag{9.10}$$

The new term contains no derivative of A'_μ. A homogeneous field now stores energy, so the zero-momentum vector mode is gapped.

With canonical scalar normalization one writes $\phi = (f + h)e^{i\pi/f}/\sqrt{2}$. Comparing the resulting vector term with $\frac{1}{2}m_A^2 A_\mu A^\mu$ gives

$$m_A = gf \tag{9.11}$$

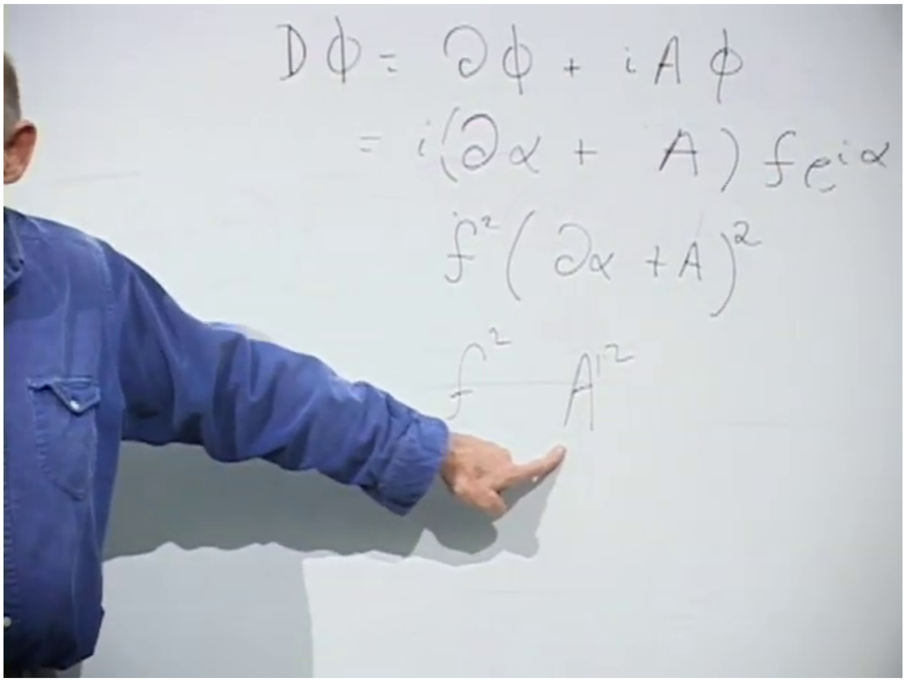

Figure 9.3: The board isolates the f^2A^2 term that signals a gauge-boson mass after the phase has been absorbed.

Lecture frame: 00:10:31.

in this abelian model. The lecture suppresses the normalization factors and summarizes the result by saying that the vector mass is set by f.

The phase mode has not vanished without accounting. Before gauging, it was a massless Goldstone boson. After gauging, it supplies the longitudinal polarization needed by a massive vector particle. The radial fluctuation, which we temporarily froze, remains as the physical Higgs excitation.

9.1.3 Why a term without derivatives means mass

Question from the audience. Why does an A'^2 term count as a mass rather than just another interaction?[a]

Response. Mass is energy at rest. For a field, a particle at rest is the quantum of a zero-momentum, infinite-wavelength mode. The derivative term gives no energy to a static homogeneous displacement, whereas $m_A^2 A'^2$ does. That nonzero zero-momentum energy is the rest-energy gap.

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[a]Lecture timestamp: 00:13:10.

The lecture develops this statement through three mechanical analogies.

- In a crystal, slowly varying displacement costs elastic energy because neighboring atoms are shifted differently. A rigid translation shifts every atom together and costs no internal energy. The acoustic phonon is therefore gapless.
- In an ideal rotationally symmetric magnet, a spatially varying spin direction costs exchange energy, but rotating every spin together does not. The associated long-wavelength spin wave is gapless.

- In a plasma, shifting the electron fluid relative to the positive background creates charge separation and a restoring electric field even for a homogeneous relative displacement. The resulting plasmon has a nonzero frequency at zero wave number.

The Higgsed gauge field behaves like the third example: the broken medium resists what would otherwise be a cost-free homogeneous displacement.

Question from the audience. Before the gauge field was introduced, did the phase α not already vibrate?[a]

Response. It did. Its energy was proportional to $f^2(\partial_\mu \alpha)^2$. A phase that varies from place to place has gradient energy and can propagate, but a uniform change of phase costs nothing. That is precisely the massless Goldstone mode.

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[a]Lecture timestamp: 00:18:24.

Question from the audience. The derivative includes a time derivative. Does actually moving the whole lattice or changing the whole field not carry kinetic energy?[a]

Response. Yes. The comparison is between static configurations, or between configurations prepared arbitrarily slowly. A lattice translating with nonzero velocity has kinetic energy. The mass test asks whether the configuration still has energy when it is at rest and spatially uniform.

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[a]Lecture timestamp: 00:22:18.

9.2 The Experimental Clue: Weak Interactions Distinguish Left and Right

The lecture now turns to the electron, muon, and quarks. Their masses also come from the Higgs field, but the mechanism looks different because a fermion mass is not a quadratic vector term. It is a coupling between two chiral fields.

9.2.1 Parity violation in beta decay

Begin with the empirical fact that the weak interaction violates parity. Quantum electrodynamics and quantum chromodynamics treat a process and its mirror image alike. The weak interaction does not. In neutron beta decay,

$$n \longrightarrow p + e^- + \bar{\nu}_e, \tag{9.12}$$

the high-energy electrons selected by the charged-current interaction have left-handed helicity: their spin points opposite to their momentum. The corresponding antineutrinos have the opposite helicity.

For an ultrarelativistic particle, define helicity by

$$h = \frac{\boldsymbol{S} \cdot \boldsymbol{p}}{|\boldsymbol{S}|\,|\boldsymbol{p}|}. \tag{9.13}$$

Spin parallel to momentum has $h = +1$; spin antiparallel has $h = -1$. The weak interaction couples to a left-chiral electron field and a left-chiral neutrino field, not to a right-chiral electron field in the same way.[1]

To expose the obstruction without the full $SU(2)_L \times U(1)_Y$ bookkeeping, imagine a fake $U(1)$ world in which only ψ_L carries the relevant charge:

$$\psi_L \longmapsto e^{i\theta}\psi_L, \qquad \psi_R \longmapsto \psi_R. \tag{9.14}$$

This is not electromagnetism: real left- and right-chiral electrons carry the same electric charge. It is a caricature of the weak interaction, in which the two chiralities sit in different electroweak representations.

Question from the audience. Does the same left–right asymmetry apply to quarks? [a]

Response. Yes, with the appropriate weak multiplets and hypercharges. Left-chiral up- and down-type quarks form weak doublets, whereas the right-chiral fields are weak singlets. Their Higgs couplings therefore face the same structural problem as the electron's.

[a] Lecture timestamp: 00:31:50.

9.3 A Dirac Mass Is Left–Right Mixing

9.3.1 From the four-component equation to two Weyl equations

A Dirac wavefunction has four components. Loosely, two describe spin and two distinguish the positive- and negative-energy solutions, with the latter reinterpreted as antiparticle degrees of freedom. In Hamiltonian form,

$$i\frac{\partial\Psi}{\partial t} = \left(-i\boldsymbol{\alpha}\cdot\boldsymbol{\nabla} + m\beta\right)\Psi. \tag{9.15}$$

The matrices α_i and β encode the relativistic relation between energy, momentum, and mass.

The chiral basis makes the role of m explicit. With $\sigma^\mu = (1, \boldsymbol{\sigma})$ and $\bar{\sigma}^\mu = (1, -\boldsymbol{\sigma})$,

$$i\bar{\sigma}^\mu\partial_\mu\psi_L = m\psi_R, \tag{9.16}$$
$$i\sigma^\mu\partial_\mu\psi_R = m\psi_L. \tag{9.17}$$

At $m = 0$, the equations are independent: a left-chiral field and a right-chiral field can propagate without talking to each other. At $m \neq 0$, each appears as the source for the other. This is the exact sense in which a Dirac mass mixes left and right.

Susskind describes this as the fermion oscillating between left and right. That language captures the coupling but should not be pushed into a literal classical picture of a point particle flipping back and forth. The precise statement is the coupled field equation (9.16)–(9.17).

Question from the audience. Are the factors of i and the signs in the two board equations consistent?[a]

Response. Not line by line on the board; an i was pulled outside inconsistently during the quick derivation. The reliable content is the coupling pattern. Equations (9.16) and (9.17) give one standard consistent convention.

[a] Lecture timestamp: 00:41:10.

[1] Editorial clarification: The lecture initially uses handedness in the helicity sense because that is visually intuitive at high energy. Gauge charges actually act on chirality, the Lorentz-representation label selected by $P_L = (1 - \gamma^5)/2$ and $P_R = (1 + \gamma^5)/2$. For a massless particle, chirality and helicity coincide for particles; for a massive particle they do not.

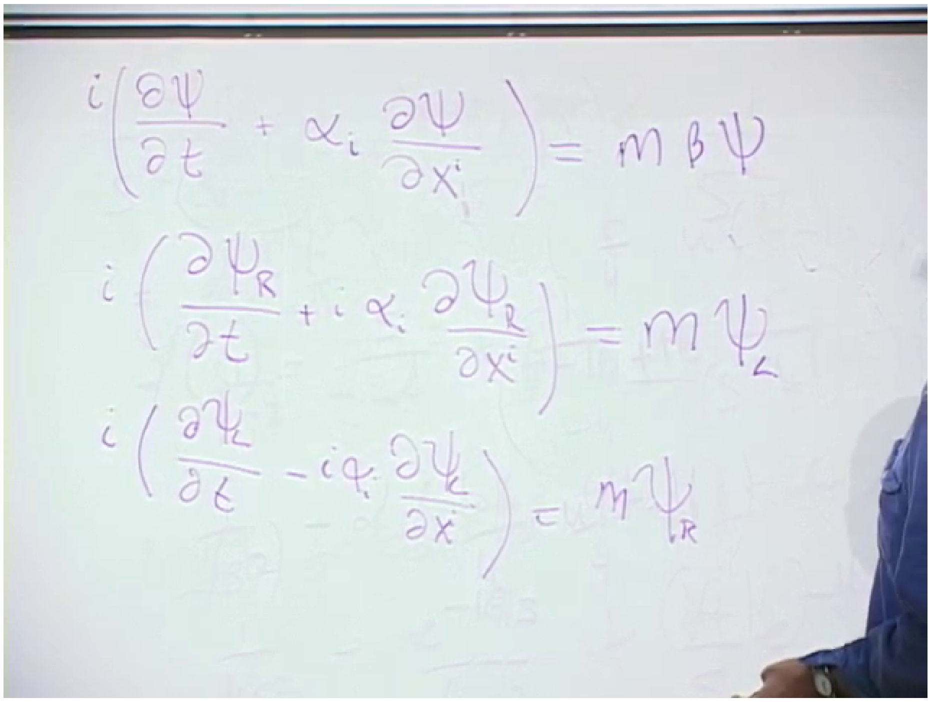

Figure 9.4: The board first writes the Dirac equation, then splits it into coupled left- and right-handed equations.

Lecture frame: 00:40:30.

9.3.2 Why the weak symmetry forbids a bare mass

The corresponding Dirac mass term in the Lagrangian is

$$\mathcal{L}_m = -m\bar{\psi}\psi = -m\left(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L\right). \tag{9.18}$$

Under the toy transformation (9.14),

$$\bar{\psi}_L\psi_R \longmapsto e^{-i\theta}\bar{\psi}_L\psi_R, \qquad \bar{\psi}_R\psi_L \longmapsto e^{i\theta}\bar{\psi}_R\psi_L. \tag{9.19}$$

Neither term is invariant. The direct left–right mixing coefficient m is therefore forbidden before symmetry breaking.

Electromagnetism alone would not cause this problem, because ψ_L and ψ_R have the same electric charge. The obstruction arises from their different weak quantum numbers.

9.3.3 Helicity questions that expose the chirality subtlety

The class now tests the intuitive handedness picture from several angles. These questions are not side issues: they reveal exactly where helicity is useful and where chirality must replace it.

Question from the audience. What would the world be like if the weak interaction emitted both electron helicities equally?[a]

Response. The immediate parity asymmetry would disappear from that process. But the real electroweak theory is chiral: only the left-chiral doublets carry the charged weak interaction. The lecture postpones a fuller account of why nature chose that structure.

[a]Lecture timestamp: 00:42:12.

Question from the audience. Suppose an electron's spin is prepared vertically and the electron is then accelerated horizontally. Is it left- or right-handed? [a]

Response. A spin state transverse to the momentum is a superposition of the two helicities. Helicity is sharp only when the spin is parallel or antiparallel to the momentum. The weak chiral projectors are nevertheless well defined for the field.

[a]Lecture timestamp: 00:42:44.

A massive particle can be slowed to rest and accelerated in the opposite direction without reversing its spin. Its helicity then changes sign. A massless particle cannot be overtaken or brought to rest by a Lorentz boost, so its helicity is invariant under proper boosts. This motivates the lecture's statement that a species with only one physical helicity must be massless. The more exact field-theory statement is that a Lorentz-invariant Dirac mass requires both chiralities.

Question from the audience. A circularly polarized photon can reflect from a mirror and reverse direction without ever coming to rest. Does that contradict the helicity argument?[a]

Response. No. The mirror is an external system that can exchange angular momentum and reverse the propagation direction. A reflection process is not a Lorentz boost that overtakes the photon. Whether a channel exists also depends on the allowed polarization states and on the material boundary conditions.

[a]Lecture timestamp: 00:45:28.

Question from the audience. Does an ordinary electron beam contain one handedness or both?[a]

Response. Both in general. A massive electron state contains both chiral components, and an unpolarized beam contains both helicities. Specialized sources can produce a polarized beam with a large helicity asymmetry.

[a]Lecture timestamp: 00:46:10.

Question from the audience. Can a beam be focused independently of its spin, or do magnetic fields separate the two polarizations?[a]

Response. Ordinary position focusing can be largely insensitive to spin, but magnetic moments couple to magnetic-field gradients. Spin-selective sources, Stern–Gerlach-like gradients, and accelerator spin dynamics can therefore prepare or separate polarizations.

[a]Lecture timestamp: 00:46:29.

Question from the audience. Is the weak difference between left and right merely an accident?[a]

Response. For the present course it is an experimental fact built into the Standard Model. It is too systematic to feel like a casual accident, but the Standard Model itself does not derive the complete pattern of chiral charges from a deeper principle.

[a]Lecture timestamp: 00:48:13.

9.4 The Yukawa Construction

9.4.1 Let the Higgs carry the missing charge

The symmetry forbids a numerical mass, but it allows an interaction with a field that supplies the missing transformation phase. Introduce a complex scalar with

$$\phi \longmapsto e^{i\theta}\phi, \qquad \phi^\dagger \longmapsto e^{-i\theta}\phi^\dagger. \tag{9.20}$$

Then

$$\mathcal{L}_Y = -y\,\bar{\psi}_L\phi\psi_R - y^*\,\bar{\psi}_R\phi^\dagger\psi_L \tag{9.21}$$

is gauge invariant. In the first term, $\bar{\psi}_L$ contributes $e^{-i\theta}$ and ϕ contributes $e^{i\theta}$; the phases cancel. The second term is its Hermitian conjugate.

Equivalently, the chiral equations acquire field-dependent mixing:

$$i\bar{\sigma}^\mu D_\mu\psi_L = y\phi\psi_R, \tag{9.22}$$

$$i\sigma^\mu\partial_\mu\psi_R = y^*\phi^\dagger\psi_L. \tag{9.23}$$

Before symmetry breaking, this is an interaction rather than a mass. A right-chiral fermion can become a left-chiral fermion only while emitting or absorbing the scalar quantum required to balance the charge. The coefficient y is a Yukawa coupling, not the gauge charge g.

Question from the audience. Is it legitimate simply to put ϕ and $\phi^\dagger$ on the right-hand sides of the two equations? [a]

Response. It is legitimate precisely because their transformation phases repair the mismatch. The decisive test is not the visual form of the equation but gauge invariance of the interaction $\bar{\psi}_L\phi\psi_R$ + h.c..

[a]Lecture timestamp: 00:51:55.

9.4.2 The vacuum converts interaction into mass

Now identify ϕ with the Higgs field and let it settle at nonzero vacuum magnitude. In the lecture's simplified normalization,

$$\phi \simeq f, \tag{9.24}$$

so (9.21) contains

$$\mathcal{L}_Y \supset -yf\,\bar{\psi}_L\psi_R - y^*f\,\bar{\psi}_R\psi_L. \tag{9.25}$$

After a harmless phase redefinition when appropriate, the fermion mass is

$$m_f = yf. \tag{9.26}$$

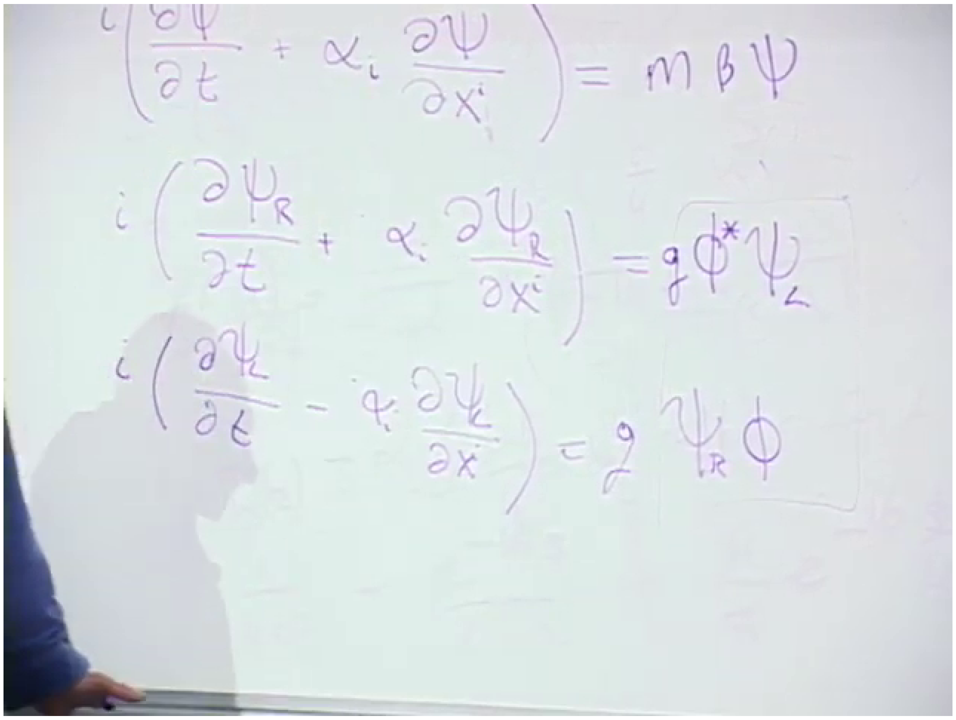

Figure 9.5: The bare mass on the right-hand sides has been replaced by the charged scalar field. The exact board signs are schematic; the invariant structure is given in (9.21).

Lecture frame: 00:51:30.

This is the fermionic counterpart of the vector result $m_A = gf$. The same Higgs vacuum that supplies a longitudinal polarization to the weak gauge bosons supplies the missing weak quantum numbers needed to join left and right fermions.

In the canonically normalized Standard Model convention,

$$H(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \qquad v \simeq 246 \text{ GeV}, \tag{9.27}$$

and therefore

$$m_f = \frac{y_f v}{\sqrt{2}}. \tag{9.28}$$

The lecture's f and the conventional $v/\sqrt{2}$ differ only by this choice of normalization.

Question from the audience. If only the product yf is a fermion mass, how can the coupling y and vacuum scale f be determined separately? [a]

Response. The weak gauge-boson masses determine the vacuum scale once the gauge couplings are known. A fermion's measured mass then determines its Yukawa coupling. In the full theory, $m_W = gv/2$, $m_Z = v\sqrt{g^2 + g'^2}/2$, and $y_f = \sqrt{2}m_f/v$.

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[a] Lecture timestamp: 00:55:36.

Question from the audience. Is the weak-boson mass about 900 GeV? [a]

Response. No; the lecture immediately corrects the slip to roughly 90 GeV. More precisely, $m_W \simeq 80$ GeV and $m_Z \simeq 91$ GeV, while the underlying vacuum scale is $v \simeq 246$ GeV.

[a]Lecture timestamp: 00:57:43.

9.4.3 One vacuum scale, many Yukawa couplings

Every charged fermion species has its own Yukawa coupling:

$$m_i = y_i f \qquad \text{(lecture normalization).} \tag{9.29}$$

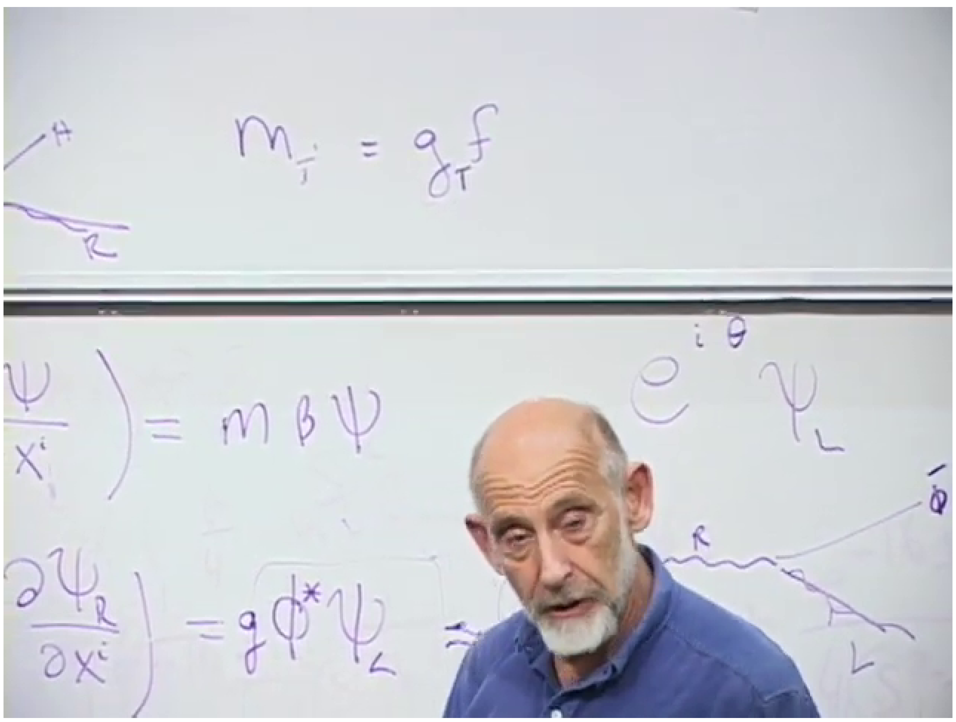

Figure 9.6: The board summarizes the species dependence as $m_i = g_i f$; the g_i written there are the Yukawa couplings called y_i here.

Lecture frame: 01:16:30.

For the electron, the lecture estimates

$$y_e \sim \frac{0.5\ \text{MeV}}{10^2\ \text{GeV}} \sim 10^{-5}. \tag{9.30}$$

Using the canonical value $v = 246$ GeV gives the more precise $y_e \simeq 2.9 \times 10^{-6}$. The top quark lies at the opposite end:

$$y_t = \frac{\sqrt{2} m_t}{v} \simeq 1. \tag{9.31}$$

Thus the Higgs mechanism explains how a gauge-invariant mass can exist, but it does not explain the enormous hierarchy among the y_i. The observed fermion masses merely translate that hierarchy into measured numbers.

Question from the audience. Does the top quark really have a Yukawa coupling of order one?[a]

Response. Yes. Its mass is comparable to the electroweak scale, so its coupling is not a tiny perturbative number like the electron's. In the modern normalization y_t is close to unity.

[a]Lecture timestamp: 00:59:09.

9.4.4 Restoring the physical Higgs fluctuation

Freezing the scalar at its vacuum value hid the radial excitation. Restore it schematically:

$$\phi(x) \simeq f + h(x). \tag{9.32}$$

The Yukawa factor becomes

$$y\phi \simeq yf + yh = m_f + yh. \tag{9.33}$$

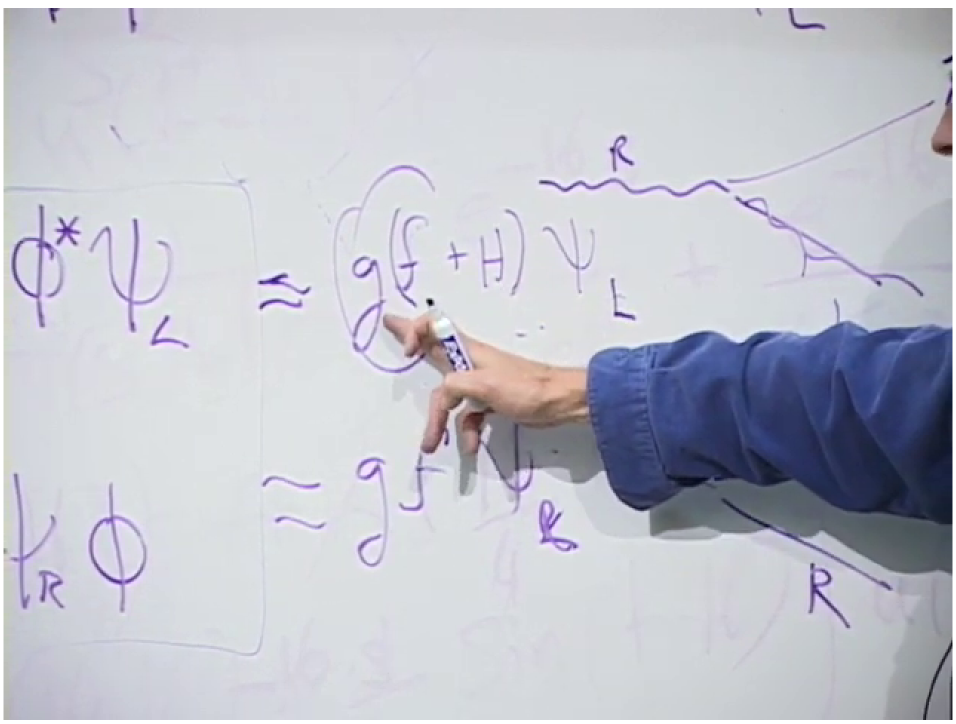

Figure 9.7: The board restores the Higgs fluctuation inside the Yukawa term: the constant part produces the fermion mass and the fluctuating part produces Higgs emission or absorption.

Lecture frame: 01:01:15.

The first term in (9.33) is the mass. The second is a vertex in which a Higgs boson couples to the fermion. In canonical Standard Model notation,

$$\mathcal{L}_Y \supset -m_f\bar{\psi}\psi - \frac{m_f}{v}h\bar{\psi}\psi. \tag{9.34}$$

The coupling is proportional to the fermion mass. Subject to kinematics, color factors, phase space, and competing channels, the Higgs therefore couples more strongly to heavier fermions.

Question from the audience. Are coupling constants themselves probabilities? [a]

Response. They enter quantum amplitudes. A simple rate is often proportional to the modulus square of an amplitude and hence to a coupling squared, but phase space, spin sums, interference, loops, and multiple vertices also matter. A coupling should not be identified directly with a probability.

[a]Lecture timestamp: 01:03:35.

Question from the audience. Does this argument assume that there is only one Higgs field?[a]

Response. It describes the minimal Standard Model, which has one complex Higgs doublet. Three of its four real components are absorbed by $W^{\pm}$ and Z, leaving one physical neutral scalar. Extensions can contain more doublets or other scalar multiplets. Supersymmetric models typically require two Higgs doublets and therefore contain several physical Higgs states.

[a]Lecture timestamp: 01:03:59.

That historical expectation can now be compared with experiment. [2]

9.5 Antiparticles, Beams, and the Meaning of Chirality

Before turning to neutrinos, the class returns to several operational questions about antiparticles and accelerator beams.

Question from the audience. What handedness does the antiparticle of a left-handed electron have?[a]

Response. A left-chiral electron field annihilates left-helicity electrons and creates right-helicity positrons in the ultrarelativistic limit. This is one place where particle versus antiparticle and chirality versus helicity must be stated carefully.

[a]Lecture timestamp: 01:06:05.

Question from the audience. Can an electron and positron collide to make a Z boson?[a]

Response. Yes, when their center-of-mass energy matches the allowed channel. Electron–positron colliders studied the Z resonance with great precision. Charge and the other quantum numbers are conserved by the full process.

[a]Lecture timestamp: 01:07:13.

Question from the audience. Are positrons all produced through one special weak process?[a]

Response. No. Pair production is a standard electromagnetic route. For example, an energetic electron can radiate a virtual photon, and that photon can produce an e^-e^+ pair in the field of another particle. The lecture sketches this with a Feynman diagram.

[a]Lecture timestamp: 01:07:37.

[2]Editorial clarification: This lecture was recorded before Higgs discovery. ATLAS and CMS announced a Higgs-like particle in 2012, now measured near 125 GeV. Its observed couplings are broadly consistent with the mass-proportional Standard Model pattern. The statement that it decays preferentially to the heaviest particles always carries the kinematic caveat: an on-shell $t\bar{t}$ pair is too heavy for a 125 GeV Higgs.

Question from the audience. Can a linear accelerator select the speed or energy of its electrons?[a]

Response. Yes. The accelerating fields and timing select an energy range. Once electrons are ultrarelativistic their speed is already extremely close to c, so additional accelerator energy mostly increases momentum and total energy rather than visibly increasing speed.

[a]Lecture timestamp: 01:09:24.

Question from the audience. Can an accelerator prepare a beam of one polarization? [a]

Response. Yes, although it is technically demanding. One can prepare spin-polarized electrons at low energy with a spin-dependent source and then accelerate them while controlling spin precession. Modern colliders quantify the resulting beam polarization rather than treating it as perfectly pure.

[a]Lecture timestamp: 01:09:47.

9.6 Neutrino Mass and the Majorana Possibility

9.6.1 Why neutrino mass forced an extension

The original minimal Standard Model included only left-chiral neutrino fields and therefore predicted massless neutrinos. Neutrino oscillations show that at least two neutrino mass eigenvalues differ and that neutrino masses are nonzero, although tiny compared with charged-fermion masses.

Question from the audience. How small are the neutrino masses, and where is their right-handed component?[a]

Response. Oscillations determine mass-squared differences, not the absolute masses. The relevant splittings correspond to scales of roughly 0.009 eV and 0.05 eV. A Dirac description adds sterile right-chiral neutrinos; a Majorana description instead relates the missing chirality to charge-conjugate neutrino fields.

[a]Lecture timestamp: 01:10:38.

For a Dirac neutrino one adds a new field ν_R and repeats the Yukawa construction:

$$\mathcal{L}_{\nu,\mathrm{Dirac}} = -y_\nu \bar{L}\,\widetilde{H}\,\nu_R + \mathrm{h.c.}, \qquad m_\nu = \frac{y_\nu v}{\sqrt{2}}. \tag{9.35}$$

The required y_ν is extraordinarily small.

There is another logical possibility because the neutrino carries no electric charge. A left-chiral neutrino can be coupled to the charge conjugate of itself, which has the opposite chirality:

$$\nu_L \longleftrightarrow (\nu_L)^c_R. \tag{9.36}$$

A mass eigenstate built this way is a Majorana fermion: it is its own antiparticle, up to phase conventions. The analogous electron–positron mixing is forbidden because it would violate exact electric-charge conservation.

Question from the audience. Why is particle–antiparticle mixing allowed for a neutrino but not for an electron?[a]

Response. An electron and a positron have opposite electric charge, so a mass term that mixes them as the same one-particle state would violate the unbroken electromagnetic symmetry. A neutrino is electrically neutral. If no exact conserved lepton number distinguishes it from its antiparticle, a Majorana mass is allowed.

[a]Lecture timestamp: 01:13:43.

The name honors Ettore Majorana, the young Italian physicist who formulated this kind of real fermion representation and disappeared in 1938. The lecture pauses over the historical mystery: his fate was never established. [3]

9.7 Why the Familiar Spectrum Clusters Near the Electroweak Scale

9.7.1 The common scale and its important qualifications

Question from the audience. Why was the Higgs mass expected to be reachable at the weak scale?[a]

Response. The curvature of the Higgs potential sets the radial Higgs mass, while the same vacuum scale f sets the W and Z masses. Without an extreme scalar self-coupling, one therefore expected the Higgs mass to be of the same broad order as the weak scale. The observed 125 GeV value indeed lies there.

[a]Lecture timestamp: 01:14:48.

The lecture's striking summary is that the elementary masses of the minimal Standard Model trace back to one broken vacuum:

$$m_W = \frac{1}{2} g v, \tag{9.37}$$

$$m_Z = \frac{1}{2}\sqrt{g^2 + g'^2}\, v, \tag{9.38}$$

$$m_f = \frac{1}{\sqrt{2}} y_f v. \tag{9.39}$$

If v vanished while the dimensionless couplings were held fixed, the weak gauge bosons and charged elementary fermions would be massless.

That statement must not be enlarged beyond its domain. The photon and gluons remain massless because an unbroken gauge symmetry protects them. Most of the proton and neutron mass is QCD binding energy, not the sum of their quark Higgs masses. Neutrino mass requires an extension of the minimal model. The Higgs mass itself depends on the scalar potential and is not simply another Yukawa mass.

[3]Editorial clarification: A gauge-invariant Majorana mass for the active Standard Model neutrino is not a renormalizable bare term before electroweak breaking. At low energy it can arise from the dimension-five Weinberg operator $(LH)(LH)/\Lambda$, or through a seesaw involving heavy sterile neutrinos. Whether neutrinos are Dirac or Majorana particles remains experimentally unsettled.

9.7.2 Particles whose mass need not wait for symmetry breaking

Suppose a new fermion had left- and right-chiral components with identical gauge quantum numbers. Then $\bar{\Psi}_L \Psi_R + \text{h.c.}$ would already be gauge invariant. Such a *vectorlike* fermion could possess a bare mass M unrelated to v:

$$\mathcal{L}_{\text{vectorlike}} = -M\bar{\Psi}\Psi. \tag{9.40}$$

There is no electroweak symmetry obstruction to $M \gg v$.

This motivates the lecture's broad naturalness argument. Particles protected by gauge or chiral symmetry remain massless until the Higgs vacuum breaks that protection, so their masses are tied to the relatively small scale v. Particles with no such protection may naturally be much heavier and therefore absent from the observed low-energy spectrum. What we see may be a selection effect: the accessible particles are the ones whose masses were forced to wait for electroweak symmetry breaking.

Question from the audience. Would particles with independent, much larger masses play no role in reality?[a]

Response. They would play little role in ordinary low-energy laboratory physics because they are hard to produce, but they could still shape high-energy history or cosmology. The lecture offers dark matter as a possible place where a heavier hidden spectrum might first reveal itself.

[a] Lecture timestamp: 01:19:39.

The 2010 discussion reflects the then-popular expectation of weak-scale or somewhat heavier dark matter. That remains a useful model class, not an established fact. Dark matter has been detected gravitationally, but its particle identity and the origin of its mass remain unknown.

9.8 Closing Questions: Cosmic Rays, Inertia, Gravity, and Chirality

9.8.1 Mass is not high laboratory energy

Question from the audience. What is the mass of a cosmic ray? [a]

Response. A cosmic ray may be a proton, an atomic nucleus, an electron, or a photon; its invariant mass is the mass of that particle or system. What is spectacular is its laboratory energy, which for the highest events reaches roughly 10^{20} to 10^{21} eV. High energy is not the same as high invariant mass.

[a] Lecture timestamp: 01:20:57.

A cosmic ray strikes a target particle that is approximately at rest. For a projectile proton of laboratory energy $E \gg m_p$,

$$s = (p_1 + p_2)^2 \simeq 2m_p E + 2m_p^2, \qquad \sqrt{s} \simeq \sqrt{2m_p E}. \tag{9.41}$$

Only the square root of the projectile energy becomes center-of-mass energy. For equal head-on beams of energy E_{beam},

$$\sqrt{s} \simeq 2E_{\text{beam}}. \tag{9.42}$$

This is why a collider is far more efficient than a fixed target at converting controlled beam energy into new-particle mass.

Question from the audience. Why can CERN's collider be a more potent probe than a single much more energetic particle hitting matter at rest? [a]

Response. In a fixed-target collision, momentum conservation carries most of the laboratory energy forward in the motion of the final state. In a symmetric collider the total spatial momentum can vanish, leaving the beam energy available as center-of-mass energy. The very highest cosmic rays can still exceed the LHC in $\sqrt{s}$, but they arrive unpredictably and cannot provide collider luminosity or a controlled detector environment.

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[a] Lecture timestamp: 01:21:50.

9.8.2 Rest mass and resistance to acceleration

Question from the audience. Are inertia and mass essentially the same thing? [a]

Response. At rest, invariant mass is both rest energy through $E_0 = mc^2$ and the parameter governing the low-velocity response to a force. Special relativity explains why those definitions belong to one quantity. For moving bodies it is clearer to keep invariant mass fixed and describe acceleration using relativistic energy and momentum.

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[a] Lecture timestamp: 01:22:41.

The invariant relation

$$E^2 = p^2c^2 + m^2c^4 \tag{9.43}$$

contains both meanings. Near rest,

$$E = mc^2 + \frac{p^2}{2m} + \cdots , \tag{9.44}$$

so the same m that gives the rest energy appears in the kinetic response. At high speed the relation between force and acceleration depends on whether the force is parallel or perpendicular to the velocity. This directional dependence is why modern treatments avoid a velocity-dependent *relativistic mass* and reserve mass for the Lorentz invariant m.

9.8.3 What gravity couples to

Question from the audience. Once the Higgs gives a particle mass, does it then begin to couple to gravity?[a]

Response. Its rest energy contributes to gravity, but mass is not the whole source. Photons are massless and are nevertheless deflected by gravity. In general relativity the source is the stress–energy tensor: energy density, momentum flow, pressure, and stress. For slowly moving matter the dominant contribution is its rest-mass energy.

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[a] Lecture timestamp: 01:23:58.

The bending of light near the Sun makes the point operationally. A photon responds to spacetime curvature despite having $m = 0$, and its energy and momentum also contribute to the gravitational field. For two nearly static massive bodies, energy is dominated by mc^2, so Newtonian gravity can be written in terms of their masses without displaying the more general stress–energy statement.

Question from the audience. Is a high-energy particle harder to accelerate in the same way in every direction?[a]

Response. No. The change of momentum for a force parallel to the motion differs from the change for a transverse force. It is correct that a more energetic particle is harder to deflect, but assigning it one enlarged moving mass hides this directional structure.

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[a]Lecture timestamp: 01:25:20.

9.8.4 An electron at rest and the promised distinction

Question from the audience. How can an electron at rest be called left- or right-handed?[a]

Response. It cannot be assigned helicity at rest because helicity uses the direction of momentum. A rest spinor can still be decomposed into left- and right-chiral components. For a massive electron both components are present, and the mass term couples them.

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[a]Lecture timestamp: 01:26:32.

Question from the audience. If an electron at rest has no helicity, can it still participate in a weak interaction?[a]

Response. Yes. The weak interaction couples to chirality, not to a frame-dependent helicity label. At high energy helicity is a convenient approximation to chirality; near rest that shortcut fails. This is the subtle distinction the lecture had postponed.

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[a]Lecture timestamp: 01:27:18.

9.9 Historical End-of-Lecture Questions

9.9.1 The pre-discovery Higgs window

In 2010 the Higgs had not yet been seen. Precision electroweak observables, especially loop-sensitive relations among the W, Z, top, and other parameters, favored a comparatively light Higgs within the Standard Model. The lecture quotes a likely upper region near 160 GeV, well below v but not separated from it by an order of magnitude.

Question from the audience. What evidence then suggested that the Higgs was lighter than the 250 GeV vacuum scale?[a]

Response. High-precision measurements could be compared with radiative corrections whose values depend logarithmically on the Higgs mass. The resulting global fits favored a light Higgs. This was indirect evidence, not a direct mass measurement; the later direct value is about 125 GeV.

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[a]Lecture timestamp: 01:28:28.

Question from the audience. Could several minima of the Higgs potential produce different values of f, perhaps tied to different particle generations? [a]

Response. Different minima can support very different low-energy physics, but there is no simple inference from the observed generations to multiple Higgs minima. The lecture defers the broader subject of multiple vacua to a later course rather than pretending to settle it here.

[a] Lecture timestamp: 01:29:28.

Question from the audience. Was the Higgs expected to be the first new particle found at the LHC?[a]

Response. Not necessarily. Its visibility depends on production rates and decay channels, some of which can be difficult or invisible. Before LHC data, supersymmetric or other new particles could plausibly have appeared first. Historically, the Higgs was discovered and no supersymmetric particle has yet been established.

[a] Lecture timestamp: 01:31:32.

9.9.2 When a neutral particle is its own antiparticle

Question from the audience. Do the photon and gluon lack antiparticles because they are massless?[a]

Response. No. Masslessness is not the criterion. A photon is its own antiparticle because no conserved charge distinguishes a photon from an antiphoton. Each gluon is also a self-conjugate gauge quantum in the appropriate color representation, although gluons themselves carry color and the color labels mix under charge conjugation. A neutron is electrically neutral but has a distinct antineutron because baryon number and its quark content distinguish them.

[a] Lecture timestamp: 01:32:12.

The neutrino returns as the subtle case. It has weak interactions, and the ultrarelativistic states called neutrino and antineutrino interact with opposite helicities. If an exact lepton number distinguishes them, they can form a Dirac particle with a distinct antiparticle. If lepton number is not exact and the mass is Majorana, the massive particle is its own antiparticle even though its two helicity states behave differently in weak processes.

Question from the audience. Do weak charge and lepton number prove that neutrinos and antineutrinos must be distinct?[a]

Response. Not conclusively. Weak interactions distinguish the helicity states, but a Majorana mass can combine a neutrino field with its charge conjugate. The decisive issue is whether an exact conserved quantum number forbids that mixing. Total lepton number is not imposed as an exact gauge symmetry by the Standard Model.

[a] Lecture timestamp: 01:32:46.

Ordinary flavor oscillations require a separate caution. [4]

[4] Editorial clarification: Ordinary observed flavor oscillations mix ν_e, ν_μ, ν_τ of the same neutrino or antineutrino sector; they do not by themselves demonstrate neutrino–antineutrino conversion. Lepton-number-violating processes such as neutrinoless double-beta decay would provide direct evidence for Majorana mass, but none has yet been confirmed.

Question from the audience. Does the Standard Model explain why electric charges are quantized?[a]

Response. The lecture acknowledges the question but does not develop it. Within the Standard Model, gauge invariance and anomaly cancellation strongly constrain consistent hypercharge assignments; a deeper explanation often invokes grand unification or other ultraviolet structure.

[a] Lecture timestamp: 01:33:39.

9.10 What the Lecture Has Established

The gauge-boson and fermion stories now line up.

1. A derivative-only gauge theory has no energy gap for a homogeneous vector potential. Gauge symmetry forbids a bare vector mass.
2. A charged scalar with nonzero vacuum magnitude contributes $g^2 f^2 A_\mu A^\mu$. Its angular Goldstone mode becomes the longitudinal polarization of a massive weak gauge boson.
3. A Dirac mass couples ψ_L and ψ_R. Because the weak interaction assigns them different gauge quantum numbers, a bare left–right mass term is forbidden.
4. A Yukawa interaction inserts the Higgs field between the two chiralities and is gauge invariant. Replacing the Higgs by its vacuum value gives $m_f = y_f f$, while retaining its fluctuation gives the physical Higgs–fermion coupling.
5. Neutrino mass obeys the same left–right logic but admits both Dirac and Majorana realizations because the neutrino is electrically neutral.
6. Particles without gauge or chiral protection could possess masses unrelated to the electroweak scale and may lie far above present experiments.

The unifying statement is therefore not that the Higgs is a substance that mechanically drags on particles. It is that the electroweak vacuum has a nonzero scalar order parameter. That order parameter changes which gauge-invariant terms are possible in the low-energy equations. For vectors it opens a zero-momentum gap; for fermions it permits left–right mixing. The different masses then measure different couplings to the same broken vacuum.

Chapter 10

The Higgs Scale, Running Couplings, and the Hierarchy Problem

Overview. The final lecture begins by looking beyond the Standard Model toward unification, supersymmetry, and the gauge-hierarchy problem. To see why the Higgs field sits at the center of those questions, we first reconstruct the Dirac Lagrangian and show that a fermion mass mixes left and right chirality. The chiral weak interaction forbids that mass as a bare term; a gauge-invariant Yukawa coupling supplies it only after the Higgs field acquires a vacuum value. We then derive that vacuum value from the Higgs potential, ask why every known elementary mass is tied to this one small scale, and use superconductivity as a physical analog. After the break, the lecture develops renormalization from conductor and dielectric screening, carries the argument into QED and QCD, follows the gauge couplings toward unification, and derives the Planck scale from short-distance gravity. The electroweak scale then returns as the one unexplained small number.

10.1 The Question Waiting Beyond the Standard Model

The Standard Model is not presented here as a closed catalogue. At the start of the evening I want to say where the next course will go. The theory has difficulties rather than logical paradoxes: the gauge hierarchy, the possible role of supersymmetry, and the question of unification. The three gauge factors

$$SU(3)_c \times SU(2)_L \times U(1)_Y \tag{10.1}$$

describe strong, weak, and hypercharge interactions separately. Could they be subgroups of one larger group? An often studied possibility is $SU(5)$, in which quarks and leptons occupy common multiplets. Whether nature uses that particular construction is a different question. The important clue is that the apparently separate structures fit remarkably neatly into a larger mathematical one.

The Large Hadron Collider was built in part to expose whatever physics settles these questions. In 2010 its most immediate quarry was the Higgs boson. Finding a Higgs could never mean only adding one particle to a table. A Higgs is identified by an interlocking set of properties: its vacuum value gives masses to the weak vector bosons and fermions, while its fluctuation couples to those same particles with definite strengths. A resonance with the wrong couplings would not be the Standard Model Higgs. [1]

[1]Editorial clarification: The lecture predates the 2012 discovery. The observed approximately 125 GeV scalar has,

The route into that argument begins not with the Higgs potential but with the Dirac equation. We need to understand exactly what a fermion mass does before asking where it comes from.

10.2 From the Dirac Equation to Its Lagrangian

10.2.1 The older Dirac notation

In the Hamiltonian notation used when the Dirac equation was first introduced, its structure is

$$i\partial_t\psi = \left(-i\alpha^i\partial_i + \beta m\right)\psi. \tag{10.2}$$

Signs and factors of i depend on conventions, and the live board temporarily suppresses some of them. The invariant content is the separation between terms containing derivatives and a mass term containing none.

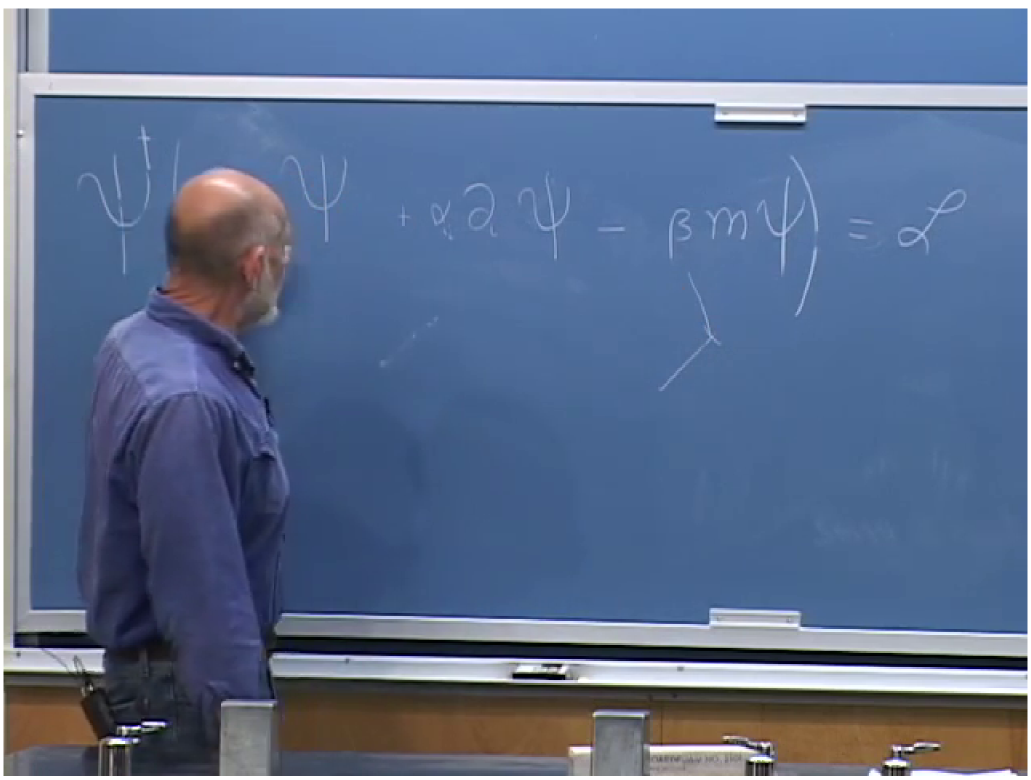

Figure 10.1: The Dirac equation and Lagrangian in the older α, β notation. The board is being used structurally rather than as a fixed sign convention.

Lecture frame: 00:05:52.

To obtain an action whose variation gives the Dirac equation, multiply the equation by $\psi^\dagger$. Moving every term to one side removes the need for an equal sign, prompting the evening's small mathematical joke: once an equation has been written as an expression that vanishes, the equal sign has done its

within present precision, the spin, parity, and coupling pattern expected of the Standard Model Higgs. No low-energy supersymmetric particles have been observed as of 2026. These later facts update the historical setting without changing the argument developed in the room.

work. The corresponding first-order Lagrangian may be written schematically as

$$\mathcal{L} = \psi^\dagger \Big(i\partial_t + i\alpha^i\partial_i - \beta m\Big)\,\psi, \tag{10.3}$$

with the understanding that a consistent metric convention fixes the relative signs.

The derivative terms move an excitation from one spacetime point to a nearby one and rotate its spinor components in the process. In a lattice picture they are hopping terms. By contrast, the mass term absorbs and re-emits the fermion at the same point. It has no derivative. This local operation will shortly reveal itself as a repeated conversion between two kinds of handedness.

Question from the audience. Should a γ^0 be placed after $\psi^\dagger$?[a]

Response. Not in the noncovariant form just written. But the question points directly to the covariant notation. Define the Dirac adjoint by $\bar\psi = \psi^\dagger\gamma^0$; then the γ^0 is indeed built into the object that stands on the left.

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[a] Lecture timestamp: 00:05:59.

10.2.2 The covariant form

Set

$$\bar\psi \equiv \psi^\dagger\beta = \psi^\dagger\gamma^0, \qquad \gamma^0 \equiv \beta, \qquad \gamma^i \equiv \beta\alpha^i. \tag{10.4}$$

Because $\beta^2 = 1$, multiplication by β converts freely between the two notations. The time derivative becomes $\bar\psi\gamma^0\partial_0\psi$, the spatial derivatives become $\bar\psi\gamma^i\partial_i\psi$, and the result is

$$\boxed{\mathcal{L}_D = \bar\psi\,(i\gamma^\mu\partial_\mu - m)\,\psi.} \tag{10.5}$$

This notation exposes the Lorentz character of each bilinear. $\bar\psi\psi$ is a scalar, whereas $\bar\psi\gamma^\mu\psi$ is a four-vector. The compact expression is not merely prettier than the α, β form; it tells us how the pieces transform.

Question from the audience. Was the spinor at the far left of the compact expression meant to be $\bar\psi$?[a]

Response. Yes. The missing bar is restored on the board. It matters: $\bar\psi$ is not $\psi^\dagger$, but $\psi^\dagger\gamma^0$.

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[a] Lecture timestamp: 00:09:39.

10.3 Chirality and What a Dirac Mass Actually Does

Besides $\gamma^0, \gamma^1, \gamma^2, \gamma^3$, define

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3, \qquad (\gamma^5)^2 = 1. \tag{10.6}$$

The factor of i shown here is the standard convention for the mostly minus Minkowski metric. The eigenvalues are $+1$ and -1, and the corresponding projectors are

$$\begin{aligned} P_L &= \frac{1-\gamma^5}{2}, \quad P_R = \frac{1+\gamma^5}{2}, \\ \psi_L &= P_L\psi, \qquad \psi_R = P_R\psi. \end{aligned} \tag{10.7}$$

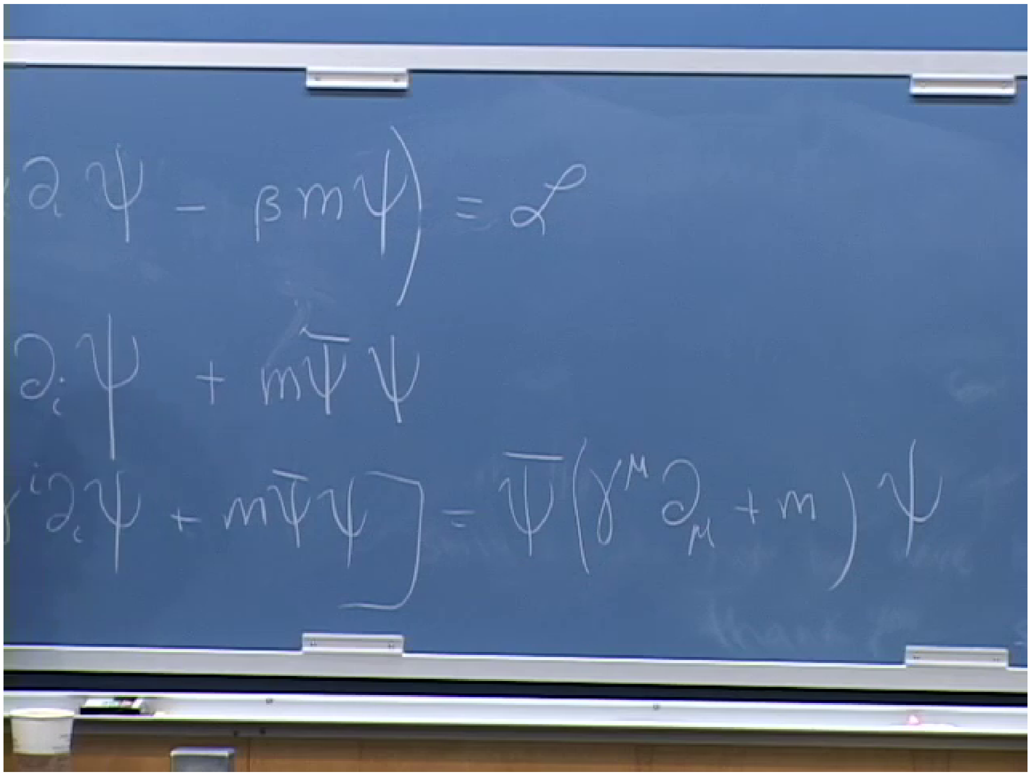

Figure 10.2: The covariant rewrite: $\bar{\psi} = \psi^\dagger\gamma^0$, $\gamma^i = \beta\alpha^i$, and the compact Dirac Lagrangian.
Lecture frame: 00:10:17.

Question from the audience. Why is the extra matrix called γ^5 when there is no γ^4? And does its eigenvalue describe helicity?[a]

Response. The name is historical: it is the additional matrix formed from the product of the four spacetime gamma matrices. Its eigenvalue labels chirality, not generally helicity. For a massless particle the two notions can be identified appropriately, which is why the language is easy to blur, but chirality is the Lorentz-representation label needed by the weak gauge theory.

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[a]Lecture timestamp: 00:10:42.

The kinetic term preserves chirality. One useful way to see this is that $\{\gamma^\mu, \gamma^5\} = 0$, so it separates into

$$\bar{\psi} i\gamma^\mu\partial_\mu\psi = \bar{\psi}_L i\gamma^\mu\partial_\mu\psi_L + \bar{\psi}_R i\gamma^\mu\partial_\mu\psi_R. \tag{10.8}$$

The scalar bilinear behaves differently:

$$\bar{\psi}\psi = \bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L. \tag{10.9}$$

Thus a Dirac mass is a local chirality-flipping operation:

$$\psi_L \longleftrightarrow \psi_R. \tag{10.10}$$

A massless fermion may propagate with definite chirality. A massive one cannot remain in one chiral sector, because the mass term continually mixes the two. It may seem surprising that this conversion has anything to do with inertia, but it is precisely the relativistic field-theory form of fermion rest mass.

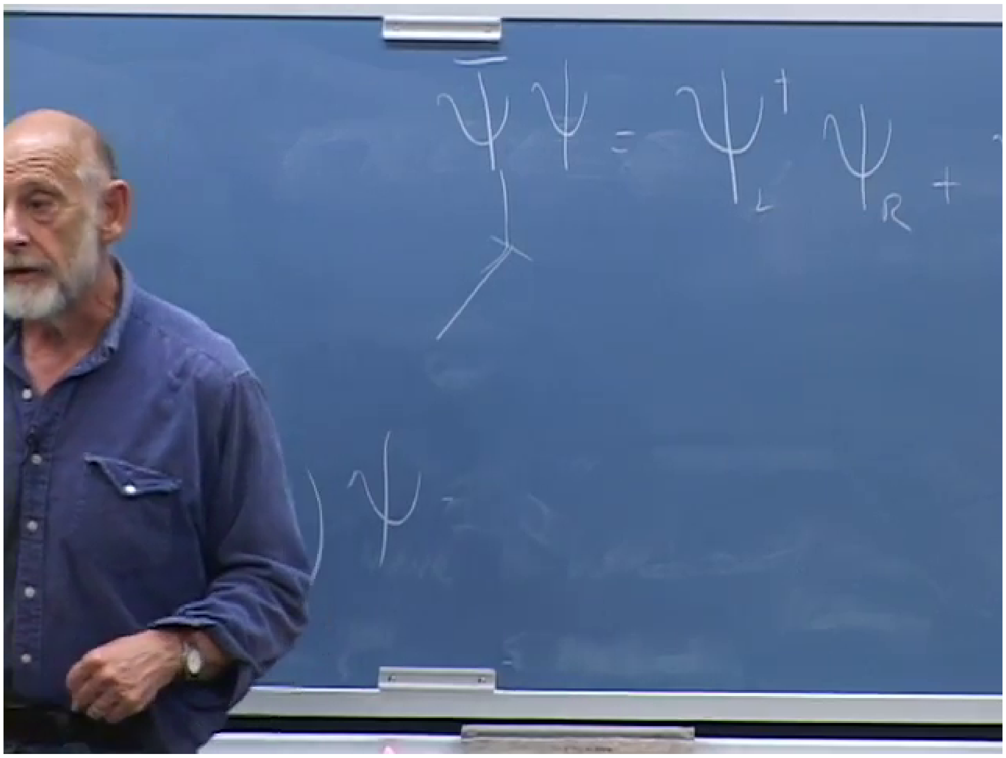

Figure 10.3: The board decomposes the scalar bilinear into left-to-right and right-to-left pieces.
Lecture frame: 00:15:38.

10.4 Why the Weak Symmetry Forbids a Bare Fermion Mass

The charged weak current is purely left-chiral. The lecture first names the Z, then corrects the statement to the $W^{\pm}$: Z couplings are chiral and unequal, but not simply absent for every right-chiral fermion. The clean statement is

$$\mathcal{L}_{\rm cc} = -\frac{g}{\sqrt{2}}\left(\bar{u}_L\gamma^\mu d_L + \bar{\nu}_L\gamma^\mu e_L\right) W^+_\mu + \text{h.c.} \tag{10.11}$$

The left-handed fermions transform as $SU(2)_L$ doublets; the right-handed charged fermions transform as singlets.

Now compare those transformation rules with (10.9). A term such as $\bar{e}_L e_R$ absorbs an object carrying one set of electroweak quantum numbers and emits one carrying another. By itself it is not gauge invariant. The familiar Dirac mass is therefore illegal as a bare term in the electroweak Lagrangian.

This is the right order of explanation. Fermion masses are not first inserted and then cosmetically associated with the Higgs. Gauge symmetry first removes the obvious mass term. The Higgs field is then forced to carry the missing quantum numbers.

10.4.1 The Higgs as the charge carrier

Let the Higgs transform as an $SU(2)_L$ doublet with hypercharge $+\frac{1}{2}$:

$$H = \begin{pmatrix} H^+ \\ H^0 \end{pmatrix}, \qquad \widetilde{H} = i\sigma^2 H^*. \tag{10.12}$$

The lecture keeps the group indices schematic and calls the field ϕ. In that notation, a Yukawa term has the form

$$\mathcal{L}_Y = -G_Y \left(\bar{\psi}_L \phi \psi_R + \bar{\psi}_R \phi^\dagger \psi_L \right). \tag{10.13}$$

The Higgs field carries away the weak charge that a naked chirality flip would fail to conserve.

The complete one-generation Standard Model expression makes the same bookkeeping explicit:

$$\mathcal{L}_Y = -y_d \bar{Q}_L H d_R - y_u \bar{Q}_L \widetilde{H} u_R - y_e \bar{L}_L H e_R + \text{h.c.} \tag{10.14}$$

Each term is gauge invariant, and each fermion species carries its own dimensionless Yukawa coupling.

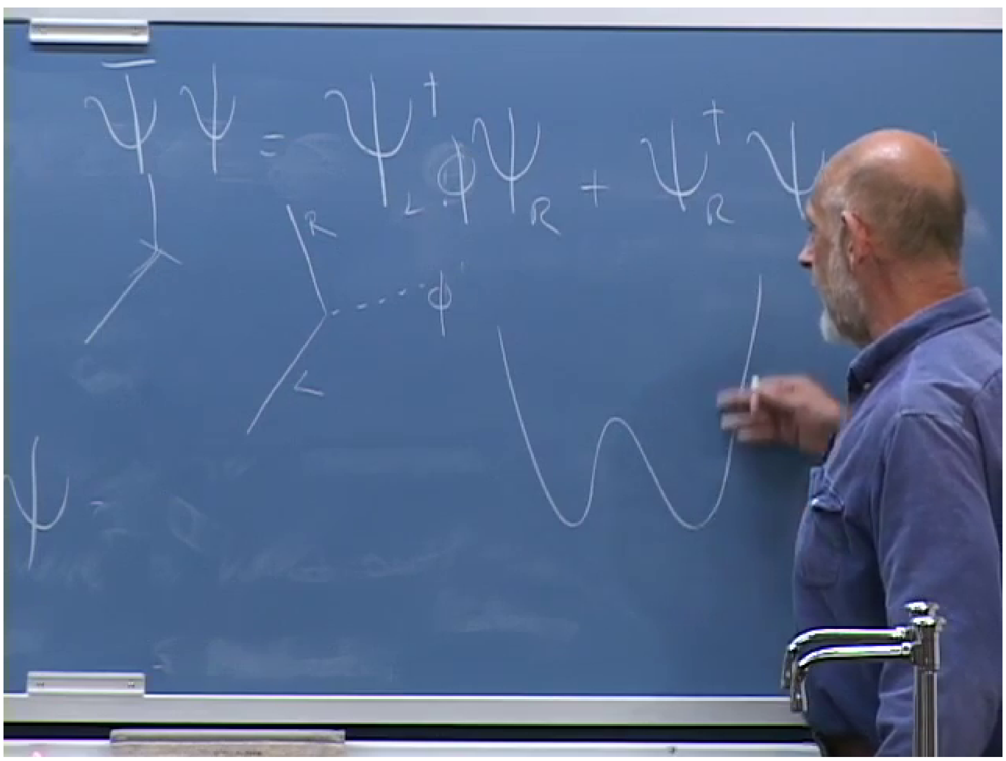

Figure 10.4: The Higgs-assisted chirality flip and the symmetry-breaking potential appear together on the board.

Lecture frame: 00:18:54.

10.5 One Interaction Becomes Both Mass and Higgs Coupling

Nature chooses a vacuum in which the neutral Higgs component does not vanish. In the lecture's schematic real-field notation,

$$\phi = F + \mathcal{H}, \tag{10.15}$$

where F is the nonfluctuating vacuum shift and $\mathcal{H}$ is the propagating Higgs fluctuation. Substitution into (10.13) gives

$$\begin{aligned}\mathcal{L}_Y = &-G_Y F\left(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L\right)\\ &-G_Y \mathcal{H}\left(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L\right).\end{aligned} \tag{10.16}$$

The first line is a Dirac mass and the second is an interaction with a physical Higgs quantum:

$$m_f = G_Y F \qquad \text{in the lecture's normalization.} \tag{10.17}$$

In the conventional Standard Model normalization,

$$\begin{aligned}H(x) &= \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v+h(x)\end{pmatrix},\\ m_f &= \frac{y_f v}{\sqrt{2}},\\ \mathcal{L}_{hff} &= -\frac{m_f}{v}h\bar{f}f,\end{aligned} \tag{10.18}$$

with $v \simeq 246\,\text{GeV}$. The lecture's rough $F \sim 200\,\text{GeV}$ is this same electroweak scale with normalization details suppressed.

The measured W and Z masses determine the scale F. The measured fermion masses then determine the Yukawa couplings. The electron's coupling is tiny; the top quark's is of order unity. If this were all the mechanism did, we would have put in one unknown number per fermion and read the same information back out.

But the second line of (10.16) is new experimental content. It says that a fermion can emit or absorb a Higgs and, read in the crossed channel, that a Higgs can decay into a fermion-antifermion pair. Schematically,

$$\begin{aligned}\mathcal{M}(h\to f\bar{f}) &\propto y_f,\\ \Gamma(h\to f\bar{f}) &= N_c\frac{y_f^2 m_h}{16\pi}\\ &\quad\times\left(1-\frac{4m_f^2}{m_h^2}\right)^{3/2},\end{aligned} \tag{10.19}$$

at tree level. Here $N_c = 3$ for quarks and 1 for leptons. A larger fermion mass means a larger Yukawa coupling, provided the decay is kinematically allowed. This does not mean every heavy channel dominates: color factors, phase space, loop-induced channels, and decays to vector bosons also matter. It means the fermionic coupling itself is proportional to mass.

Question from the audience. How would one actually predict a Higgs decay time without doing every technical detail?[a]

Response. The interaction supplies a Hamiltonian matrix element. Draw the corresponding Feynman diagram, calculate its transition amplitude, square the amplitude, and sum or integrate over all allowed final states. The result is the decay rate. The lifetime is the inverse total width, $\tau = \hbar/\Gamma_{\text{tot}}$. This is the same quantum mechanical logic used in particle calculations since the early development of quantum field theory.

[a] Lecture timestamp: 00:25:32.

At the time of the lecture, this Higgs phenomenology was still prediction rather than direct observation. The W and Z self-interactions had already provided stringent tests of the gauge structure; the Higgs and its direct couplings remained to be found. The historical uncertainty belongs in the record. So does the later outcome: a Higgs-like particle was found, and measurements of its couplings are now tests of the same mass-generating structure developed here.

10.6 The Potential That Chooses F

The vacuum shift is not chosen by hand after writing the Yukawa coupling. It is determined by the Higgs potential. For one real cross-section through the full doublet, take

$$V(\phi) = -\frac{\mu^2}{2}\phi^2 + \frac{\lambda}{4}\phi^4, \qquad \mu^2 > 0, \quad \lambda > 0. \tag{10.20}$$

Near $\phi = 0$, the negative quadratic term makes an upside-down parabola. Calling μ an imaginary mass is shorthand for saying that the origin is unstable; it does not mean that the physical Higgs excitation about the true vacuum has an imaginary mass.

The quadratic term alone would drive the field without bound and leave no lowest-energy state. The positive quartic term eventually dominates and turns the potential upward. An odd power would spoil the toy $\phi \mapsto -\phi$ symmetry. A quartic is the first even stabilizing interaction and, in four spacetime dimensions, the highest polynomial scalar interaction with a dimensionless coupling.

Differentiate:

$$\begin{aligned} \frac{dV}{d\phi} &= -\mu^2\phi + \lambda\phi^3 \\ &= \phi\left(-\mu^2 + \lambda\phi^2\right). \end{aligned} \tag{10.21}$$

The stationary point $\phi = 0$ is the unstable top. The two minima satisfy

$$\boxed{F^2 = \frac{\mu^2}{\lambda}}, \qquad F = \frac{\mu}{\sqrt{\lambda}} \tag{10.22}$$

for this real-field convention.

The factors 1/2 and 1/4 were inserted precisely so they disappear upon differentiation. Near the origin ϕ^4 is smaller than ϕ^2; at large field it grows faster and stabilizes the system. That simple competition produces the double well.

The dimensions also settle a momentary verbal slip in the transcript. λ is dimensionless in four spacetime dimensions, while μ and F have dimensions of mass. Expanding $\phi = F + \mathcal{H}$, the curvature of the potential gives

$$m_{\mathcal{H}}^2 = \left.\frac{d^2V}{d\phi^2}\right|_{\phi=F} = -\mu^2 + 3\lambda F^2 = 2\mu^2 = 2\lambda F^2. \tag{10.23}$$

In the full Standard Model notation, $V(H) = -\mu^2 H^\dagger H + \lambda(H^\dagger H)^2$, one likewise has $m_h^2 = 2\lambda v^2$. The measured Higgs mass and v imply λ near 0.13 at the electroweak scale. The lecture's 2010 estimate was intentionally rough; the structural claim was that λ is not an absurdly tiny number.

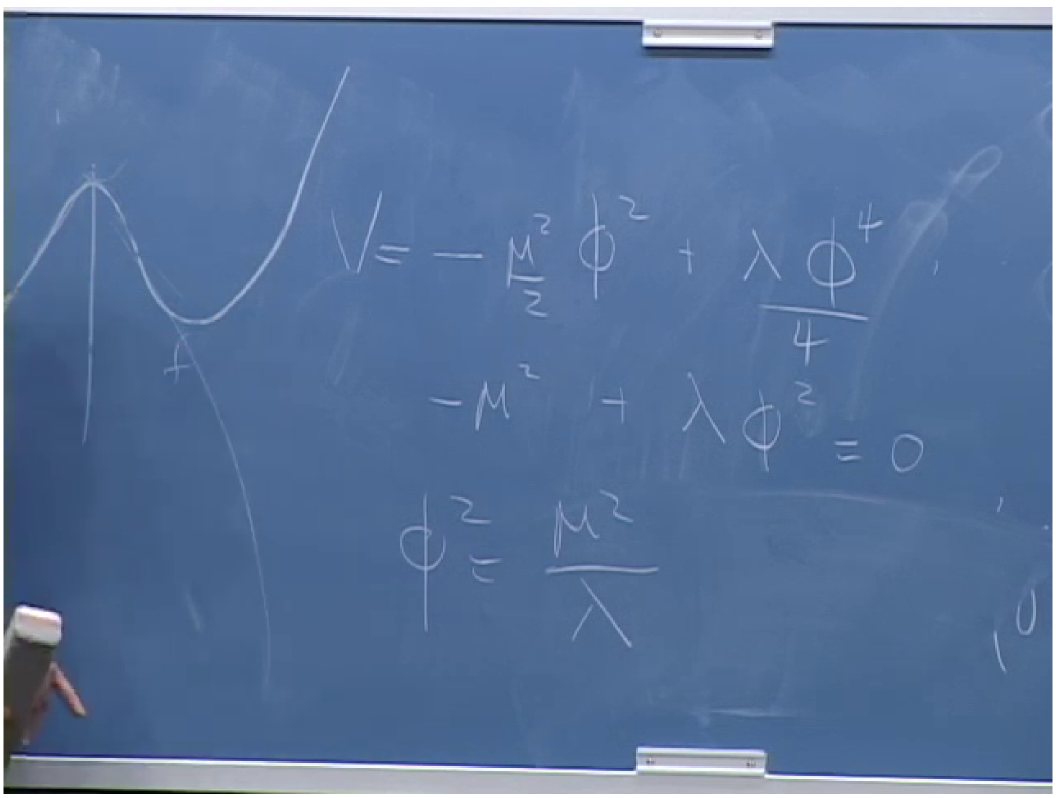

Figure 10.5: The completed minimization on the board: $V = -\mu^2\phi^2/2 + \lambda\phi^4/4$, $V' = 0$, and $\phi^2 = \mu^2/\lambda$.

Lecture frame: 00:34:32.

10.7 Why Is Everything We See Tied to This Scale?

The same F controls the W and Z masses and, multiplied by different Yukawa couplings, all charged-fermion masses. That empirical concentration raises a more interesting question than the algebra: why should all known elementary particles borrow their masses from the same symmetry-breaking phenomenon?

There is no theorem that every possible particle must. Imagine a vectorlike fermion whose left- and right-chiral components transform in the same way under the gauge group. Then $\bar{\psi}_L \psi_R + \text{h.c.}$ can be gauge invariant without a Higgs. A complete gauge singlet can likewise carry a mass unrelated to electroweak breaking. Such particles could naturally be much heavier than the particles tied to F. The particles we have discovered may be light enough to see precisely because symmetry would make them massless when F vanishes.

The electroweak scale is therefore the exceptional number. The speculative picture is that generic unprotected masses lie near a much larger fundamental scale, while particles protected by chiral gauge symmetry are dragged down with F.

Question from the audience. Is the Planck mass one candidate for the fundamental mass scale with which F should be compared?[a]

Response. Yes. Another is the scale suggested by gauge-coupling unification. Both will emerge from arguments later in the lecture, even though no particles of those enormous masses have been directly observed.

[a]Lecture timestamp: 00:38:43.

Question from the audience. What does it mean to say that the other scales are about fifteen orders of magnitude above F, and what particles correspond to them?[a]

Response. The electroweak scale is of order 10^2 GeV, while unification is suggested near 10^{16} GeV and the Planck scale lies near 10^{19} GeV. We have not produced particles at those masses. We infer the scales indirectly: one from extrapolating measured gauge couplings and the other from combining quantum mechanics, relativity, and Newton's constant.

[a]Lecture timestamp: 00:39:54.

10.8 A Charged Condensate in the Laboratory

Before the break, an audience question asks for an ordinary condensed matter analog of the Higgs mechanism. The answer develops in stages rather than by analogy alone.

Question from the audience. Beyond ultracold Bose condensates, is there a common solid-state example of a Higgs-like mechanism?[a]

Response. Yes. A neutral bosonic field can acquire a nonzero expectation value in a Bose condensate or superfluid. If the condensed object is charged, the same basic phenomenon alters the gauge field. Superconductivity supplies the familiar example: charged Cooper pairs condense, and electromagnetic fields acquire a finite penetration length inside the material.

[a]Lecture timestamp: 00:40:50.

At molecular resolution a helium atom can be treated as one effective bosonic particle with a field

Φ_{He}. Empty space corresponds to zero field; a superfluid phase has $\langle\Phi_{\text{He}}\rangle \neq 0$ and spontaneously breaks the relevant global phase symmetry. Helium atoms are neutral, so this is not yet the gauge-field mass mechanism.

In a superconductor, electrons form weakly bound Cooper pairs. The pair field is schematically an electron bilinear,

$$\Delta(x) \sim \langle\psi_\downarrow(x)\psi_\uparrow(x)\rangle, \tag{10.24}$$

and carries electric charge $2e$. When Δ condenses, the electromagnetic gauge field responds exactly as a gauge field coupled to a charged order parameter must. Static magnetic fields decay over the London penetration depth:

$$B(x) \propto e^{-x/\lambda_L}. \tag{10.25}$$

The inverse penetration length plays the role of an effective photon mass inside the medium. It is not a vacuum photon mass; it is a property of the superconducting phase.

Question from the audience. Are the two electrons in a Cooper pair not separated, even lying on opposite sides of the Fermi surface?[a]

Response. They are not a tiny molecule localized at one point. A Cooper pair is extended and is naturally described in momentum space by correlated states near opposite momenta. For the symmetry argument, however, the pair acts as a single bosonic charged order parameter. Its condensation is what matters.

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[a] Lecture timestamp: 00:43:13.

Historically this relationship is more than a teaching metaphor. Superconductivity and Anderson's analysis of gauge fields in a broken phase helped make the relativistic Higgs mechanism intelligible. The particle-physics construction was developed independently in relativistic field theory, but the shared mathematics is real.

10.9 Bare Parameters and Measured Parameters

After the break, we return to the Lagrangian. It contains parameters called masses and charges, but those input parameters are rarely identical to quantities measured without qualification. High-frequency modes interact with low-frequency modes and dress them. The systematic relation between parameters defined at one resolution and observables at another is renormalization.

Electric charge gives a picture that can be built without calculation. At large separation, Coulomb's law defines the observed charge:

$$V(r) \sim \frac{e_1 e_2}{4\pi r}, \qquad F(r) \sim \frac{e_1 e_2}{4\pi r^2}. \tag{10.26}$$

The lecture suppresses 4π and writes e^2/r or e^2/r^2. What matters is that the coefficient inferred from a force measurement can depend on the distance at which the system is probed.

10.9.1 A conductor: complete screening

Place a positive test charge in an ideal conductor. Its electric field drives mobile negative charge toward it and positive charge away. Current flows until the local source is surrounded by an opposite cloud and no field remains in the bulk outside that cloud; compensating charge is pushed to the remote boundary. A negative source produces the opposite rearrangement. Two dressed sources far apart can then exert essentially no force, even though the charges inserted microscopically were nonzero.

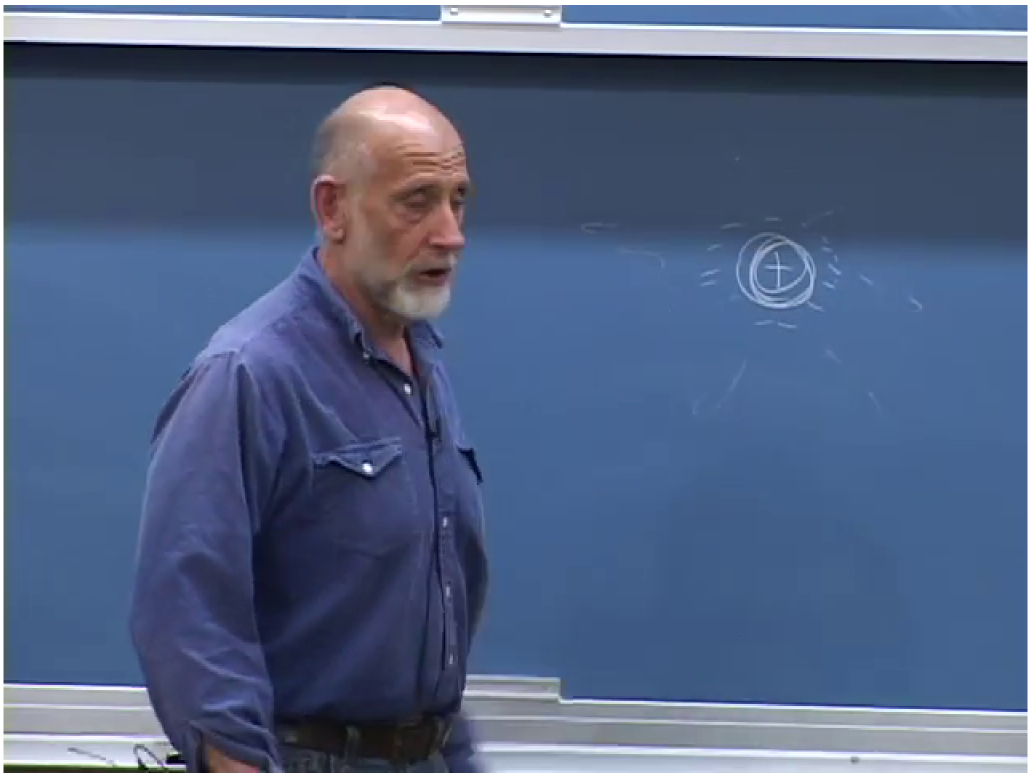

Figure 10.6: A source charge surrounded by an oppositely charged screening cloud in the conductor discussion.

Lecture frame: 00:50:24.

Question from the audience. Is this rearrangement what is meant by screening?[a]

Response. Yes. At distances larger than the cloud, the inserted charge is screened. Bring another probe inside the cloud and some of the screening can no longer lie between source and probe; the underlying charge then appears stronger.

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[a] Lecture timestamp: 00:51:25.

The cloud has a characteristic scale set by the dynamics and density of the mobile charges. At separations r smaller than that scale, the probe penetrates the cloud. One may therefore define a running charge $e(r)$ through

$$F(r) \equiv \frac{e^2(r)}{4\pi r^2}. \tag{10.27}$$

For the idealized conductor, $e(r)$ approaches zero at very long distance and the microscopic value at very short distance.

10.9.2 A dielectric: partial screening

In an insulator, electrons are bound to atoms rather than free to cross the sample. A crude but useful model pictures each electron cloud attached to its ion by a spring. Without an external field, the average positive and negative charge centers coincide. An applied field shifts them slightly apart and induces a dipole.

Around a negative source, nearby electron clouds shift away and leave their positive ions slightly closer; around a positive source they shift the other way. The bound charges cannot flow far enough to neutralize the source completely, so the long-distance field is reduced by the dielectric constant rather than eliminated:

$$V_{\mathrm{medium}}(r) = \frac{1}{\epsilon_r}\frac{e_1 e_2}{4\pi\epsilon_0 r}. \tag{10.28}$$

At distances inside the polarization cloud, the probe again sees more of the bare source. Complete and partial screening are two versions of the same scale-dependent idea.

10.10 The QED Vacuum as a Polarizable Medium

Quantum electrodynamics has no material atoms in its vacuum, but it has charged quantum fields. Electron-positron fluctuations contribute loops to photon propagation. In the older Dirac-sea language, the electric field distorts the filled negative-energy states; in modern language, vacuum polarization dresses the photon propagator. The medium picture says that a virtual pair is polarized: its positive and negative components shift in opposite directions.

A positive source is surrounded by a small excess of negative polarization charge, and a negative source by positive polarization charge. The charge measured far away is therefore smaller than the parameter one would associate with an unresolved source. Unlike an ordinary material, the vacuum contains fluctuations over a continuum of momenta. Low-energy pairs correspond to longer distance structure; high-energy pairs probe shorter distances. There is no final cloud radius at which the process simply stops.

As two electrons are brought closer together, the largest polarization clouds cease to fit between them first. Bringing them closer peels away successively smaller clouds. The effective electric coupling grows logarithmically with inverse distance, or equivalently with energy:

$$\alpha(Q) \simeq \frac{\alpha(\mu)}{1 - \dfrac{2\alpha(\mu)}{3\pi}\ln\left(\dfrac{Q}{\mu}\right)} \tag{10.29}$$

for one charged Dirac fermion well above its threshold at leading order. Additional charged species change the coefficient when their thresholds are crossed.

Electron scattering gives the operational version. Higher collision energy means shorter de Broglie wavelength and smaller closest approach. The amplitude must therefore use the coupling appropriate to that momentum transfer, not one fixed number for every scale.

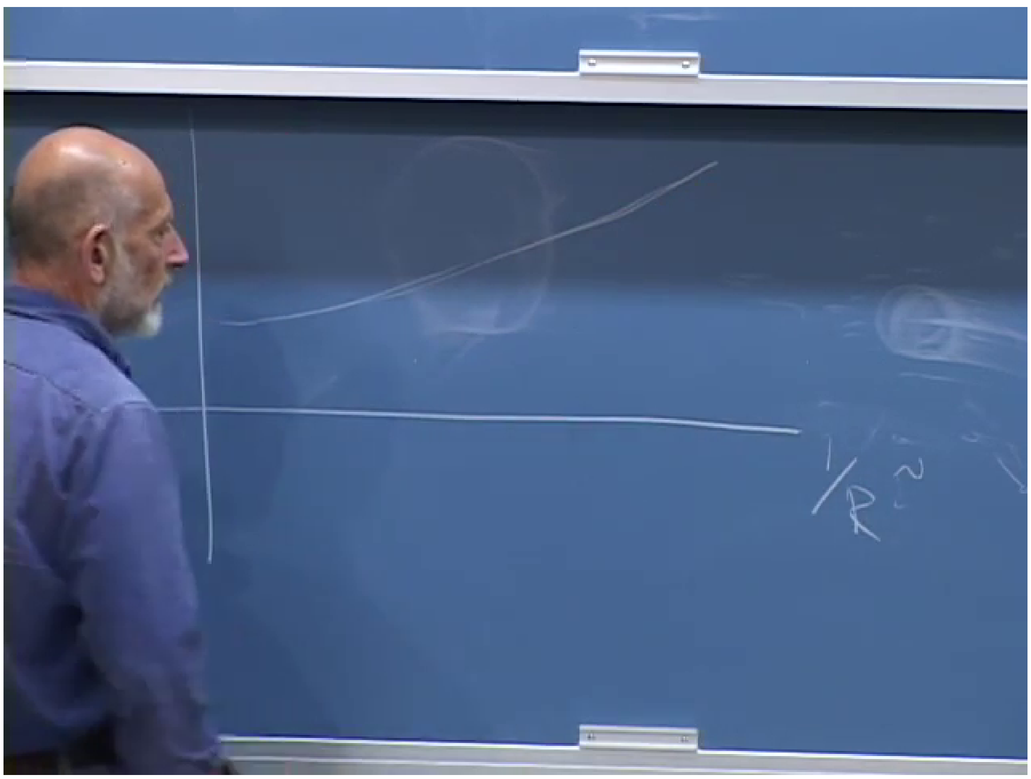

Figure 10.7: The QED effective charge rises slowly as the horizontal scale $1/R$, and therefore the probing energy, increases.

Lecture frame: 01:02:13.

Question from the audience. At what distance does the running first become measurable?[a]

Response. Electron vacuum polarization becomes relevant around the electron Compton scale, $\lambda_C = \hbar/(m_e c)$, roughly 4×10^{-11} cm for the reduced wavelength. The effect is small and logarithmic, so precision is required. Heavier charged particles join the running at their own thresholds.

[a]Lecture timestamp: 01:03:54.

Question from the audience. Does speaking of the smallest distance assume that we already know what happens there?[a]

Response. No. We extrapolate the theory only while its particle content and interactions remain valid. New physics can change the running. Experiment does not tell us that QED remains the complete description to arbitrarily short distance.

[a]Lecture timestamp: 01:04:49.

Question from the audience. For gluons, is the scale dependence the opposite? Why? [a]

Response. Yes. Quark loops screen color charge in a way analogous to charged matter, but gluons themselves carry color and interact nonlinearly. Their contribution has the opposite sign and dominates for the observed number of quark flavors. The result is antiscreening and asymptotic freedom.

[a]Lecture timestamp: 01:05:03.

Question from the audience. How could one determine a bare electric charge if doing so requires an infinitely small distance?[a]

Response. One does not measure a unique regulator-independent bare charge directly. One defines a parameter at a chosen short-distance cutoff or renormalization scale, then adjusts it so a measured low-energy quantity remains fixed. The scale-dependent renormalized coupling is the useful physical object.

[a]Lecture timestamp: 01:05:28.

Question from the audience. Does the QED charge approach a finite value at infinite energy?[a]

Response. Not in perturbative QED extrapolated by itself. Its positive beta function makes the coupling continue to grow logarithmically and leads formally to a Landau pole at a fantastically high scale. Long before treating that formal pole as physical, one expects the description to be embedded in electroweak theory and ultimately altered by further physics.

[a]Lecture timestamp: 01:05:39.

10.10.1 What renormalization changes and what it preserves

Choose a cutoff length a, call the corresponding input $e(a)$, and calculate a fixed long-distance observable. A different cutoff a' is equally acceptable if the input is changed to $e(a')$ so the same observable is reproduced:

$$\frac{d}{d\ln\mu}\,\mathcal{O}_{\text{physical}} = 0, \qquad \mu\frac{de}{d\mu} = \beta(e). \tag{10.30}$$

That compensating flow is the renormalization group.

Question from the audience. Must the word bare refer specifically to the Planck length?[a]

Response. No. One may terminate the description at any chosen cutoff and call the parameter there bare. Changing the cutoff while changing that parameter appropriately leaves accessible physics unchanged. The Planck length is a plausible place where ordinary quantum field theory must be replaced, not a required convention for renormalization.

[a] Lecture timestamp: 01:06:20.

Question from the audience. Is this procedure a matter of zooming in and attenuating the description?[a]

Response. It is zooming in conceptually. We ask how the parameters of a finer-grained description must change so that predictions at a fixed, experimentally accessible wavelength remain the same.

[a] Lecture timestamp: 01:07:15.

Question from the audience. Does the same running phenomenon happen with gravity?[a]

Response. Gravity has scale dependence too, but not simply the same dielectric-screening mechanism. For the present lecture a dimensional argument is enough: localization raises the energy, energy gravitates, and the resulting dimensionless gravitational strength grows rapidly with energy.

[a] Lecture timestamp: 01:08:14.

10.11 QCD: Antiscreening, Flux Tubes, and Asymptotic Freedom

At hadronic distances, chromoelectric field lines do not spread spherically like an electromagnetic Coulomb field. Gluon self-interactions squeeze them into a narrow tube between a quark and antiquark. If the tube has approximately fixed energy per unit length σ, then

$$V_{q\bar{q}}(r) \simeq -\frac{C_F \alpha_s(r)}{r} + \sigma r. \tag{10.31}$$

At large r, the linear term dominates. Stretching the pair creates more of the tube, like pulling a strand of taffy, and the energy grows rather than tending to zero. This is the confinement side of the story.

At distances shorter than the tube radius, the potential returns approximately to a Coulomb form, but its coupling decreases with energy. At one loop,

$$\alpha_s(Q) = \frac{4\pi}{\left(11 - \frac{2}{3}n_f\right)\ln(Q^2/\Lambda^2_{\mathrm{QCD}})}. \tag{10.32}$$

For $n_f < 17$, the coefficient is positive and $\alpha_s(Q) \to 0$ as $Q \to \infty$. QCD is weak at very short distance and strong at long distance. This is the precise running-coupling version of the lecture's opposite behavior.

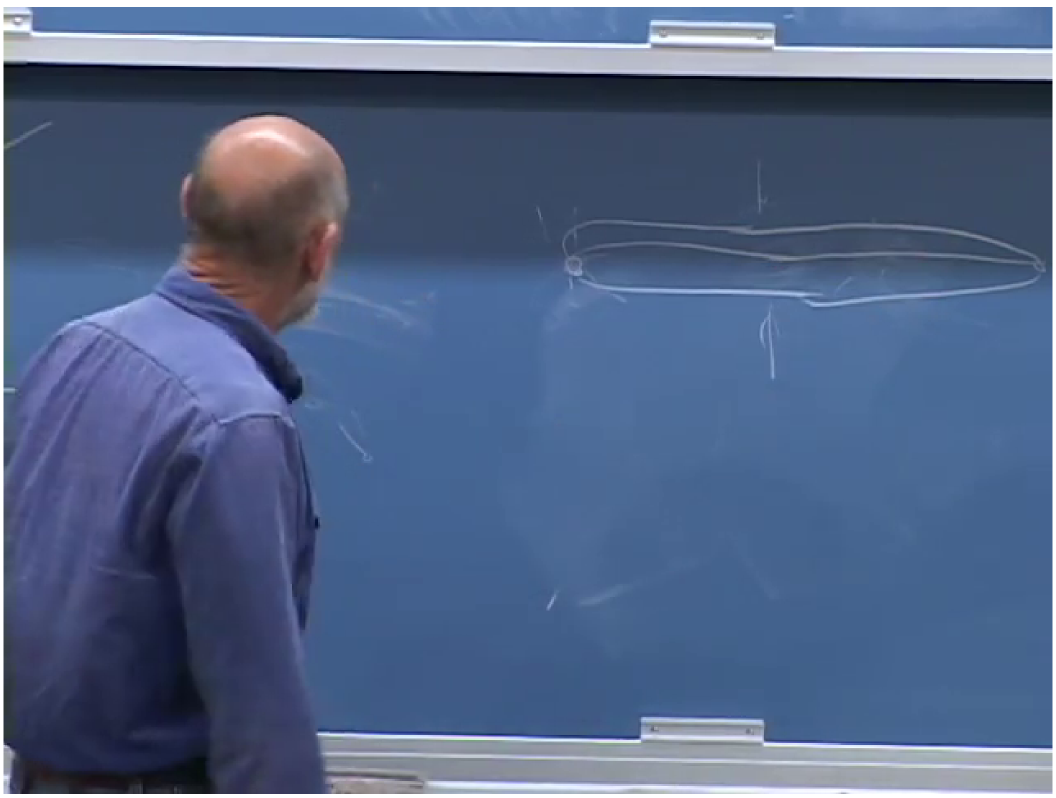

Figure 10.8: Chromoelectric field lines are drawn as a tube joining separated color sources rather than as a freely spreading Coulomb field.

Lecture frame: 01:09:56.

Question from the audience. What happens to the weak coupling between the QED and QCD curves?[a]

Response. Its running is intermediate and depends on which electroweak factor is being followed. Above the electroweak scale the proper quantities are the $U(1)_Y$ and $SU(2)_L$ couplings. One rises and the other falls slowly. Together with the falling strong coupling, their extrapolated values come surprisingly close at high energy.

[a]Lecture timestamp: 01:11:53.

10.12 Running Toward Unification

Measure the three gauge couplings near the weak scale and evolve them with the particle content of a proposed theory. Because every beta function is logarithmic, an enormous change in energy produces only a moderate change in coupling. The Standard Model trajectories approach one another but do not meet exactly. With the low-energy particle content of the minimal supersymmetric Standard Model, the one-loop trajectories meet much more closely near

$$M_{\mathrm{GUT}} \sim 10^{16}\,\mathrm{GeV}. \tag{10.33}$$

The plot is evidence, not proof. Thresholds from undiscovered particles, higher-loop corrections, and details of a unified theory shift the curves. The 2010 lecture emphasizes the approximately percent-level convergence in a supersymmetric spectrum because it made weak-scale supersymmetry especially attractive. The absence of superpartners at collider energies now means that the simplest version of that spectrum is not realized as originally expected; precision unification remains model dependent.

Why regard the vicinity of a meeting point as fundamental? It is where the input may become simplest. Three unrelated low-energy couplings could be different dressed descendants of one high-energy coupling associated with a single gauge group. This is the numerical clue behind the earlier $SU(5)$ question.

Gravity appears on the same logarithmic sketch as a curve that rises much faster. It does not cross the gauge curves at exactly the GUT scale, but it becomes order one within a few decades of energy. On a logarithmic axis, that proximity is suggestive: 10^{16}, 10^{17}, 10^{18}, and 10^{19} GeV are vastly closer to one another than any is to the electroweak scale.

10.13 Why Gravity Strengthens at Short Distance

10.13.1 Energy, not rest mass alone, gravitates

Newton's force between two slowly moving masses is

$$F_G = \frac{G m_1 m_2}{r^2}. \tag{10.34}$$

For a relativistic dimensional estimate, replace each rest mass by the corresponding energy divided by c^2:

$$F_G(r) \sim \frac{G E_1 E_2}{c^4 r^2}. \tag{10.35}$$

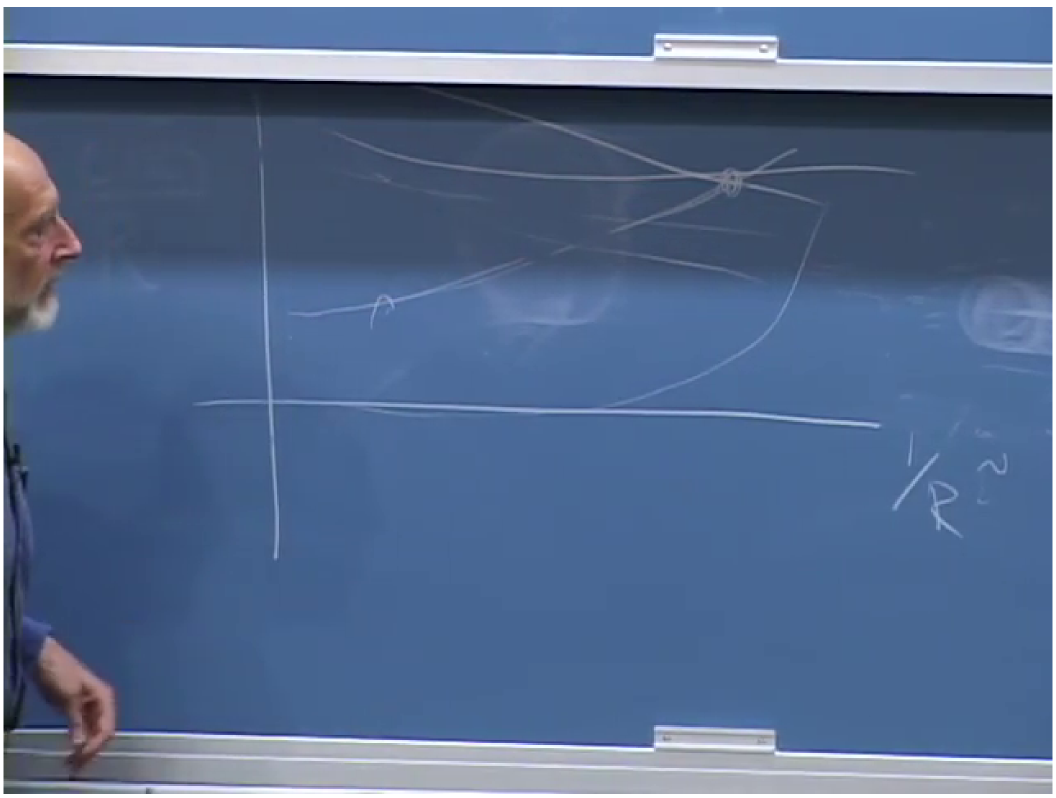

Figure 10.9: The board's qualitative coupling plot: the strong curve falls, the electroweak curves evolve slowly toward it, and gravity rises steeply near the far ultraviolet.

Lecture frame: 01:16:29.

This is not a substitute for general relativity; stress, momentum flow, and pressure also source gravity. It is the correct scaling estimate for the question at hand.

Localize each particle to a region of size r. The uncertainty principle requires

$$p \sim \frac{\hbar}{r}. \tag{10.36}$$

At sufficiently small r, the momentum is ultrarelativistic, so

$$E \simeq pc \sim \frac{\hbar c}{r}. \tag{10.37}$$

Substituting two such energies into (10.35) gives

$$F_G(r) \sim \frac{G\hbar^2}{c^2 r^4}. \tag{10.38}$$

The corresponding potential-energy scale is

$$U_G(r) \sim -\frac{G\hbar^2}{c^2 r^3}. \tag{10.39}$$

This is the distinction corrected during the live calculation: the potential scales as $1/r^3$, while the force scales as $1/r^4$.

Question from the audience. For an ultrarelativistic particle, is the energy $p^2/(2m)$?[a]

Response. No. That is the nonrelativistic formula. From $E^2 = p^2c^2 + m^2c^4$, the limit $pc \gg mc^2$ gives $E \simeq pc$. Combined with $p \sim \hbar/r$, the localization energy is $\hbar c/r$.

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[a]Lecture timestamp: 01:22:18.

Question from the audience. Did the factors of c cancel, and was the claimed $1/r^3$ law a force law?[a]

Response. The board calculation is corrected in real time. The two effective masses contribute E_1E_2/c^4; with $E_i \sim \hbar c/r$, two powers of c remain in the denominator. The resulting force is $G\hbar^2/(c^2r^4)$. The $1/r^3$ expression belongs to the potential energy.

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[a]Lecture timestamp: 01:23:38.

Question from the audience. The gravitational force under discussion is attractive, correct?[a]

Response. Yes. The equations above display magnitudes. The Newtonian gravitational potential carries a minus sign for positive energies, so the force is attractive in this estimate.

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[a]Lecture timestamp: 01:24:42.

Question from the audience. Is gravity gaining relative strength because localization raises energy, while electric charge itself does not gain the same kinematic factor?[a]

Response. Exactly. Energy is the gravitational charge. Squeezing a quantum state raises its momentum and energy, so it also raises what sources gravity. Electric charge is dimensionless and does not acquire that direct $1/r$ factor, apart from its much slower logarithmic renormalization.

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[a]Lecture timestamp: 01:25:02.

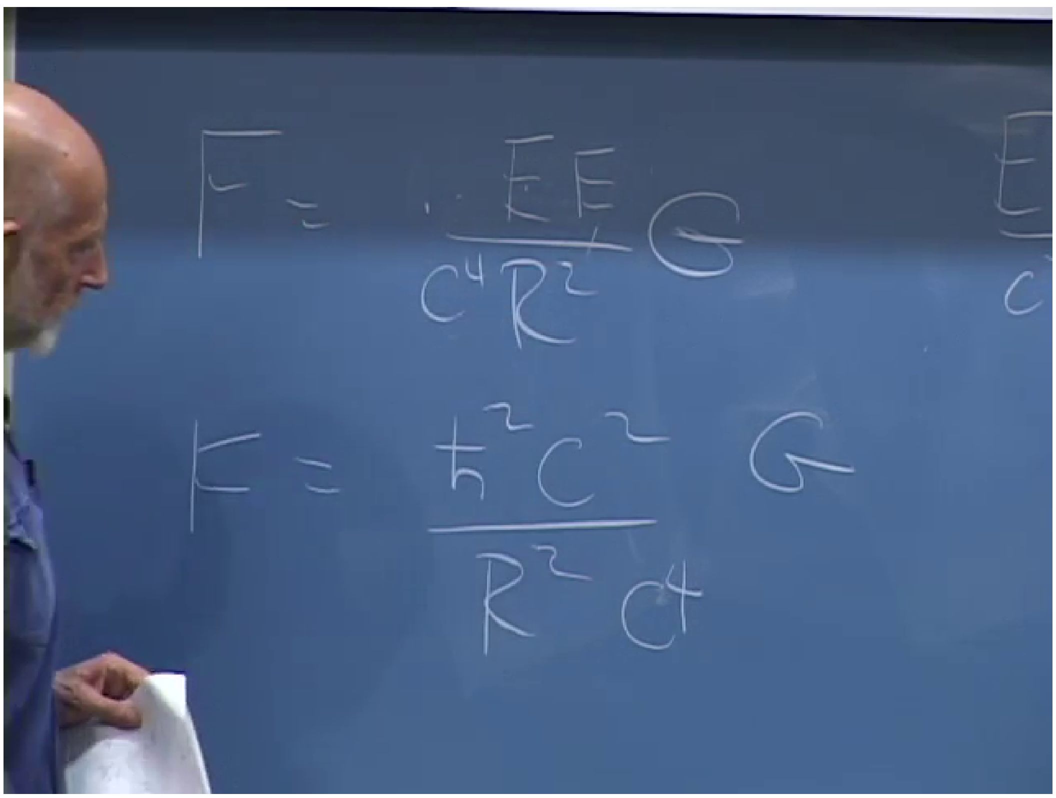

Figure 10.10: The board substitutes relativistic energies into Newton's law while checking the powers of c and r.

Lecture frame: 01:23:29.

10.13.2 The Planck crossing

The electromagnetic force between charges of order one in natural units has magnitude

$$F_{\mathrm{EM}}(r) \sim \alpha \frac{\hbar c}{r^2}. \tag{10.40}$$

Equating it to (10.38) gives

$$r^2 \sim \frac{G\hbar}{\alpha c^3}. \tag{10.41}$$

The lecture sets the dimensionless gauge factor to order one to expose the fundamental gravitational length:

$$\boxed{\ell_P = \sqrt{\frac{\hbar G}{c^3}}} \simeq 1.616 \times 10^{-35}\,\mathrm{m}. \tag{10.42}$$

Keeping the actual α moves the equality with electromagnetism by an order-one logarithmic-scale factor, $r \sim \ell_P/\sqrt{\alpha}$. The fundamental statement is cleaner in natural units:

$$\begin{aligned} \alpha_G(E) &\equiv \frac{GE^2}{\hbar c^5} = \left(\frac{E}{E_P}\right)^2, \\ E_P &= \sqrt{\frac{\hbar c^5}{G}} \simeq 1.22 \times 10^{19}\,\mathrm{GeV}. \end{aligned} \tag{10.43}$$

Gravity becomes strong when E approaches E_P.

Question from the audience. Was G or the electric coupling being set equal to one in the comparison?[a]

Response. The electric coupling was being treated as an order-one number for the rough estimate. Newton's constant must remain: it is the dimensional quantity that determines the Planck scale.

[a]Lecture timestamp: 01:27:02.

Question from the audience. The intermediate powers of r, $\hbar$, and c do not appear dimensionally consistent. What is the corrected calculation?[a]

Response. Use energies from the start. Localization gives $E \sim \hbar c/r$, hence $U_G \sim GE^2/(c^4 r) = G\hbar^2/(c^2 r^3)$. Compare this with $U_{\mathrm{EM}} \sim \alpha\hbar c/r$. Setting α to order one yields $r^2 = G\hbar/c^3$. The class and board arrive at $r = \sqrt{G\hbar/c^3}$, the Planck length.

[a]Lecture timestamp: 01:27:47.

10.14 The Hierarchy Problem Returns

The gauge couplings become comparable near a very high scale, and gravity becomes strong near the Planck scale. These enormous scales now look less like arbitrary large numbers and more like places where a microscopic description might simplify. Against them, the electroweak vacuum value F is anomalously small:

$$\frac{v}{E_P} \sim 2 \times 10^{-17} \tag{10.44}$$

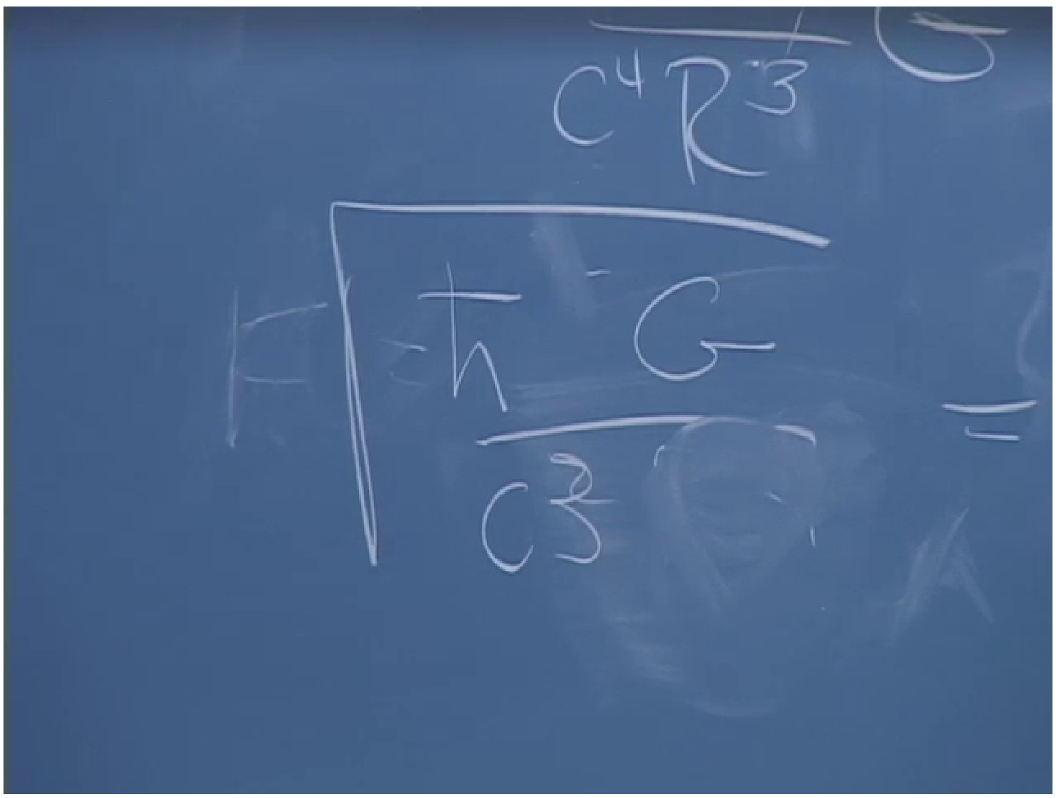

Figure 10.11: The corrected endpoint of the live dimensional check: $r = \sqrt{\hbar G/c^3}$.

Lecture frame: 01:32:16.

in units with $c = 1$. Using a GUT scale instead of the Planck scale changes the precise ratio but not the puzzle.

Once F is small, the masses of the W, Z, charged leptons, and quarks follow because gauge symmetry makes their masses proportional to it. Particles without that protection can remain heavy. The outstanding question is therefore not why every known mass is independently tiny. It is why the Higgs sector supplies this one tiny symmetry-breaking scale.

Question from the audience. If one could calculate F, would all the other masses then be calculable?[a]

Response. It would determine the overall electroweak scale and, with the gauge and Yukawa couplings supplied, the corresponding particle masses. The remaining problem is that an ordinary ultraviolet-sensitive calculation tends to drive the Higgs mass parameter toward the high fundamental scale unless a symmetry or dynamical mechanism protects it.

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[a]Lecture timestamp: 01:34:39.

The word fine-tuned anticipates that sensitivity. In a cutoff description, a scalar mass parameter receives corrections of the rough form

$$\delta\mu^2 \sim \frac{C}{16\pi^2}\Lambda^2, \tag{10.45}$$

where C combines gauge, Yukawa, and scalar couplings and Λ marks the scale at which the effective theory is replaced. If Λ is near a GUT or Planck scale, obtaining $\mu^2 \sim v^2$ appears to require a very precise cancellation between the input and radiative contributions. This is the naturalness form of the gauge-hierarchy problem. It is an editorial completion of the calculation the lecture promises for the following quarter, not a claim that a particular regulator artifact is itself observable.

Question from the audience. Should the Planck length be visualized as the final screening-cloud radius?[a]

Response. No. The dielectric picture explains gauge-charge renormalization, but the Planck argument used localization energy and the dimensional gravitational coupling. There is no literal gravitational screening cloud in this derivation.

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[a]Lecture timestamp: 01:35:03.

10.14.1 The closing questions about interactions

The last minutes return to the Higgs potential and to what terms a Lagrangian may contain.

Question from the audience. Could the Higgs potential include ϕ^6 and still remain symmetric?[a]

Response. Yes. Even powers preserve the toy $\phi \mapsto -\phi$ symmetry, whereas ϕ^5 would break it. In four dimensions the coefficient of ϕ^6 has negative mass dimension, so the term is suppressed by a high scale, for example $c_6\phi^6/\Lambda^2$. It is a valid effective interaction even though it is not power-counting renormalizable.

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[a]Lecture timestamp: 01:35:20.

Question from the audience. Do the powers of the Higgs field represent creation and annihilation processes, so that ϕ^4 predicts Higgs-Higgs scattering?[a]

Response. Yes. After expanding about the vacuum, powers of the quantized field generate vertices. The quartic term contains a four-Higgs vertex, and the shifted potential also produces cubic interactions. Knowing their coefficients predicts processes such as $hh \to hh$, subject to the usual diagrammatic calculation.

[a] Lecture timestamp: 01:35:52.

Question from the audience. Must a fundamental Lagrangian be polynomial in its fields, with no power above four?[a]

Response. For a perturbatively renormalizable four-dimensional scalar theory, operators of dimension at most four form the traditional finite list. Modern effective field theory is less restrictive: higher dimension operators are allowed and encode short-distance physics, but their effects are suppressed by powers of the high scale. Thus the occasional exception is essential to the old rule.

[a] Lecture timestamp: 01:36:09.

Question from the audience. Were self-interaction terms omitted from the discussion of running couplings?[a]

Response. They were included conceptually. Gluon self-interactions are exactly what reverse the sign relative to QED and produce antiscreening. Higgs self-interactions likewise contribute to the running of scalar and other Standard Model parameters.

[a] Lecture timestamp: 01:36:44.

Question from the audience. Are there cross-terms in which different interactions affect one another's running?[a]

Response. Yes. At higher loop order, beta functions contain mixed products of gauge, Yukawa, and scalar couplings. Perturbation theory remains useful while those dimensionless couplings are sufficiently small, because successive loop orders are correspondingly suppressed.

[a] Lecture timestamp: 01:37:23.

Question from the audience. What exactly was meant by saying that F is finely tuned?[a]

Response. The complete answer is deferred in the lecture. The issue is the instability of a light scalar scale against much larger ultraviolet contributions, summarized by (10.45). Supersymmetry was one proposed way to enforce cancellations, but the hierarchy problem itself does not select a confirmed solution.

[a] Lecture timestamp: 01:37:44.

Question from the audience. Can the renormalization group itself be explained now? [a]

Response. Not in the remaining minutes. Its operative content has already appeared: change the resolution, let the couplings flow, and preserve fixed physical predictions. A full treatment belongs to the next stage of the course.

[a] Lecture timestamp: 01:37:56.

That is where the sequence properly leaves us. The Higgs mechanism has made the Standard Model internally coherent: it preserves the chiral gauge symmetry, gives masses to weak bosons and fermions, and predicts physical Higgs interactions. Renormalization then teaches us how those interactions change with scale and hints that gauge forces may share a unified origin. Gravity supplies a second high scale by an independent argument. None of those successes explains why the Higgs vacuum value is so far below both. The final lecture therefore does not close the subject; it identifies the small number on which the next subject must concentrate.